

ThrustOne Theory Manual

Physics, Models and Interpretation

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ThrustOne Theory Manual

Purpose and scope

This book explains the physical models behind ThrustOne: the balances they solve, the assumptions that close those balances, and the meaning of their outputs. It is intended for a reader who wants to understand a calculation rather than obtain a number without its context. Familiarity with elementary thermodynamics, mechanics and calculus is helpful; the opening chapters establish the conventions needed to connect those subjects to propulsion calculations.

The first edition addresses 49 explicitly bounded calculation modes across nine engine families. Some modes describe complete idealized cycles. Others describe a component, a geometry construction, a chemical state or a segment of a flowpath. A family name does not imply that all of its physical processes are implemented. In particular, a combined-cycle submodel is not a vehicle-level transition analysis, and a nozzle reference contour is not a solved wall-pressure field.

The user guide explains where to enter values, choose a branch and export a result. This theory manual explains why those inputs are needed and what can be concluded from the result. The two documents serve different purposes. Field inventories and operating instructions do not replace the derivations here; theoretical background does not create an additional software capability.

How the book is organized

Chapters 1–4 establish control boundaries, static and total states, normalization and freestream conditions. These conventions recur throughout the book. Chapters 5–8 explain mixture properties, equilibrium and finite-rate chemistry. They distinguish a frozen composition from constant heat capacity, and an equilibrium state from a time-dependent chemical process.

Chapters 9–15 develop turbojet and turbofan cycles, components, mixing and reference-based off-design calculations. Chapters 16–22 cover rocket performance, chemical expansion, reference geometry, parameter studies and established-burning solid-motor models. Chapters 23–29 connect shocks and duct balances to ramjet and scramjet flowpaths, including variable properties and finite-rate species transport.

Chapters 30–32 explain the bounded combined-cycle submodels. Their differing connections matter: explicit independent branch allocation, momentum-based entrainment and turbine-to-fan shaft transfer are distinct closures. Chapters 33–34 explain numerical results and defensible parameter comparisons. The appendices provide notation, normalization, domain summaries and a mode-to-chapter directory.

Reading paths

For a first propulsion course, begin with Chapters 1–4, then select the relevant engine family. Read Chapter 33 before interpreting a failed solve or a state close to a sonic boundary. A reader who understands mass and energy conservation but is new to propulsion will usually benefit most from following one complete station chain before exploring many model options.

For turbine cycles, follow Chapters 9–10 and 13–14 before studying component maps and off-design closure in Chapters 11–12 and 15. A design-point pressure ratio and a pressure ratio solved from a reference engine are not interchangeable inputs. Mixed-flow calculations also require a more careful mass-flow basis than a single-stream cycle.

For rocket calculations, read Chapters 16–17 before sizing in Chapter 18 or contour construction in Chapter 19. Thermochemical state, performance scale and reference geometry answer different questions. Read Chapter 20 when comparing altitude, pressure, expansion ratio or mixture-ratio results. Solid-propellant readers should then use Chapters 21–22, which explicitly exclude ignition and tail-off modeling.

For reacting flows, read Chapters 5–8, then the reconstruction in Chapter 26 and the flowpath formulation in Chapter 29. This order is important: a species enthalpy reference affects energy closure, and a chemically changing mixture cannot be interpreted using one fixed gas constant. Chapters 23–24 supply the shock, friction, heat and mass-source background.

For combined cycles, use Chapters 30–32 together with the underlying turbine, ramjet and mixing chapters. Compare the control diagrams before comparing performance. An identical word such as “bypass” can denote a prescribed allocation in one model and a mass ratio determined by a balance in another.

Four questions to carry into every calculation

The first question is what crosses the control boundary. Captured air, on-board primary material, fuel, oxidizer, heat and shaft work enter different balances. A rocket and an airbreathing engine do not have the same inlet momentum debit.

The second is which state is specified. Static pressure acts on a surface; total pressure describes an isentropic reference state. Chamber static temperature and stagnation temperature coincide only under an appropriate velocity approximation. An apparently small label difference can change the calculation.

The third is what closes the problem. A prescribed exit pressure, a fixed area, a selected Mach branch, an energy constraint and a fitted inlet relation are different closures. One must not silently substitute one for another while comparing results.

The fourth is what the output is divided by. Specific thrust may use core

air, total captured air or another stream. Specific impulse may use all injected mass rather than fuel alone. A ratio without its denominator is an incomplete physical statement.

Common conventions

Equations use SI units unless a local example explicitly identifies another unit. Temperatures in thermodynamic relations are absolute. Pressures are absolute unless stated otherwise; a gauge pressure is not a valid direct replacement in a pressure ratio. Internal energy and enthalpy may be negative under a formation-inclusive reference convention. Their signs alone do not determine physical admissibility.

A subscript t usually denotes a total state, a subscript 0 a freestream reference, and a superscript star a critical or sonic reference defined locally. Neither station numbering nor star notation is globally interchangeable across every model. The chapter's definition controls its meaning.

The positive flow direction is downstream through the engine. Thrust is normally reported as the useful reaction-force magnitude opposite that flow direction. When a model permits negative net force, the sign indicates an unfavorable result under its accounting. Other modes explicitly reject nonpositive performance; the applicable chapter states the distinction.

Common symbols are listed in Appendix A. Some established symbols have more than one use: F may be thrust or a momentum function; s may be entropy or specific thrust; β may denote a shock angle or a flow ratio. Local definitions and units resolve those collisions. The notation reference is a guide to reading, not permission to substitute symbols across equations without checking their definitions.

Worked examples and evidence

Examples use original inputs and explanations. A simple accounting example may isolate mass flow, energy or force without constructing a complete engine. Such an example is labeled accordingly. It demonstrates the

balance being discussed and must not be mistaken for an independently confirmed App operating point.

Where a numerical value has a documented local reference, the internal editorial record preserves its actual location and qualifications. Independent source evidence, direct illustrative arithmetic and application-generated workflow results have different roles. The book does not turn a value calculated by the application into proof of that same application's correctness.

Derivations use mathematical relations and original exposition. Diagrams establish control boundaries, station conventions or geometric constructions. They are not reproductions of source illustrations and do not imply that the application solves every feature shown as background context.

Model hierarchy and limits

An ideal constant-property calculation can reveal a useful relation with very few inputs. A variable-property calculation changes how the state is recovered. An equilibrium calculation changes the composition constraint. A finite-rate calculation adds chemical evolution. None of these changes automatically adds spatial transport, wall cooling, shocks, multi-phase physics or a complete hardware design.

Greater mathematical detail is useful when it answers the intended question and its extra inputs are meaningful. It is not a universal ordering from an inferior model to a superior one. A complicated mechanism with unsuitable coefficients can provide a less defensible conclusion than a simple model used within its stated scope.

An interrupted flow history is presented as a valid prefix, not a completed nozzle or engine. A sampled maximum is identified as a sampled result, not a continuous optimum. A geometrically plausible contour is distinguished from an optimized flow solution. These distinctions are part of the result, rather than qualifications to remove when presenting a favorable number.

The equations in this book support interpretation of the current calcula-

tion modes. They do not constitute component qualification, operating approval, structural certification or a guarantee of a real engine's performance. The productive use of the models is to formulate a clear question, preserve its assumptions and report the answer with the same boundaries.

1. Control volumes and engine performance

1.1 Begin with the physical boundary

An engine performance calculation begins by choosing a boundary, not by choosing a thrust equation. The boundary identifies which mass streams enter and leave, where pressures act, and which mechanical or thermal interactions are included. Two calculations may use individually correct equations and still disagree because they describe different boundaries.

For an airbreathing engine, a useful stationary control volume in the engine frame begins upstream of the inlet and ends at the exhaust plane. Captured air enters at the flight speed relative to the engine. Fuel enters through a separate supply. A turbofan has either separate core and bypass exit planes or one common mixed-flow exit. A rocket has no captured atmospheric-air stream: its propellant enters from onboard supplies. A standalone nozzle has an upstream state supplied by the user and does not, by itself, define the thrust of an entire vehicle.

Draw these boundaries before reducing the equations. Mark each inlet and exit area, the outward surface normal, static pressure, mass flow and axial velocity. Static pressure belongs in the surface-force balance. Total pressure is a useful state variable for losses and expansion, but it is not the pressure exerted on a plane in the momentum equation.

Throughout this chapter, the positive exhaust direction is downstream rel-

ative to the engine. We report the positive magnitude of the useful reaction force on the engine in the opposite direction. A signed force-vector formulation is equally valid, provided every normal and velocity uses the same convention.

1.2 Mass conservation and the choice of reference flow

The integral mass balance is

$$\frac{d}{dt} \int_{CV} \rho dV + \int_{CS} \rho \mathbf{V} \cdot \mathbf{n} dA = 0.$$

For steady, one-dimensional streams this reduces to

$$\sum_{\text{out}} \dot{m}_j = \sum_{\text{in}} \dot{m}_i.$$

For a simple turbojet without bleed or auxiliary discharge,

$$\dot{m}_e = \dot{m}_a + \dot{m}_f = (1 + f)\dot{m}_a, \quad f = \frac{\dot{m}_f}{\dot{m}_a}.$$

The factor $1 + f$ is a mass balance, not an efficiency correction. Neglecting it is an explicit ideal-cycle approximation. It must not disappear merely because fuel flow is small. Its effect also depends on exhaust speed, so a percentage error in mass flow need not produce the same percentage error in net thrust after inlet momentum is subtracted.

For a separate-flow turbofan, define core air flow $\dot{m}_c$, bypass air flow $\dot{m}_b$, and bypass ratio $\alpha = \dot{m}_b/\dot{m}_c$. Total captured air is

$$\dot{m}_a = (1 + \alpha)\dot{m}_c.$$

Fuel-air ratio normally refers to core air, $f = \dot{m}_f/\dot{m}_c$. Therefore fuel per total captured air is $f/(1 + \alpha)$. A comparison that divides total-air-based

specific thrust into core-air-based fuel ratio overstates fuel consumption by a factor of $1 + \alpha$.

For a rocket, total propellant flow includes oxidizer and fuel. With mixture ratio $MR = \dot{m}_{ox}/\dot{m}_f$, the fuel flow is $\dot{m}_p/(1 + MR)$. Rocket specific impulse uses total propellant consumption, not fuel consumption alone. This distinction becomes especially important when comparing a rocket with an airbreathing engine whose oxidizer comes from the atmosphere.

1.3 Momentum flux is not thrust

The axial momentum balance for a steady control volume equates net external force on the fluid to the change in axial momentum flux. For uniform streams, each momentum-flux contribution is $\dot{m}V$. Pressure forces also act on each control surface. To compare an exhaust plane with its surrounding atmosphere, the relevant excess-pressure contribution is $(p_e - p_a)A_e$.

For one captured air stream and one exhaust, with negligible axial momentum supplied through the fuel line and with installation drag excluded, the customary uninstalled net thrust is

$$F_N = \dot{m}_e V_e - \dot{m}_a V_0 + (p_e - p_a)A_e.$$

The assumptions matter. If an injected stream has appreciable axial momentum at the chosen boundary, its inlet momentum belongs in the balance. If an engine model includes multiple exits, the exhaust terms must be summed. If a propulsion installation includes spillage, external nacelle forces or other drag, an uninstalled cycle calculation does not automatically account for them.

Gross thrust is the exhaust contribution,

$$F_G = \dot{m}_e V_e + (p_e - p_a)A_e,$$

and inlet momentum drag for the simple captured-air boundary is $\dot{m}_a V_0$. Thus $F_N = F_G - \dot{m}_a V_0$. Calling gross thrust “net thrust” silently removes the cost of accelerating captured air from its incoming momentum state.

A different quantity appears in one-dimensional duct and mixer calculations:

$$J = pA + \dot{m}V.$$

This is a stream momentum function. It combines absolute-pressure force and momentum flux at one station. It is useful for connecting stations across a selected duct model. It is not, in general, an engine thrust. In particular,

$$F_G = J_e - p_a A_e,$$

and a net airbreathing calculation must still account for its inlet boundary. ThrustOne’s F/G state-reconstruction input uses a momentum function for F; it must not be populated with a whole-engine thrust simply because both quantities have units of newtons.

1.4 Pressure mismatch and the expansion condition

When $p_e = p_a$, the pressure-thrust term vanishes. This is the fully expanded condition, not a general property of nozzles. A design calculation may prescribe this condition and recover the exit area. A fixed-area nozzle in a changing atmosphere generally cannot remain fully expanded at every altitude.

For an underexpanded exit, $p_e > p_a$, the excess-pressure term is positive. For an attached overexpanded calculation, $p_e < p_a$, it is negative. That algebra alone does not establish whether an actual overexpanded nozzle remains attached. Boundary-layer separation, shock structure and asymmetric loads require additional physical information. A reference contour and an ideal exit state do not supply that information.

In the simplest fixed-internal-state altitude comparison,

$$\frac{\partial F_G}{\partial p_a} = -A_e.$$

This is a partial derivative holding the nozzle exit state and area fixed. It is not a universal engine-altitude law. It ceases to describe the full variation when the internal state, mass flow or flow topology responds to ambient pressure. When interpreting an App sweep, identify what the selected mode actually keeps fixed before assigning a trend to engine physics.

1.5 Worked example: why the reference flow matters

Consider an illustrative airbreathing stream with the following quantities. These are invented round-number inputs for understanding the accounting, not specifications of a real engine.

Quantity	Value
Captured air flow	80 kg/s
Fuel flow	2 kg/s
Flight speed	250 m/s
Exhaust speed	750 m/s
Exit static pressure	100 kPa
Ambient static pressure	95 kPa
Exit area	0.30 m ²

Mass conservation gives an exhaust flow of 82 kg/s. The three thrust terms are

$$\dot{m}_e V_e = 82(750) = 61,500 \text{ N},$$

$$\dot{m}_a V_0 = 80(250) = 20,000 \text{ N},$$

$$(p_e - p_a)A_e = (100,000 - 95,000)(0.30) = 1,500 \text{ N}.$$

Gross thrust is therefore 63,000 N and net thrust is 43,000 N. Specific thrust based on captured air is

$$F_s = \frac{43,000}{80} = 537.5 \text{ N s/kg}.$$

Dividing by the exhaust mass flow instead gives approximately 524.39 N s/kg. That value is not an arithmetic error; it describes a different normalization. A result label that merely says “specific thrust” without identifying its reference mass flow is incomplete.

The fuel-air ratio is $2/80 = 0.025$, so thrust-specific fuel consumption can be calculated either directly or through consistently normalized quantities:

$$TSFC = \frac{2}{43,000} = \frac{0.025}{537.5} \approx 4.65116 \times 10^{-5} \text{ kg/(N s)}.$$

If fuel mass addition were incorrectly omitted while all other supplied values were retained, the momentum term would fall by 1,500 N, giving 41,500 N net thrust. In this example that error happens to equal the pressure-thrust contribution. Omitting either one alone would happen to give the same 41,500 N result; omitting both would give 40,000 N. Agreement between two numbers can therefore conceal different accounting errors. Comparisons should inspect the terms, not only the final force.

As a second thought experiment, hold the exit state and geometry fixed but raise ambient pressure to 100 kPa. Pressure thrust becomes zero and net thrust becomes 41,500 N. This illustrates a controlled change in one boundary condition; it does not claim that a physical nozzle would retain all its internal states under an arbitrary ambient-pressure change.

1.6 Consumption, impulse and power describe different questions

Specific thrust measures force obtained per unit reference mass flow. Thrust-specific fuel consumption measures fuel flow required per unit force. Specific impulse measures thrust relative to a specified consumable mass flow and standard gravity:

$$I_{sp} = \frac{F}{\dot{m}_{\text{consumable}} g_0}, \quad g_0 = 9.80665 \text{ m/s}^2.$$

The consumable is normally total propellant for a rocket and fuel for a fuel-based airbreathing impulse measure. For a mixed injected stream, the appropriate denominator depends on the model's stated convention. A number in seconds is not comparable across engines until that convention is known.

Useful propulsive power is $F_N V_0$ for the uninstalled force convention used above. At zero flight speed it is zero even when static thrust is substantial. Thus zero propulsive power at a static condition is not evidence that a thrust calculation failed. Efficiency describes a rate of useful energy delivery; force describes momentum exchange. They answer different engineering questions.

Overall efficiency based on fuel chemical power requires a consistent heating-value convention and a compatible model boundary. Relations that split overall efficiency into thermal and propulsive factors often assume fully expanded streams. They should not be applied automatically to a pressure-mismatched nozzle without accounting for the associated pressure work. The App's omission of an efficiency under an unsupported convention is more informative than a plausible-looking number with an undefined denominator.

1.7 Species and elements: two different conservation statements

Total mass conservation remains valid when chemical reactions change the composition. Conservation of each species does not. A mixture can lose molecular hydrogen and molecular oxygen while gaining water, with no loss of total mass. Therefore a balance that treats every inlet species as a conserved passive tracer describes frozen transport or nonreacting mixing, not chemical equilibrium.

Let $\dot{n}_i$ be the molar flow of species i , in kmol/s, and $\dot{\xi}_r$ the rate of progress of reaction r , also in kmol/s for its chosen stoichiometric equation. With

signed stoichiometric coefficients ν_{ir} , positive for products, a steady reactor balance can be written

$$\dot{n}_{i,out} - \dot{n}_{i,in} = \sum_r \nu_{ir} \dot{\xi}_r.$$

Changing the scale of a reaction equation changes its numerical rate-of-progress convention. Species production, rather than the numerical value of $\dot{\xi}_r$ alone, is the physical quantity that must remain consistent.

Let a_{ki} count atoms of element k in one molecule of species i . A chemically balanced reaction satisfies

$$\sum_i a_{ki} \nu_{ir} = 0 \quad \text{for every element } k.$$

Multiplying the species balances by a_{ki} and summing over species eliminates the reaction sources:

$$\sum_i a_{ki} \dot{n}_{i,out} = \sum_i a_{ki} \dot{n}_{i,in}.$$

Elements are conserved even though the individual species are not. If the molar masses W_i are constructed consistently from the same elemental inventory, multiplying by W_i also recovers total mass conservation. In an unsteady reactor, the corresponding accumulation terms must be retained; in a duct with injection, the injected element flows appear on the inlet side.

These equations explain the different questions asked by three App model classes. Frozen-property calculations evaluate a composition supplied by the user. Equilibrium calculations find a composition compatible with an elemental inventory and the selected thermodynamic constraints. Finite-rate calculations evolve species according to a specified reaction mechanism and finite elapsed time or flow distance. A conserved element balance alone cannot distinguish a chemically equilibrated mixture from a reacting or frozen mixture: all three may satisfy it.

An algebraic equilibrium composition is therefore not proof that a short combustor has enough residence time to reach equilibrium. Conversely, an incomplete finite-rate reaction does not violate thermodynamics merely because its composition differs from equilibrium. Chapters 5–8 develop these distinctions and the data conventions needed to use them.

1.8 Energy and entropy complete the control-volume description

The momentum balance defines force, but it does not determine an exhaust speed from temperature, fuel and component losses. For that purpose the energy balance must be joined to thermodynamic state relations. Define heat transfer $\dot{Q}$ as positive into the fluid and shaft power $\dot{W}_s$ as positive out of the fluid. For steady streams,

$$\dot{Q} - \dot{W}_s = \sum_{out} \dot{m} \left(h + \frac{V^2}{2} + gz \right) - \sum_{in} \dot{m} \left(h + \frac{V^2}{2} + gz \right).$$

Enthalpy $h = u + p/\rho$ contains the flow work required to carry material across the control surface. This is why adding a further pressure-work term to an enthalpy-based stream balance would count flow work twice. In this statement h and $V^2/2$ have units of J/kg; multiplication by kg/s gives watts. Gravitational potential is usually negligible across an engine's internal height, but neglecting it is a geometric approximation, not a property of all flow systems.

Write $h_t = h + V^2/2$ after neglecting height changes. An adiabatic stationary nozzle without shaft work then obeys $h_{t,e} = h_{t,in}$, so

$$V_e^2 = V_{in}^2 + 2(h_{in} - h_e).$$

A decrease in static enthalpy provides the kinetic-energy increase. A compressor instead receives shaft work: $\dot{W}_s < 0$, raising total enthalpy in an adiabatic single stream. A turbine extracts work, giving a positive $\dot{W}_s$

and a drop in total enthalpy. Connecting turbine and compressor balances requires the mechanical-efficiency convention and the correct mass flow through each component. It is not sufficient to equate their temperature drops when the fluids or flows differ.

For a combustor with air and fuel streams, the reference state of enthalpy is decisive. With thermochemical species enthalpies that include formation enthalpy, an idealized steady adiabatic balance is

$$\dot{m}_a h_{t,a} + \dot{m}_f h_{t,f} = (\dot{m}_a + \dot{m}_f) h_{t,p}.$$

Chemical conversion changes the species composition and hence the mixture enthalpy-temperature relation. It does not require an additional heating-value source in this particular equation. Adding the fuel heating value as well would count chemical energy twice. A simplified cycle formulation based on sensible enthalpies may instead represent released chemical energy through an explicit heating-value term. The two conventions can describe compatible physics, but their terms cannot be intermixed. Chapter 10 follows the actual conventions used by the variable-property cycle model.

Energy conservation also does not determine irreversibility. Entropy supplies a further constraint:

$$\sum_{out} \dot{m}s - \sum_{in} \dot{m}s = \sum_j \frac{\dot{Q}_j}{T_{b,j}} + \dot{S}_{gen}, \quad \dot{S}_{gen} \geq 0.$$

Here $T_{b,j}$ is the absolute boundary temperature associated with each heat interaction, and $\dot{S}_{gen}$ has units of W/K. An adiabatic single-stream device can preserve total enthalpy while generating entropy and losing total pressure. An inlet shock is an important example. Thus an unchanged total temperature, under an appropriate fixed-composition property model, does not imply an unchanged total pressure.

For multiple streams, do not replace the entropy-flow statement with a comparison of a single unweighted inlet and outlet entropy. Mixing

changes both the flow weights and composition. The component chapters use these control-volume balances to distinguish an ideal expansion from a lossy inlet, a nonreacting mixer from a combustor, and a shaft-work interaction from heat addition.

1.9 What propulsive efficiency rewards

A simple fully expanded, single-stream model exposes an important design tradeoff. Neglect fuel mass addition and external installation losses, and let the inlet and exit mass flows both be $\dot{m}$. Then

$$F_N = \dot{m}(V_e - V_0), \quad \dot{E}_{jet} = \frac{\dot{m}}{2}(V_e^2 - V_0^2).$$

The useful propulsive power is $F_N V_0$. Dividing by the increase in stream kinetic-energy flux gives

$$\eta_p = \frac{F_N V_0}{\dot{E}_{jet}} = \frac{2V_0(V_e - V_0)}{(V_e - V_0)(V_e + V_0)} = \frac{2V_0}{V_e + V_0},$$

for a positive-thrust state with $V_e > V_0 \geq 0$. The cancelled factor is also the factor that determines thrust. At $V_e = V_0$, the original force and power expressions both vanish for fixed mass flow. The limiting value of one for η_p is not a finite-thrust operating point. Maintaining a prescribed positive thrust while taking the velocity increment to zero requires an unbounded mass flow in this simplified model.

This result explains why accelerating more air through a smaller velocity increment can improve propulsive efficiency. It does not establish that an indefinitely large bypass ratio is best. Larger capture area, nacelle drag, component weight, fan limits and other installation effects are outside this elementary derivation. In ThrustOne, the ideal trend is useful for understanding the cycle, while the absence of a full installation model must remain explicit.

For an original illustrative comparison, take $V_0 = 200$ m/s and require 20 kN net thrust. An ideal stream with $V_e = 600$ m/s requires 50 kg/s and

adds 8 MW of jet kinetic power. A stream with $V_e = 400$ m/s requires 100 kg/s and adds 6 MW. Both deliver 4 MW of useful propulsive power. Their propulsive efficiencies are therefore 0.50 and approximately 0.667. The second stream produces the same force with a smaller kinetic-energy increment but twice the air flow. No fuel consumption follows until a cycle establishes the fuel power needed to produce that jet power.

If a compatible thermal efficiency is defined as $\eta_{th} = \dot{E}_{jet}/(\dot{m}_f LHV)$, then multiplication gives

$$\eta_o = \eta_p \eta_{th} = \frac{F_N V_0}{\dot{m}_f LHV}.$$

This identity follows from consistent definitions, not from an independent physical law. An inappropriate intermediate jet-power definition cannot be repaired by multiplying two efficiency values. Pressure-mismatched exits, additional power exports and changing reference conventions require the energy accounting to be stated before using the decomposition.

1.10 Worked example: a two-exit engine

Consider a separate-flow engine with core air flow 20 kg/s, bypass ratio 4, fuel flow 0.6 kg/s and flight speed 200 m/s. Let the fully expanded core and bypass exit speeds be 700 and 300 m/s. There is no bleed, no separate power export and no installation drag in this example. We prescribe exit states to study accounting; these values are not claimed to be the solution of a specified thermodynamic cycle.

The bypass flow is 80 kg/s and the total captured air flow is 100 kg/s. The core exit includes fuel, so it carries 20.6 kg/s. The common whole-engine force is the sum of both exhaust contributions minus the captured-air momentum:

$$F_N = 20.6(700) + 80(300) - 100(200) = 18,420 \text{ N}.$$

Equivalently, one may subtract each stream's own inlet momentum. The

core contribution is $20.6(700) - 20(200) = 10,420$ N; the bypass contribution is $80(300) - 80(200) = 8,000$ N. Their sum gives the same whole-engine result. Subtracting the full 100 kg/s inlet contribution from each stream would count inlet momentum drag twice.

On a total-air basis, $F_s = 184.2$ N s/kg. On a core-air basis it is 921 N s/kg. Their ratio is five, exactly $1 + \alpha$. The fuel-air ratio based on core air is $f = 0.03$, whereas fuel per total air is 0.006. Consequently,

$$TSFC = \frac{0.6}{18,420} = \frac{0.006}{184.2} = \frac{0.03}{921} \approx 3.25733 \times 10^{-5} \text{ kg}/(\text{N s}).$$

Mixing the numerator from the core convention with the denominator from the total-air convention would give five times this consumption. The arithmetic would look ordinary; only an explicit flow definition exposes the mistake.

Now compare an independent rocket accounting example: 10 kg/s total propellant, exit speed 2,500 m/s, exit area 0.05 m², exit pressure 150 kPa and ambient pressure 100 kPa. Its thrust under the stated ideal nozzle boundary is 27,500 N, comprising 25,000 N momentum thrust and 2,500 N pressure thrust. The effective exhaust velocity is $F/\dot{m}_p = 2,750$ m/s, which exceeds the actual exit speed because pressure also contributes to force. Specific impulse is approximately 280.42 s. Calling 2,750 m/s the actual flow velocity would transfer the pressure-force contribution into the kinetic-energy calculation incorrectly.

1.11 Closing an internal duct problem

Figure 1.1 distinguishes the whole-engine boundary from a boundary that encloses only a duct. For a steady single stream with no mass addition, gravity or other axial body force, let $F_{w,x}$ be the wall force on the fluid, positive downstream. The momentum equation is

$$p_1 A_1 - p_2 A_2 + F_{w,x} = \dot{m}(V_2 - V_1),$$

or, after collecting terms,

$$J_2 - J_1 = F_{w,x}.$$

The reaction of the fluid on the wall has the opposite sign. In a frictionless constant-area duct with no axial wall-pressure resultant, this reduces to $J_2 = J_1$. In a variable-area duct, the inclined-wall pressure contributes to $F_{w,x}$; friction contributes another force. A constant momentum function cannot therefore be assumed for every nozzle or source-term segment. Mass injection adds momentum terms associated with the incoming stream as well.

The relation illustrates why a pair of station states does not automatically define an entire propulsion system. A station momentum function does not contain the external ambient subtraction appropriate to a complete nozzle, nor the captured-air momentum of a complete engine. The wall reaction of an internal component also need not equal the final vehicle propulsive force.

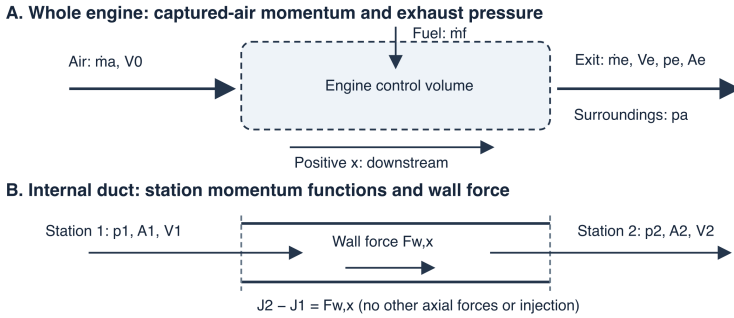


Figure 1: Figure 1.1. Original control-boundary diagrams. The upper boundary defines uninstalled engine performance; the lower one defines a force balance on the fluid inside a duct. The wall-force arrow indicates the positive convention, not its predicted sign.

A solvable component problem must specify enough independent information to determine its outlet. For example, continuity, momentum and energy alone do not supply an equation of state, a wall-friction law, a

chemical composition or a nozzle branch. Conversely, prescribing geometry, all inlet and outlet states, heat addition and wall force independently may overconstrain the problem. Chapters 2, 24 and 26 show how particular closures turn the same balances into different computable models.

1.12 Applying the chapter in ThrustOne

Before comparing two outputs, identify the selected mode, reference mass flow, fuel or propellant denominator, exit-pressure closure and included boundary forces. The same engine label can contain a complete cycle model and a standalone component model with different boundaries.

Use exported SI inputs to recover the physical state behind a result. A changed display unit does not change the control volume. A changed model, nozzle closure or normalization does. The purpose of the following chapters is to make those differences explicit before deriving the individual model equations.

The following distinctions apply to the current implementation. They are physical interpretation rules; they do not require the reader to memorize an export schema.

Model family or input	Interpretation needed for comparison
Turbojet cycle	Specific thrust is per captured air flow; an absolute force requires an air-flow scale. Afterburning consumption includes the fuel added downstream.
Separate-flow turbofan	Whole-engine specific thrust uses total captured air. Fuel-air ratio uses core air. A bypass-stream specific contribution may use bypass air, so stream quantities must be weighted before summation.
Variable-property turbofan	Core-based fuel and total-air-based specific thrust remain distinct, including the optional mixed-flow path.

Model family or input	Interpretation needed for comparison
Rocket performance	Thrust has no captured-air momentum subtraction. Impulse uses total propellant, and effective exhaust velocity includes any pressure-thrust contribution.
Calibrated scramjet full engine	Gross exhaust force and captured-air momentum give net force. The reported injected-mass impulse uses all modeled injection, not fuel alone.
Earlier constant-property scramjet flowpath	A net result requires captured-air information. A generic force label must be interpreted together with the mode and supplied boundary state.
F/G state reconstruction	F is a station momentum function. Constant-property G uses $c_p T + V^2/2$; NASA9 frozen G uses formation-inclusive enthalpy. These G values are not interchangeable.

Two efficiency conventions require particular care. The constant-property turbojet and turbofan calculations retain a thrust-power overall efficiency when an exit is not fully expanded, but suppress the thermal/propulsive kinetic-energy split in that condition. The variable-property cycle implementation reports no such efficiency outputs even with fully expanded exits: it uses formation-enthalpy combustion and does not supply the compatible heating-value denominator used by the other model. A missing metric is therefore not equivalent to a zero metric or proof of failed calculation.

The balances in this chapter determine relationships among states and streams. Subsequent chapters supply the properties, losses, geometry and root choices needed to determine those states. That separation makes a meaningful comparison possible: first establish that both calculations describe the same boundary, then compare how their model assumptions predict the flow through it.

2. Static, stagnation and sonic states

2.1 A flowing state contains more than temperature and pressure

At a station in an engine, static temperature and static pressure describe the local thermodynamic state of a specified gas. Velocity supplies the additional information needed to describe its motion relative to the engine. A gas can therefore have the same static temperature and pressure at two stations yet carry different kinetic energy and different mass flux through equal areas.

The term *static* does not mean that the gas is stationary. It distinguishes local thermodynamic properties from properties obtained by imagining a particular process that brings the moving gas to rest. A fluid element moving at high speed still has a static temperature, density and pressure. In the notation of this book, an unsubscripted T or p is static unless the selected model states otherwise; the subscript t denotes a stagnation or total state.

Mass flow requires geometry as well as state. For a uniform one-dimensional stream normal to its section,

$$\dot{m} = \rho V A.$$

Neither $T/p/M$ nor $T/p/V$ alone fixes an absolute mass flow without an area. Conversely, prescribed mass flow can be used to recover area after the thermodynamic state and speed have been determined. This difference separates a state-reconstruction task from a nozzle-sizing task even when both display the same station quantities.

The following derivations assume a steady stream, negligible gravitational-height change and an ideal gas. Additional restrictions are stated as they are introduced. The distinction between a frozen gas and an equilibrium reacting mixture is deferred until Section 2.10; it cannot be hidden in the choice of a numerical gamma.

2.2 Thermally and calorically perfect descriptions

An ideal gas with fixed composition obeys

$$p = \rho RT, \quad dh = c_p(T) dT, \quad c_p(T) - c_v(T) = R.$$

Here R is the mixture-specific gas constant in J/(kg K). A thermally perfect description permits temperature-dependent heat capacity while retaining ideal-gas behavior. A calorically perfect description additionally holds c_p , c_v and $\gamma = c_p/c_v$ constant. It follows that

$$c_p = \frac{\gamma R}{\gamma - 1}, \quad c_v = \frac{R}{\gamma - 1}.$$

These are connected properties, not three arbitrary independent constants. Supplying mutually inconsistent values of R , c_p and gamma would cause the energy equation and equation of state to describe different gases. The App's constant-property models may derive one property from the others; a comparison with an external calculation must follow that same convention.

For a fixed composition the entropy differential is

$$ds = \frac{c_p(T)}{T} dT - R \frac{dp}{p}.$$

It separates thermal and pressure contributions. With constant c_p , integration between two states gives

$$s_2 - s_1 = c_p \ln \frac{T_2}{T_1} - R \ln \frac{p_2}{p_1}.$$

Setting this change to zero produces an isentropic relation. It does not make an arbitrary real flow isentropic: a shock, friction or mixing can generate entropy while conserving total enthalpy. The state relation must be used with the physical process, not as a replacement for it.

2.3 Stagnation enthalpy comes from energy conservation

For an adiabatic stream with no shaft work, Chapter 1 gives

$$h_t = h + \frac{V^2}{2} = \text{constant along the modeled stream.}$$

The associated stagnation temperature is defined by $h(T_t) = h_t$ for the same frozen composition. Under the constant- c_p approximation,

$$h(T_t) - h(T) = c_p(T_t - T) = \frac{V^2}{2},$$

and therefore

$$T_t = T + \frac{V^2}{2c_p}.$$

This temperature relation follows from energy; it does not require the actual deceleration to be reversible. A stationary adiabatic normal shock

preserves total enthalpy even though it generates entropy. For the same calorically perfect gas this means total temperature is unchanged while total pressure falls.

A hot gas can possess considerable static enthalpy but modest kinetic energy. A colder gas moving rapidly can have the same total enthalpy. A nozzle uses this exchange to accelerate the stream; a diffuser converts kinetic energy into static enthalpy. Neither component creates energy simply by changing velocity.

For a variable-cp gas the correct relation is instead

$$\int_T^{T_t} c_p(\theta) d\theta = \frac{V^2}{2}.$$

The temperature on the upper limit is an unknown. Replacing the integral with $c_p(T)(T_t - T)$ is a further approximation. Formation-enthalpy constants cancel between states with unchanged composition, but they cannot generally be discarded when species compositions differ. Chapters 5 and 10 make the reacting and frozen reference conventions explicit.

Stagnation properties also depend on the velocity reference frame. A freestream that has one total enthalpy relative to the ground has another relative to a fast-moving engine. The cycle calculations use the engine frame. Rotating blade relative-total properties introduce a different frame and should not be substituted into an absolute-frame nozzle state.

2.4 Stagnation pressure requires an isentropic reference process

Stagnation pressure is the pressure reached by bringing the local flow to rest isentropically, with the selected composition constraint. It is a constructed state property; it is not generally the pressure that an actual lossy stopping process delivers.

For a calorically perfect gas, set $s_t = s$ in the entropy relation:

$$0 = c_p \ln \frac{T_t}{T} - R \ln \frac{p_t}{p}.$$

Rearrangement gives

$$\frac{p_t}{p} = \left(\frac{T_t}{T} \right)^{c_p/R} = \left(\frac{T_t}{T} \right)^{\gamma/(\gamma-1)}.$$

At zero velocity, $T_t=T$ and $p_t=p$. At nonzero velocity, the same gas has $T_t>T$ and $p_t>p$. Substituting total pressure into $\rho = p/(RT)$ while leaving the temperature static mixes two states and overestimates the local density. The static mass-flow equation must use a mutually compatible static density and speed.

If a fixed-composition adiabatic device preserves T_t but generates entropy, its inlet and outlet total states satisfy

$$s_2 - s_1 = -R \ln \frac{p_{t,2}}{p_{t,1}}.$$

A positive entropy change then corresponds to a total-pressure ratio below one. Total-pressure recovery can describe a loss even when total temperature is constant. Inlet recovery and combustor pressure-loss parameters should be interpreted in this way, rather than as direct changes in temperature or energy.

A pressure probe that creates a shock in a supersonic stream measures a post-shock stopping pressure. That measured value is not the upstream isentropic total pressure unless the shock loss is accounted for. This chapter defines the state quantities; it does not provide a probe-correction model or claim that the App reconstructs an arbitrary measurement installation.

2.5 Mach number and sonic reference ratios

For a frozen ideal-gas composition, the local acoustic speed follows from an isentropic infinitesimal disturbance:

$$a_f^2 = \left(\frac{\partial p}{\partial \rho} \right)_{s,Y} = \gamma(T)RT, \quad M_f = \frac{V}{a_f}.$$

The subscript Y indicates that composition is held fixed. Under constant properties, writing $V^2 = M^2 \gamma RT$ in the stagnation-temperature relation gives

$$\tau(M) \equiv \frac{T_t}{T} = 1 + \frac{\gamma - 1}{2} M^2.$$

Temperature, pressure and density ratios then become

$$\frac{T}{T_t} = \tau^{-1}, \quad \frac{p}{p_t} = \tau^{-\gamma/(\gamma-1)}, \quad \frac{\rho}{\rho_t} = \tau^{-1/(\gamma-1)}.$$

The starred state is the sonic state on the same constant-property isentrope, with $M=1$:

$$\frac{T^*}{T_t} = \frac{2}{\gamma + 1}, \quad \frac{p^*}{p_t} = \left(\frac{2}{\gamma + 1} \right)^{\gamma/(\gamma-1)}.$$

For $\gamma=1.4$ these are approximately 0.833333 and 0.528282. They are ratios of static sonic properties to total properties. They are not ratios to a general inlet static state, and they change with γ . In particular, 0.833333 follows directly from $2/2.4$; a corrupted table or OCR value near it must not override the defining equation.

A star does not always identify a physical throat present in the geometry. A subsonic duct can be assigned a hypothetical sonic reference area for its existing mass flow and total state. That reference becomes the actual minimum-area section only when the assumptions and boundary conditions establish a choked smooth-nozzle solution.

2.6 Deriving the maximum mass flux

Define axial mass flux $j_m = \dot{m}/A = \rho V$. This notation avoids confusion with the G input used elsewhere for total specific enthalpy. Substituting the ideal-gas law, acoustic speed and static-to-total ratios gives

$$j_m = \frac{p_t}{\sqrt{T_t}} \sqrt{\frac{\gamma}{R}} M \left(1 + \frac{\gamma-1}{2} M^2 \right)^{-\frac{\gamma+1}{2(\gamma-1)}}.$$

For fixed gas, p_t and T_t , only the last two factors vary with Mach number. Put $b = (\gamma - 1)/2$ and differentiate the logarithm with respect to M :

$$\frac{d \ln j_m}{dM} = \frac{1}{M} - \frac{\gamma + 1}{2(\gamma - 1)} \frac{2bM}{1 + bM^2} = \frac{1 - M^2}{M(1 + bM^2)}.$$

The derivative is positive for $0 < M < 1$, zero at $M = 1$ and negative for $M > 1$. Thus the sonic state maximizes mass flux at fixed total conditions. The critical value is

$$j_m^* = \frac{p_t}{\sqrt{T_t}} \sqrt{\frac{\gamma}{R}} \left(\frac{2}{\gamma + 1} \right)^{\frac{\gamma+1}{2(\gamma-1)}}.$$

An increase in velocity on the supersonic branch is accompanied by a sufficiently large density decrease that mass flux falls. Faster flow does not automatically mean more mass through the same area when total conditions are held fixed.

The formula also exposes its scale dependence: doubling p_t doubles the critical mass flux, while multiplying T_t by four halves it if gas properties remain unchanged. In a reacting chamber, temperature and properties can change together, so the latter relation is a controlled constant-property comparison, not a complete combustion trend.

Choking means that a selected throat has reached this limiting mass flux for its upstream total state. Further changes downstream cannot simply demand a larger steady mass flow through that same section without chang-

ing the upstream state or the physical flow pattern. It does not mean that every station is sonic or that the entire engine stops responding to operating conditions.

2.7 Area–Mach inversion and the two roots

For the same stream and isentropic total conditions,

$$\dot{m} = A j_m = A^* j_m^*.$$

Taking the ratio gives

$$\frac{A}{A^*} = \frac{1}{M} \left[\frac{2}{\gamma + 1} \left(1 + \frac{\gamma - 1}{2} M^2 \right) \right]^{\frac{\gamma + 1}{2(\gamma - 1)}}.$$

This function reaches its minimum value of one at $M=1$. For a ratio greater than one, it has a subsonic root and a supersonic root. The specified area ratio by itself cannot choose between them. A ratio below one has no root under these assumptions. An algorithm that silently returns one branch has added a physical assumption without informing the reader.

The differential form explains the geometry. Continuity gives $d\rho/\rho + dV/V + dA/A = 0$. For an inviscid isentropic stream without added mass or heat, momentum gives $dp = -\rho V dV$, and the acoustic relation gives $dp = a^2 d\rho$. Combining them yields

$$\frac{d\rho}{\rho} = -M^2 \frac{dV}{V}, \quad \frac{dA}{A} = (M^2 - 1) \frac{dV}{V}.$$

A subsonic flow accelerates through a contraction, whereas a supersonic flow accelerates through an expansion. Smooth passage through $M=1$ requires an area extremum in this restricted model; a converging-diverging nozzle supplies the familiar minimum. The relation does not guarantee that a supplied back pressure will establish the supersonic solution downstream. Shock-containing and entirely subsonic solutions require the appropriate boundary treatment.

Heat transfer, friction and injection alter this differential equation. Consequently the common statement “Mach one occurs at the minimum area” is not a general theorem for the source-term ducts used in other App modes. A sonic event in such a model is a compatibility condition, not merely a geometric feature.

2.8 Worked example: the same area supports different states

Use $\gamma=1.4$, $R=287 \text{ J/(kg K)}$, $p_t=300 \text{ kPa}$ and $T_t=600 \text{ K}$. Let a choked reference throat have area 0.004 m^2 . The constant-property model gives

$$T^* = 600 \frac{2}{2.4} = 500 \text{ K}, \quad p^* \approx 158.485 \text{ kPa}.$$

The critical mass flux is approximately $495.023 \text{ kg/(m}^2 \text{ s)}$, so the mass flow is

$$\dot{m} = 0.004(495.023) \approx 1.98009 \text{ kg/s}.$$

Select a downstream area ratio of 1.6875, giving $A=0.00675 \text{ m}^2$. The supersonic area root is $M=2$. At that state,

$$\begin{aligned} T &= \frac{600}{1 + 0.2(2)^2} = 333.333 \text{ K}, \\ p &= 300 \left(\frac{333.333}{600} \right)^{3.5} \approx 38.3414 \text{ kPa}, \\ V &= 2\sqrt{1.4(287)(333.333)} \approx 731.938 \text{ m/s}. \end{aligned}$$

Using static density gives approximately 0.400781 kg/m^3 . Multiplying density, speed and exit area recovers 1.98009 kg/s .

The subsonic root for the same area ratio is approximately 0.372244. It has a much higher pressure and density, and a much lower speed:

Quantity	Subsonic area root	Supersonic area root
Mach number	0.372244	2.000000
Static temperature, K	583.820	333.333
Static pressure, kPa	272.627	38.3414
Speed, m/s	180.291	731.938
Density, kg/m ³	1.62708	0.400781
Mass flux, kg/(m ² s)	293.347	293.347

The two states share total conditions, mass flux and area ratio, but their pressures are very different. They are alternative algebraic branches, not two simultaneous exit states. Connecting a selected root to an actual throat and prescribed ambient pressure requires a complete nozzle-flow solution. In particular, this table is not a model of a normal shock between the two states: a physical shock would change total pressure and would not connect them on the same isentrope.

2.9 Near-sonic inversion is sensitive

Let $M = 1 + \delta$. Expanding the area relation around its minimum gives

$$\frac{A}{A^*} = 1 + \frac{2}{\gamma + 1} \delta^2 + O(\delta^3).$$

There is no linear term. The two branches approach each other as the square root of the area excess:

$$M \approx 1 \pm \sqrt{\frac{\gamma + 1}{2} \left(\frac{A}{A^*} - 1 \right)}.$$

For gamma=1.4, an area-ratio excess of 0.0001 corresponds to a leading-order Mach deviation of about 0.0109545. An area displayed as 1.000 to only three decimal places can therefore conceal a physically relevant difference between slightly subsonic and slightly supersonic states.

Direct substitution provides another illustration: $M=0.99$ gives $A/A^*=1.000083802$, while $M=1.01$ gives 1.000082876 . The small

asymmetry comes from terms beyond the quadratic approximation. Rounding both ratios to one would erase both the deviation and the branch.

This is conditioning, not automatically a defective root solver. The slope of the forward area function vanishes at the minimum, so its inverse becomes sensitive. A solver should retain the selected branch and sufficient internal precision; a report should not imply more information than the input area ratio supports. Chapters 26 and 33 discuss the related distinction between a valid limiting state, an unbracketed root and a terminated integration.

2.10 Frozen variable-property expansion

For a frozen ideal-gas mixture, R remains constant with temperature but c_p and γ may change. Specify the total state and composition. A trial static temperature supplies enthalpy and entropy, from which the isentropic pressure and speed follow:

$$\ln \frac{p_t}{p} = \frac{1}{R} \int_T^{T_t} \frac{c_p(\theta)}{\theta} d\theta, \quad V(T) = \sqrt{2[h(T_t) - h(T)]}.$$

Then $\rho(T) = p(T)/(RT)$, $j_m(T) = \rho(T)V(T)$, and the local frozen Mach number follows from the local heat-capacity ratio. This sequence replaces a single global γ power law. The property data must support every trial state, not only the inlet and final accepted exit.

An original analytic example makes the distinction visible without relying on a species database. Let a hypothetical frozen ideal gas have $R=287$ J/(kg K) and

$$c_p(T) = 1000 + 0.1T \quad \text{J/(kg K)}$$

over the chosen illustrative interval. The coefficient 0.1 has units J/(kg K²). Between $T=600$ K and $T_t=900$ K,

$$h_t - h = 1000(900 - 600) + 0.05(900^2 - 600^2) = 322,500 \text{ J/kg}.$$

Therefore $V \approx 803.119 \text{ m/s}$. At the static state $c_p = 1060 \text{ J/(kg K)}$, $\gamma \approx 1.37128$ and the local frozen acoustic speed is approximately 485.937 m/s , giving $M \approx 1.65272$. The exact pressure ratio for this hypothetical property law is

$$\frac{p_t}{p} = \exp \left[\frac{1000 \ln(900/600) + 0.1(900 - 600)}{287} \right] \approx 4.55990.$$

If c_p were frozen at its 600 K value for the entire deceleration, the same kinetic energy would instead give $T_t \approx 904.245 \text{ K}$ and a pressure ratio of approximately 4.54904 . The point is not that this particular difference is always large. The approximation has changed both the energy integral and entropy integral, and no single local γ guarantees that both are represented consistently. The App's NASA-property models use their actual property functions, not this illustrative linear law.

For fixed composition, maximizing the isentropic mass flux still leads to the local frozen sonic condition. Along the isentrope, energy gives $V dV = -dh = -dp/\rho$, while $dp = a_f^2 d\rho$. It follows that

$$\frac{d(\rho V)}{d\rho} = V - \frac{a_f^2}{V},$$

which vanishes at $V = a_f$. This is a local property result; it does not restore the constant- γ area formula. The critical temperature and pressure must still be found using the variable-property enthalpy and entropy relations.

In an equilibrium expansion, composition is allowed to change along the path. The path derivative used to identify maximum mass flux is then different from the frozen acoustic derivative. A reported frozen Mach number need not equal one at the equilibrium critical state. ThrustOne

distinguishes the equilibrium critical mass-flux closure from its local frozen_mach diagnostic; Chapter 17 develops the chemical and sonic conventions in detail.

2.11 What choking does and does not determine

A convergent nozzle connected to a reservoir provides a simple boundary example. If downstream pressure is sufficiently high, the nozzle can remain subsonic, with exit static pressure matching the downstream condition in the ideal model. As back pressure falls to the critical ratio, the minimum-area exit becomes sonic. Further lowering ambient pressure does not lower the internal exit pressure in the ideal choked convergent model; instead the jet is underexpanded outside the exit.

This distinction prevents an invalid combination of assumptions: setting the exit of a convergent nozzle to $M=1$ while also assigning it an arbitrary pressure below the critical static pressure at unchanged total conditions overdetermines the state. A calculation that fully expands to a specified low pressure may be sizing an exit area for a converging-diverging nozzle rather than predicting a fixed convergent nozzle.

A supersonic area-root calculation has a similarly limited meaning. It determines a candidate isentropic exit state from area ratio and total conditions. It does not automatically determine whether the physical nozzle has an internal shock, separated flow or a start transient. Those features require further equations and information. The selected App mode must tell the reader which boundary problem is actually closed.

The constant-property cycle, frozen-property nozzle and equilibrium rocket nozzle share conservation laws but impose different closures and property assumptions. Their results should be compared by matching total state, composition, area and ambient convention, not simply by matching a label such as “nozzle exit Mach.”

2.12 Relating the derivation to the selected App mode

The same physical relation appears under different boundary conditions in the App. The distinctions below are essential to choosing an interpretable calculation.

Calculation	State/branch actually determined	Interpretation limit
Explicit-gas rocket performance	Supersonic area–Mach exit root, with a sonic limit at unity area ratio	Ambient pressure changes the pressure-thrust term and warnings; it does not automatically reposition a shock or select a subsonic exit.
Finite-contraction chamber plane	Subsonic area–Mach root from a supplied static chamber plane, then conversion to total conditions	This is an independent finite-contraction calculation, not an automatic correction applied to all rocket modes.
Constant-property component nozzle	Expansion to a prescribed exit static pressure	Its critical-pressure-ratio/choking flag is diagnostic; it does not clamp the exit Mach to one or size an area.
Frozen-property component nozzle	Entropy and enthalpy closure at the prescribed exit pressure	A state calculation without fixed-geometry ambient matching or a separation solution.
Separate-flow turbofan off-design	Fixed-throat convergent-nozzle constraints within reference-normalized matching	Available total-to-ambient pressure ratio is capped at the sonic ratio for the internal exit state; pressure mismatch remains in the force accounting.

Calculation	State/branch actually determined	Interpretation limit
Frozen variable-property nozzle profile	Sonic or supersonic inlet, nondecreasing area, conserved entropy and total enthalpy	A supersonic inlet need not be the throat. The permitted branch never crosses through a sonic singularity into a subsonic solution.
Frozen thermochemical rocket nozzle	Chamber composition established first, then a frozen sonic throat and downstream expansion	Requires ambient pressure below the model's throat pressure; it does not solve the alternative unchoked branch.
Equilibrium rocket nozzle	Critical state at an interior maximum of equilibrium mass flux, then a post-critical expansion branch	Local frozen Mach is a diagnostic, not the equilibrium sonic definition. A selected freeze area is an input, not a predicted kinetic freeze location.

Temperature-domain checks in variable-property calculations are physical-data constraints. A requested expansion that would fall below the mixture's common data range is rejected instead of extrapolating the species fits. The existence of a constant-gamma answer outside that interval does not authorize the variable-property model to continue there.

When reading a result, first identify whether its boundary was specified by exit pressure, area ratio, an inlet state and increasing area, or reference-matched convergent geometry. Only then interpret its Mach number, mass flow and pressure-thrust contribution. The algebra of an isentrope is common to these problems; their boundary conditions decide which particular state it describes.

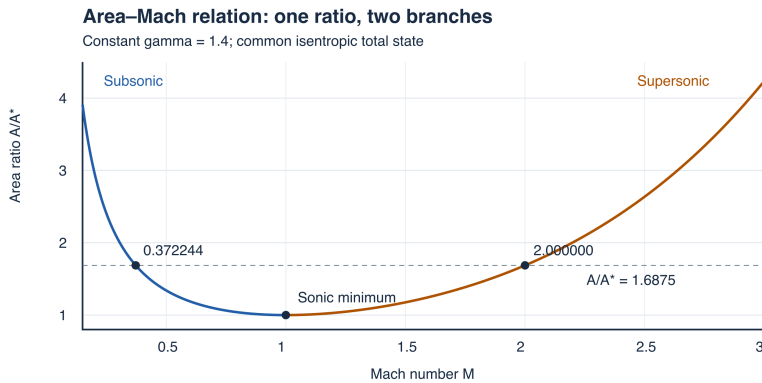


Figure 2: Figure 2.1. Original constant-gamma area–Mach curve for $\gamma=1.4$. A single ratio above unity intersects the subsonic and supersonic branches. The example ratio 1.6875 intersects at approximately $M=0.372244$ and $M=2$; the branches coalesce at the sonic minimum.

3. Performance normalization and similarity

3.1 Why one physical engine needs several coordinate systems

A compressor running at the same shaft speed does not necessarily operate at the same aerodynamic condition when its inlet temperature changes. The blade speed can remain fixed while the acoustic speed changes. Similarly, the same air mass flow passing through a component at lower inlet pressure occupies a larger volume and may require a different flow velocity. Raw rpm and kg/s therefore do not isolate the similarity of the internal flow.

Normalization removes selected scales so that meaningful comparisons become easier. It does not remove the underlying physics. A corrected coordinate can compensate for inlet temperature and pressure while leaving Reynolds-number effects, tip clearance, gas composition and geometry changes unresolved. Two points close together on a corrected map are not automatically interchangeable engines.

Three different operations must be kept separate. A dimensionless quantity has no physical units, such as a pressure ratio. A corrected quantity retains units while referring a measured value to a chosen temperature and pressure convention. A reference-normalized quantity is a ratio to one selected operating point. A corrected mass flow in kg/s, a mass-flow parameter containing dimensional gas constants, and a flow ratio relative

to a design point are related but are not the same object.

The distinction matters in ThrustOne because component-map coordinates, standalone component states and reference-based off-design calculations coexist. External maps may use rpm, a fraction of design corrected speed, or a percentage. ThrustOne expects its map coordinates and dependent tables to be dimensionless and design-point normalized; it does not perform the physical temperature/pressure correction on import. An external map must first be expressed in that convention.

3.2 Obtaining flow and speed similarity groups

Consider geometrically similar rotating components with characteristic diameter D . Let inlet total pressure and temperature be p_t and T_t , and let the working gas have fixed R and γ . The acoustic velocity scale is proportional to $\sqrt{RT_t}$. A density scale is proportional to $p_t/(RT_t)$. Their product with an area scale gives a mass-flow scale:

$$\rho_{scale} V_{scale} D^2 = \frac{p_t}{RT_t} \sqrt{RT_t} D^2 = \frac{p_t D^2}{\sqrt{RT_t}}.$$

A corresponding dimensionless flow group is

$$\Pi_m = \frac{\dot{m} \sqrt{RT_t}}{p_t D^2}.$$

Let N denote shaft revolutions per second. Blade speed is proportional to ND , so a dimensionless speed group is

$$\Pi_N = \frac{ND}{\sqrt{RT_t}}.$$

Using angular velocity instead of revolutions per second introduces the constant 2π ; using rpm introduces a factor of 60. Such constants are harmless only when they are carried consistently. An rpm coordinate cannot

be inserted into a formula calibrated with revolutions per second without conversion.

For a fixed component diameter and fixed gas, holding these groups constant gives the familiar proportionalities

$$\dot{m} \frac{\sqrt{T_t}}{p_t} = \text{constant}, \quad \frac{N}{\sqrt{T_t}} = \text{constant}.$$

The first result also follows from the mass-flux relation in Chapter 2 for fixed Mach number and area. A compressor does not have one uniform Mach number everywhere, but the scaling motivates useful correlating coordinates for similar internal flow patterns.

Additional dimensionless parameters remain. Reynolds number involves viscosity, speed, density and length. Relative clearance involves a clearance-to-diameter ratio. Gas-property changes affect gamma and the thermodynamic conversion from work to pressure ratio. Consequently a family of maps at matched corrected flow and speed may still show changes in efficiency or stability boundaries. In this book, similarity is a stated approximation or a data correlation, never a claim that two dimensional scales determine every feature of a real compressor.

3.3 Corrected quantities retain a reference convention

Choose reference temperature T_{ref} and pressure p_{ref} . At the inlet station relevant to the component, define

$$\theta = \frac{T_t}{T_{ref}}, \quad \delta = \frac{p_t}{p_{ref}}.$$

The usual fixed-gas corrected mass flow and corrected speed are

$$\dot{m}_c = \dot{m} \frac{\sqrt{\theta}}{\delta}, \quad N_c = \frac{N}{\sqrt{\theta}}.$$

Because θ and δ are dimensionless, corrected mass flow has the same units as actual mass flow and corrected speed has the same units as actual speed. These definitions can be inverted:

$$\dot{m} = \dot{m}_c \frac{\delta}{\sqrt{\theta}}, \quad N = N_c \sqrt{\theta}.$$

The pressure and temperature belong to the chosen component inlet, not automatically to the atmosphere. In a turbojet, the compressor inlet total state follows the inlet's ram rise and pressure recovery. In a turbofan, the high-pressure compressor inlet follows the fan or booster path. Correcting every spool with freestream static conditions would mix incompatible reference stations.

A turbine and compressor on one shaft share actual mechanical speed, but their corrected speeds usually differ because their inlet total temperatures differ. If the same T_{ref} is used for both,

$$\frac{N_{c,t}}{N_{c,c}} = \sqrt{\frac{T_{t,c}}{T_{t,t}}}.$$

There is no mechanical contradiction: the two coordinates describe the same shaft motion relative to different local acoustic scales.

The selected reference conditions are part of a data definition. They should accompany a map or case documentation. A corrected value without its inlet station and reference temperature/pressure is an incomplete specification, even if the number is finite and lies inside a tabulated axis range.

3.4 Worked example: preserving an aerodynamic coordinate

Use the illustrative convention $T_{ref}=288$ K and $p_{ref}=100$ kPa. Suppose a reference component point has inlet totals equal to these values, actual

mass flow 20 kg/s and actual speed 10000 rpm. Its corrected values are then 20 kg/s and 10000 rpm.

Now change the inlet totals to 360 K and 80 kPa while asking for the same corrected coordinates. The ratios are $\theta=1.25$ and $\delta=0.8$. Therefore,

$$\dot{m} = 20 \frac{0.8}{\sqrt{1.25}} \approx 14.3108 \text{ kg/s},$$

$$N = 10000 \sqrt{1.25} \approx 11180.34 \text{ rpm}.$$

The hotter stream requires a higher actual speed to maintain the same corrected speed. The reduced pressure and higher temperature reduce the actual mass flow corresponding to the selected corrected flow. Feeding the old actual values into the new state would move the operating point on the map.

For example, retaining 10000 rpm at 360 K gives a corrected speed of approximately 8944.27 rpm under this convention. A controller holding actual speed fixed and a calculation holding corrected speed fixed are imposing different conditions. Neither is intrinsically wrong; they answer different questions.

These calculations do not establish that the same point is physically attainable at both conditions. They presume the same gas, geometry and usable map correlation. Temperature limits, Reynolds-number changes and shaft constraints can invalidate the supposed correspondence. The example isolates the coordinate transformation so those additional decisions remain visible.

3.5 Changing the correction convention does not change the engine

Suppose the same map data are re-expressed using $T_{ref,2}$ and $p_{ref,2}$ rather than $T_{ref,1}$ and $p_{ref,1}$. Applying the definitions to the same actual state gives

$$\frac{\dot{m}_{c,2}}{\dot{m}_{c,1}} = \sqrt{\frac{T_{ref,1}}{T_{ref,2}} \frac{p_{ref,2}}{p_{ref,1}}}, \quad \frac{N_{c,2}}{N_{c,1}} = \sqrt{\frac{T_{ref,2}}{T_{ref,1}}}.$$

For the preceding 20 kg/s and 10000 rpm corrected point, changing the convention to 300 K and 101.325 kPa gives approximately 19.8556 kg/s and 10206.21 rpm. The physical engine has not changed. Every point and every query must be transformed consistently before the new coordinates can be used.

A design-normalized map may instead use $\widehat{N}_c = N_c/N_{c,R}$. A value of 0.95 then means 95% of the reference corrected speed. It does not mean 0.95 rpm, and it does not necessarily mean 95% of reference mechanical speed. With $T_t/T_{t,R} = 1.21$, a mechanical speed ratio of 0.95 corresponds to

$$\frac{N_c}{N_{c,R}} = \frac{N}{N_R} \sqrt{\frac{T_{t,R}}{T_t}} = \frac{0.95}{1.1} \approx 0.863636.$$

The corrected speed ratio has changed more because the inlet is hotter. Conversely, holding a corrected speed ratio at 0.95 under these temperature conditions would imply a mechanical speed ratio of 1.045. Confusing the two could conceal a shaft-speed exceedance in a real application, even though a simple map interpolation produces a valid number.

3.6 Pressure, temperature and work ratios describe different effects

A component total-pressure ratio is $\pi = p_{t,out}/p_{t,in}$; its total-temperature ratio is $\tau = T_{t,out}/T_{t,in}$. These definitions do not include an efficiency by themselves. A compressor model supplies the relation between them.

For a calorically perfect gas and an isentropic compressor, $\tau_s = \pi^{(\gamma-1)/\gamma}$. With an isentropic efficiency defined as ideal work divided by actual work,

$$\eta_c = \frac{\tau_s - 1}{\tau - 1}, \quad \tau = 1 + \frac{\pi^{(\gamma-1)/\gamma} - 1}{\eta_c}.$$

The actual specific work is $\Delta h_t = c_p T_{t,in}(\tau - 1)$. Thus the same pressure ratio and efficiency require more work per kilogram when the inlet total temperature is higher. A pressure-ratio match alone is not a shaft-power match.

A polytropic efficiency uses a different differential process convention and produces a different finite pressure-to-temperature relation. One must not replace isentropic efficiency with a polytropic number in the equation above. Chapters 11 and 15 derive the component conventions; here the important distinction is between a coordinate definition and the physical relation that connects coordinates.

For a fixed-geometry compressor family, a simplified work-coefficient assumption may give

$$\Delta h_t = K N^2,$$

where K contains geometry, units and the assumed work coefficient. This is an operating-region approximation, not a universal compressor law. Combining it with the constant-property isentropic-efficiency relation and dividing by a reference point gives

$$\left(\frac{N}{N_R} \right)^2 = \frac{T_{t,in}}{T_{t,in,R}} \frac{\pi^{(\gamma-1)/\gamma} - 1}{\pi_R^{(\gamma-1)/\gamma} - 1} \frac{\eta_{c,R}}{\eta_c}.$$

If efficiency is unchanged, the last factor is one. The inlet-temperature ratio disappears only after converting this mechanical-speed ratio into a corrected-speed ratio. This derivation explains why a speed estimate in a reference-normalized cycle can resemble a map relation without actually using a measured map.

Changing gas composition or c_p changes the derivation as well. A variable-property compressor requires an enthalpy calculation, and

the assumed constant K may no longer represent the same blade loading. Similarity relations should therefore be restricted to the gas and component approximation under which they were obtained.

3.7 A reference point replaces constants, not missing equations

Referencing is useful when a constant is difficult to identify separately but an operating point is known. For a choked section with fixed area, gas and discharge convention, Chapter2 gives

$$\dot{m} = C \frac{p_t}{\sqrt{T_t}}.$$

The constant C includes the area and critical-flow factor. Comparing with a reference state eliminates it:

$$\frac{\dot{m}}{\dot{m}_R} = \frac{p_t}{p_{t,R}} \sqrt{\frac{T_{t,R}}{T_t}}.$$

If the section area changes, an area ratio must be restored. If gamma or R changes, the critical-flow factors must also be restored. If the section becomes unchoked, the Mach-dependent mass-flow function replaces the fixed sonic factor. A reference value does not authorize a model to ignore those changes.

As a simple original example, a fixed choked section with total-pressure ratio 0.7 and total-temperature ratio 1.2 has a mass-flow ratio of approximately 0.639010. Multiplying a reference flow of 25 kg/s gives about 15.9752 kg/s. The reduction is stronger than the pressure ratio alone because the gas is hotter. This applies to the mass passing through the selected section. Converting hot-gas flow to captured air requires the fuel-addition and bleed assumptions, rather than silently equating the two flows.

A full off-design engine must satisfy multiple constraints simultaneously: compatible component mass flows, shaft power balances, nozzle capacity, specified geometry and an operating control such as fuel flow or turbine temperature. A reference point can supply constants in these equations. It cannot replace the equations themselves. If a model assumes two sections remain choked, its result is conditional on that assumption even when every reference-normalized residual is small.

3.8 Map coordinates and physical operating lines

A component map relates performance quantities over an operating domain. In the App's representation, a dimensionless corrected-speed axis and a beta axis locate a point, while normalized corrected flow, pressure-ratio multiplier and efficiency multiplier are dependent tables. Beta is an ordering coordinate along a tabulated line; it is not inherently a blade angle, a bypass ratio or a physical control setting.

This representation permits interpolation without requiring every speed line to have the same corrected-flow values. It also means that a requested corrected flow is not simply the second map coordinate. In general an inverse query would search for beta locations whose interpolated output matches a target quantity, and a nonmonotone map could have multiple inverse roots. The current App implements a narrower inverse: it requires consistently monotone pressure-ratio rows and solves for beta at a prescribed normalized speed and pressure-ratio multiplier. It rejects a non-monotone map for this inverse task rather than providing a multiple-root selector. It does not invert corrected flow.

A component's operating line is generated by the rest of the engine and its controls. Interpolating a map at a prescribed speed and beta does not solve for that operating line. Neither does drawing a smooth curve through available points establish a surge boundary from first principles. A surge-pressure-ratio array, where supplied, is additional normalized information. The App reports the pressure-ratio indicator $\pi_{\text{surge}}/\pi_{\text{operating}} - 1$. That is neither a mass-flow-distance margin nor a dynamic stability prediction.

Out-of-range coordinates require new physical information, not just a more convenient interpolation formula. The App rejects extrapolation from the supplied map domain. That refusal is materially different from predicting that the real component cannot operate anywhere beyond the table: the table may be incomplete, while the App has no justified model for the unprovided region.

3.9 Using normalized results without overstating similarity

Before comparing two runs, identify the correction inlet station, reference temperature and pressure, gas model, geometry, reference operating point, and whether a displayed speed is actual, corrected or reference-normalized. Chapter1's flow-basis rules still apply: a corrected core flow is not automatically corrected total inlet air, and a core-based specific work is not a whole-engine power.

Data provenance must be carried by the case or accompanying map documentation. A file that preserves the numeric map axes but omits their engineering meaning can round-trip perfectly while remaining ambiguous. A successful import proves that a data structure was read, not that its corrected coordinates match the intended reference convention.

Reference-normalized turbine-engine modes are useful because they expose changes around a specified design state without demanding a complete measured component set. Their limits should be stated positively and precisely: fixed-area or choked assumptions, supplied reference states, chosen efficiency relations and the matching equations actually solved. They should not be described as arbitrary engine simulation, a validated stability envelope or a general substitute for measured maps.

The following chapters use these distinctions to connect atmospheric inputs, component states and engine cycles. A normalization is successful when it makes the model's assumptions easier to inspect, rather than hiding them behind apparently universal curves.

3.10 Interpreting the App’s map multipliers

A small original example shows why a normalized map output is not itself a physical pressure ratio. Suppose the standalone compressor calculation supplies a baseline total-pressure ratio of 10 and a baseline isentropic efficiency of 0.85. At the selected dimensionless corrected speed and beta, let the map return flow multiplier 0.90, pressure-ratio multiplier 1.10 and efficiency multiplier 0.96. The component uses

$$\pi_c = 10(1.10) = 11.0, \quad \eta_c = 0.85(0.96) = 0.816.$$

The flow ordinate 0.90 remains a dimensionless corrected-flow ratio. It is not 0.90 kg/s and cannot be turned into an absolute flow without the missing design-flow scale and correction convention. If an inverse query is intended to find this point, its target map pressure-ratio multiplier is 1.10, not the component’s physical pressure ratio 11.0.

If the normalized surge pressure ratio at this speed is 1.20, the displayed pressure-ratio margin is

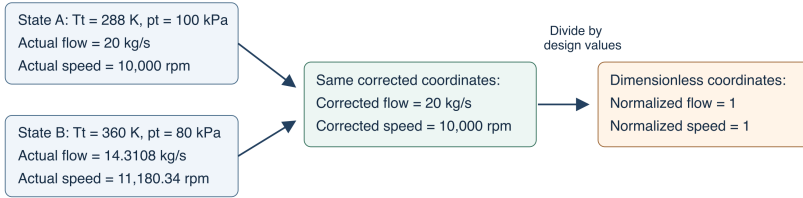
$$\frac{1.20}{1.10} - 1 \approx 0.090909.$$

The common normalization cancels in this ratio. A margin of about 9.09% under this convention says nothing by itself about transient surge onset or the distance in mass flow to a physical stability boundary. It is a useful interpretation of supplied map data, not a replacement for such data.

The analytic demonstration-map generator selects a design point and normalizes its dependent tables there. It illustrates the workflow; it is not a measured manufacturer’s map. The generic beta axis is only required to be finite and strictly increasing. Although the demonstration generator uses beta from 0 to 1, that interval does not establish a universal physical meaning for beta.

The cycle-level reference models use another set of quantities. Their mechanical spool-speed ratios include the appropriate inlet total-temperature

Corrected is not yet design-normalized



Correction convention for both states: $T_{ref} = 288 \text{ K}$, $p_{ref} = 100 \text{ kPa}$. Gas and geometry held fixed.

Figure 3: Figure 3.1. Original coordinate transformation. Different dimensional inlet conditions can produce the same corrected coordinates; design-point normalization is an additional operation. The App's map consumes normalized coordinates rather than reconstructing them from rpm and mass flow.

factors in the work relation. They are inferred speed ratios under a closed-form reference model, not measured shaft speeds and not the normalized corrected-speed coordinates of the standalone component map. The single-spool turbojet also reports a corrected-flow ratio obtained from its own reference closure; that output is not automatically the ordinate of an independently supplied map.

The distinction prevents a tempting but invalid shortcut: taking a cycle's mechanical speed ratio, inserting it directly as map corrected speed, and treating the returned flow multiplier as a completed whole-engine match. A coupled match would require compatible reference stations and scales, component flow continuity, shaft-power constraints and nozzle conditions to be solved together. The present map lookup and reference-cycle modes are separate bounded calculations.

4. Freestream and inlet bookkeeping

An engine calculation begins before air crosses its physical inlet. The flight condition establishes a static thermodynamic state, the speed of the incoming stream, and the energy and momentum carried toward the engine. The inlet then changes that state. Keeping these operations distinct prevents an atmospheric pressure from being mistaken for compressor entry pressure, or an inlet pressure loss from being applied twice.

This chapter develops the route from a flight specification to inlet boundary conditions. The atmosphere supplies an optional relation between altitude, pressure and temperature. Gas properties turn Mach number into velocity. A stagnation calculation accounts for ram heating. An inlet model supplies losses or a downstream state. Finally, a chosen capture area converts incoming mass flux into mass flow. Each operation requires its own assumptions.

The reader should be able to reconstruct a resolved flight state, explain which quantities are independently specified, and identify when two superficially similar input routes describe different physical conditions. Chapter 2 supplies the detailed stagnation derivation; Chapter 3 supplies corrected-flow conventions. The calibrated variable-property inlet receives its full treatment in Chapter 27.

4.1 Stations before components

Use station 0 for the undisturbed incoming stream and station 2 for a representative inlet exit or compressor entry. At station 0, distinguish static pressure p_0 from total pressure p_{t0} . The subscript 0 identifies a location; it does not itself mean stagnation. Likewise, T_0 is the local static temperature and T_{t0} is the temperature associated with bringing that stream to rest adiabatically without shaft work.

For a calorically perfect ideal gas,

$$\rho_0 = \frac{p_0}{RT_0}, \quad a_0 = \sqrt{\gamma RT_0}, \quad V_0 = M_0 a_0.$$

These relations require composition through R , and the heat-capacity model through γ . Mach number alone does not specify speed. Two streams at the same Mach number and different temperatures have different velocities even when their composition is identical.

An adiabatic inlet without shaft work preserves total enthalpy while dissipative mechanisms increase entropy. Its static pressure may increase as velocity decreases, while total pressure decreases. These statements are compatible: static pressure measures one part of the local state, while total pressure also reflects the stream's recoverable mechanical potential.

A cycle inlet can be represented by total-pressure recovery without resolving its internal geometry. A flowpath inlet instead needs a downstream static state, velocity and area relationship. The amount of information returned by these two models differs. A recovery factor alone cannot locate shocks, predict boundary-layer separation or determine an exit Mach number.

4.2 Standard atmosphere as a closure

An altitude is a geometric coordinate. It becomes a thermodynamic state only after an atmospheric model has been selected. A standard atmosphere provides a reproducible reference profile; it does not measure the weather encountered by a vehicle.

ThrustOne’s standard-atmosphere helper uses a constant-molecular-weight approximation over geometric altitudes from 0 to 86 km. Its sea-level values are

$$T_{\text{SL}} = 288.15 \text{ K}, \quad p_{\text{SL}} = 101325 \text{ Pa}, \quad R = 287.05287 \text{ J kg}^{-1} \text{ K}^{-1}.$$

The profile uses geopotential altitude H internally. With geometric altitude z and reference Earth radius $R_E = 6356766 \text{ m}$,

$$H = \frac{R_E z}{R_E + z}, \quad z = \frac{R_E H}{R_E - H}.$$

This transformation allows a constant reference gravitational acceleration $g_0 = 9.80665 \text{ m s}^{-2}$ in the hydrostatic equations. Under the corresponding inverse-square gravity approximation,

$$\frac{dH}{dz} = \frac{R_E^2}{(R_E + z)^2}, \quad g(z) dz = g_0 dH.$$

Geometric and geopotential altitude are close near sea level, but their difference grows with height. Layer boundaries below are geopotential values; entering a listed boundary as geometric altitude does not place the calculation exactly on that boundary.

Geopotential layer base, km	Base temperature, K	Lapse rate, K/km
0	288.15	−6.5
11	216.65	0
20	216.65	+1.0
32	228.65	+2.8
47	270.65	0
51	270.65	−2.8
71	214.65	−2.0

The lapse rate is dT/dH , with its sign included. The final layer is used only up to the supported upper boundary. This is a piecewise atmospheric

reference, not a promise that constant gas properties or an engine model remain physically adequate throughout the full altitude range.

Hydrostatic pressure in a layer

A stationary atmosphere obeys

$$\frac{dp}{dH} = -\rho g_0 = -\frac{pg_0}{RT}.$$

Within a layer beginning at H_b , write

$$T = T_b + L(H - H_b).$$

For nonzero lapse rate L , integration gives

$$\ln \frac{p}{p_b} = -\frac{g_0}{RL} \ln \frac{T}{T_b}, \quad p = p_b \left(\frac{T}{T_b} \right)^{-g_0/(RL)}.$$

For an isothermal layer, integrate with $T = T_b$:

$$p = p_b \exp \left[-\frac{g_0(H - H_b)}{RT_b} \right].$$

The exponential relation is also the limit of the lapse relation as L approaches zero. Substituting a zero lapse rate directly into the power-law expression creates a division by zero; the separate expression is part of the model's definition.

Pressure at a new layer base is inherited from the preceding layer. It is not reset to sea-level pressure. This propagation preserves pressure continuity; the specified base temperatures preserve temperature continuity. The slope of temperature can change abruptly even though temperature itself remains continuous. Consequently, derivatives of downstream quantities may change at layer boundaries without indicating a solver defect.

4.3 Dynamic pressure is not stagnation-pressure rise

Dynamic pressure is defined by

$$q = \frac{1}{2} \rho_0 V_0^2.$$

For the constant-property ideal gas above,

$$q = \frac{\gamma}{2} p_0 M_0^2.$$

Temperature cancels because density varies inversely with temperature while squared speed at fixed Mach varies directly with temperature. This cancellation is conditional on the selected gas model. With temperature-dependent heat capacities, γ itself changes with temperature.

The corresponding inversions are

$$p_0 = \frac{2q}{\gamma M_0^2}, \quad M_0 = \sqrt{\frac{2q}{\gamma p_0}}.$$

The first inversion requires nonzero Mach number. At $M_0 = 0$, every finite positive static pressure gives $q = 0$; the pair $M_0 = 0, q = 0$ cannot determine pressure. Positive q paired with zero Mach is inconsistent. The second inversion is well defined at zero dynamic pressure when static pressure is already known.

Total pressure follows the compressible stagnation relation,

$$\frac{p_{t0}}{p_0} = \left(1 + \frac{\gamma - 1}{2} M_0^2 \right)^{\gamma/(\gamma-1)}.$$

Expanding this relation for small Mach number gives $p_{t0} - p_0 = q + O(M_0^4)$. Thus dynamic pressure approximates the stagnation-pressure rise at low Mach number. It is not their exact difference at general Mach

number. Using $p_{t0} = p_0 + q$ at supersonic speed discards the compressible pressure relation even if every other part of the cycle uses it correctly.

Dynamic pressure is useful as a loading coordinate. It does not independently establish a vehicle's structural limit, thermal limit or engine operability boundary. Those limits require additional models or supplied constraints.

4.4 What each flight input route determines

The generic turbojet flight adapter resolves the following routes before the applicable cycle calculation. This adapter is not a universal input resolver for every engine family; component and dual-spool modes have separate handling.

Route	Supplied quantities	Resolution
Mach and ambient state	M_0, T_0, p_0	Retain the supplied ambient state
Altitude and Mach	Geometric z, M_0	Obtain T_0, p_0 from the standard atmosphere
Mach and dynamic pressure	M_0, q	Compute pressure, invert the standard atmosphere, then obtain standard temperature
Altitude and dynamic pressure	Geometric z, q	Obtain standard temperature and pressure, then compute Mach

The generic dynamic-pressure routes use $\gamma = 1.4$ to resolve the flight condition. This choice is independent of an editable cold-gas ratio used later in a cycle. Changing a cycle property does not automatically redefine the flight adapter's dynamic-pressure convention. For a comparison involving a different gas-property model, reconstruct the physical ambient state and velocity consistently rather than assuming that matching labels guarantees identical conditions.

The Mach-and-dynamic-pressure route imposes more than the algebraic pressure relation. It also imposes the standard-atmosphere temperature associated with that pressure. A manually entered ambient temperature is therefore not an independent degree of freedom on this route.

Pressure decreases monotonically through the supported standard atmosphere. The App uses that monotonicity to invert pressure to geometric altitude within its supported interval. A pressure outside that interval cannot be assigned a supported standard altitude. Such rejection does not mean the pressure is thermodynamically impossible; it means the chosen atmospheric closure does not represent it.

The altitude-and-dynamic-pressure route permits zero dynamic pressure and resolves zero Mach. The Mach-and-dynamic-pressure route requires both quantities to be positive because pressure inversion would otherwise be underdetermined or inconsistent. Different boundary behavior follows from the different unknowns.

4.5 Ram heating and inlet recovery

Define the ram temperature ratio

$$\tau_r = 1 + \frac{\gamma - 1}{2} M_0^2.$$

Then

$$T_{t0} = T_0 \tau_r, \quad p_{t0} = p_0 \tau_r^{\gamma/(\gamma-1)}.$$

These are freestream total conditions before an inlet loss is applied. For an adiabatic, no-work constant-property inlet,

$$T_{t2} = T_{t0}, \quad p_{t2} = \pi_d p_{t0}.$$

Here π_d is total-pressure recovery. Multiplying static pressure by recovery instead produces the wrong station quantity. Reducing total tempera-

ture by the same factor would incorrectly remove energy from an adiabatic inlet.

The current constant-property turbojet cycle forms its nonideal inlet ratio from the supplied recovery factor and an empirical Mach-dependent ram-recovery factor:

$$\pi_d = \pi_{d,\text{supplied}} \eta_r(M_0),$$

$$\eta_r(M) = \begin{cases} 1, & M \leq 1, \\ 1 - 0.075(M - 1)^{1.35}, & 1 < M < 5, \\ 800/(M^4 + 935), & M \geq 5. \end{cases}$$

The ideal cycle setting uses an inlet ratio of one. The empirical correlation summarizes a loss allowance for a cycle calculation. It does not compute a particular shock arrangement, spill flow or inlet unstart. Nor should its piecewise join be interpreted as a newly predicted physical discontinuity.

For a fixed-composition ideal gas at unchanged total temperature, the entropy change is

$$\Delta s = -R \ln \pi_d.$$

Recovery below one therefore corresponds to positive entropy production. This provides a physical interpretation of the loss: the total enthalpy remains available in the energy balance, but less pressure potential remains for subsequent components. It also explains why a compressor pressure ratio must multiply p_{t2} , not the loss-free p_{t0} .

4.6 Capture area and incoming mass flow

For a uniform freestream normal to a capture plane,

$$\dot{m}_a = \rho_0 V_0 A_c.$$

The area A_c belongs to the upstream streamtube entering the engine. It need not equal a geometric inlet lip area under conditions with spillage, external compression or a nonuniform incoming flow. Assigning an area to a calculation is not equivalent to predicting the streamtube that a physical installation captures.

For a steady inlet with no bleed or added mass,

$$\rho_0 V_0 A_c = \rho_2 V_2 A_2.$$

Defining contraction ratio as $CR = A_c/A_2$ gives

$$CR = \frac{\rho_2 V_2}{\rho_0 V_0}, \quad A_2 = \frac{A_c}{CR}.$$

This is a continuity statement. It does not independently determine the exit state. Energy, entropy or a loss closure, and an appropriate branch selection are still required.

If an inlet slows the flow, density usually rises through compression; whether the area decreases depends on their combined effect on mass flux. Interpreting a velocity decrease alone as evidence for a larger exit area ignores density. The same warning applies when comparing cold ambient air with heated downstream gas.

A specific-thrust result also does not determine absolute thrust until its normalization and mass flow are established. Use the air-flow basis of the selected engine model. In particular, fuel addition changes downstream mass flow, while bypass engines may distinguish core air from total captured air, as developed in Chapter 1.

4.7 Worked flight-to-inlet calculation

Consider an illustrative constant-property stream at geometric altitude 10,000 m and Mach 2. Use standard air, $\gamma = 1.4$, supplied inlet recovery 0.95, and an independently assigned capture area of 0.20 m². This

example demonstrates bookkeeping; it does not establish the design of a physical inlet.

First convert altitude:

$$H = \frac{6356766(10000)}{6356766 + 10000} = 9984.2934 \text{ m.}$$

The state lies in the first atmospheric layer. Consequently,

$$T_0 = 288.15 - 0.0065H = 223.2521 \text{ K,}$$

$$p_0 = 101325 \left(\frac{223.2521}{288.15} \right)^{9.80665/(0.0065 \cdot 287.05287)} \approx 26499.87 \text{ Pa.}$$

Using unrounded intermediate values gives the following chain:

Quantity	Result	Role
Static density	0.413510 kg/m ³	Incoming mass per volume
Speed of sound	299.532 m/s	Converts Mach to speed
Flight speed	599.063 m/s	Momentum and kinetic energy
Dynamic pressure	74.1996 kPa	Loading coordinate
Freestream total temperature	401.854 K	Inlet total-energy state
Freestream total pressure	207.347 kPa	Loss-free stagnation pressure
Combined inlet recovery	0.878750	0.95×0.925
Inlet-exit total pressure	182.206 kPa	Compressor boundary pressure
Captured air flow	49.5438 kg/s	Based on the assigned capture area

At this Mach number, $p_{t0} - p_0 \approx 180.847$ kPa, much greater than $q \approx 74.200$ kPa. Their difference is a compressibility effect, not an inlet loss. The inlet loss subsequently reduces total pressure by approximately 25.141 kPa.

The same standard-atmosphere condition can be specified by Mach 2 and the computed dynamic pressure: the algebra gives the original static pressure, and atmospheric inversion returns the original altitude and temperature. Altitude 10,000 m with that dynamic pressure likewise recovers Mach 2. Small differences from entering rounded values are expected.

If corrected mass flow is needed at compressor entry, Chapter 3's correction must use T_{t2} and p_{t2} . Using loss-free freestream total pressure would reduce the reported corrected flow and conceal the inlet loss. A reference station is part of the definition of a corrected quantity.

4.8 A nonstandard atmosphere is a different case

Now prescribe $T_0 = 250$ K, $p_0 = 20$ kPa, and Mach 2 directly, with the same constant gas properties and capture area. The dynamic pressure is

$$q = \frac{1.4}{2}(20000)(2^2) = 56000 \text{ Pa.}$$

The speed is approximately 633.935 m/s and the captured mass flow is 35.3348 kg/s. This state is fully specified without assigning a standard altitude.

Entering only Mach 2 and 56 kPa dynamic pressure into the generic atmospheric route recovers 20 kPa static pressure, but it obtains temperature from the standard profile. It therefore does not preserve the independently supplied 250 K state. Equal Mach and dynamic pressure are insufficient to guarantee equal velocity or mass flow when atmospheric temperature is not the same.

At fixed p_0 , M_0 , composition and constant γ ,

$$V_0 \propto \sqrt{T_0}, \quad \rho_0 \propto T_0^{-1}, \quad \dot{m}_a \propto T_0^{-1/2}$$

for fixed capture area. Warming the stream raises speed but lowers captured flow; dynamic pressure remains unchanged. A study at fixed dynamic pressure therefore does not hold every physically important inlet quantity fixed. State explicitly whether altitude, static pressure, temperature, Mach number or capture area is constrained when interpreting a trend.

4.9 The separate calibrated inlet convention

The calibrated inlet uses frozen variable-property air with nitrogen and oxygen mole fractions 0.79 and 0.21. Its current implementation requires the NASA9 property representation. It computes sound speed and enthalpy from that gas model, rather than treating the generic flight adapter's constant $\gamma = 1.4$ as a universal convention.

Its altitude routes are bounded to geometric altitudes from 0 to 70 km. In the Mach-and-dynamic-pressure route, it searches altitude using dynamic pressure evaluated with its own gas properties. In the altitude-and-dynamic-pressure route, it derives Mach from the dynamic pressure of a unit-Mach state at that altitude. Both procedures preserve the calibrated model's property convention.

The inlet solution requires supersonic freestream flow. Its reference condition sets an entropy-loss relation, and that relation is then used with total-enthalpy conservation and continuity to obtain downstream states. A constant loss coefficient calibrated from one reference condition is a modeling assumption, not a measured inlet map spanning every flight condition.

The full calibrated scramjet flowpath additionally requires a supersonic inlet exit. A mathematically available subsonic exit cannot silently replace that required branch. The separate inlet model and the complete engine model therefore have different admissibility conditions.

In that flowpath, the captured air flow is computed from freestream density, Mach, local frozen sound speed and the supplied capture area. Inlet exit area follows from contraction ratio. This arrangement connects absolute geometry to mass flow, but it still does not predict external spillage

or aircraft installation effects.

Do not multiply the calibrated inlet's pressure recovery by the generic turbojet ram correlation merely because both are called inlet recovery. Each belongs to a separate loss closure. Combining them without re-defining the model counts losses twice and changes the meaning of the calibration.

Flight inputs are resolved before inlet losses

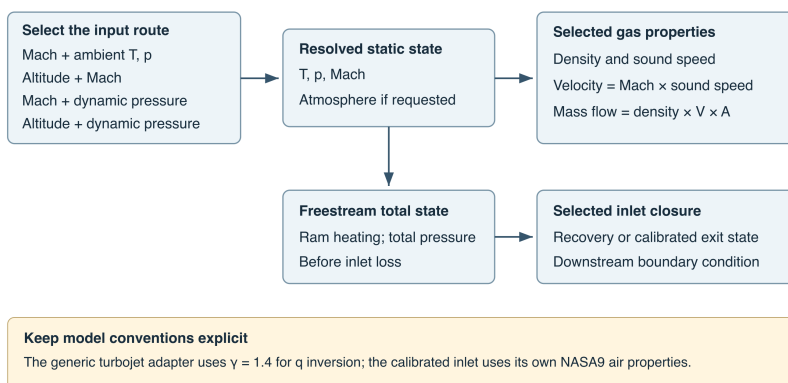


Figure 4: Flight-state resolution and inlet bookkeeping

4.10 Reading a resolved result

A useful flight-state report identifies the input route, the resolved static state, the property convention, and the inlet closure. An altitude label alone omits the atmospheric assumption. A Mach label alone omits the speed of sound. A pressure label without its static or total qualification omits the energy-state definition.

Before comparing two runs, reconstruct those definitions. The most informative discrepancies are often structural: a standard-temperature substitution, a recovery applied to the wrong pressure, a capture area changed between cases, or a different sound-speed convention. These can alter a comparison even when both numerical solutions have converged accu-

rately.

The chapter's equations provide a short route from independent inputs to interpretable inlet conditions. They also identify what remains outside the result: weather prediction, installation drag, spillage, internal shock layout and general inlet operability. Later engine chapters build on the resolved state and add only the component closures supported by their particular models.

5. Frozen mixtures and property polynomials

A fixed composition does not imply a fixed heat capacity. A stream can retain the same species proportions while its molecular energy storage changes strongly with temperature. Frozen thermodynamics describes that situation: the composition is prescribed, while enthalpy, heat capacities, entropy, density and sound speed are evaluated at the requested state.

The distinction matters in propulsion because temperature changes can be large. A single heat-capacity ratio may offer a useful simplified cycle, but a more detailed temperature calculation requires integrated property relations. Replacing a constant gamma with a local value in every familiar power law does not perform those integrations.

This chapter develops the property model used by the frozen-gas calculation and shared by several flowpath models. It explains composition normalization, polynomial evaluation, reference energies, entropy of mixing and temperature inversion. Chemical equilibrium changes the composition and is treated in Chapter 6. Finite-rate chemistry changes composition over a specified time or distance and is treated in Chapters 7–8 and 29.

5.1 Specifying composition without specifying batch size

Let n_i be an amount of species i , W_i its molecular weight in kg/kmol, and $n = \sum_i n_i$. The mole fraction is

$$x_i = \frac{n_i}{n}, \quad \sum_i x_i = 1.$$

All amounts must use a common basis. Their absolute scale is irrelevant to an intensive property calculation: amounts 79 and 21 describe the same mixture as 0.79 and 0.21. The App accepts nonnegative relative molar amounts, merges repeated entries for the same recognized species, and normalizes their sum. At least one amount must be positive. Negative amounts and unknown species do not define an admissible mixture.

The mixture molecular weight follows from total mass divided by total molar amount:

$$\overline{W} = \frac{\sum_i n_i W_i}{\sum_i n_i} = \sum_i x_i W_i.$$

The mass fraction is therefore

$$Y_i = \frac{n_i W_i}{\sum_j n_j W_j} = \frac{x_i W_i}{\overline{W}}.$$

Conversely, supplied mass fractions imply

$$x_i = \frac{Y_i/W_i}{\sum_j Y_j/W_j}, \quad \overline{W} = \left(\sum_i \frac{Y_i}{W_i} \right)^{-1}.$$

These transformations explain why mass percentages cannot be entered unchanged in a molar-composition field. The error becomes especially large when light hydrogen is mixed with heavier species. Normalization

only makes the entries sum to one; it cannot determine that the wrong physical basis was supplied.

A species with exactly zero amount makes no thermodynamic contribution. In the current property calculation, recognized zero-amount entries are omitted after normalization, including from the active temperature-range intersection. An unknown species name is still rejected even if its entered amount is zero. This separates a valid absent component from an unrecognized component definition.

5.2 Molar and mass-specific quantities

Use an overbar for a molar quantity, such as $\bar{h}_i$ in J/kmol. Unbarred h denotes J/kg. The universal gas constant in the App is

$$R_u = 8314.46261815324 \text{ J kmol}^{-1} \text{ K}^{-1}.$$

The mixture mass-specific gas constant is

$$R = \frac{R_u}{\bar{W}}.$$

For an ideal mixture with frozen composition,

$$\begin{aligned} \bar{c}_p &= \sum_i x_i \bar{c}_{p,i}, & \bar{h} &= \sum_i x_i \bar{h}_i, \\ c_p &= \frac{\bar{c}_p}{\bar{W}} = \sum_i Y_i c_{p,i}, & h &= \frac{\bar{h}}{\bar{W}} = \sum_i Y_i h_i. \end{aligned}$$

Either molar or mass-weighted expressions are valid when the weights and property units match. A mole-weighted sum of species values already expressed in J/kg generally gives the wrong mixture value.

The ideal-gas relation is $p = \rho RT$. Since $h = u + RT$ and composition is fixed,

$$c_v = c_p - R, \quad \gamma = \frac{c_p}{c_v}, \quad \rho = \frac{p}{RT}, \quad a_f = \sqrt{\gamma RT}.$$

The subscript on a_f identifies frozen sound speed. The thermal response used in this derivative holds species proportions fixed. An equilibrium acoustic response can include composition changes and need not equal the frozen response. Likewise, the heat capacity of an equilibrium path is not obtained merely by evaluating frozen heat capacities at its current composition: the changing species amounts also contribute to the energy derivative.

Within this ideal frozen model, pressure changes density and entropy but does not change enthalpy or heat capacity at a fixed temperature. Real-gas pressure dependence, liquid phases and condensation are outside the property calculation.

5.3 NASA9: one heat-capacity fit and two integration constants

A property record consists of a temperature interval, a reference pressure and polynomial coefficients. The NASA9 representation can be written with coefficients $a_1, \dots, a_9$:

$$\frac{\bar{c}_p^\circ(T)}{R_u} = a_1 T^{-2} + a_2 T^{-1} + a_3 + a_4 T + a_5 T^2 + a_6 T^3 + a_7 T^4.$$

The superscript circle denotes the species standard-state function. Temperature is in kelvin; the coefficient units are those required by this expression. Changing the input temperature unit without changing the coefficients destroys the fit.

Enthalpy follows from $d\bar{h}/dT = \bar{c}_p$. Integrating each term gives

$$\frac{\bar{h}^\circ(T)}{R_u} = -\frac{a_1}{T} + a_2 \ln T + a_3 T + \frac{a_4 T^2}{2} + \frac{a_5 T^3}{3} + \frac{a_6 T^4}{4} + \frac{a_7 T^5}{5} + a_8.$$

Equivalently,

$$\frac{\bar{h}^\circ(T)}{R_u T} = -\frac{a_1}{T^2} + \frac{a_2 \ln T}{T} + a_3 + \frac{a_4 T}{2} + \frac{a_5 T^2}{3} + \frac{a_6 T^3}{4} + \frac{a_7 T^4}{5} + \frac{a_8}{T}.$$

At fixed reference pressure, $d\bar{s}^\circ/dT = \bar{c}_p^\circ/T$. Integrating once more yields

$$\frac{\bar{s}^\circ(T)}{R_u} = -\frac{a_1}{2T^2} - \frac{a_2}{T} + a_3 \ln T + a_4 T + \frac{a_5 T^2}{2} + \frac{a_6 T^3}{3} + \frac{a_7 T^4}{4} + a_9.$$

The logarithms use the temperature convention of the fitted coefficients. The integration constants carry the selected enthalpy and entropy references. They are not optional numerical decorations. Deleting a_8 preserves the derivative $d\bar{h}/dT$ while changing chemical energy differences; deleting a_9 preserves the entropy derivative while changing chemical potentials.

The representation contains nine coefficients because seven terms describe heat capacity and two constants establish the integrated reference states. It does not mean that heat capacity is a ninth-degree polynomial. The inverse powers and logarithms are essential parts of the representation.

Each interval has its own complete coefficient set. The current NASA9 records use a lower interval from 200 to 1000 K and an upper interval from 1000 to 6000 K; the upper set is selected at exactly 1000 K. Coefficient changes approximate the same species functions across a broad range. Small fit inconsistencies at a join should not be interpreted as a phase transition, nor should a polynomial be extended past its recorded upper limit simply because it still returns a number.

5.4 NASA7 and database consistency

The hydrogen–oxygen mechanism uses NASA7 records. With coefficients $b_1, \dots, b_7$, its heat capacity is

$$\frac{\bar{c}_p^\circ}{R_u} = b_1 + b_2T + b_3T^2 + b_4T^3 + b_5T^4.$$

The integrated functions are

$$\frac{\bar{h}^\circ}{R_u T} = b_1 + \frac{b_2 T}{2} + \frac{b_3 T^2}{3} + \frac{b_4 T^3}{4} + \frac{b_5 T^4}{5} + \frac{b_6}{T},$$

$$\frac{\bar{s}^\circ}{R_u} = b_1 \ln T + b_2 T + \frac{b_3 T^2}{2} + \frac{b_4 T^3}{3} + \frac{b_5 T^4}{4} + b_7.$$

The App evaluates these through the same algebraic representation by placing two zero coefficients before the seven NASA7 coefficients. This is an exact embedding of the NASA7 functional form, not a refit and not a conversion to a different physical dataset.

Property set	Included species	Recorded pressure reference	Temperature qualifications
General NASA9 set	Ar, CH4, CO, CO2, H, H2, H2O, N, N2, NO, O, O2, OH	100,000 Pa	200–6000 K for the included records
Hydrogen–oxygen NASA7 set	Ar, H, H2, H2O, H2O2, HO2, N2, O, O2, OH	101,325 Pa	Reactive species: 200–3500 K; Ar and N2: 300–5000 K

A database choice changes more than the number of coefficients. It changes the available species, fitted functions, reference conventions and valid intervals. The standalone frozen-property mode uses the

general NASA9 set. The NASA7 set is associated with the supported hydrogen–oxygen mechanism where selected; it is not an arbitrary replacement for every model’s property data. In particular, calibrated inlet/engine calculations require their NASA9 convention.

No single common range covers every imaginable mixture. For an actual set of active species, the usable interval is

$$T_{\min} = \max_i T_{\min,i}, \quad T_{\max} = \min_i T_{\max,i}.$$

For example, an active NASA7 mixture of H₂, O₂ and N₂ is restricted to 300–3500 K. Pure H₂ in that database has a 200 K lower boundary; adding positive N₂ narrows it to 300 K. A later reacting calculation may need species that were absent initially, so its admissible domain must account for its own species requirements. An initial-mixture property range alone does not guarantee that an entire reactive trajectory is supported.

5.5 Ideal mixing entropy and the pressure term

For ideal gases, each species contributes entropy at its partial pressure $p_i = x_i p$:

$$\bar{s}(T, p, x) = \sum_i x_i \left[\bar{s}_i^\circ(T) - R_u \ln \left(\frac{x_i p}{p_i^\circ} \right) \right].$$

The reference pressure p_i° must match the species record. When every record shares one reference pressure p° , this separates into

$$\bar{s} = \sum_i x_i \bar{s}_i^\circ(T) - R_u \sum_i x_i \ln x_i - R_u \ln(p/p^\circ).$$

The middle term is the ideal entropy of mixing. It is nonnegative because every nonzero mole fraction is at most one. The last term represents the pressure effect: isothermal compression reduces entropy. The mass-specific value is $s = \bar{s}/\bar{W}$.

For fixed composition between two states,

$$s_2 - s_1 = \int_{T_1}^{T_2} \frac{c_p(T)}{T} dT - R \ln \frac{p_2}{p_1}.$$

The mixing term cancels because composition is unchanged. It must remain when comparing different compositions, such as in a mixing or equilibrium calculation. Removing it because it cancels along one frozen path would corrupt those other problems.

A zero mole fraction contributes the limiting value $x \ln x \rightarrow 0$. The implementation omits zero components and splits the logarithm into $\ln x_i + \ln(p/p_i^\circ)$ for active species. This avoids forming a tiny partial-pressure product that could underflow before taking a logarithm. It does not remove the physical mixing term or turn a trace species into a finite concentration floor.

Changing the reference pressure also requires changing the standard-state entropy convention if the physical entropy is to stay unchanged. Retaining the old entropy coefficients while substituting a new pressure reference shifts the result. This is why the difference between 1 bar and 1 atmosphere belongs in the property definition, even when its numerical effect seems modest.

5.6 Formation enthalpy and sensible heating

A species enthalpy can be expressed schematically as

$$\bar{h}_i(T) = \Delta_f \bar{h}_i^\circ(T_r) + \int_{T_r}^T \bar{c}_{p,i}(\vartheta) d\vartheta,$$

with the reference temperature and elemental conventions consistent with the dataset. The polynomial integration constant carries that reference information. Species with negative formation enthalpy can have negative enthalpy over part of the supported range. Negative enthalpy is not negative temperature and does not itself make the state invalid.

For a frozen stream of unchanged composition, the reference contribution cancels from $h_2 - h_1$. This is why a constant-cp approximation can sometimes reproduce a temperature rise even when it uses a different zero of enthalpy. In a reacting stream, however, species amounts change. The difference between product and reactant formation terms is then part of the energy balance.

If the energy equation already uses formation-inclusive species enthalpies, adding a fuel heating value as a second chemical-energy source counts reaction energy twice. A heating-value cycle and a formation-enthalpy chemistry calculation are different accounting formulations. Each can be meaningful, but their terms cannot be combined casually.

Similarly, a supplied total enthalpy

$$h_t = h(T, x) + \frac{V^2}{2}$$

uses the same enthalpy reference as the property model. A numerical value previously formed as $c_p T + V^2/2$ is not automatically transferable. The reference offset affects an absolute target even when it has no effect on a frozen temperature difference. This point is particularly important for the state-reconstruction modes developed in Chapter 26.

5.7 Inverting an enthalpy or entropy target

The direct property problem supplies temperature and pressure. The enthalpy inversion instead supplies composition, pressure and a mass-specific enthalpy target, and seeks temperature from

$$f(T) = h(T, x) - h_{\text{target}} = 0.$$

For an admissible frozen ideal gas, $dh/dT = c_p > 0$. Enthalpy is therefore increasing within the supported fit intervals. The App brackets the calculation by the common temperature range, evaluates its endpoint enthalpies, and rejects a target outside them. It then uses bounded bisection

to locate the temperature. Its finite iteration count is a numerical procedure, not a statement that the fitted thermodynamic data have that many digits of physical accuracy.

Pressure remains a required positive input for the reported state, even though ideal-gas enthalpy at fixed composition does not depend on it. A pressure change should affect the returned density and entropy, not the temperature recovered from a fixed enthalpy target.

The shared property layer also supports entropy inversion at a supplied pressure. The derivative $(\partial s / \partial T)_p = c_p / T$ is positive over an admissible positive-temperature range, so the same bracketing reasoning applies. Unlike enthalpy inversion, changing the specified pressure generally changes the temperature required to recover the same entropy.

A target outside the bracket does not authorize extrapolation. It may indicate an incorrect unit, an incompatible enthalpy reference, or a real physical state outside the available data. These cases have different engineering meanings, even if all produce a refusal to return an extrapolated temperature.

5.8 Worked composition and entropy example

Consider 79 relative molar units of N₂ and 21 of O₂ in the general database. Their molecular weights are 28.0134 and 31.9988 kg/kmol. Normalization gives $x_{N_2} = 0.79$ and $x_{O_2} = 0.21$, followed by

$$\overline{W} = 0.79(28.0134) + 0.21(31.9988) = 28.850334 \text{ kg/kmol.}$$

The mass fractions and gas constant are

$$\begin{aligned} Y_{N_2} &= 0.76708249, \\ Y_{O_2} &= 0.23291751, \\ R &= 288.19294 \text{ J kg}^{-1} \text{ K}^{-1}. \end{aligned}$$

Thus 21% oxygen by moles corresponds to approximately 23.29% by mass. The gas constant differs from the standard-atmosphere value in Chapter 4 because this explicit binary mixture has a different molecular-weight definition. The difference should be retained when moving between these models.

The ideal mixing contribution is

$$s_{\text{mix}} = -R[0.79 \ln 0.79 + 0.21 \ln 0.21] = 148.1187 \text{ J kg}^{-1} \text{ K}^{-1}.$$

This is the mixing contribution, not the complete entropy of the gas. The species standard-state entropies and the pressure term are still required for the complete value.

If pressure is doubled at unchanged temperature and composition,

$$\Delta s = -R \ln 2 = -199.7601 \text{ J kg}^{-1} \text{ K}^{-1}.$$

Density doubles, while enthalpy, heat capacities and frozen sound speed remain unchanged. This contrasting case separates the ideal pressure effect from temperature and composition effects without requiring a second polynomial evaluation.

5.9 Worked property and inverse example

Argon provides a transparent example in the lower NASA9 interval. Its recorded coefficients reduce to $\bar{c}_p/R_u = 2.5$, and the enthalpy integration constant gives

$$R = 208.1321372 \text{ J kg}^{-1} \text{ K}^{-1}, \quad c_p = 520.3303431 \text{ J kg}^{-1} \text{ K}^{-1},$$

$$h = c_p(T - 298.15 \text{ K}), \quad \gamma = 5/3.$$

At 600 K and 150 kPa the direct calculation yields

Quantity	Illustrative value
Enthalpy	157.061714 kJ/kg
Constant-volume heat capacity	312.198206 J/(kg K)
Density	1.201160 kg/m ³
Frozen sound speed	456.215012 m/s

For the inverse calculation, supply $h = 157061.714$ J/kg, the same pure composition and a positive pressure. The analytical lower-interval inverse is

$$T = 298.15 + \frac{h}{520.3303431} \approx 600 \text{ K.}$$

The App uses its general bracketed property inversion rather than an argon-specific shortcut. The simple analytical result makes the unit and reference convention visible: entering 157.061714 as J/kg would describe a state close to the enthalpy reference temperature, not 600 K.

Heating this gas from 600 to 900 K requires an enthalpy rise of approximately 156.099103 kJ/kg. Isentropically doubling its pressure from the 600 K state instead gives

$$c_p \ln(T_2/600) = R \ln 2, \quad T_2 = 600 2^{0.4} = 791.704746 \text{ K.}$$

Both calculations remain within the lower interval, where the constant- c_p reduction is exact for the recorded fit. That reduction must not be generalized to every species or temperature interval. A variable- c_p mixture requires the integrated functions developed above.

5.10 From a property result to a flow model

The standalone **Frozen Composition · Variable-Specific-Heat Properties** mode evaluates the prescribed composition directly or recovers tem-

perature from its enthalpy target. Its returned molecular weight, gas constant, heat capacities, gamma, enthalpy, entropy, density and sound speed describe one thermodynamic state. It does not predict whether the mixture would react, how fast it would react, or which composition would be in equilibrium.

A frozen nozzle adds mass continuity, total-enthalpy conservation and an entropy or loss condition. A frozen component adds a particular efficiency or pressure closure. A reacting flowpath instead updates species amounts while maintaining its energy and momentum conventions. The property model supplies constitutive values to those equations; it does not supply the missing component closure.

When reading a result, retain the composition basis, database, reference pressure, enthalpy convention and supported temperature interval alongside the displayed temperature and pressure. Those definitions allow the number to be reused correctly. The later thermochemistry and flowpath chapters rely on this same distinction between a local property evaluation and the physical process that connects two states.

6. Gas-phase equilibrium

A chemical-equilibrium calculation determines a composition compatible with a specified thermodynamic constraint and a conserved elemental inventory. It does not merely balance one reaction equation. It compares all admissible compositions within the selected species set and finds the state with no remaining thermodynamic tendency to change along an allowed reaction direction.

ThrustOne provides two related problems. The temperature–pressure mode fixes temperature and pressure, then determines composition. The constant-pressure energy mode fixes pressure and the incoming enthalpy plus supplied heat, then determines both temperature and composition. Their common chemical model is an ideal gas with a bounded set of 13 species. A converged result is equilibrium within that set; it does not establish that the set contains every important real species or phase.

Chapter 5 developed the formation-inclusive enthalpy and ideal mixing entropy used here. This chapter adds elemental constraints, chemical potentials and the energy closure. Reaction rates and residence times enter later chapters, not this equilibrium calculation.

6.1 The inventory that reactions cannot change

Let n_i be the amount of species i , and let A_{ki} count atoms of element k in one molecule of that species. The elemental inventory is

$$b_k = \sum_i A_{ki} n_i, \quad An = b.$$

For example, using columns H₂, O₂, H₂O, H, O and OH gives

$$A = \begin{bmatrix} 2 & 0 & 2 & 1 & 0 & 1 \\ 0 & 2 & 1 & 0 & 1 & 1 \end{bmatrix},$$

where the rows are hydrogen and oxygen. An initial amount of 1.2 H₂ and 0.7 O₂ contains 2.4 hydrogen atom units and 1.4 oxygen atom units. Any admissible equilibrium composition must retain both quantities, regardless of how much water, atomic gas or hydroxyl it contains.

Molecular amount need not be conserved. Dissociating one molecule into two changes total molecular count without changing the elemental inventory or total mass. Consequently, equilibrium mole fractions must be formed from the final total amount $N = \sum_i n_i$, not from the initial total.

The App derives inventory from the declared initial species. It then admits database species whose constituent elements all have positive inventories. Supplying hydrogen and oxygen does not authorize production of a nitrogen species; nitrogen is absent from that problem. Conversely, a species need not be present initially to appear among the products when its elements are available.

At fixed temperature and pressure, two initial mixtures with the same elemental inventory define the same equilibrium problem. Their chemical preparation history is not an additional constraint. In the energy-closure mode, their incoming enthalpies may differ, so identical inventories alone do not guarantee identical final states.

6.2 What is minimized at fixed temperature and pressure

For a homogeneous ideal gas mixture, the total Gibbs energy is

$$G = \sum_i n_i \mu_i,$$

where the chemical potential μ_i is the partial molar Gibbs energy. With a common standard pressure p° ,

$$\mu_i(T, p, x) = \bar{g}_i^\circ(T) + R_u T \ln \left(x_i \frac{p}{p^\circ} \right), \quad \bar{g}_i^\circ = \bar{h}_i^\circ - T \bar{s}_i^\circ.$$

The logarithmic term includes both mixing and pressure. It follows directly from the ideal entropy expression in Chapter 5. Omitting mixing entropy removes the $\ln x_i$ contribution and changes the composition selected by the calculation.

At fixed T , p and elemental inventory, equilibrium minimizes G over non-negative species amounts. The imposed temperature implies that heat can be exchanged as needed to maintain that temperature. A TP equilibrium result therefore does not, by itself, predict the temperature of an adiabatic combustor.

For the ideal mixture, the composition-dependent Hessian has quadratic form

$$\delta n^T \nabla^2 G \delta n = R_u T \left[\sum_i \frac{(\delta n_i)^2}{n_i} - \frac{(\sum_i \delta n_i)^2}{N} \right] \geq 0.$$

The inequality follows from the weighted Cauchy–Schwarz inequality. It expresses the convex character of the ideal-mixture problem within its positive domain. The scale direction must still be constrained by inventory; a freely scalable amount is not an independent physical equilibrium branch. This thermodynamic structure is useful, but it does not make finite-precision iteration infallible or make a restricted species set complete.

6.3 Reaction directions and dimensionless equilibrium constants

A reaction direction ν changes composition through an extent ξ :

$$n = n^{(0)} + \nu\xi.$$

Take products as positive and reactants as negative. Element conservation requires $A\nu = 0$. For an interior equilibrium, differentiation at fixed temperature and pressure gives

$$\frac{dG}{d\xi} = \sum_i \nu_i \mu_i = 0.$$

Dividing by $R_u T$ and inserting the chemical potentials yields

$$\sum_i \nu_i \frac{\bar{g}_i^\circ}{R_u T} + \ln \prod_i \left(x_i \frac{p}{p^\circ} \right)^{\nu_i} = 0.$$

Define the dimensionless reaction quotient and equilibrium constant by

$$Q_p = \prod_i \left(x_i \frac{p}{p^\circ} \right)^{\nu_i}, \quad K_p(T) = \exp \left[- \sum_i \nu_i \frac{\bar{g}_i^\circ(T)}{R_u T} \right].$$

Then equilibrium requires $Q_p = K_p$. A pressure ratio is essential inside this definition. Substituting a numerical pressure in pascals into a formula whose constant assumes atmospheres changes the result whenever $\Delta\nu = \sum_i \nu_i \neq 0$.

For a fixed reaction definition, K_p depends on temperature through the standard-state functions. Pressure still affects composition through Q_p , even though it does not change $K_p(T)$. At fixed temperature, increasing pressure generally opposes dissociation reactions that increase molecular count. This is a consequence of the pressure exponent, rather than an extra empirical rule imposed on the solver.

The App retains reaction directions to evaluate chemical stationarity after solving for element potentials. Those directions are thermodynamic constraints, not a kinetic mechanism. Redundant directions can check the same equilibrium from different stoichiometric combinations without describing additional reaction rates.

6.4 Positivity, bounds and trace species

For a single reaction, nonnegative species amounts bound its extent:

$$\xi \geq \max_{\nu_i > 0} \left(-\frac{n_i^{(0)}}{\nu_i} \right), \quad \xi \leq \min_{\nu_i < 0} \left(-\frac{n_i^{(0)}}{\nu_i} \right).$$

These inequalities prevent consuming more reactant than is available. In a multireaction system, the feasible set is the intersection of all elemental and nonnegativity constraints; independently pushing each reaction to completion does not generally locate the coupled equilibrium.

Trace species matter because chemical potentials depend on logarithms. A small mole fraction can contribute a large negative $\ln x_i$, even while contributing negligibly to bulk molecular weight. Rounding such a value to zero before evaluating chemical affinity can therefore destroy an otherwise balanced equilibrium calculation.

This does not mean every imaginable species must have a positive amount. Species requiring absent elements are excluded first. For the active ideal-gas species set, the numerical method represents positive trace amounts and rejects a solution if necessary amounts fall below its representable range. A printed value that rounds to zero is also different from a mathematically exact zero; reports should preserve sufficient notation to distinguish them.

The general reaction-coordinate routine in the code uses bounded reaction extents and logarithmic endpoint interpolation. The current 13-species gas mode instead uses a coupled element-potential solver. These are different numerical routes to the same ideal equilibrium conditions; the ex-

istence of the general routine should not be mistaken for the algorithm driving every gas-equilibrium result.

6.5 Element potentials reduce the coupled problem

Introduce one Lagrange multiplier λ_k for each active element and minimize

$$\mathcal{L} = G - \sum_k \lambda_k \left(\sum_i A_{ki} n_i - b_k \right).$$

Stationarity with respect to an active species amount gives

$$\mu_i = \sum_k A_{ki} \lambda_k.$$

Writing $\ell_k = \lambda_k / (R_u T)$ and $g_i^\circ = \bar{g}_i^\circ / (R_u T)$, the species mole fractions satisfy

$$\ln x_i = \sum_k A_{ki} \ell_k - g_i^\circ - \ln(p/p^\circ).$$

The unknowns are the element potentials plus total molecular amount. They are determined by

$$N \sum_i A_{ki} x_i = b_k \quad \text{for each element,} \quad \sum_i x_i = 1.$$

This formulation handles species amounts spanning many orders of magnitude without repeatedly subtracting almost exhausted reactant inventories. The implementation normalizes the initial total amount for conditioning, uses logarithmic elemental balances and normalization, and restores the amount scale in the result. It evaluates sums of exponentials with a shifted logarithmic sum to limit overflow and underflow.

Newton-like corrections solve the coupled residual equations. Damping and shorter trial steps are used when an unrestricted step does not reduce the residual. These are numerical safeguards, not combustion efficiencies. The solver has a finite iteration budget and refuses a nonconverged composition rather than returning its last iterate as equilibrium.

Multiplying all initial amounts by a common positive factor leaves the equilibrium mole fractions unchanged, while scaling all final amounts and elemental inventories by the same factor. This scale invariance is a useful interpretation of relative molar inputs. It does not imply that pressure, temperature or the elemental ratios can be rescaled without changing composition.

6.6 What the residuals actually say

Three separate conditions are needed to interpret a computed composition:

Condition	Meaning	What it does not establish
Element conservation	The final atoms match the initial inventory	Chemical equilibrium
Mole-fraction normalization and positivity	The mixture is a valid composition	Correct thermodynamic data
Chemical stationarity	Allowed reaction directions have negligible Gibbs derivative	Completeness of the species or phase model

The displayed equilibrium residual is the largest absolute dimensionless chemical-potential sum over the active checking directions,

$$r_{\text{eq}} = \max_r \left| \sum_i \nu_{ir} \frac{\mu_i}{R_u T} \right|.$$

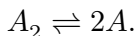
For one direction this is $|\ln(Q_p/K_p)|$. Its scale depends on the chosen stoichiometric coefficients: doubling every coefficient doubles this residual even though the equilibrium composition is unchanged. A small residual is therefore a numerical consistency statement for the defined model, not a universal percentage error in every species amount.

The present gas mode requires this affinity residual below 10^{-8} and separately checks each positive elemental inventory to a relative tolerance of 10^{-8} . Its internal logarithmic closure uses a tighter residual threshold. These tolerances describe acceptance of the numerical equations; they do not imply eight-digit accuracy for real combustion products or property measurements.

An iteration count measures solver work. It is not a reaction time. More iterations do not imply slower chemistry, and a rapidly converged equilibrium does not imply that a flowing combustor has enough residence time to attain it.

6.7 Worked dissociation example with an explicit model boundary

To expose the pressure and extent algebra, consider a hypothetical two-species gas with the single reaction



Assume one initial molar unit of A_2 , no inert gas, and an illustrative dimensionless $K_p = 0.16$ at a fixed temperature. This is an analytical teaching model, not a fitted hydrogen or oxygen equilibrium and not a replacement for the App's 13-species result.

Let α be the fraction dissociated. Then

$$n_{A_2} = 1 - \alpha, \quad n_A = 2\alpha, \quad N = 1 + \alpha,$$

$$x_{A_2} = \frac{1 - \alpha}{1 + \alpha}, \quad x_A = \frac{2\alpha}{1 + \alpha}.$$

The elemental inventory is $2n_{A_2} + n_A = 2$, independently of α . The reaction quotient is

$$Q_p = \frac{x_A^2}{x_{A_2}} \frac{p}{p^\circ} = \frac{4\alpha^2}{1 - \alpha^2} \frac{p}{p^\circ}.$$

Solving $Q_p = K_p$ gives the physical nonnegative root

$$\alpha = \sqrt{\frac{K_p}{4p/p^\circ + K_p}}.$$

At $p = p^\circ$, $\alpha = 0.1961161$, while $x_A = 0.3279216$. The dissociation fraction and atomic mole fraction are different because dissociation changes total molecular count. Substituting the result back into the quotient gives 0.16, and the atom balance remains exactly two units analytically.

Pressure ratio	Dissociation fraction α	Atomic mole fraction x_A
0.25	0.3713907	0.5416264
1	0.1961161	0.3279216
4	0.0995037	0.1809975

The pressure trend follows with no temperature change: lower pressure favors the side with more molecules. Adding an inert gas at fixed total pressure would change the partial pressures and this algebra. Holding volume instead of pressure would define a different problem. Neither alteration can be represented by reusing the table unchanged.

6.8 From specified temperature to an energy closure

An adiabatic combustion temperature is unknown. For a constant-pressure process with no shaft work and negligible kinetic and potential energy changes, the mass-specific energy balance is

$$h_{\text{eq}}(T, p, b) = h_{\text{in}} + q,$$

where q is supplied heat per unit incoming mixture mass. Positive q adds energy. All species enthalpies use the formation-inclusive convention from Chapter 5.

The App's HP mode obtains h_{in} from the declared gaseous mixture at one common inlet temperature. At each trial product temperature, it solves TP equilibrium afresh and evaluates the enthalpy of that equilibrium composition. The outer root function is

$$f(T) = h_{\text{eq}}(T, p, b) - h_{\text{in}} - q.$$

Thus an HP calculation contains two coupled levels: chemical closure at a trial temperature, and energy closure across temperatures. Evaluating the original frozen composition at every trial temperature would solve a different, nonreacting heating problem.

The declared lower and upper temperatures must bracket the enthalpy target. Both endpoint chemical calculations must succeed. A failed intermediate equilibrium is reported as an energy-root evaluation failure; it is not skipped to construct an apparently smooth enthalpy curve through missing states. The current outer calculation uses bounded bisection and requires an enthalpy residual within the larger of 0.01 J/kg and $10^{-9}|h_{\text{target}}|$.

With $q = 0$, the calculation is adiabatic gas-phase equilibrium. No extra heating value is added: reaction energy is already encoded in the species reference enthalpies. Liquid injection, vaporization enthalpy and separate inlet-stream temperatures are not implicit features of this standalone closure.

6.9 An energy-closure example that can be reconstructed exactly

Pure argon isolates the HP temperature closure without introducing a chemical-data comparison. Supply pure Ar at 450 K and 200 kPa, with $q = 130000$ J/kg, and choose a product-temperature bracket of 500–800 K. In the included species set, an inventory containing only argon permits only argon, so its equilibrium mole fraction remains one.

Within this temperature interval, the lower NASA9 argon record gives

$$c_p = 520.3303431 \text{ J kg}^{-1}\text{K}^{-1}, \quad h = c_p(T - 298.15 \text{ K}).$$

The incoming enthalpy and target are

$$h_{\text{in}} = 79012.1626 \text{ J/kg}, \quad h_{\text{target}} = 209012.1626 \text{ J/kg}.$$

The bracket endpoint enthalpies are approximately 105028.6798 and 261127.7827 J/kg, so they enclose the target. The analytical root is

$$T = 450 + \frac{130000}{520.3303431} = 699.841282 \text{ K}.$$

This example demonstrates the same incoming-energy and temperature-bracket definitions used by the HP mode. It does not establish the accuracy of a reactive flame-temperature prediction. In a reacting mixture, dissociation and recombination change the equilibrium enthalpy curve, so replacing the curve with this constant- c_p formula would be inappropriate.

The contrast with TP mode is immediate. Specifying pure argon at 699.841282 K and 200 kPa in TP mode reproduces the final composition and properties, but TP mode does not determine that 130 kJ/kg was required to reach it from 450 K. That conclusion uses the inlet state and energy balance.

6.10 Thermal response along equilibrium

The equilibrium mixture's reported heat capacity is evaluated with its composition frozen at the returned state. If Y_i^{eq} varies with temperature, the total derivative of equilibrium enthalpy at fixed pressure is

$$\frac{dh_{\text{eq}}}{dT} = \sum_i Y_i^{\text{eq}} c_{p,i} + \sum_i h_i \frac{dY_i^{\text{eq}}}{dT}.$$

The first term is the frozen mixture heat capacity. The second term represents the energy effect of changing composition. It can be important in dissociation regions and cannot be reconstructed from the displayed frozen gamma alone.

Consequently, an equilibrium result can supply composition and local frozen properties to a downstream frozen nozzle, but a nozzle that remains in equilibrium must repeatedly update composition and enforce its own energy and entropy constraints. Chapter 17 distinguishes those paths. Neither is a finite-rate chemistry prediction merely because the composition differs from the inlet mixture.

6.11 Scope of the two App modes

The TP and HP gas-equilibrium modes use the general NASA9 set: Ar, CH₄, CO, CO₂, H, H₂, H₂O, N, N₂, NO, O, O₂ and OH. They contain no ions, condensed carbon, liquid water or general hydrocarbon species library. Temperature support is bounded by the records and the selected energy-root interval; numerical existence inside a polynomial range does not establish physical adequacy of the omitted-species assumptions.

A carbon-rich mixture can therefore produce a mathematically consistent gas-only result even where condensed carbon would matter in a broader model. A very hot state can likewise require ionization outside the current species set. These are model limitations, not residual failures. The reported conservation and affinity checks cannot detect physics that was never admitted to the calculation.

For a usable interpretation, retain the selected mode, pressure, inventory, species set, and either specified temperature or inlet enthalpy plus supplied heat. Read the equilibrium residual together with the energy residual when applicable. A converged state then has a precise meaning: it solves the declared ideal-gas chemical constraints, and, in HP mode, the declared energy balance. Whether a real engine approaches that state additionally depends on mixing, reaction rates, heat transfer and residence time.

7. Elementary and pressure-dependent rates

Equilibrium identifies a limiting composition; kinetics supplies the speed and direction of local chemical change. A finite-rate model starts from a reaction list, temperature and species concentrations. It returns production and consumption rates that a reactor or flow solver integrates over time or distance. The rate calculation alone does not determine temperature, residence time or the motion of the gas.

This chapter develops the mass-action and pressure-dependent expressions used by ThrustOne. The same rate evaluator supports the constant-temperature reactor and finite-rate flowpaths. Chapter 8 explains the reactor constraint, while Chapter 29 explains spatial transport through a reacting flowpath. Keeping those constraints separate from the reaction law makes a rate coefficient transferable only where its units, species definitions and thermodynamic references remain consistent.

7.1 Reaction progress and species production

Write reaction r as

$$\sum_i \nu'_{ir} X_i \rightleftharpoons \sum_i \nu''_{ir} X_i, \quad \nu_{ir} = \nu''_{ir} - \nu'_{ir}.$$

Reactant and product coefficients are positive on their respective sides;

the net coefficient is positive for net production. Let q_r be the net volumetric rate of reaction progress in $\text{kmol}/(\text{m}^3 \text{ s})$. The production rate of species i is

$$\dot{\omega}_i = \sum_r \nu_{ir} q_r.$$

The mass source is $W_i \dot{\omega}_i$, in $\text{kg}/(\text{m}^3 \text{ s})$. Reaction progress is therefore not the same quantity as the rate of disappearance of an arbitrary reactant: the latter includes that reactant's stoichiometric coefficient and a sign.

For the illustrative reaction $\text{H}_2 + \text{O} \rightarrow \text{H} + \text{OH}$, one progress unit consumes one unit each of H_2 and O and creates one each of H and OH . Hydrogen and oxygen atoms balance separately. Multiplying the net source vector by an element-count row gives zero, so a sum of balanced reactions preserves every element regardless of their instantaneous rates.

The App checks the declared reaction coefficients for element conservation before compiling the rates. It merges repeated terms of the same species on each side and rejects identical net sides. It requires recognized species and positive finite side coefficients. This protects the algebraic inventory; it does not establish that an entered reaction is an experimentally justified elementary step.

7.2 Concentration and reaction order

The concentration convention is

$$C_i = \frac{n_i}{V} \quad [\text{kmol}/\text{m}^3], \quad C_{\text{tot}} = \sum_i C_i = \frac{p}{R_u T}.$$

For an elementary mass-action reaction, the forward progress rate is

$$q_f = k_f(T) \prod_i C_i^{\nu'_i}.$$

The current implementation uses the entered reactant-side coefficients as reaction orders. It does not provide a separate arbitrary reaction-order field. A globally balanced combustion equation is not automatically an elementary rate law: choosing its stoichiometric coefficients as exponents is a modeling decision that requires kinetic support.

If the total forward order is $m = \sum_i \nu'_i$, the coefficient units are

$$[k_f] = (\text{m}^3/\text{kmol})^{m-1} \text{s}^{-1}.$$

A first-order coefficient has units s^{-1} ; a second-order coefficient has units $\text{m}^3/(\text{kmol s})$. The numerical coefficient changes when the concentration unit changes. For example, numerical concentrations in mol/cm^3 are 10^{-3} times the corresponding numerical values in kmol/m^3 . Converting an m -order coefficient from a mol/cm^3 convention to the App's convention multiplies its numerical value by $10^{-3(m-1)}$. This follows by converting both the rate and every concentration factor, not by changing a unit label.

For a species with positive order, zero concentration makes that directional rate zero. The reverse rate may still be nonzero if products are present. A negative concentration is not an admissible extension of a fractional power; it is rejected before rate evaluation.

7.3 Modified Arrhenius temperature dependence

The supported temperature law is

$$k_f = AT^\beta \exp\left(-\frac{E_a}{R_u T}\right).$$

Temperature is in kelvin, R_u is in $\text{J}/(\text{kmol K})$, and E_a must be in J/kmol . An activation energy stated in J/mol must be multiplied by 1000 before it is paired with this value of R_u . The exponent must be dimensionless.

The pre-exponential factor has the coefficient units divided by K^β . Thus A does not always have the same dimensions as a conventional first- or

second-order rate coefficient. The App's unit description includes this temperature-power factor.

Taking a logarithm gives

$$\ln k_f = \ln A + \beta \ln T - \frac{E_a}{R_u T},$$

and hence

$$\frac{d \ln k_f}{d \ln T} = \beta + \frac{E_a}{R_u T}.$$

This derivative explains why relatively small temperature changes can strongly affect rates. A fitted negative activation energy is also possible in this representation; it is not automatically an input error. The App requires finite Arrhenius parameters and a nonnegative pre-exponential factor. Whether a particular fit is appropriate over a proposed temperature interval is a separate question.

For two temperatures, a useful comparison is

$$\frac{k_f(T_2)}{k_f(T_1)} = \left(\frac{T_2}{T_1} \right)^\beta \exp \left[\frac{E_a}{R_u} \left(\frac{1}{T_1} - \frac{1}{T_2} \right) \right].$$

The factor A cancels. This compares coefficients only: if concentrations change with temperature, the full rate changes by additional concentration factors. At fixed pressure, for example, total concentration decreases as temperature rises.

7.4 Reverse rates from thermodynamics

For a reversible mass-action step,

$$q = q_f - q_b = k_f \prod_i C_i^{\nu_i'} - k_b \prod_i C_i^{\nu_i''}.$$

At equilibrium the directional rates are equal. Therefore

$$\frac{k_f}{k_b} = K_c, quad K_c = \prod_i (C_i^{eq})^{\nu_i}.$$

Here K_c is the dimensional concentration quotient in the selected concentration units. Chapter 6's K_p is dimensionless. Since $p_i = C_i R_u T$, their relation is

$$K_c = K_p \left(\frac{p^\circ}{R_u T} \right)^{\Delta\nu}, \quad \Delta\nu = \sum_i \nu_i.$$

The concentration factor has units kmol/m^3 . It supplies the dimensions needed when the forward and reverse orders differ. Setting $k_b = k_f/K_p$ without this conversion is correct only for the special case $\Delta\nu = 0$ under these definitions.

The App obtains Gibbs functions from the selected thermodynamic database and forms

$$\ln K_c = - \sum_i \nu_i \frac{\bar{g}_i^\circ}{R_u T} + \Delta\nu \ln \left(\frac{p^\circ}{R_u T} \right).$$

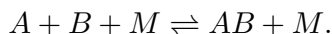
It evaluates the species Gibbs functions at a common 100,000 Pa state. The NASA7 records' own reference pressure is handled inside the property calculation, so this operation includes the required entropy adjustment; it does not simply relabel their coefficient reference pressure. The reverse coefficient then uses the same effective forward coefficient divided by this thermodynamic K_c .

This construction enforces thermodynamic consistency for the included reversible steps. It does not prove the accuracy of their forward rates. Multiplying both directional coefficients by the same factor changes relaxation time while preserving their ratio and equilibrium composition.

Near equilibrium, two large directional rates may nearly cancel. The implementation evaluates logarithmic rates and uses an exponential-difference expression based on `expm1` to reduce cancellation error. An overflowing rate is rejected rather than interpreted as instantaneous completion. Numerical reformulation expands the range of reliable evaluation; it does not supply a missing stiff integration method.

7.5 Third-body collisions

Some reactions require a collision partner to remove or supply energy. A schematic recombination is



The collider appears unchanged in the net species balance, but its concentration affects the rate. Define

$$[M]_{\text{eff}} = \sum_i \epsilon_i C_i,$$

where ϵ_i is the efficiency of species i as a collider relative to the convention of the rate fit. The forward progress rate for a simple third-body step is

$$q_f = k_3(T)[M]_{\text{eff}} \prod_i C_i^{\nu'_i}.$$

The collider adds one concentration power to the overall rate. Consequently, $[k_3] = (\text{m}^3/\text{kmol})^m/\text{s}$, before the Arrhenius temperature-power adjustment.

The App uses efficiency one for species not listed in an enabled third-body efficiency map. An explicit zero efficiency removes that species' collider contribution; it does not remove the species from the mixture. An empty but present efficiency map therefore enables an all-unity collider term,

whereas an absent map without falloff leaves an ordinary mass-action coefficient.

Do not add a fictitious database species named M. Collider effects are represented by the efficiency sum. Including a real species on both sides of a reaction introduces its concentration through the side exponents; additionally enabling a third-body term can then count an extra factor unless that is deliberately part of the supplied model.

7.6 Reduced pressure and Lindemann falloff

A falloff reaction transitions between low- and high-pressure rate laws. Let $k_0(T)$ be its low-pressure coefficient and $k_\infty(T)$ its high-pressure coefficient. Define the dimensionless reduced pressure

$$P_r = \frac{k_0[M]_{\text{eff}}}{k_\infty}.$$

The basic Lindemann effective coefficient is

$$k_{\text{eff}} = k_\infty \frac{P_r}{1 + P_r}.$$

At $P_r \ll 1$, $k_{\text{eff}} \approx k_0[M]_{\text{eff}}$. At $P_r \gg 1$, it approaches k_∞ . The low-pressure coefficient therefore has one additional inverse-concentration unit compared with the high-pressure coefficient. Both have independently supplied Arrhenius parameters.

The main reaction parameters represent k_∞ when falloff is enabled; the nested low-pressure parameters represent k_0 . After forming k_{eff} , the code multiplies by the explicit reactant concentration powers. It does not multiply by the collider sum a second time. The collider is already present through P_r .

Reduced pressure is not the physical pressure in pascals. Its value also depends on temperature, mixture efficiencies and the two rate fits. Two

gases at the same pressure can occupy different positions on the falloff curve.

7.7 Troe broadening

Troe broadening modifies the transition shape through a factor F :

$$k_{\text{eff}} = k_{\infty} \frac{P_r}{1 + P_r} F.$$

The supported center expression is

$$F_{\text{cent}} = (1 - a) \exp(-T/T_3) + a \exp(-T/T_1) + \exp(-T_2/T),$$

where the final term is omitted when T_2 is not supplied. The parameter a lies between zero and one; the characteristic temperatures are positive. This a is a fit parameter, unrelated to sound speed.

Using base-10 logarithms, define

$$\begin{aligned} c &= -0.4 - 0.67 \log_{10} F_{\text{cent}}, \\ n &= 0.75 - 1.27 \log_{10} F_{\text{cent}}, \\ u &= \log_{10} P_r + c. \end{aligned}$$

The broadening formula is

$$\log_{10} F = \frac{\log_{10} F_{\text{cent}}}{1 + [u/(n - 0.14u)]^2}.$$

The implementation evaluates an equivalent scaled form to avoid an unnecessary overflow when the ratio in the denominator becomes large. Its parameters and temperature convention must come from the same rate model as the low- and high-pressure coefficients. Supplying arbitrary

valid positive temperatures merely creates a computable curve; it does not calibrate a physical reaction.

The Lindemann factor alone has simple asymptotes. With Troe enabled, the actual coefficient also includes its broadening expression over the fitted domain. Statements about a fitted curve should retain that qualifier rather than assuming that every parameter set behaves like an idealized sketch at unlimited pressure.

7.8 Worked elementary-rate calculation

Consider the balanced step $\text{H}_2 + \text{O} \rightarrow \text{H} + \text{OH}$ with illustrative forward parameters

$$\begin{aligned} A &= 2.0 \times 10^7 \text{ m}^3 \text{ kmol}^{-1} \text{ s}^{-1} \text{ K}^{-1/2}, \\ \beta &= 0.5, \\ E_a &= 6.0 \times 10^7 \text{ J/kmol}. \end{aligned}$$

These values are chosen for arithmetic demonstration, not as a recommended hydrogen mechanism. At 1000 K,

$$\begin{aligned} k_f &= 2.0 \times 10^7 \sqrt{1000} \exp[-60000000/(8314.462618 \times 1000)] \\ &= 4.64529 \times 10^5 \text{ m}^3/(\text{kmol s}). \end{aligned}$$

With $C_{\text{H}_2} = 0.002$ and $C_{\text{O}} = 0.001 \text{ kmol/m}^3$,

$$q_f = k_f C_{\text{H}_2} C_{\text{O}} = 0.929058 \text{ kmol}/(\text{m}^3 \text{ s}).$$

For an irreversible step, each reactant source is -0.929058 and each product source is $+0.929058$ in the same units. The hydrogen atom source is $2(-q_f) + q_f + q_f = 0$, and the oxygen atom source is $-q_f + q_f = 0$. Mass conservation follows when the molecular weights use consistent atomic masses.

Raising temperature to 1200 K while holding concentrations fixed multiplies the coefficient and forward rate by approximately 3.646925. Holding pressure and mole fractions fixed instead would also reduce both reactant concentrations by a factor 1000/1200, so the rate ratio would include that factor squared. A temperature-sensitivity statement must identify which quantities are held fixed.

If reversibility is enabled, the net source additionally requires product concentrations and the thermodynamic reverse coefficient. The forward number alone no longer equals the net progress rate.

7.9 Worked collider and falloff comparison

At a selected temperature, suppose a mixture has three collider contributions: 0.003 kmol/m³ with efficiency one, 0.001 kmol/m³ with efficiency two, and 0.002 kmol/m³ with efficiency zero. Then

$$[M]_{\text{eff}} = 0.003 + 2(0.001) + 0(0.002) = 0.005 \text{ kmol/m}^3.$$

Its total concentration is 0.006 kmol/m³, so replacing the effective collider concentration by total concentration would change the modeled rate.

For an illustrative unimolecular falloff step, take evaluated coefficients $k_{\infty} = 10^6 \text{ s}^{-1}$ and $k_0 = 2 \times 10^8 \text{ m}^3/(\text{kmol s})$. The resulting reduced pressure is one, and the Lindemann coefficient is $5 \times 10^5 \text{ s}^{-1}$. For reactant concentration 0.0002 kmol/m³, the forward progress rate is 100 kmol/(m³ s). This deliberately rapid example says nothing about whether an explicit time integrator can cover a requested duration economically.

For a Troe comparison at 1000 K, use illustrative $a = 0.5$, $T_3 = 1000 \text{ K}$, $T_1 = 10000 \text{ K}$, and omit T_2 . Then $F_{\text{cent}} = 0.63635843$, the supported broadening expression gives $F = 0.65468982$ at $P_r = 1$, and

$$k_{\text{eff}} = 327344.909 \text{ s}^{-1}.$$

The forward rate becomes approximately 65.4690 kmol/(m³ s). The change arises from the explicitly selected broadening model. It should

not be described as a new collision efficiency or applied again through another multiplier.

7.10 Model inputs, numerical limits and physical meaning

ThrustOne supports 1–50 declared reactions with elementary side-power mass action, reversible rates derived from the chosen thermodynamic data, optional third-body efficiencies, and Lindemann or Troe falloff. A zero main pre-exponential factor disables that reaction. When a collider-dependent coefficient has zero effective collider concentration, it contributes no progress rate. Unsupported or nonfinite rates are refused.

The included hydrogen–oxygen option contains 10 thermodynamic species and 29 reaction entries. It is a bounded mechanism option, not a general hydrocarbon mechanism loader or a complete reacting-flow simulation package. Its thermodynamic data use the NASA7 representation described in Chapter 5. The general demonstration $\text{H}_2 \rightarrow 2 \text{H}$ with a unit first-order coefficient is an algorithm example and must not be interpreted as calibrated dissociation kinetics.

Rates computed here are local chemical source terms. They omit molecular diffusion, turbulent mixing, heat transfer and transport coefficients. A reaction may be thermodynamically favorable yet remain slow; it may be locally fast yet limited by mixing in a real combustor. Choosing a rate law, computing its source terms and resolving a complete engine are therefore distinct steps. The next chapter supplies one specific time-evolution constraint and explains how to read a completed or interrupted chemical history.

8. Reactor constraints and chemical timescales

A reaction mechanism does not determine a unique temperature or pressure history. Those histories depend on the physical constraints imposed on the reacting mixture. A rigid insulated vessel, a rigid temperature-controlled vessel and a constant-pressure combustor can use the same chemical rates and still produce different trajectories.

ThrustOne's finite-rate reactor is homogeneous, closed to mass, constant in volume and externally maintained at a specified temperature. It integrates species concentrations and reconstructs pressure from their changing total. It does not integrate an adiabatic temperature rise. This chapter derives that model, explains its numerical stopping behavior, and explains the reported thermostat diagnostic. The energy balance must follow the stated rigid-vessel constraint; a constant-pressure heat load is a different quantity.

8.1 Choose the control boundary first

Take a rigid vessel of volume V , containing a spatially uniform ideal gas. No mass enters or leaves. Ignore bulk kinetic energy, potential energy, surface reactions and mechanical stirring work. Species can interconvert through the supplied reaction list, while an external thermal boundary maintains a constant temperature T .

With concentration $C_i = n_i/V$, the species equations are

$$\frac{dC_i}{dt} = \dot{\omega}_i(T, C) = \sum_r \nu_{ir} q_r(T, C).$$

There is no dilution term because volume is fixed, and no inlet or outlet term because the vessel is closed. Chapter 7 supplies the local reaction-rate expressions. An open stirred reactor would require additional mass-exchange terms and a residence-time definition; it is not obtained merely by assigning a longer integration duration here.

A prescribed temperature removes the temperature differential equation from the solved system. The external heat exchange needed to enforce that prescription becomes a diagnostic. For an adiabatic vessel, temperature would instead be an unknown coupled to composition through internal-energy conservation. Those are different mathematical systems.

8.2 Initial pressure sets the concentration scale

Let the entered relative molar amounts be a_i , with normalized initial fractions $x_{i0} = a_i / \sum_j a_j$. The initial concentrations are

$$C_{i0} = x_{i0} \frac{p_0}{R_u T}.$$

Thus the relative amounts set composition, while initial pressure and temperature set absolute concentration. Multiplying every relative amount by ten does not increase the physical concentration. Increasing initial pressure at fixed temperature and composition does.

This distinction matters because reaction rates can depend on concentration to different powers. A second-order rate scales quadratically with a uniform concentration increase at an unchanged temperature, while a first-order rate scales linearly. Falloff and collider efficiencies add further dependencies described in Chapter 7.

The reactor does not need a numerical vessel volume to compute concentrations or pressure. Doubling the vessel volume at the same initial state

doubles total mass and total heat-transfer rate, but leaves the concentration trajectory and heat-transfer rate per unit volume unchanged. Absolute equipment loads require a separately known volume.

8.3 Pressure can change at constant temperature

At every time,

$$p(t) = R_u T \sum_i C_i(t).$$

Differentiating at constant temperature gives

$$\frac{dp}{dt} = R_u T \sum_i \dot{C}_i = R_u T \sum_r \Delta \nu_r q_r.$$

Reactions that increase molecular count can raise pressure even though temperature is fixed. Reactions that reduce molecular count can lower it. In a mechanism with several reactions, the net effect is the weighted sum of their progress rates.

Total mass density is different:

$$\rho = \sum_i W_i C_i, \quad \frac{d\rho}{dt} = \sum_i W_i \dot{C}_i = 0$$

for balanced reactions and consistent molecular weights. Pressure change therefore does not imply a mass leak. At constant temperature and density, changing mean molecular weight changes the mixture gas constant and hence pressure.

The displayed mole fraction is

$$x_i(t) = \frac{C_i(t)}{\sum_j C_j(t)}.$$

Its variation reflects both the species concentration and the changing denominator. A mole fraction falling by half does not necessarily mean that half the original amount of that species remains. Use concentration or the analytical inventory relation when reasoning about conversion.

8.4 The thermostat energy balance

For a closed rigid vessel with no other work, the first law is

$$\frac{dU}{dt} = \dot{Q}.$$

Let positive $\dot{Q}$ mean heat added to the gas. Since

$$\frac{U}{V} = \sum_i C_i \bar{u}_i(T),$$

its derivative is

$$\frac{\dot{Q}}{V} = \sum_i \bar{u}_i \dot{\omega}_i + \left(\sum_i C_i \bar{c}_{v,i} \right) \frac{dT}{dt}.$$

For the isothermal constraint, $dT/dt = 0$, so the required volumetric heat-transfer rate is

$$\dot{q}_V = \sum_i \bar{u}_i \dot{\omega}_i.$$

Formation-inclusive internal energies are required. An exothermic reaction at fixed temperature generally requires heat removal, giving a negative $\dot{q}_V$ under this convention. Endothermic dissociation can require positive heat input. No additional fuel heating-value term should be added to the species formation-energy accounting.

For an ideal gas, $\bar{u}_i = \bar{h}_i - R_u T$. Consequently,

$$\dot{q}_V = \sum_i \bar{h}_i \dot{\omega}_i - R_u T \sum_i \dot{\omega}_i = \sum_i \bar{h}_i \dot{\omega}_i - \frac{dp}{dt}.$$

Pressure per time and power per volume have the same dimensions: Pa/s equals W/m³. The correction is nonzero when the net molecular production rate is nonzero.

Interpreting the App thermostat diagnostic

The result named `thermostat_heat_rate` reports the rigid-vessel heat-transfer rate per unit volume at the final returned state. Define the enthalpy-source sum

$$S_h = \sum_i \bar{h}_i \dot{\omega}_i.$$

The reported value is $S_h - R_u T \sum_i \dot{\omega}_i$, in W/m³. Positive values denote heat supplied to the mixture; negative values denote heat removal. It is an instantaneous final-state rate, not the time-integrated energy supplied during the calculation.

An enthalpy-source sum alone would omit the pressure-production term. That difference matters whenever the reaction changes total molecular amount, even though the vessel volume and temperature are fixed. The diagnostic does not feed back into the concentration integration: the external thermostat imposes temperature independently. Neither a small integration error nor a conserved mass would justify substituting a constant-pressure energy balance for the rigid-vessel balance.

8.5 What an adiabatic calculation would require

Setting heat exchange to zero in the first law gives

$$\frac{dT}{dt} = - \frac{\sum_i \bar{u}_i \dot{\omega}_i}{\sum_i C_i \bar{c}_{v,i}}.$$

Temperature then changes the rates, and the changing rates alter composition and temperature again. This coupling can produce rapid thermal acceleration. It is not present in the externally isothermal reactor.

A constant-temperature species history can reveal kinetic conversion and pressure evolution under thermal control. It cannot show an adiabatic ignition-temperature rise, determine an adiabatic flame temperature or establish heat-release-driven thermal runaway. The HP equilibrium calculation in Chapter 6 solves a limiting energy-constrained state, while an adiabatic finite-rate history would require the coupled differential equation above. Neither is silently activated by selecting a chemical mechanism in the current reactor.

8.6 Chemical timescales and stiffness

For an irreversible first-order loss $dC/dt = -kC$, the characteristic decay time is $\tau = 1/k$. After one such time, the concentration is e^{-1} of its initial value. For a coupled mechanism, there is usually no single chemical timescale valid for all species and all parts of the trajectory.

A local estimate $C_i/|\dot{\omega}_i|$ can be useful away from zeros, but it becomes misleading near equilibrium: a small net source may result from cancellation between very large directional rates. The linearized source Jacobian then provides fast relaxation scales that are invisible in the small net source alone.

Stiffness arises when fast stable modes constrain an explicit time step much more strongly than the slower evolution of interest would suggest. Tightening a user tolerance does not remove stiffness. Extending the requested duration can increase the number of required steps by orders of magnitude even when the later plotted composition appears nearly flat.

A flow comparison often uses a Damköhler-type ratio $Da = \tau_{\text{flow}}/\tau_{\text{chem}}$. Large values suggest chemistry has time to relax relative to the chosen flow scale; small values suggest limited conversion. This is background interpretation, not a universal criterion computed by the homogeneous reactor. A duration is not a combustor residence time until a separate flow and mixing model justifies that correspondence.

8.7 The explicit Heun step

For $dC/dt = f(C)$ at prescribed temperature, the App forms an Euler prediction and a corrected Heun state:

$$k_1 = f(C_n), \quad C_E = C_n + \Delta t k_1,$$

$$k_2 = f(C_E), \quad C_H = C_n + \frac{\Delta t}{2}(k_1 + k_2).$$

The predictor–corrector difference supplies a local step-size indicator. The implementation scales each species difference by the largest of its old concentration, corrected concentration and an initial-total-concentration floor:

$$e = \max_i \frac{|C_{H,i} - C_{E,i}|}{\max(|C_{n,i}|, |C_{H,i}|, 10^{-10}C_{\text{tot},0})}.$$

The floor prevents vanishing trace species from requiring a meaningless relative error against zero. It does not replace their concentrations with the floor or discard their rates.

If the indicator exceeds the selected tolerance, the attempted step is reduced and retried. Nonfinite or negative predicted or corrected concentrations also cause step reduction. The code does not accept a negative concentration and then clip it to zero; clipping would alter elemental inventories and could conceal a step-size failure.

Accepted steps advance time and update the history. The next step changes within bounded growth or reduction factors. The method remains explicit; adaptivity does not turn it into an implicit stiff solver. A finite local error indicator also does not certify an equally small global error over an arbitrarily long history.

8.8 A reconstructable chemical history

Use the deliberately illustrative irreversible reaction



with pure initial H₂ at 1200 K and 80 kPa. This is an algorithm demonstration with a chosen constant rate, not a calibrated hydrogen dissociation mechanism. It differs from the default demonstration coefficient so that the example's timescale is explicit.

The initial concentration is

$$C_0 = \frac{80000}{8314.462618(1200)} = 0.0080181570 \text{ kmol/m}^3.$$

The exact species solution is

$$C_{\text{H}_2} = C_0 e^{-kt}, \quad C_{\text{H}} = 2C_0(1 - e^{-kt}).$$

Hence

$$C_{\text{tot}} = C_0(2 - e^{-kt}), \quad p = p_0(2 - e^{-kt}), \quad x_{\text{H}_2} = \frac{e^{-kt}}{2 - e^{-kt}}.$$

Time, s	H2 concentration, kmol/m ³	H mole fraction	Pressure, kPa
0	0.00801816	0	80.0000
0.10	0.00656471	0.3069059	94.5015
0.25	0.00486326	0.5647334	111.4775
0.50	0.00294972	0.7746003	130.5696

At 0.50 s, the fraction of original H₂ remaining is $e^{-1} = 0.3678794$, while its mole fraction is only 0.2253997. The distinction arises entirely

from the increase in molecular count. The hydrogen atom concentration remains $2C_{H2} + C_H = 2C_0$, and mass density is unchanged.

The pressure approaches twice its initial value as the imposed one-way reaction exhausts H₂. This asymptote is a consequence of the artificial irreversible model. A real reversible mechanism at the same temperature would generally approach a different composition and would require a thermodynamically consistent reverse rate.

For this example, the difference between an enthalpy-source sum and the rigid-vessel thermostat rate is

$$S_h - \dot{q}_V = R_u T k C_{H2} = p_0 k e^{-kt}.$$

It equals approximately 97044.9 W/m³ at 0.25 s. The difference can be computed without knowing the absolute species enthalpies, because it depends only on molecular production and temperature. The App subtracts this pressure-production term. The example demonstrates why the selected energy constraint changes a heat-load calculation even when the species history is identical.

8.9 One step explains the error indicator

For the same first-order loss, normalize the initial reactant concentration to one. A trial step of 0.10 s has $k\Delta t = 0.2$. Euler predicts

$$C_E/C_n = 1 - 0.2 = 0.8,$$

while Heun gives

$$C_H/C_n = 1 - 0.2 + \frac{0.2^2}{2} = 0.82.$$

The exact exponential factor is 0.81873075. The predictor–corrector difference is 0.02, whereas the Heun error against the exact solution is ap-

proximately 0.00126925. The indicator is therefore not numerically identical to the corrected state's true error.

Halving the step gives Euler 0.9 and Heun 0.905. Their difference falls to 0.005, one quarter of the previous difference, consistent with its leading quadratic dependence on step size. This motivates a square-root step adjustment based on the tolerance ratio. The actual multicomponent calculation takes the largest scaled difference across species and also applies positivity requirements.

These analytical values explain the integration method; they are not a claim that the App uses a fixed 0.10 s step or that every mechanism has this smooth error behavior. In a stiff region, positivity or stability may force much smaller steps.

8.10 Completion, conservation and displayed histories

The requested duration is the target end time. Completion means that accepted integration reaches that time. The App also limits accepted steps and total attempted steps; a sequence of rejected steps can exhaust the attempt budget without advancing the state appreciably. A time step too small to change the floating-point time is another stopping condition.

A returned incomplete history contains the accepted prefix and its actual final time. It must not be read as a prediction at the requested end time. Conversely, completed $\backslash$ = $\backslash$ 1 means that the integration interval was covered; it does not mean chemical equilibrium was reached. An irreversible example can complete its interval while continuing to react, and a reversible system can be near equilibrium long before the requested duration ends.

The reactor checks final mass density against its initial value and reports a relative mass residual. Balanced sources also imply elemental conservation analytically, but the displayed mass residual alone is not a separate numerical report for every element. A small conservation residual does not establish accurate reaction timing: both an inaccurate and an accurate integrator can preserve a linear invariant.

The plotted history is sampled from accepted states to keep the result manageable, with the final accepted state retained. Plot samples are therefore not a complete record of every numerical step. A straight segment joining samples does not resolve an unrecorded rapid event, and the absence of a narrow feature in the reduced plot is not proof that no such feature occurred.

8.11 Included mechanism and model boundaries

The hydrogen–oxygen option supplies the bounded 10-species, 29-reaction mechanism described in Chapter 7. The current reactor evaluates thermodynamic properties for every species in the selected database at its prescribed temperature. Thus its effective property domain can be narrower than the range of only the initially populated species: the NASA7 mechanism’s common interval is 300–3500 K, while the general NASA9 set spans 200–6000 K. These are data limits, not a claim of mechanism accuracy across the entire interval.

The reactor excludes spatial gradients, diffusion, turbulent mixing, liquid vaporization, wall chemistry and adiabatic thermal feedback. It also does not supply a general stiff integration method. The supported rates and supplied concentrations define a homogeneous kinetic task, and its result must be interpreted within that task.

A useful record retains temperature, initial pressure and composition, mechanism and thermodynamic database, duration, actual returned time, completion status and conservation diagnostic. Interpret the thermostat metric as the final-state rigid-vessel heat-transfer rate derived in Section 8.4. Preserve its sign and volumetric units; multiply by the vessel volume only when an actual total heat-transfer rate is required.

9. Turbojet design-point derivation

A turbojet converts part of the energy added by fuel into compressor work and leaves the remainder available for exhaust acceleration. The turbine is not assigned an independent useful power output in this model: its work is determined by the compressor load and shaft efficiency. The nozzle then converts the remaining total-state potential into jet velocity at a specified exit pressure.

The design-point problem supplies flight conditions, compressor pressure ratio, turbine-entry total temperature, gas properties and component loss parameters. It returns a station chain, fuel–air ratio, exhaust state and thrust normalized by incoming air flow. Optional air mass flow supplies the dimensional scale. It does not select a compressor map operating point or solve a fixed nozzle’s back-pressure response.

This chapter covers the unaugmented constant-property design-point mode. Chapter 10 adds afterburning and distinguishes variable-property cycles. Chapters 11–12 address components and reference-constrained off-design problems.

9.1 Stations and specified quantities

Use stations 2, 3, 4 and 5 for inlet exit, compressor exit, burner exit/turbine entry and turbine exit. Station 9 identifies the nozzle exit. The station table in this App reports **total** temperature and pressure, including at labels

0 and 9. Static freestream inputs and static nozzle results are separate quantities; a row labeled 9 must not be read as static exit temperature merely because it names the exit plane.

The cold side uses constant $c_{p,c}$ and γ_c ; the hot side uses constant $c_{p,h}$ and γ_h . Their gas constants are derived consistently:

$$R_c = c_{p,c} \frac{\gamma_c - 1}{\gamma_c}, \quad R_h = c_{p,h} \frac{\gamma_h - 1}{\gamma_h}.$$

These are supplied effective gas properties. The cycle does not calculate a species composition or a smooth temperature-dependent heat capacity. Its heating-value energy convention must remain separate from Chapter 5's formation-inclusive chemistry.

The exit-pressure input is

$$r_e = \frac{p_0}{p_9}.$$

This orientation matters. Values below one mean exit pressure exceeds ambient pressure; values above one mean exit pressure is below ambient. Full expansion corresponds to one. It is not the nozzle total-to-static pressure ratio.

9.2 Inlet total state

From the resolved static freestream state,

$$a_0 = \sqrt{\gamma_c R_c T_0}, \quad V_0 = M_0 a_0,$$

$$\tau_r = 1 + \frac{\gamma_c - 1}{2} M_0^2, \quad \pi_r = \tau_r^{\gamma_c/(\gamma_c - 1)}.$$

The inlet exit has

$$T_{t2} = T_0 \tau_r, \quad p_{t2} = p_0 \pi_r \pi_d.$$

Chapter 4 explains how the nonideal cycle combines the supplied inlet recovery with its empirical Mach-dependent recovery factor. The inlet preserves total temperature in this adiabatic no-work approximation, while losing total pressure. Applying recovery to both quantities would incorrectly remove energy.

The inlet calculation does not determine captured mass flow from a lip geometry. Air flow remains an optional supplied dimensional scale in this cycle. Its specific quantities can be computed without a mass-flow value.

9.3 Compressor work and polytropic efficiency

Let $\pi_c = p_{t3}/p_{t2}$. For an infinitesimal isentropic compression of a constant-property gas,

$$\frac{dT_{ts}}{T_t} = \frac{\gamma_c - 1}{\gamma_c} \frac{dp_t}{p_t}.$$

Compressor polytropic efficiency e_c compares that ideal infinitesimal temperature rise with the actual rise. Integration gives

$$\tau_c = \frac{T_{t3}}{T_{t2}} = \pi_c^{(\gamma_c - 1)/(\gamma_c e_c)}.$$

Therefore

$$T_{t3} = T_{t2} \tau_c, \quad p_{t3} = p_{t2} \pi_c, \quad w_c = c_{p,c}(T_{t3} - T_{t2}).$$

The work w_c is per kilogram of incoming air. Lower polytropic efficiency produces a greater temperature rise and more required work for the same pressure ratio.

The corresponding finite pressure-ratio isentropic efficiency is

$$\eta_c = \frac{\pi_c^{(\gamma_c-1)/\gamma_c} - 1}{\tau_c - 1}.$$

The App's supplied compressor efficiency in this cycle is polytropic, while the derived η_c is an output. They must not be interchanged. At pressure ratio one, compressor work vanishes and the finite-ratio expression becomes a zero-over-zero form; the solver omits that derived efficiency instead of reporting an arbitrary numerical value.

9.4 Burner fuel balance

Let $f = \dot{m}_f/\dot{m}_a$, and neglect fuel inlet sensible enthalpy in the heating-value model. With burner efficiency η_b and heating value Q_R , the non-ideal burner energy balance per unit air mass is

$$c_{p,c}T_{t3} + f\eta_b Q_R = (1 + f)c_{p,h}T_{t4}.$$

The factor $1 + f$ accounts for the fuel mass joining the hot stream. Solving explicitly,

$$f = \frac{c_{p,h}T_{t4} - c_{p,c}T_{t3}}{\eta_b Q_R - c_{p,h}T_{t4}}.$$

A positive numerator requires added energy relative to the compressor outlet. A positive denominator requires that the useful fuel energy exceed the prescribed outgoing enthalpy carried by each additional unit of fuel mass. The solver checks both conditions. An arbitrary turbine-entry temperature is not achievable merely because the field accepts a positive number.

The pressure closure is independently supplied:

$$p_{t4} = \pi_b p_{t3}.$$

Burner pressure loss does not determine burner efficiency or vice versa. One modifies the available expansion pressure; the other modifies fuel required for the specified temperature under the chosen energy model. Neither predicts flame stability, emissions, combustor length or mixing completeness.

Because the hot and cold heat capacities are independently fixed, comparing temperatures alone is not enough to assess the energy-rise requirement. The model compares $c_{p,h}T_{t4}$ with $c_{p,c}T_{t3}$. These effective enthalpy expressions are not a species-resolved chemistry calculation.

9.5 Turbine work closes the shaft

The turbine supplies compressor power through mechanical efficiency η_m :

$$\eta_m(1+f)c_{p,h}(T_{t4} - T_{t5}) = c_{p,c}(T_{t3} - T_{t2}).$$

Hence

$$T_{t5} = T_{t4} - \frac{c_{p,c}(T_{t3} - T_{t2})}{\eta_m(1+f)c_{p,h}}, \quad \tau_t = T_{t5}/T_{t4}.$$

The required temperature drop follows from shaft load, rather than from an independently entered turbine pressure ratio. If the load would require a nonpositive turbine-exit total temperature, the cycle is rejected.

Turbine polytropic efficiency e_t relates the actual infinitesimal temperature decrease to an isentropic decrease. Integrating gives

$$\tau_t = \pi_t^{e_t(\gamma_h-1)/\gamma_h}, \quad \pi_t = \tau_t^{\gamma_h/[e_t(\gamma_h-1)]}.$$

The pressure leaving the turbine is $p_{t5} = \pi_t p_{t4}$. For the same required work and temperature drop, lower turbine efficiency demands a larger pressure drop, leaving less expansion potential for the nozzle.

The derived finite-ratio turbine isentropic efficiency is

$$\eta_t = \frac{1 - \tau_t}{1 - \tau_t^{1/e_t}}.$$

It is distinct from the input e_t . Mechanical efficiency is also distinct: it accounts for the fraction of turbine work delivered to the compressor, rather than the thermodynamic pressure cost of producing that work.

9.6 Nozzle expansion at a supplied exit pressure

For the unaugmented cycle,

$$T_{t9} = T_{t5}, \quad p_{t9} = \pi_n p_{t5}.$$

The supplied π_n represents total-pressure loss. The remaining expansion is evaluated from total to specified static pressure. Define

$$NPR = \frac{p_{t9}}{p_9} = r_e \pi_r \pi_d \pi_c \pi_b \pi_t \pi_n.$$

A positive jet solution requires $NPR > 1$. Then

$$T_9 = T_{t9} NPR^{-(\gamma_h - 1)/\gamma_h},$$

$$M_9 = \sqrt{\frac{2}{\gamma_h - 1} [NPR^{(\gamma_h - 1)/\gamma_h} - 1]}, \quad V_9 = \sqrt{2c_{p,h}(T_{t9} - T_9)}.$$

The current cycle permits the exit Mach implied by the supplied pressure; it does not cap it at one as a fixed convergent nozzle model would. A supersonic result therefore implies a suitable expansion path within this idealized nozzle closure, not a prediction that a particular convergent nozzle can achieve it.

No exit area is independently imposed to match ambient pressure. The cycle can report an overexpanded or underexpanded pressure contribution, but it does not resolve internal shocks, flow separation or side loads. Those omissions matter when choosing a physically plausible exit-pressure input.

9.7 Thrust and fuel consumption

The uninstalled thrust balance is

$$F = \dot{m}_a[(1 + f)V_9 - V_0] + (p_9 - p_0)A_9.$$

Using continuity at the exit,

$$\frac{A_9}{\dot{m}_a} = \frac{1 + f}{\rho_9 V_9} = \frac{(1 + f)R_h T_9}{p_9 V_9},$$

so specific thrust becomes

$$F_s = \frac{F}{\dot{m}_a} = (1 + f)V_9 - V_0 + \frac{(1 + f)R_h T_9}{V_9}(1 - r_e).$$

This form eliminates an independently supplied exit area from the thrust expression. The pressure term is positive for $r_e < 1$ and negative for $r_e > 1$. At full expansion it vanishes.

Thrust-specific fuel consumption is

$$TSFC = \frac{\dot{m}_f}{F} = \frac{f}{F_s},$$

in kg/(N s). A numerical value in mg/(N s) is one million times the SI value. If air flow is supplied, $F = \dot{m}_a F_s$; scaling flow alone leaves the normalized station solution and TSFC unchanged. The App requires positive net specific thrust for this mode rather than reporting a positive fuel-consumption number for a nonpropulsive result.

At full expansion, the kinetic-energy increase per unit air flow is

$$\Delta e_k = \frac{(1 + f)V_9^2 - V_0^2}{2}.$$

The nonideal cycle can report $\eta_{th} = \Delta e_k / (fQ_R)$ and $\eta_p = V_0 F_s / \Delta e_k$. Outside full expansion, the current App omits this decomposition because it does not include a corresponding pressure-work term in the kinetic-energy denominator. It retains overall efficiency $V_0 F_s / (fQ_R)$. Chapter 1 explains the interpretation, including zero propulsive power at zero flight speed despite positive static thrust.

9.8 What the ideal setting changes

The ideal setting is a particular air-standard approximation. It uses cold-side heat capacity and gamma throughout, unit component recoveries and efficiencies, complete expansion, and neglects added fuel mass in turbine work and exhaust momentum. Fuel consumption is still estimated from the heating value:

$$f = \frac{c_p(T_{t4} - T_{t3})}{Q_R}, \quad T_{t5} = T_{t4} - (T_{t3} - T_{t2}).$$

Thus enabling ideal mode is not exactly equivalent to setting every loss parameter to one in the nonideal fuel-mass model. The approximation in $1 + f$ and the hot-gas property convention also change. The App requires $p_0/p_9 = 1$ for this ideal cycle.

Using $\tau_\lambda = T_{t4}/T_0$, the ideal full-expansion relations reduce to

$$\frac{T_9}{T_0} = \frac{\tau_\lambda}{\tau_r \tau_c}, \quad \frac{V_9^2}{2c_p T_0} = \tau_\lambda - \tau_r(\tau_c - 1) - \frac{\tau_\lambda}{\tau_r \tau_c}.$$

At fixed flight state and turbine-entry temperature, differentiating the last expression with respect to τ_c gives a stationary point at

$$\tau_c = \frac{\sqrt{\tau_\lambda}}{\tau_r}.$$

This idealized velocity optimum is meaningful only if it lies in the admissible compressor and heating domain. It is not a universal engine optimum: it neglects losses, fuel mass, component maps and other design objectives. The nonideal model must retain its full station coupling when evaluating a compressor-ratio trend.

9.9 Worked nonideal design point

Choose an original illustrative state with $M_0 = 0.8$, $T_0 = 250$ K and $p_0 = 40$ kPa. Supply $\pi_c = 8$, $T_{t4} = 1500$ K, $c_{p,c} = 1005$, $c_{p,h} = 1150$ J/(kg K), $\gamma_c = 1.4$, $\gamma_h = 1.33$, and $Q_R = 43$ MJ/kg. Use inlet recovery 0.97, burner pressure ratio 0.95, nozzle pressure ratio 0.98, polytropic efficiencies 0.90 and 0.92, burner efficiency 0.98 and mechanical efficiency 0.99. Select full expansion and an air flow of 20 kg/s.

At subsonic flight the empirical ram factor is one, so $\pi_d = 0.97$. The ram temperature ratio is 1.128, and the compressor temperature ratio is 1.93506356. Sequential evaluation gives:

Total-state station	Temperature, K	Pressure, kPa
Inlet exit 2	282.000	59.1444
Compressor exit 3	545.688	473.155
Burner exit 4	1500.000	449.497
Turbine exit 5	1273.817	219.665
Nozzle total state 9	1273.817	215.272

The burner balance gives $f = 0.0291125482$. The turbine work balance gives $\tau_t = 0.84921118$, followed by $\pi_t = 0.48869123$. These values show that much of the compressor's pressure rise is consumed by the turbine while it supplies shaft work.

The nozzle total-to-static pressure ratio is 5.38180297. Its static exit temperature is 838.974845 K, exit Mach is 1.77234908, and velocity is

1000.06822 m/s. The corresponding freestream velocity is 253.61388 m/s. Because full expansion removes the pressure term,

$$\begin{aligned} F_s &= (1.0291125482)(1000.06822) - 253.61388 \\ &= 775.56887 \text{ N s/kg.} \end{aligned}$$

The dimensional thrust is approximately 15.5114 kN and TSFC is 3.7537025×10^{-5} kg/(N s), or 37.5370 mg/(N s). These are direct arithmetic results from the stated simplified equations, not measured engine performance or a component-map design.

The station table's 1273.817 K at row 9 and the static exit value 838.975 K describe different states and must both be retained correctly. A report that replaced one with the other would obscure the nozzle energy conversion.

9.10 Interpreting trends and refusals

Increasing compressor pressure ratio raises compressor work and temperature. At a fixed turbine-entry temperature it tends to reduce required burner heat, but it also increases the turbine work demand and changes nozzle pressure potential. Specific thrust need not increase monotonically with pressure ratio. The ideal stationary point above illustrates why an apparent tradeoff exists even before losses are included.

Reducing burner or nozzle pressure recovery can reduce exhaust velocity without changing the earlier shaft temperature balance in this prescribed design-point closure. Reducing mechanical efficiency instead increases required turbine temperature drop. These distinct responses help identify which assumption drives a result.

The solver rejects nonfinite states, invalid efficiencies, incompatible burner energy, nonpositive available turbine temperature, insufficient nozzle pressure ratio and nonpositive net specific thrust. A refusal should be interpreted through the first failed balance rather than treated automatically as a missing implementation. Conversely, a numerically accepted result remains a bounded cycle estimate: it does not establish

compressor operability, turbine cooling feasibility, nozzle separation behavior, installation drag or a manufacturable engine geometry.

10. Afterburning and variable-property cycles

Afterburning adds fuel downstream of the turbine. It can increase the energy available to the nozzle without requiring that the additional heat pass through the turbine blades. It also consumes fuel, adds exhaust mass and introduces total-pressure loss. Its net effect depends on all three changes, not on afterburner temperature alone.

A second distinction concerns thermodynamics. A constant-property heating-value model prescribes a temperature and estimates fuel required to reach it. The variable-property model in ThrustOne prescribes gaseous fuel amounts and uses formation-inclusive equilibrium chemistry to obtain chamber temperature. These are different closures. A change between them is more than replacing one heat capacity with a temperature-dependent value.

This chapter develops both formulations and their current implementation boundaries. The unaugmented station chain is in Chapter 9; shared property and equilibrium relations are in Chapters 5–6. Turbofan mixing details are deferred to Chapters 13–14.

10.1 Two fuel streams, one normalization

Take one kilogram of incoming core air as the accounting basis. Let f denote main fuel mass per unit core air, and f_{AB} additional afterburner

fuel on that same basis. The mass crossing the turbine is $1 + f$. The mass leaving an active afterburner is $1 + f + f_{AB}$.

Station 6 denotes afterburner entry and station 7 its exit. In this simplified turbojet chain, station 6 receives the turbine-exit state without an additional interconnecting loss, so $T_{t6} = T_{t5}$ and $p_{t6} = p_{t5}$. The afterburner pressure ratio is

$$\pi_{AB} = p_{t7}/p_{t6}.$$

Additional fuel is not normalized by the already enlarged turbine flow. If a fuel ratio $\hat{f}_{AB}$ is instead defined per kilogram of afterburner inlet gas, conversion requires $f_{AB} = (1 + f)\hat{f}_{AB}$. Confusing the two bases changes both energy and momentum accounting.

In the prescribed design-point model, the compressor state, main fuel ratio and turbine work closure are established before afterburning. Added downstream fuel does not contribute to the mass flow extracting turbine work. This is a model assumption about an independently closed upstream chain; a real fixed-geometry engine can experience upstream changes through back-pressure and matching interactions that this calculation does not solve.

10.2 Constant-property afterburner energy balance

Let $c_{p,AB}$ be the effective exhaust heat capacity, η_{AB} the afterburner combustion efficiency and Q_R the heating value. Neglect supplied fuel sensible enthalpy as in Chapter 9. The balance is

$$(1 + f)c_{p,h}T_{t6} + f_{AB}\eta_{AB}Q_R = (1 + f + f_{AB})c_{p,AB}T_{t7}.$$

Collecting terms in the added fuel gives

$$f_{AB} = (1 + f) \frac{c_{p,AB}T_{t7} - c_{p,h}T_{t6}}{\eta_{AB}Q_R - c_{p,AB}T_{t7}}.$$

The numerator is the increase in effective specific enthalpy, multiplied by the incoming mass. The denominator accounts for the enthalpy carried out by the fuel mass itself. A positive prescribed temperature alone does not guarantee a valid denominator.

The current constant-property branch requires the afterburner outlet temperature and its effective enthalpy to be no lower than at entry. It rejects a nonpositive fuel-energy denominator. If different hot and afterburner heat capacities are supplied, the piecewise $c_p T$ convention can introduce an enthalpy-reference jump; the App warns about that limitation. It does not determine a composition that explains the change in heat capacity.

In the ideal air-standard approximation, heat capacities are the same throughout, efficiencies and recoveries are one, and fuel mass is neglected in the flow balances. The corresponding estimate is

$$f_{AB} = c_p(T_{t7} - T_{t6})/Q_R.$$

This approximation should not be confused with the nonideal fuel-mass expression evaluated at unit efficiencies.

10.3 The nozzle must receive the afterburner energy

With nozzle recovery π_n , the downstream total state should be

$$T_{t9} = T_{t7}, \quad p_{t9} = p_{t5} \pi_{AB} \pi_n.$$

For a constant-property expansion to the selected static exit pressure,

$$T_9 = T_{t7} \left(\frac{p_9}{p_{t9}} \right)^{(\gamma_{AB}-1)/\gamma_{AB}},$$

$$V_9 = \sqrt{2c_{p,AB}(T_{t7} - T_9)}.$$

The corresponding thrust per unit core air is

$$F_s = (1 + f + f_{AB})V_9 - V_0 + \frac{(1 + f + f_{AB})R_{AB}T_9}{V_9}(1 - p_0/p_9).$$

Both fuel streams contribute to consumption:

$$TSFC = (f + f_{AB})/F_s.$$

Pressure loss reduces expansion potential, while heating raises available enthalpy and added fuel increases exhaust mass. A rise in nozzle velocity is not alone a measure of fuel economy. The fully expanded efficiency qualifications of Chapter 9 remain applicable.

Keeping one consistent nozzle state

The constant-property calculation uses the afterburner outlet total temperature to recover nozzle static temperature and exhaust velocity. The same total state appears in the station chain. Afterburner pressure loss changes the expansion pressure ratio, and added fuel changes the exhaust mass factor. All three effects belong to one nozzle closure.

Using the unaugmented turbine-exit temperature with the augmented nozzle pressure and mass would create an inconsistent state. In particular, the reported exit Mach and static temperature would fail to reconstruct the station total temperature. A plausible-looking thrust value or successful calculation status would not reveal the omitted thermal energy by itself.

The dry and zero-temperature-rise limits can hide such an error. An active-afterburner example with a substantial temperature rise is therefore necessary when interpreting the energy added downstream of the turbine.

10.4 Worked afterburner balance

Extend Chapter 9's original nonideal example. Its main fuel ratio is 0.0291125482, turbine-exit total temperature is 1273.816775 K, and dry nozzle total pressure is 215.272119 kPa. Retain the hot-gas heat capacity

1150 J/(kg K) and gamma 1.33 for the afterburner. Specify $T_{t7} = 1800$ K, $\eta_{AB} = 0.96$, $\pi_{AB} = 0.94$, the same 43 MJ/kg heating value and full expansion to 40 kPa.

The energy balance gives

$$f_{AB} = 0.0158818420, \quad f + f_{AB} = 0.0449943902.$$

The new nozzle total-to-static pressure ratio is 5.05889479. Using the afterburner temperature consistently yields

$$T_9 = 1203.876692 \text{ K}, \quad V_9 = 1170.932795 \text{ m/s},$$

$$F_s = 970.004322 \text{ N s/kg}.$$

These values are direct arithmetic consequences of the stated simplified balances. Substituting the dry turbine-exit temperature would instead give approximately 851.954625 K, 985.029413 m/s and 775.736331 N s/kg for the same pressure and mass factors. That deliberately inconsistent comparison isolates the effect of omitting the afterburner heat from the nozzle-energy evaluation.

The App uses the augmented total temperature in this closure. The comparison explains why total-to-static consistency is essential: pressure recovery, fuel mass and thermal energy must all refer to the same operating condition. It is not a separate permissible approximation for an active afterburner.

10.5 Variable properties change the independent inputs

The variable-property turbojet uses dry air with mole fractions 0.79 N₂ and 0.21 O₂, and a selected gaseous fuel H₂, CH₄ or CO. Main fuel–air ratio and fuel temperature are inputs. Turbine-entry temperature becomes an equilibrium energy-closure result rather than a prescribed target.

Definition	Constant-property cycle	Variable-property cycle
Main combustion input	Target turbine-entry temperature	Gaseous fuel ratio and feed temperature
Chemical energy	Heating value and burner efficiency	Species formation-inclusive enthalpy
Compression/turbine efficiency	Polytropic input	Explicit isentropic input
Composition	Not solved	Equilibrium in chambers, frozen downstream
Exit condition	Supplied ambient/exit pressure ratio	Full expansion to ambient pressure

The variable-property input values are not direct replacements for the identically named constants in the other branch. In particular, using the same numerical efficiency does not preserve component work when one definition is polytropic and the other isentropic.

The inlet obtains total enthalpy from $h_{t0} = h_0 + V_0^2/2$, inverts it for total temperature, and obtains total pressure from entropy. It applies the supplied inlet recovery directly. It does not automatically add the constant-property turbojet's empirical Mach recovery correlation.

10.6 Compressor and main equilibrium chamber

For a supplied compressor pressure ratio, first find the isentropic outlet from

$$s(T_{t3s}, p_{t3}, x_a) = s(T_{t2}, p_{t2}, x_a).$$

Then the actual work and enthalpy are

$$w_c = \frac{h_{t3s} - h_{t2}}{\eta_c}, \quad h_{t3} = h_{t2} + w_c.$$

Temperature is recovered from $h(T)$; a local gamma power law is not substituted for this integral property calculation.

On the unit-core-air basis, the incoming air amounts are $x_{a,i}/\overline{W}_a$ kmol and the fuel amount is f/W_f kmol. The adiabatic main chamber target is

$$h_{4,\text{target}} = \frac{h_a(T_{t3}) + fh_f(T_f)}{1 + f}.$$

At $p_{t4} = \pi_b p_{t3}$, the solver finds an equilibrium composition and temperature whose mass-specific enthalpy matches that target. Air and fuel need not have the same feed temperature. Internally, the shared equilibrium helper's common-temperature reference is adjusted to reproduce the two actual feed enthalpies. That adjustment is bookkeeping, not external heating of the chamber.

There is no additional LHV term or unspecified combustion efficiency. The model assumes equilibrium of the admitted gas species and excludes vaporization and finite reaction time. A fuel-rich input does not guarantee a temperature rise proportional to added fuel; equilibrium composition and the available oxidizer determine the resulting state within the restricted species set.

10.7 Frozen turbine and pressure recovery

After main-chamber equilibrium, composition is frozen through the turbine. For a turbojet with compressor load w_c ,

$$h_{t5} = h_{t4} - \frac{w_c}{(1 + f)\eta_m}.$$

The inversion of this enthalpy gives actual T_{t5} . Turbine isentropic efficiency defines the ideal enthalpy decrease associated with the same pressure change:

$$h_{t5s} = h_{t4} - \frac{h_{t4} - h_{t5}}{\eta_t}.$$

Recover T_{t5s} , then determine pressure from equal entropy between the inlet and ideal outlet. The actual exit at that pressure has the larger entropy associated with its smaller enthalpy drop. The two temperatures are calculation states used to apply an efficiency definition, not two simultaneously existing exhaust streams.

For the single-spool turbofan extension, shaft load additionally includes fan work on core and bypass air. That normalization is developed in Chapter 13. The turbojet case has zero bypass and no fan pressure rise.

10.8 Variable-property afterburning

The optional afterburner adds a specified f_{AB} per kilogram of core air, using the same selected fuel species with its own feed temperature. The main-chamber products are carried as actual kmol per unit core air, not merely as a vector of fractions. This retains their mass $1 + f$ before new fuel is added.

At $p_{t7} = \pi_{AB} p_{t5}$, the new equilibrium target is

$$h_{7,\text{target}} = \frac{(1 + f)h_5 + f_{AB}h_{f,AB}}{1 + f + f_{AB}}.$$

The chamber solve determines T_{t7} and new composition. Neither the main shaft work nor the main fuel ratio is recomputed from the added downstream mass. Temperature-root bounds are explicit, and every trial equilibrium must converge.

When the optional afterburner has zero added fuel, the implemented branch retains the incoming temperature and composition while applying its pressure loss. It does not re-equilibrate the turbine's frozen products in this special case. Thus enabling a zero-fuel afterburner is a pressure-loss element, not necessarily a perfect no-change switch. The deliberately different zero-fuel closure should be retained when interpreting comparisons.

For positive added fuel, this model does not impose a universal temperature-rise promise. Dissociation, feed enthalpy and the chemical

inventory affect the energy root. The label afterburner describes the flowpath location and supplied fuel; it does not replace the energy calculation with a prescribed temperature rise.

10.9 Variable-property exhaust and diagnostics

The resulting composition is frozen again through the nozzle. Apply nozzle pressure recovery at unchanged total enthalpy, then solve

$$s(T_9, p_0, x_7) = s(T_{t7}, p_{t9}, x_7),$$

$$V_9 = \sqrt{2[h(T_{t7}, x_7) - h(T_9, x_7)]}.$$

For the turbojet, $F_s = (1 + f + f_{AB})V_9 - V_0$ and $TSFC = (f + f_{AB})/F_s$. Full expansion eliminates pressure thrust in this branch. It does not establish compatibility with a fixed exit area or solve a separated nozzle.

The variable-property cycle's station 9 is explicitly labeled as a **static exit state**, unlike the total-state row 9 in the constant-property cycle. Preserving these labels is essential when comparing station tables. The result also distinguishes main, afterburner and total fuel ratios; the main `fuel_air_ratio` field is not their sum.

The implementation returns main-burner, shaft and afterburner energy residuals, plus an afterburner mass residual and chamber equilibrium residuals. Energy balances on this unit-core-air basis have units J/kg of core air, not automatically watts. Multiplying by actual core flow gives a power residual. A small residual establishes closure of these equations, not independent accuracy of a real afterburner model.

No thermal or propulsive efficiency is currently reported by this variable-property cycle, even though its streams are fully expanded. The implementation has not selected a separate fuel chemical-power normalization for those efficiencies. A reader should not import a heating-value denominator silently to manufacture a field the model does not define.

10.10 Comparing models without changing the question

A useful comparison must align fuel basis, feed state, component efficiency definition, pressure losses and nozzle closure. Holding turbine-entry temperature fixed in the constant-property cycle while holding fuel ratio fixed in the variable-property cycle generally describes different experiments. One result cannot be used as a direct correction factor for the other without first matching the independent constraints.

For an energy-accounting illustration, suppose a unit-core-air main chamber receives air enthalpy 400 kJ/kg and gaseous fuel enthalpy 100 kJ/kg at $f = 0.02$. Its target is $(400 + 0.02 \times 100)/1.02 = 394.117647$ kJ/kg of products. If turbine exit enthalpy is 250 kJ/kg and the afterburner adds 0.01 kg of fuel at 100 kJ/kg, its target is $(1.02 \times 250 + 0.01 \times 100)/1.03 = 248.543689$ kJ/kg. The smaller mass-specific target does not by itself imply a colder product: composition and formation enthalpies determine the temperature corresponding to that target. These supplied enthalpies demonstrate normalization only, not an asserted physical flame state.

The present variable-property engine excludes cooling flows, condensed phases, liquid fuel preparation, finite-rate chemistry and unrestricted map matching. Mixed turbofan exhaust adds further compatibility equations described in Chapter 14. The constant-property afterburner instead retains prescribed heat capacities and an explicit heating-value balance. Keeping these boundaries visible allows each model to be interpreted at its stated level of physical detail.

11. Turbojet components and maps

An independent component calculation isolates one thermodynamic transformation. It is useful for examining a pressure loss, efficiency definition or nozzle expansion without requiring a complete engine to close. The same independence limits its meaning: a turbine result does not automatically satisfy a compressor load, and a compressor map query does not automatically match an engine operating point.

ThrustOne's turbojet component mode contains inlet, compressor, burner, turbine and nozzle calculations. Four also support frozen variable-property thermodynamics; the reacting burner uses its separate equilibrium closure instead. This chapter explains the independent unknowns and the optional compressor map. The full-cycle balances are developed in Chapter 9, and corrected-coordinate definitions in Chapter 3.

11.1 One component, one closure

The total inlet state is normally supplied directly. The component's defining ratio, temperature or efficiency then determines its outlet. For an inlet, the starting state is instead the static freestream and Mach number. A nozzle additionally needs the static pressure to which it expands.

Component	Main prescribed change	Principal result
Inlet	Flight state and recovery	Recovered total state
Compressor	Total-pressure ratio and efficiency	Temperature rise and specific work
Burner	Outlet total temperature and fuel-energy assumptions	Fuel ratio and pressure loss
Turbine	Total-temperature ratio and efficiency	Work and required pressure decrease
Nozzle	Recovery and exit static pressure	Exit temperature, velocity and Mach

Work in these independent calculations is specific energy, J/kg of the gas crossing the component. Converting it to shaft power requires a mass flow and, where applicable, a mechanical-efficiency convention. A burner adds mass; a compressor does not. Those facts must be handled explicitly when combining component results into an engine.

The common state fields `tt_in`, `pt_in`, `tt_out` and `pt_out` are total quantities. Nozzle static fields are returned separately. A zero total-temperature change across an adiabatic nozzle is consistent with a large static-temperature decrease and velocity increase.

11.2 Inlet recovery is not a compressor pressure ratio

The constant-property inlet first forms freestream total temperature and pressure from static conditions using the relations of Chapter 2. It then multiplies total pressure by the supplied recovery and empirical Mach recovery described in Chapter 4. It does not perform shaft work or increase total temperature beyond the incoming stagnation value.

A compressor pressure ratio greater than one describes pressure rise supported by work input. Inlet recovery at most one describes loss from a recoverable total state. Both may accompany a static pressure rise, but their energetic meanings differ.

The frozen variable-property inlet obtains the stagnation state from enthalpy and entropy functions. It applies the directly supplied recovery, without adding the constant-property empirical Mach factor. Entering identical recovery numbers in the two component variants does not necessarily give identical total-pressure loss at supersonic flight.

11.3 Compressor efficiency conversions

For constant heat capacities, define $a = (\gamma - 1)/\gamma$. A compressor with pressure ratio π_c and polytropic efficiency e_c has

$$\tau_c = \pi_c^{a/e_c}, \quad \eta_c = \frac{\pi_c^a - 1}{\pi_c^{a/e_c} - 1}.$$

The isentropic efficiency compares the finite ideal and actual temperature rises. Conversely, with a known isentropic efficiency,

$$\tau_c = 1 + \frac{\pi_c^a - 1}{\eta_c}, \quad w_c = c_p T_{t,in} (\tau_c - 1).$$

These expressions are two consistent routes to the same state only when the efficiency conversion is performed at the same pressure ratio. Using an isentropic efficiency converted at one ratio and applying it at another is a distinct empirical model. That distinction becomes important in the map branch below.

At pressure ratio one without a map, the implementation returns unchanged total state and zero work. It omits an undefined finite-rise efficiency. Values slightly above one produce small work, and the code uses exponential-difference functions to avoid subtracting nearly equal powers unnecessarily.

In the variable-property component, efficiency is explicitly isentropic. Find the entropy-matched outlet at the specified pressure, divide its enthalpy rise by efficiency, and invert the resulting actual enthalpy for temperature. A single locally evaluated gamma cannot replace this procedure over a finite temperature change.

11.4 Burner and turbine subproblems

The constant-property burner uses

$$f = \frac{c_{p,h}T_{t,out} - c_{p,c}T_{t,in}}{\eta_b Q_R - c_{p,h}T_{t,out}}, \quad p_{t,out} = \pi_b p_{t,in}.$$

The component requires an outlet temperature above inlet temperature, a positive denominator and a nonnegative resulting fuel ratio. Its heat_release is $f\eta_b Q_R$ per unit incoming air mass. It is not a heat-transfer rate in watts without a flow scale, nor is it a finite-rate chemical source.

A frozen variable-property burner is deliberately not supplied: frozen composition cannot describe combustion. The gas-phase energy closure in Chapter 6 is the appropriate supported reacting problem. Merely changing heat capacity in the burner subproblem does not silently activate that chemistry.

The independent turbine takes an actual temperature ratio τ_t and polytropic efficiency e_t . It returns

$$\pi_t = \tau_t^{\gamma/[e_t(\gamma-1)]}, \quad w_t = c_p T_{t,in} (1 - \tau_t),$$

$$\eta_t = \frac{1 - \tau_t}{1 - \tau_t^{1/e_t}}.$$

Unlike the full engine, this subproblem does not obtain τ_t from compressor work. The supplied ratio fixes work, and efficiency determines its pressure cost. At $\tau_t = 1$, the total state is unchanged and zero work is returned.

For frozen variable properties, the actual enthalpy decrease follows from the supplied temperature ratio. Dividing it by isentropic efficiency gives the ideal decrease associated with the required pressure ratio. Entropy then determines that pressure. This is the inverse component problem developed in Chapter 10, without an engine shaft-load constraint.

11.5 Nozzle criticality and the missing geometry constraint

The independent constant-property nozzle uses

$$p_{t,out} = \pi_n p_{t,in}, \quad p_9 = p_0 / r_e,$$

where $r_e = p_0 / p_9$ is the supplied static-pressure ratio. From $NPR = p_{t,out} / p_9 > 1$, it computes static temperature, Mach and velocity. The critical total-to-static pressure ratio is

$$NPR_* = \left(\frac{\gamma + 1}{2} \right)^{\gamma/(\gamma-1)}.$$

The output `is_choked` indicates whether the specified ratio reaches this critical threshold. It does not impose a fixed convergent exit at Mach one. The solver continues to return the Mach number implied by the supplied static pressure, including supersonic values, and does not size throat or exit area in this component result.

A user must therefore distinguish a criticality indicator from a geometric boundary condition. A convergent nozzle with a fixed throat/exit would require a different closure after choking. An internal shock or separation pattern also requires information absent from this component problem.

The frozen nozzle component solves entropy at the requested static pressure and obtains velocity from enthalpy drop. Its boundary permits equal exit static and available total pressure, yielding zero speed. The constant-property variant rejects $NPR \leq 1$. This difference in zero-flow boundary handling is part of current behavior, not evidence that one branch predicts a flowing jet at equal pressures.

11.6 What the compressor map contains

The optional map stores a rectangular grid. Rows correspond to normalized corrected speed and columns to an auxiliary coordinate β . At each

point it contains normalized corrected flow, a pressure-ratio multiplier and an efficiency multiplier. All are dimensionless; they are not raw rpm, kg/s or an independently complete compressor specification.

Chapter 3 explains how physical corrected flow and speed require explicit reference conventions. The map record itself does not supply dimensional reference flow or temperature and pressure metadata. A map value of one therefore means a normalization point, not one kilogram per second.

The β coordinate parametrizes a line through the tabulated surface. Its numerical value does not directly mean flow, pressure ratio or surge margin. The user supplies speed and beta to select a point; the tables provide the associated three outputs.

The implementation accepts only compressor maps, with at least two points on each strictly increasing finite axis and matching positive tables. It checks dimensions and finite values, not whether the data came from a physical compressor test. A synthetic demonstration map is available, but its smooth appearance is not experimental validation.

11.7 How a map point enters the component balance

Let the baseline pressure ratio be $\pi_{c,b}$, baseline polytropic efficiency $e_{c,b}$, and sampled map multipliers r_π and r_η . The current component first converts the baseline efficiency at the baseline pressure ratio:

$$\eta_{c,b} = \frac{\pi_{c,b}^a - 1}{\pi_{c,b}^{a/e_{c,b}} - 1}.$$

It then forms

$$\pi_c = \pi_{c,b} r_\pi, \quad \eta_c = \eta_{c,b} r_\eta,$$

and calculates

$$\tau_c = 1 + \frac{\pi_c^a - 1}{\eta_c}.$$

This scales the baseline **isentropic** efficiency, not the baseline polytropic efficiency. Recomputing an unscaled polytropic law at the new pressure ratio would describe a different model. The returned baseline `e_c` value must not be read as proof that the final mapped state follows that same polytropic efficiency.

The mapped pressure ratio must exceed one and the resulting efficiency must lie in $(0, 1]$. Map multipliers can exceed one, but the final physical efficiency cannot. The map's corrected-flow ratio is reported for interpretation; it is not automatically converted to actual mass flow or used to close an engine shaft balance.

Frozen variable-property compression rejects an attached compressor map. Its explicit isentropic component closure and the current constant-property map scaling remain separate supported paths.

11.8 Shape-preserving interpolation

For adjacent axis points x_i, x_{i+1} , define $h_i = x_{i+1} - x_i$, $t = (x - x_i)/h_i$, and secant slopes $d_i = (y_{i+1} - y_i)/h_i$. A cubic Hermite interval is

$$\begin{aligned} y(x) = & (2t^3 - 3t^2 + 1)y_i + (t^3 - 2t^2 + t)h_i m_i \\ & + (-2t^3 + 3t^2)y_{i+1} + (t^3 - t^2)h_i m_{i+1}. \end{aligned}$$

The endpoint derivatives m_i determine whether a smooth cubic introduces artificial overshoot. The App sets an interior derivative to zero when adjacent secants change sign or include a zero slope. When they share a sign, it uses a weighted harmonic mean:

$$m_i = \frac{w_1 + w_2}{w_1/d_{i-1} + w_2/d_i}, \quad w_1 = 2h_i + h_{i-1}, \quad w_2 = h_i + 2h_{i-1}.$$

Endpoint slopes use a one-sided estimate limited according to neighboring secants. With only two points, both derivatives equal the secant and the interpolant reduces to a straight line.

The map samples each speed row along beta first, then interpolates those sampled values across speed. This is a specified two-pass operation. It is not a global fitted compressor surface, and the order should not be silently exchanged when reconstructing results. Shape preservation reduces artificial wiggles relative to an unrestricted cubic; it does not supply missing data or prove aerodynamic behavior between widely separated samples.

Every query must lie within both axis bounds. The App rejects exterior points instead of clamping or extrapolating. Clamping would return a boundary value for a different requested operating point, while extrapolation would introduce behavior unsupported by the stored data.

11.9 Inverse lookup and surge interpretation

The supported inverse operation supplies normalized speed and a target **map pressure multiplier**, then seeks beta. It is not an inverse for dimensional corrected flow. Every stored pressure row must be strictly monotone in the same direction before this operation is admitted. A row with a plateau, a turning point or opposite direction causes rejection, with direct beta selection required instead.

The inverse brackets beta on its stored interval, checks the target against the forward endpoint values and bisection using the same two-pass forward interpolator. A returned point must meet the relative pressure-multiplier residual. This is a deliberately restricted inverse; it does not discover and select among arbitrary multiple compressor branches.

An optional surge-pressure curve provides the indicator

$$S_{\pi} = \frac{r_{\pi, \text{surge}}}{r_{\pi, \text{operating}}} - 1.$$

The design pressure scale cancels in this ratio. This quantity compares pressure ratios at the selected speed. It contains no additional corrected-

flow factor and is not a universal compressor surge-margin definition. A positive value does not independently establish dynamic stability, and a negative value is not an integrated surge simulation. The code reports this indicator when the curve exists; it does not use it as a complete engine operability constraint.

11.10 Worked mapped compressor

Use an illustrative baseline pressure ratio 6, polytropic efficiency 0.88, gamma 1.4, heat capacity 1005 J/(kg K), inlet total temperature 320 K and inlet total pressure 100 kPa. Suppose a supplied map point gives pressure multiplier 0.95, efficiency multiplier 0.98 and normalized corrected-flow value 0.90. Let the associated surge-pressure multiplier be 1.12.

Baseline conversion gives $\eta_{c,b} = 0.8471282884$. The mapped state therefore uses

$$\pi_c = 5.7, \quad \eta_c = 0.8301857226.$$

Its actual total temperature is

$$T_{t,out} = 320 \left[1 + \frac{5.7^{0.4/1.4} - 1}{0.8301857226} \right] = 568.324713 \text{ K}.$$

The outlet pressure is 570 kPa, and work is

$$w_c = 1005(568.324713 - 320) = 249566.336 \text{ J/kg}.$$

The pressure-only surge indicator is $1.12/0.95 - 1 = 0.17894737$, or about 17.89%. The flow value 0.90 remains a dimensionless map result. It does not make the compressor power 0.90 times its specific work without a dimensional flow scale.

These sampled map values are illustrative inputs, not a claim about a tested machine or the built-in synthetic map. The example isolates

the baseline-efficiency conversion and shows why scaling polytropic efficiency directly would produce a different temperature.

11.11 A small interpolation and inverse example

Consider two speed rows at 0.8 and 1.0, with beta endpoints 0 and 1. Let their pressure multipliers be respectively $[0.9, 0.7]$ and $[1.1, 0.9]$. Because both axes have only two points, the interpolation is linear in each pass.

At beta 0.25, the row values are 0.85 and 1.05. At speed 0.9, the second pass gives 0.95. Reversing the same calculation at speed 0.9 with target multiplier 0.95 returns beta 0.25. Both rows decrease strictly, satisfying the inverse admission rule.

A query at speed 1.1 is outside the data and is rejected. Adding an interior point that makes either pressure row turn upward would still allow a direct forward query within the grid, but would disallow the restricted inverse. This distinction separates interpolation of supplied data from an unambiguous inverse operating-point selection.

11.12 From components to an engine

Independent component results can support a hand-assembled station chain when mass bases, gas properties, efficiency definitions and static/total states are compatible. They do not themselves guarantee that one component supplies exactly the work another consumes, that nozzle area passes the resulting flow, or that a compressor map point matches shaft speed and turbine capacity.

The next chapter adds particular reference-based off-design closures. Those constraints must be read explicitly rather than inferred from the presence of a map file. A clear component calculation is valuable because it reveals each assumption separately; combining its outputs without the missing conservation and matching equations would conceal that advantage.

12. Turbojet off-design closure

A design-point calculation specifies a compressor pressure ratio. An off-design calculation asks how that ratio and the flow change when an already defined engine operates at a different flight condition or throttle setting. Answering the second question requires additional assumptions about the engine. A reference point alone is not enough unless a rule connects that point to the new condition.

The current turbojet off-design modes use reference-normalized relations associated with fixed turbine areas and choked flow. They hold turbine ratios fixed and derive compressor behavior from a normalized thermal loading. They do not match arbitrary compressor and turbine maps or integrate shaft dynamics. This chapter separates those prescribed relations from general off-design physics and derives the associated exit-area diagnostic.

12.1 Reference state versus operating state

Use subscript R for the reference condition. The single-spool reference stores ambient state, inlet ratios, compressor temperature and pressure ratios, compressor isentropic efficiency, turbine ratios, nozzle pressure ratio and air flow. The operating inputs supply current ambient conditions, Mach number, turbine-entry temperature and ambient-to-exit static-pressure ratio.

The reference is an engine definition retained across operating cases. Replacing it with each new operating state would erase the comparison being made. Conversely, changing ambient pressure in the current case does not change the saved reference pressure.

The current compressor pressure ratio is a result of the reference relation. A positive value in the generic design-input field does not independently prescribe that result in this mode. Likewise, an optional current air-flow entry is not the mass-flow closure: air flow is obtained from the reference scaling. Inputs retained for shared data structures should not be mistaken for independent degrees of freedom in every solver.

A consistent reference satisfies its own temperature/pressure/efficiency relations. For example,

$$\pi_{cR} = [1 + \eta_{cR}(\tau_{cR} - 1)]^{\gamma_c/(\gamma_c - 1)}.$$

Entering independently rounded or incompatible reference quantities can prevent exact recovery of the intended reference operating point. The App can construct a reference from a nonideal design calculation and a supplied air-flow scale. That helper checks compressor work and a critical nozzle-pressure threshold; it does not independently simulate every assumed choked station.

12.2 Why choking enables reference scaling

For a constant-property gas at a choked section,

$$\dot{m} = A_* \frac{p_t}{\sqrt{T_t}} \sqrt{\frac{\gamma}{R}} \left(\frac{2}{\gamma + 1} \right)^{(\gamma+1)/[2(\gamma-1)]}.$$

At fixed area and gas properties, the combination $\dot{m} \sqrt{T_t}/p_t$ is constant. Referencing removes the unknown dimensional area and retains a ratio between operating states.

The single-spool model assumes choking at turbine entry and nozzle throat, conventionally stations 4 and 8, with fixed turbine temperature and pressure ratios. This allows a closed compressor operating relation and a mass-flow scaling. It is a restricted family of operating states, not proof that choking persists under every supplied throttle or back pressure.

The physical turbine-entry mass flow is $(1 + f)\dot{m}_a$. An exact comparison with fixed area and composition would retain the changing fuel-mass factor along with burner loss and gas-property factors. The implemented air-flow relation adopts the reference approximation in which those additional factors are treated as constant. Its use of the newly computed fuel ratio in thrust does not make the flow-scaling approximation exact.

12.3 Thermal loading determines compressor rise

Define the thermal-loading parameter

$$\theta = \frac{T_{t4}}{T_{t2}}, \quad \theta_R = \frac{T_{t4R}}{T_{t2R}}, \quad T_{t2} = T_0 \tau_r.$$

The single-spool operating relation is

$$\tau_c - 1 = (\tau_{cR} - 1) \frac{\theta}{\theta_R}.$$

It follows that the compressor temperature rise scales with turbine-entry temperature under the fixed-ratio reference approximation. At higher inlet total temperature with unchanged turbine-entry temperature, the available normalized loading is reduced.

The reference isentropic efficiency then gives

$$\pi_c = [1 + \eta_{cR}(\tau_c - 1)]^{\gamma_c/(\gamma_c - 1)}.$$

There is no map beta search here. The resulting compressor line is specified by the reference relation itself. Chapter 11's independent map capability does not imply that this off-design solver performs map matching.

Once $T_{t3} = T_{t2}\tau_c$ is known, main fuel follows the nonideal heating-value balance of Chapter 9. The turbine instead retains

$$T_{t5} = T_{t4}\tau_{tR}, \quad p_{t5} = p_{t4}\pi_{tR}.$$

It does not recompute turbine ratios from a fresh exact shaft-work equation using every changed fuel fraction or gas property. Changes outside the assumptions of the reference family should therefore not be interpreted as generally matched engine states.

12.4 Air flow, corrected flow and speed

The implemented air-flow scaling is

$$\frac{\dot{m}_a}{\dot{m}_{aR}} = \frac{p_0\pi_r\pi_d\pi_c}{p_{0R}\pi_{rR}\pi_{dR}\pi_{cR}} \sqrt{\frac{T_{t4R}}{T_{t4}}}.$$

At otherwise identical dimensionless conditions, decreasing ambient pressure decreases dimensional mass flow and thrust while leaving specific thrust unchanged. This is one reason that a specific-performance plot cannot replace an absolute-thrust plot.

The reported corrected-flow ratio is

$$\frac{\dot{m}_c}{\dot{m}_{cR}} = \frac{\pi_c}{\pi_{cR}} \sqrt{\frac{\theta_R}{\theta}}.$$

It is a reference-normalized result, not a dimensional corrected flow in kg/s. The mechanical speed ratio is

$$\frac{N}{N_R} = \sqrt{\frac{T_{t2}}{T_{t2R}} \frac{\pi_c^{(\gamma_c-1)/\gamma_c} - 1}{\pi_{cR}^{(\gamma_c-1)/\gamma_c} - 1}}.$$

The square root is essential: the quantity inside it scales a work-like term with speed squared. This mechanical speed ratio is not the corrected-speed map coordinate discussed in Chapter 3. To form a corrected-speed ratio at the same inlet reference convention, an additional temperature correction is required.

The aliases `speed_ratio`, `n_nr` and `N_NR` refer to the same mechanical ratio. Multiple output names do not represent separate speed definitions.

12.5 Nozzle and admissibility

Available nozzle pressure ratio is

$$NPR = (p_0/p_9)\pi_r\pi_d\pi_c\pi_b\pi_{tR}\pi_n.$$

The exit velocity and pressure thrust are calculated from the specified exit static pressure using Chapter 9's relations. The model does not solve a fixed exit-area back-pressure problem; its exit area is intended to vary consistently with the requested expansion while throat area remains fixed.

The current single-spool solver checks $NPR > 1$ for a jet, but does not require the current NPR to exceed the critical choking ratio. Thus an accepted low-NPR result does not establish the choking assumption on which its reference model relies. Check that assumption when interpreting operating points. The dual-spool branch, described below, explicitly rejects an unchoked nozzle condition.

Other refusals include invalid reference values, insufficient burner energy, nonpositive computed compressor rise, nonpositive thrust and nonfinite results. Such numerical guards cannot prove that the fixed-ratio turbine approximation remains valid under all changed inputs. In particular, independently varying gas properties or loss factors can invalidate cancellations used to obtain the reference scaling.

12.6 Exit-area reconstruction

For an isentropic constant-property nozzle with fixed throat area, the mass-flow parameter gives, up to factors constant between operating and reference states,

$$\frac{A_9}{A_*} \propto \frac{NPR^{(\gamma+1)/(2\gamma)}}{\sqrt{NPR^{(\gamma-1)/\gamma} - 1}}.$$

Therefore the consistent exit-area ratio is

$$\frac{A_9}{A_{9R}} = \left(\frac{NPR}{NPR_R} \right)^{(\gamma+1)/(2\gamma)} \sqrt{\frac{NPR_R^{(\gamma-1)/\gamma} - 1}{NPR^{(\gamma-1)/\gamma} - 1}}.$$

The `area_ratio_9` diagnostic uses the pressure-ratio orientation shown above. Both factors equal one at the reference state. That normalization alone is not enough to establish the formula away from the reference: reversing the first pressure ratio would still reproduce the reference area while producing incorrect off-reference geometry.

The area diagnostic is calculated after thrust and flow and does not feed them back into the solver. It reconstructs the exit area implied by the fixed-throat, isentropic nozzle relation; it does not introduce a geometric matching calculation or validate a real nozzle under separation, shocks or unchoked operation.

12.7 Worked normalized throttle change

Consider an illustrative reference inlet total temperature of 282 K, turbine-entry temperature 1500 K, compressor temperature ratio 1.8 and reference isentropic compressor efficiency 0.85, with gamma 1.4. Choose a consistent reference pressure ratio from the compressor relation:

$$\pi_{cR} = [1 + 0.85(1.8 - 1)]^{3.5} = 6.14585750.$$

Reduce turbine-entry temperature to 1350 K at unchanged inlet total temperature and unchanged inlet pressure and recovery factors. The loading ratio is 0.9, giving

$$\tau_c = 1 + 0.8(0.9) = 1.72, \quad T_{t3} = 282(1.72) = 485.04 \text{ K},$$

$$\pi_c = [1 + 0.85(0.72)]^{3.5} = 5.31835877.$$

The predicted air-flow ratio is

$$\frac{\dot{m}_a}{\dot{m}_{aR}} = \frac{5.31835877}{6.14585750} \frac{1}{\sqrt{0.9}} = 0.91216602.$$

The mechanical speed ratio is $\sqrt{0.9} = 0.94868330$. Flow and speed do not fall in the same proportion. The flow relation combines reduced compressor pressure with the square-root turbine-temperature factor; the speed relation follows the compressor work scaling.

These numbers demonstrate the reference equations only. A complete operating point additionally needs turbine ratios, burner assumptions and a nozzle condition that remains inside the choking and thrust domain. No dimensional thrust or physical engine performance is inferred from this partial reference specification.

12.8 Two spools require two reference rises

The dual-spool mode separates low- and high-pressure compressor temperature ratios, τ_{cL} and τ_{cH} . Define

$$\Theta = \frac{T_{t4}/T_0}{T_{t4R}/T_{0R}}.$$

The implemented relations are

$$\tau_{cL} = 1 + \Theta \frac{\tau_{rR}}{\tau_r} (\tau_{cLR} - 1),$$

$$\tau_{cH} = 1 + \Theta \frac{\tau_{rR} \tau_{cLR}}{\tau_r \tau_{cL}} (\tau_{cHR} - 1).$$

The second compressor receives the first compressor's outlet. Its normalization must therefore include τ_{cL} ; treating both compressors as though they had the same inlet temperature would alter high-spool work.

Separate supplied isentropic efficiencies convert each temperature ratio into pressure ratio. Air flow scales with the product $\pi_{cL} \pi_{cH}$, the actual inlet pressure factors and the turbine-temperature square root. The reference denominator uses its stored recovery. A consistent interpretation requires that stored reference factor represent the intended reference inlet convention, especially when preparing a nonzero-Mach reference.

High- and low-turbine temperature and pressure ratios are explicit fixed inputs. Their product determines the turbine-exit state. The code does not solve new independent turbine maps or integrate two shaft accelerations. The physical rationale involves spool work and choked capacities, but the implementation uses the resulting reference relations rather than an unrestricted coupled matching solve.

12.9 Dual-spool speed and afterburner behavior

The low-spool mechanical speed ratio is formed from the low-compressor pressure-work ratio and its inlet total temperature. The high-spool ratio uses the high-compressor inlet temperature, which includes the low-compressor temperature rise:

$$\frac{N_H}{N_{HR}} = \sqrt{\frac{T_0 \tau_r \tau_{cL}}{T_{0R} \tau_{rR} \tau_{cLR}} \frac{\pi_{cH}^{(\gamma_c-1)/\gamma_c} - 1}{\pi_{cHR}^{(\gamma_c-1)/\gamma_c} - 1}}.$$

This extra temperature factor is required even though both shafts belong to the same engine. Neither speed ratio alone determines corrected-flow

matching on a component map.

Optional afterburning uses the main-fuel-plus-air mass entering the chamber and adds its own fuel mass and pressure loss. The dual-spool implementation uses afterburner temperature in its nozzle static-temperature expression. The constant-property design-point path likewise uses its augmented nozzle total temperature. Similar feature names still do not imply identical matching assumptions or numerical paths.

The dual-spool warning explicitly assumes choking at stations 4, 4.5 and 8 and variable nozzle area to isolate downstream afterburning from turbine matching. It checks the current nozzle's critical pressure ratio. It does not solve internal turbine throat states to demonstrate all of those assumptions independently.

For unchanged upstream inputs, enabling afterburning leaves reference-derived compressor ratios, core flow and shaft speed ratios unchanged, while changing nozzle pressure, temperature, exhaust mass and thrust. That is the intended variable-area reference model; it is not a prediction of what happens when fuel is added to an arbitrary fixed nozzle.

12.10 Reading a reference-family result

Both off-design modes produce a restricted steady operating relation. They omit compressor-map matching, surge dynamics, spool transients, cooling-flow schedules and general unchoked operation. The single-spool implementation additionally requires care because a simple positive-NPR guard does not enforce its choking assumption, so users must retain the stated choked-flow domain when interpreting its reconstructed exit area.

A useful report retains reference identity, current flight state and throttle, fixed turbine ratios, pressure-loss assumptions and nozzle setting. Interpret mechanical speed, corrected flow and dimensional flow separately. Do not substitute one for another because all three are expressed as ratios near one.

Preserving those distinctions makes reference normalization useful for

controlled comparisons. It also identifies where a broader engine model would need new equations or data instead of silently extending the present approximation outside its declared domain.

13. Separate-flow turbofans

A separate-flow turbofan divides incoming air between a heated core and a bypass stream. The turbine supplies work to the compression system, including the work used to accelerate bypass air through the fan. The two streams leave through separate nozzles, and their momentum and pressure contributions combine in the net thrust.

The additional stream introduces two common accounting errors: applying fuel ratio to total air when it is defined on core air, and multiplying a total-air specific thrust by core flow. It also creates a thermodynamic tradeoff. Increasing bypass flow can spread momentum addition over more air, but the fan work must still come from the core turbine.

This chapter covers the constant-property single-spool separate design mode and the separate-exhaust branch of the variable-property single-spool mode. Their compressor pressure ratios have different inlet references. A physically consistent comparison must resolve that difference before numerical inputs are reused.

13.1 Core, bypass and fuel mass

Define bypass ratio and main fuel ratio by

$$\alpha = \frac{\dot{m}_b}{\dot{m}_c}, \quad f = \frac{\dot{m}_f}{\dot{m}_c},$$

where $\dot{m}_c$ is core **air** flow and $\dot{m}_b$ bypass air flow. Total incoming air is

$$\dot{m}_a = (1 + \alpha)\dot{m}_c.$$

Without afterburning, the core nozzle carries $(1 + f)\dot{m}_c$, and the bypass nozzle carries $\alpha\dot{m}_c$. Fuel does not belong in the definition of bypass ratio.

Normalize first to one kilogram of core air. The incoming air inventory is then $1 + \alpha$, fuel is f , core exhaust is $1 + f$, and bypass exhaust is α . This basis makes shaft work and burner energy particularly clear. Divide by $1 + \alpha$ only when forming a quantity explicitly defined per unit total inlet air.

Separate exhaust means the streams need not have equal exit temperatures, velocities or static pressures. They share the ambient reference pressure used for net pressure thrust, but their two nozzle expansions are independently specified in the constant-property design model.

13.2 Constant-property compression convention

The constant-property mode defines core compressor pressure ratio as p_{t3}/p_{t2} , from inlet exit directly to core burner entry. The fan ratio defines p_{t13}/p_{t2} for the bypass path. It does not multiply the fan ratio into the supplied core ratio a second time.

Using the inlet total state from Chapter 4,

$$T_{t3} = T_{t2}\tau_c, \quad T_{t13} = T_{t2}\tau_f,$$

$$\tau_c = \pi_c^{(\gamma_c-1)/(\gamma_c e_c)}, \quad \tau_f = \pi_f^{(\gamma_c-1)/(\gamma_c e_f)}.$$

The input efficiencies here are polytropic. Derived isentropic efficiencies follow the finite-rise conversion in Chapter 11. Fan and core compressor are represented as effective compression paths; the model does not require a particular physical split between fan root and additional core stages.

The total compression work per kilogram of core air is

$$w_{\text{load}} = c_{p,c}(T_{t3} - T_{t2}) + \alpha c_{p,c}(T_{t13} - T_{t2}).$$

The first term already includes the full core compression from station 2. Adding another fan-work term for core air would count part of that path twice under this convention.

13.3 Burner and single-shaft turbine balance

Only core air enters the main burner. The nonideal fuel balance is

$$c_{p,c}T_{t3} + f\eta_b Q_R = (1 + f)c_{p,h}T_{t4},$$

which gives

$$f = \frac{c_{p,h}T_{t4} - c_{p,c}T_{t3}}{\eta_b Q_R - c_{p,h}T_{t4}}.$$

At prescribed core pressure ratio and turbine-entry temperature, this expression contains no bypass ratio. Changing bypass ratio alone therefore does not change main fuel ratio in this specific design closure. It does change how much turbine work must be extracted after combustion.

The shaft balance is

$$\eta_m(1 + f)c_{p,h}(T_{t4} - T_{t5}) = w_{\text{load}}.$$

In temperature-ratio notation, with $\tau_\lambda = c_{p,h}T_{t4}/(c_{p,c}T_0)$,

$$\tau_t = 1 - \frac{\tau_r[(\tau_c - 1) + \alpha(\tau_f - 1)]}{\eta_m(1 + f)\tau_\lambda}.$$

Turbine polytropic efficiency then determines π_t from τ_t . More bypass work lowers turbine-exit total temperature and pressure at unchanged

main-burner conditions. Eventually the chosen fan demand can leave insufficient energy or pressure for a usable core nozzle.

The solver rejects nonpositive turbine temperature ratio and nonpositive total thrust. These guards identify incompatible prescribed operating points; they do not optimize fan pressure ratio or bypass ratio automatically.

13.4 Two nozzle calculations

The constant-property core nozzle receives

$$T_{t9} = T_{t4}\tau_t, \quad p_{t9} = p_{t2}\pi_c\pi_b\pi_t\pi_{n,c}.$$

The bypass nozzle receives

$$T_{t19} = T_{t2}\tau_f, \quad p_{t19} = p_{t2}\pi_f\pi_{n,b}.$$

Each has its own prescribed ambient-to-exit static-pressure ratio. Its exit Mach, temperature and velocity follow the total-to-static equations in Chapter 9, with hot-gas properties for the core and cold-air properties for bypass flow.

These are pressure-specified expansions, not fixed-area convergent-nozzle solutions. A Mach number above one does not establish that a particular nozzle geometry supports it. Separation, internal shocks and installed nacelle forces remain outside the calculation.

The current constant-property solver evaluates both nozzle pressure conditions even when bypass ratio is zero. Thus an otherwise valid core calculation can still be rejected by an invalid unused bypass-nozzle setting. The zero-bypass limit of the governing thrust and work equations exists, but the present input guards do not fully remove that inactive calculation. Retain a valid bypass setting when using this limit; do not infer an aerodynamic failure of the core from that rejection.

13.5 Combining thrust on a consistent basis

For the two nozzles,

$$F = \dot{m}_c[(1+f)V_9 - V_0] + (p_9 - p_0)A_9 + \alpha\dot{m}_c(V_{19} - V_0) + (p_{19} - p_0)A_{19}.$$

Eliminating areas with each stream's continuity equation gives a core contribution per unit core air,

$$F_{s,c} = (1+f)V_9 - V_0 + \frac{(1+f)R_h T_9}{V_9}(1 - p_0/p_9),$$

and a bypass contribution per unit bypass air,

$$F_{s,b} = V_{19} - V_0 + \frac{R_c T_{19}}{V_{19}}(1 - p_0/p_{19}).$$

The total result on the core-air basis is $F/\dot{m}_c = F_{s,c} + \alpha F_{s,b}$. The App's principal specific thrust uses total air:

$$F_s = \frac{F_{s,c} + \alpha F_{s,b}}{1 + \alpha}.$$

Fuel consumption follows

$$TSFC = \frac{f}{F_{s,c} + \alpha F_{s,b}} = \frac{f}{(1 + \alpha)F_s}.$$

The constant-property mode's optional dimensional flow input is core air flow. Therefore net thrust is $\dot{m}_c[F_{s,c} + \alpha F_{s,b}]$. Its reported total mass flow is incoming air flow, not total exhaust including fuel. These definitions should accompany exported values.

13.6 Why bypass flow can improve momentum efficiency

For a single fully expanded stream without added mass, thrust is $\dot{m}(V_e - V_0)$, while kinetic-energy increase is $\dot{m}(V_e^2 - V_0^2)/2$. At a fixed thrust, accelerating more mass through a smaller velocity increment can reduce the required kinetic-energy increase. This is the basic motivation for bypass propulsion.

The argument does not mean that arbitrarily large bypass ratio always improves an engine. The fan requires work, inlet and nozzle losses consume pressure potential, and a real installation has size, drag and mass consequences. Only the first two effects are represented in this simplified cycle. The final design tradeoff cannot be inferred from a total-air specific-thrust value alone.

For two fully expanded streams, kinetic power per unit core air is

$$\Delta e_k = \frac{(1 + f)V_9^2 + \alpha V_{19}^2 - (1 + \alpha)V_0^2}{2}.$$

The constant-property model uses it for thermal and propulsive efficiencies. If either active stream is not fully expanded, the App omits that kinetic decomposition and retains overall thrust-power efficiency. A zero-mass bypass stream is excluded from the full-expansion efficiency condition, even though its nozzle calculation is still evaluated as noted above.

13.7 Worked separate-flow cycle

Choose an illustrative nonideal case: $M_0 = 0.8$, $T_0 = 250$ K, bypass ratio 3, fan ratio 1.5, core ratio 10 and turbine-entry temperature 1500 K. Use $c_{p,c} = 1005$, $c_{p,h} = 1150$ J/(kg K), gamma 1.4/1.33, heating value 43 MJ/kg, inlet recovery 0.97, burner pressure ratio 0.95 and both nozzle recoveries 0.98. Compressor and fan polytropic efficiencies are 0.90, turbine polytropic efficiency is 0.92, burner efficiency 0.98 and mechanical efficiency 0.99. Both streams expand to ambient pressure.

The inlet total temperature is 282 K. Direct evaluation gives:

Quantity	Value
Core compressor outlet total temperature	585.746127 K
Bypass fan outlet total temperature	320.738501 K
Main fuel/core-air ratio	0.028116421
Turbine outlet total temperature	1139.421071 K
Core nozzle pressure ratio	4.12771119
Bypass nozzle pressure ratio	2.17355642
Core exit velocity	881.574083 m/s
Bypass exit velocity	358.124321 m/s

Freestream speed is 253.613880 m/s. The core contributes 652.746911 N s/kg of core air, and bypass contributes 104.510442 N s/kg of bypass air. Therefore

$$F/\dot{m}_c = 652.746911 + 3(104.510442) = 966.278236 \text{ N s/kg.}$$

Total-air specific thrust is 241.569559 N s/kg. With core air flow 10 kg/s, total air flow is 40 kg/s, thrust is 9.662782 kN, and fuel flow is approximately 0.281164 kg/s. TSFC is 29.097645 mg/(N s).

Multiplying the principal specific thrust by 10 kg/s would understate thrust by a factor of four. Either multiply it by total air flow or multiply thrust per core flow by core air flow. These are equivalent routes when the definitions are retained.

The example is direct arithmetic for the stated model, not a measured engine. Its absolute ambient pressure can be supplied independently when a dimensional station pressure is needed; the fully expanded specific results above depend on pressure ratios and temperature rather than an absolute pressure scale in this prescribed-flow calculation.

13.8 Variable-property fan and core architecture

The variable-property single-spool mode first compresses **all** air through the fan. Core air then passes through an additional compressor. Here the compressor pressure ratio is relative to the fan exit:

$$p_{t13} = p_{t2}\pi_f, \quad p_{t3} = p_{t13}\pi_{c,additional}.$$

Its work per unit core air is

$$w_{\text{load}} = (1 + \alpha)w_f + w_{c,additional}.$$

This differs in notation from the constant-property closure because the core fan work is now explicit. In the constant-property convention it was already included in the core's station-2-to-3 work. Neither formula should be transplanted without also changing the compressor reference station.

To align pressure paths, a comparison can set the constant-property core ratio to $\pi_f\pi_{c,additional}$. This does not alone align work: the property functions and efficiency definitions also differ. The variable-property fan and compressor use explicit isentropic efficiencies and enthalpy/entropy inversions.

The gaseous main fuel ratio is specified per core air, and equilibrium determines burner temperature from the separate fuel and air feed enthalpies. The turbine extracts the combined shaft load from $(1 + f)$ core gas. Its composition is frozen after the equilibrium chamber. Chapter 10 supplies the detailed thermodynamic steps.

13.9 Variable-property normalization and optional afterburning

The variable-property mode takes **total** inlet air flow as its dimensional input and obtains core flow as $\dot{m}_a/(1 + \alpha)$. This is the opposite input convention from the constant-property mode's optional core-flow field, though the physical definitions are consistent.

For a separate core afterburner with additional ratio f_{AB} , the thrust per core air is

$$F/\dot{m}_c = (1 + f + f_{AB})V_9 + \alpha V_{19} - (1 + \alpha)V_0.$$

Both nozzles are fully expanded in this branch. Fuel flow is $(f + f_{AB})\dot{m}_c$, and TSFC is total fuel ratio divided by thrust per core air. Afterburning affects the core exhaust and adds pressure loss after the turbine; it does not heat the bypass stream or alter the already closed fan work.

When bypass ratio is zero, the variable-property implementation skips the bypass nozzle evaluation. Its fan stage can still have a supplied pressure ratio and act on core air in the turbofan configuration. Therefore zero bypass alone is not identical to the turbojet configuration unless the compression paths are also aligned.

No thermal or propulsive efficiency is currently returned by this variable-property cycle. Its energy and shaft residuals describe closure, and its species fractions describe the final frozen nozzle mixture. The absence of a selected chemical-power denominator should not be filled by importing a heating value from another model without an explicit definition.

13.10 Limits of a separate-stream comparison

Separate flow avoids a mixer compatibility problem: there is no requirement that the two internal nozzle-entry streams share one static pressure plane. Each follows its own expansion to its selected exit condition. Mixing them upstream of a common nozzle requires new area, momentum and static-pressure conditions, not an average of the two velocities.

Both design modes are single-spool steady models. They do not solve a multispool operating schedule, component-map matching, cooling extraction, installed drag or fixed-geometry nozzle response. The constant-property ideal option additionally uses unit losses and cold properties throughout while neglecting fuel mass in turbine/exhaust balances; it requires both exit pressure ratios to be one.

A useful comparison reports bypass ratio, the compressor reference stations, fuel and thrust mass bases, both nozzle conditions and the physical flow scale. Those definitions reveal whether an improvement comes from a different core cycle, more bypass mass, a changed fan load or simply a changed normalization. Chapter 14 adds the conservation conditions needed when the two streams are deliberately combined.

14. Mixing and mixed-flow cycles

A common nozzle requires a common stream. Combining separate exhaust thrusts is not equivalent to mixing those streams upstream: a mixer must reconcile mass, species, energy, momentum and static pressure. Its outlet total pressure is a consequence of these constraints, rather than a mass-weighted average of inlet total pressures.

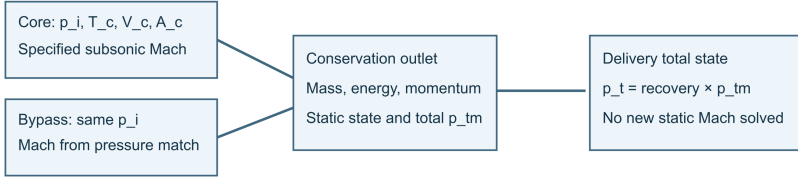
ThrustOne provides constant-property and frozen NASA9 mixing models. Both assume two subsonic incoming streams at equal static pressure and a constant total cross-sectional area. Both reconstruct a subsonic conservation outlet before applying an optional downstream total-pressure loss. This separation of planes is essential to interpreting the result.

14.1 The three distinct planes

The inlet plane contains adjacent core and bypass areas, A_c and A_b , with a shared static pressure p_i . Their temperatures, velocities, total pressures and compositions can differ. The conservation outlet has one uniform state at total area $A = A_c + A_b$. A third, conceptual delivery state applies the user-specified additional total-pressure recovery.

The conservation outlet and the delivery state must not be combined into one implied static state. The additional recovery does not feed back into the inlet matching or the momentum balance. Its downstream static Mach is not separately reconstructed by this mixer.

Mixer planes and the location of added loss



Additional downstream loss does not alter inlet pressure matching or the conservation outlet.

Figure 5: Distinct inlet, conservation and delivery planes

A field named `ideal_mixer_exit_mach` refers to the conservation outlet before the additional pressure loss. The word *ideal* here does not mean that unlike-stream mixing is reversible or isentropic. Irreversibility can already arise in the conservation-based mixing process.

14.2 Matching the incoming static pressures

The input supplies both total states and the core Mach number. For constant properties,

$$T_c = \frac{T_{tc}}{1 + (\gamma_c - 1)M_c^2/2}, \quad p_i = p_{tc}(T_c/T_{tc})^{\gamma_c/(\gamma_c-1)}.$$

The bypass Mach follows from its total pressure and that required common static pressure:

$$M_b = \sqrt{\frac{2}{\gamma_b - 1} [(p_{tb}/p_i)^{(\gamma_b-1)/\gamma_b} - 1]}.$$

A positive-flow bypass must have sufficient total pressure and a resulting Mach strictly below one. If its total pressure is below the required static pressure, an isentropic incoming state cannot be formed. If the required Mach is supersonic, it lies outside this two-subsonic-inlet model.

Reducing downstream mixer recovery cannot repair either incompatibility, because it acts at a different plane.

After velocities and densities are known, continuity supplies areas:

$$A_c = \frac{\dot{m}_c}{\rho_c V_c}, \quad A_b = \frac{\dot{m}_b}{\rho_b V_b}.$$

Thus the input Mach and total states determine the required inlet geometry for the supplied flows. The model does not accept an arbitrary second inlet area and independently guarantee pressure matching.

For frozen NASA9 properties, the core static temperature is found from $h(T_c) + M_c^2 a_f(T_c)^2/2 = h_{tc}$, and entropy gives static pressure. The bypass static temperature is found at that pressure with its own total entropy; its velocity follows from enthalpy drop. These integrations replace constant-gamma powers while retaining the same physical matching condition.

14.3 Conserved mixture quantities

Total mass flow is $\dot{m} = \dot{m}_c + \dot{m}_b$. With adiabatic walls and no shaft work,

$$h_{tm} = \frac{\dot{m}_c h_{tc} + \dot{m}_b h_{tb}}{\dot{m}}.$$

For a constant-area mixer without an imposed external axial force, the incoming axial momentum-plus-pressure flux is

$$J = p_i A + \dot{m}_c V_c + \dot{m}_b V_b.$$

The uniform outlet must satisfy

$$\dot{m} = \rho_m V_m A, \quad h_m + V_m^2/2 = h_{tm}, \quad p_m A + \dot{m} V_m = J.$$

The quantity J is not net engine thrust. It contains internal pressure forces and incoming momentum at the selected mixer boundary. Engine thrust requires a control volume extending to the external incoming and outgoing streams.

For constant effective properties, use mass-weighted c_p and R , then form $\gamma = c_p/(c_p - R)$. Averaging the two gammas directly does not generally reproduce this thermodynamic relation. Mixed total temperature is $T_{tm} = h_{tm}/c_p$ under the shared $c_p T$ convention; it is a simple mass average only if the two heat capacities are equal.

For frozen species mixtures, sum actual species molar flows,

$$\dot{n}_i = \frac{\dot{m}_c x_{ic}}{\bar{W}_c} + \frac{\dot{m}_b x_{ib}}{\bar{W}_b}.$$

Normalize their sum to obtain outlet mole fractions. Averaging mole fractions by mass flow would be wrong when molecular weights differ. No equilibrium chemistry is solved inside the frozen mixer, even if the combined mixture could react physically.

14.4 Deriving the constant-property velocity roots

Let $B = J/\dot{m}$. The momentum equation gives $p_m A/\dot{m} = B - V_m$, while continuity and the ideal-gas law give $p_m A/\dot{m} = RT_m/V_m$. Using $T_m = T_{tm} - V_m^2/(2c_p)$,

$$\left(1 - \frac{R}{2c_p}\right) V_m^2 - BV_m + RT_{tm} = 0.$$

Define $a = 1 - R/(2c_p)$. The discriminant is $D = B^2 - 4aRT_{tm}$. A negative discriminant means there is no real outlet satisfying the declared momentum, energy and area conditions. It is not fixed by silently replacing the root with Mach one.

The lower-velocity root is evaluated in the cancellation-resistant form

$$V_m = \frac{2RT_{tm}}{B + \sqrt{D}}.$$

The solver then recovers static temperature, pressure and Mach and checks the subsonic branch and conservation residuals. The constant-property implementation permits a tiny numerical tolerance around Mach one; the variable-property mixer explicitly requires an exit below one. Neither model supplies a supersonic mixer branch or a shock-train transition.

The NASA9 calculation instead calls the frozen flux-state reconstruction with mass, area, total enthalpy and momentum flux, explicitly selecting its subsonic branch. Chapter 26 develops that more general inversion.

14.5 Total-pressure recovery after conservation

At the reconstructed outlet, total temperature comes from the conserved total enthalpy. Total pressure is obtained by isentropically stagnating that outlet state. For constant properties,

$$p_{tm} = p_m (T_{tm}/T_m)^{\gamma/(\gamma-1)}.$$

For NASA9 properties, the same operation uses entropy differences. Only after this calculation does the additional loss apply:

$$p_{t,delivered} = \pi_{m,extra} p_{tm}.$$

The conserved Mach and momentum residual still describe the preceding outlet. Treating the delivered pressure as though it belonged to the same static state would mix two definitions. No extra temperature drop accompanies this adiabatic total-pressure loss.

14.6 Original mixer example

Choose two illustrative constant-property streams with $c_p = 1000 \text{ J/(kg K)}$, $R = 250 \text{ J/(kg K)}$, hence $\gamma = 4/3$. Let core and bypass flow be 2 and 3 kg/s, total temperatures 800 and 400 K, and both inlet Mach numbers 0.4 at common static pressure 100 kPa. Their matching total pressures are equal because gamma and Mach are equal; their velocities are not.

The core static temperature is 779.220779 K and velocity 203.858877 m/s. Bypass static temperature is 389.610390 K and velocity 144.149994 m/s. Continuity gives areas 0.01911177 and 0.02027109 m². Consequently,

$$A = 0.03938286 \text{ m}^2, \quad h_{tm} = 560000 \text{ J/kg}, \quad J = 4778.453994 \text{ N}.$$

The subsonic conservation root gives $V_m = 174.309256 \text{ m/s}$, $T_m = 544.808142 \text{ K}$, $p_m = 99.203244 \text{ kPa}$ and $M_m = 0.40903388$. Its total temperature is 560 K and total pressure is 110.739772 kPa. Applying an additional recovery of 0.96 gives delivered total pressure 106.310181 kPa.

These are direct analytical arithmetic results for an original teaching case. They demonstrate why outlet static pressure need not equal the matched inlet pressure, and why an averaged total temperature alone is insufficient to recover outlet velocity or pressure. They are not a measured mixer-performance dataset.

14.7 Embedding the mixer in a dual-spool cycle

The constant-property mixed turbofan places the fan before the core-bypass split. The fan therefore acts on all incoming air. Its high-pressure compressor acts only on core air, downstream of the fan. Let the core-air flow be the accounting unit and let α denote bypass air divided by core air. The main fuel ratio f is also based on core air.

For cold-gas exponent $e_c = (\gamma_c - 1)/\gamma_c$, the fan and high-pressure compressor give

$$T_{t13} = T_{t2} \pi_f^{e_c/\eta_{pf}}, \quad T_{t3} = T_{t13} \pi_{cH}^{e_c/\eta_{pc}}.$$

Here π_{cH} is an additional pressure ratio after the fan. Thus $p_{t4} = p_{t2} \pi_f \pi_{cH} \pi_b$, rather than an expression in which the fan pressure rise disappears from the core path. The burner balance is

$$f = \frac{c_{ph} T_{t4} - c_{pc} T_{t3}}{\eta_b Q - c_{ph} T_{t4}}.$$

The high-pressure turbine drives the high-pressure compressor; the low-pressure turbine drives the fan. With separate mechanical efficiencies,

$$T_{t45} = T_{t4} - \frac{c_{pc}(T_{t3} - T_{t13})}{\eta_{mH}(1 + f)c_{ph}},$$

$$T_{t5} = T_{t45} - \frac{(1 + \alpha)c_{pc}(T_{t13} - T_{t2})}{\eta_{mL}(1 + f)c_{ph}}.$$

The factor $1 + \alpha$ in the second equation is essential: core air also passes through the fan. Conversely, only the hot core stream passes through these turbines. Assigning bypass mass to turbine power would overstate the available shaft work.

For hot-gas exponent $e_h = (\gamma_h - 1)/\gamma_h$, the model uses turbine polytropic efficiency to infer the two pressure ratios:

$$\pi_{tH} = (T_{t45}/T_{t4})^{1/(e_h \eta_{pt})}, \quad \pi_{tL} = (T_{t5}/T_{t45})^{1/(e_h \eta_{pt})}.$$

The mixer receives core mass $1 + f$ at $T_{t5}, p_{t4} \pi_{tH} \pi_{tL}$, and bypass mass α at $T_{t13}, p_{t2} \pi_f$. Positive turbine temperatures alone do not guarantee an admissible engine. These two streams must also satisfy the static-pressure and subsonic conditions of Sections 14.2–14.4. A thermodynamically possible compressor–turbine calculation can therefore end with an incompatible mixer.

This calculation closes shaft power at a prescribed design point. It does not match two rotating component maps, determine shaft speeds, or solve a fixed installed mixer geometry. The mixer areas are consequences of the chosen states, flows and core Mach. A sequence of design-point calculations may imply a different area at every point.

14.8 Post-mixer afterburning and fuel bases

The incoming mass at the afterburner is

$$\mu = 1 + f + \alpha$$

per unit core air. Let f_{ab} be added afterburner fuel per unit core air, h_{tm} the mixed total enthalpy, and $h_{t7} = c_{p,ab}T_{t7}$ the target hot-gas total enthalpy in the effective constant-property convention. The energy balance is

$$\mu h_{tm} + \eta_{ab} f_{ab} Q = (\mu + f_{ab}) h_{t7},$$

hence

$$f_{ab} = \mu \frac{c_{p,ab}T_{t7} - c_{pm}T_{tm}}{\eta_{ab}Q - c_{p,ab}T_{t7}}.$$

Leaving out μ changes the fuel reference to incoming mixed gas. That can be a useful ratio, but it cannot then be added directly to main-chamber f . The App reports both fuel additions on the core-air basis.

The denominator must be positive, the required enthalpy increase non-negative, and the target temperature cannot be below the incoming mixed temperature in this mode. Because the effective heat capacity may change across the afterburner, temperature ordering and enthalpy ordering are separate checks. This is an effective sensible-enthalpy model; it does not calculate an equilibrium species inventory.

As an original bookkeeping example, continue the 5 kg/s mixer case with incoming total enthalpy 560 kJ/kg. Specify an afterburner gas with $c_p = 1200 \text{ J/(kg K)}$, target total temperature 1000 K, $Q = 43 \text{ MJ/kg}$ and efficiency 0.98. Direct substitution gives added fuel 0.0781632 kg/s. Relative to the 5 kg/s incoming mixture, the ratio is 0.0156326. If the 2 kg/s hot core stream contains main fuel, neither of these numbers is automatically the added fuel per kilogram of core air: the original core-air flow must first be recovered. This example deliberately preserves the distinction between a stand-alone mixer and an engine's accounting basis.

Total pressure delivered to the nozzle is the conservation-derived mixed total pressure multiplied successively by the extra mixer recovery, afterburner pressure ratio when enabled, and nozzle pressure ratio. The nozzle uses the post-afterburner total temperature when afterburning is active. Applying its pressure loss upstream in the mixer matching calculation would define a different model.

14.9 Common-nozzle thrust

Write $r_e = p_0/p_9$, $R_9 = c_{p9}(\gamma_9 - 1)/\gamma_9$, and p_{t9} for total pressure after all specified downstream losses. The nozzle pressure ratio is $p_{t9}/p_9 = p_{t9}r_e/p_0$. In the constant-property mode,

$$T_9 = T_{t7}(p_9/p_{t9})^{(\gamma_9-1)/\gamma_9}, \quad V_9 = \sqrt{2c_{p9}(T_{t7} - T_9)}.$$

Let $\mu_e = 1 + f + \alpha + f_{ab}$. Eliminating exit area through continuity gives thrust per unit core-air flow:

$$\frac{F}{\dot{m}_{a,c}} = \mu_e V_9 - (1 + \alpha)V_0 + \frac{\mu_e R_9 T_9}{V_9}(1 - r_e).$$

The pressure term has the sign expected physically: an exit pressure above ambient gives positive pressure thrust. Specific thrust displayed on the total-air basis is this result divided by $1 + \alpha$. With a user-specified total-air flow, multiply total-air specific thrust by that flow to obtain force. Fuel consumption is

$$\text{TSFC} = \frac{f + f_{ab}}{F/\dot{m}_{a,c}}.$$

Nozzle exit pressure is prescribed here; it is not the result of a supplied convergent–divergent nozzle geometry. A computed supersonic exit does not establish that a physical installed nozzle starts or remains attached. Thermal and propulsive efficiencies based on the kinetic-energy decomposition are provided only for full expansion. Overall efficiency uses thrust power divided by fuel chemical power, as discussed in Chapter 1.

14.10 The variable-property mixed branch

The NASA9 mixed-exhaust option belongs to the variable-property single-spool turbofan model. It is not the constant-property dual-spool calculation with a different value of gamma. The fan acts on all air, the core compressor follows the fan, and the shaft load combines both contributions. Component efficiencies use the variable-property mode’s isentropic convention. Main combustion determines equilibrium products from the prescribed fuel feed; subsequent turbine expansion retains that composition.

The frozen mixer combines the turbine-exit products with bypass air by species molar flow. It conserves their actual NASA9 total enthalpies, then reconstructs a subsonic outlet. No reaction is silently inserted at this mixing step. The resulting total pressure, after the additional recovery, is the pressure available to downstream combustion or expansion.

If post-mixer combustion is enabled, fuel is added on the core-air basis. Using formation-inclusive enthalpies, its target specific enthalpy is

$$h_{t7} = \frac{\mu h_{tm} + f_{ab} h_f}{\mu + f_{ab}}.$$

The equilibrium calculation solves for the temperature and composition consistent with this enthalpy, the combined elemental inventory and the specified afterburner pressure. Do not add a lower heating value to this

expression: the chemical reference energies already carry the reaction energy. The available fuel selections and bounded equilibrium species set remain those of the variable-property mode; mixing does not enlarge that chemistry model.

At zero post-mixer fuel, the implementation preserves the incoming frozen mixture and temperature while applying the specified pressure loss. It does not run an extra equilibrium relaxation merely because the afterburner option is present. With mixed exhaust selected, a separate pre-mixer core afterburner input is rejected; the supported location is downstream of mixing.

Finally, the variable-property nozzle expands the resulting frozen composition to ambient pressure. Its exit state is static, whereas the mixer delivery and afterburner states are total states. Its common-exhaust momentum balance is the fully expanded form of Section 14.9. The mode's total-air input and core-air fuel basis must still be kept distinct.

14.11 Interpreting refusal and comparing designs

A simple incompatible case follows directly from the mixer example. Keep the core conditions that establish an inlet static pressure of 100 kPa, but set bypass total pressure to 95 kPa with positive bypass flow. An adiabatically accelerated bypass stream cannot have static pressure above its total pressure. The computed bypass Mach would therefore not be real. Raising an arbitrary downstream mixer recovery cannot repair this upstream incompatibility.

Setting bypass total pressure exactly to 100 kPa gives zero bypass speed at that matching pressure. Positive mass flow would then require unbounded area. This limiting case is also inadmissible, even though the pressure inequality alone no longer asks for an imaginary Mach. At the other extreme, a sufficiently high bypass total-to-static ratio implies a sonic or supersonic bypass inlet, outside the declared subsonic mixer branch.

These refusals identify a conflict among chosen cycle states, flow split and mixer assumptions. They do not establish that every physical mixer at those engine conditions is impossible. Changing core Mach changes the

shared inlet static pressure; changing component pressure ratios changes the entering total states; a different geometry or flow regime would require a different model.

For comparisons, hold the accounting basis and external flight condition fixed. Compare fuel flow and net thrust on the same total-air flow, inspect whether the inferred mixer area changes, and retain the same nozzle-expansion assumption. Equal total-air specific thrust from two runs says nothing by itself about equal engine size or equal shaft speed.

Conservation residuals describe the algebraic closure, not the accuracy of turbulent mixing or an experimental loss correlation. The stand-alone mixers use their supplied dimensional mass flows. The constant-property cycle calls its mixer on a unit-core-air flow basis, so its raw mixer residuals belong to that normalized calculation. The variable-property cycle scales the embedded extensive mixer quantities to the actual core flow. Small residuals from these two outputs should not be compared without accounting for this distinction.

The chapter's useful physical boundary is therefore explicit: the App can connect specified subsonic streams through a one-dimensional conservation mixer and use the delivered total state in a bounded engine cycle. It does not predict mixing length, temperature pattern, wall heat transfer, acoustics, or installed mixer drag.

15. Turbofan components and off-design

A component calculation asks what happens across one specified device. An off-design calculation asks which component states can coexist when the engine's reference constraints are retained. The second problem cannot be solved by choosing every component outlet independently: shaft power and flow capacity couple the choices.

ThrustOne exposes both problems. The turbofan component mode provides fan, booster, bypass-nozzle and two-stream mixer subproblems. Its separate-flow off-design mode uses a reference-normalized dual-spool approximation with convergent exhaust nozzles. It does not use the optional component maps to solve an operating line. Keeping these two capabilities distinct is necessary when interpreting the word matching.

15.1 Boundaries of the component problems

The fan and booster selections both call the shared compressor thermodynamics. Their different names specify their intended engine location, not different aerodynamic blade models. A fan normally processes all incoming air in the cycles considered here; a core booster processes core air. The independent component calculation does not infer either mass allocation or shaft coupling.

For a constant-property compressor with pressure ratio π , polytropic efficiency η_p , inlet total temperature T_{ti} , and $e = (\gamma - 1)/\gamma$,

$$T_{to} = T_{ti} \pi^{e/\eta_p}, \quad w = c_p(T_{to} - T_{ti}).$$

Specific work w is measured in J/kg of gas crossing that component. If a fan requires 60 kJ/kg and processes 40 kg/s, its fluid power is 2.4 MW. If a booster requires the same specific work but processes only 10 kg/s, its fluid power is 0.6 MW. A shared temperature rise does not imply a shared shaft load.

The frozen variable-property option instead uses an isentropic outlet enthalpy and the specified isentropic efficiency, then inverts enthalpy for the actual outlet temperature. Chapter 11 supplies that derivation and the optional constant-property compressor-map scaling. The same map domain, normalized-coordinate and no-extrapolation restrictions apply when that shared compressor calculation is reached through the fan or booster selection. A frozen-property calculation does not silently reuse that constant-property map option.

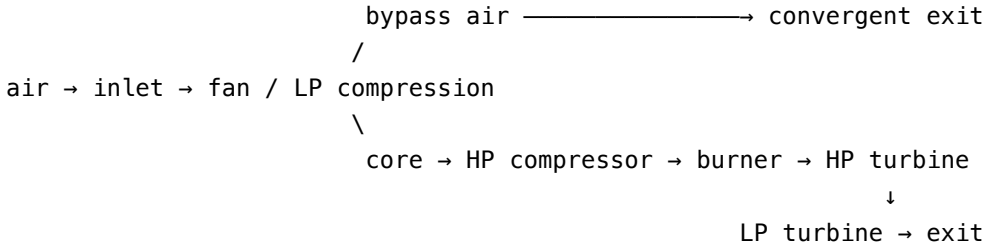
The bypass-nozzle component also delegates to the common nozzle solver. It uses the specified total state, pressure recovery and requested exit pressure. This independent calculation is not restricted to the convergent-nozzle law used by the off-design engine below. A supersonic result from a prescribed-pressure component nozzle must not be inserted into a convergent-exit interpretation.

For the mixer selection, the constant-property inputs or the supplied variable-property mixer input determine which conservation solver runs. Both match inlet static pressures and select a subsonic outlet as developed in Chapter 14. The mixer requires two complete stream descriptions; a fan pressure ratio alone cannot determine them.

15.2 The reference engine and its flowpath

The off-design model assumes equal fan and low-pressure-compressor outlet states. Thus the core proceeds through a low-pressure compression rise equal to the fan rise, followed by an additional high-pressure compressor. The bypass stream leaves after the fan. The high-pressure

turbine powers the high-pressure compressor, and the low-pressure turbine powers the fan and the equivalent low-pressure core compression.



HP shaft: HP turbine ↔ HP compressor

LP shaft: LP turbine ↔ fan and equivalent LP core compression

Reference data include the flight state, turbine inlet temperature, component pressure and temperature ratios, bypass ratio, two nozzle exit Mach numbers and total-air mass flow. Current inputs supply the new flight state and turbine inlet temperature, together with retained gas properties, efficiencies and pressure losses. High-pressure turbine temperature and pressure ratios are fixed inputs. They are not discovered from a turbine map.

The two high-pressure turbine stations are assumed choked to support the normalized closure. This is a modeling assumption: the program does not reconstruct internal turbine throat areas and confirm choking from local measured geometry. The two exhaust nozzles, in contrast, may become unchoked and are explicitly switched between subsonic and sonic exit states.

Use a subscript R for reference values, $\tau = T_{to}/T_{ti}$ for component total-temperature ratios, and $\pi = p_{to}/p_{ti}$ for total-pressure ratios. Define

$$\tau_r = 1 + \frac{\gamma_c - 1}{2} M_0^2, \quad \tau_\lambda = \frac{c_{ph} T_{t4}}{c_{pc} T_0}, \quad H = \frac{\tau_\lambda / \tau_r}{\tau_{\lambda R} / \tau_{rR}}.$$

The dimensionless H compares turbine inlet thermal level to incoming total-air thermal level. It is not simply the ratio of the two turbine inlet temperatures when the flight state changes.

15.3 Convergent-nozzle closure

Let $q = p_t/p_0$ denote the total-to-ambient pressure ratio after nozzle losses. Forward flow requires $q > 1$. For a convergent nozzle, the exit is its minimum-area plane, giving the critical ratio

$$q_* = \left(\frac{\gamma + 1}{2} \right)^{\gamma/(\gamma-1)}.$$

The actual total-to-exit pressure ratio is $q_e = \min(q, q_*)$, so

$$M_e = \sqrt{\frac{2}{\gamma-1}(q_e^{(\gamma-1)/\gamma} - 1)}, \quad \frac{p_0}{p_e} = \frac{q_e}{q}.$$

Below choking, $p_e = p_0$ and $M_e < 1$. Above choking, $M_e = 1$, while exit pressure remains above ambient. The model retains the resulting pressure thrust. It does not continue an isentropic expansion to a supersonic exit inside a convergent nozzle.

For an original example, take $\gamma = 1.4$, $c_p = 1000 \text{ J/(kg K)}$, and $T_t = 360 \text{ K}$. At $q = 1.5$, direct arithmetic gives $M_e = 0.783659$, $T_e = 320.6201 \text{ K}$ and $V_e = 280.6419 \text{ m/s}$, with no pressure thrust. At $q = 3$, the critical ratio is 1.892929, giving $M_e = 1$, $T_e = 300 \text{ K}$ and $V_e = 346.4102 \text{ m/s}$. Now $p_0/p_e = 0.630976$. Further increasing total pressure at fixed total temperature increases mass capacity and pressure thrust, rather than raising this exit Mach above one.

15.4 How fixed flow areas create coupling

For a fixed gas, mass flow can be written

$$\dot{m} = \frac{Ap_t}{\sqrt{T_t}} \sqrt{\frac{\gamma}{R}} \Phi(M), \quad \Phi(M) = M \left(1 + \frac{\gamma-1}{2} M^2 \right)^{-\frac{\gamma+1}{2(\gamma-1)}}.$$

Constants cancel when comparing the same gas and station with its reference. This is why the implementation's mass-flow function omits $\sqrt{\gamma/R}$. Comparing different gases would require restoring it.

The bypass/core flow ratio becomes

$$\alpha = \alpha_R \frac{\pi_{cH,R}}{\pi_{cH}} \sqrt{H \frac{\tau_{fR}}{\tau_f} \frac{\Phi_c(M_{19})}{\Phi_c(M_{19R})}}.$$

This relation combines bypass-nozzle capacity with the assumed choked core capacity. It makes bypass ratio an output. Holding it independently fixed would overconstrain this model. The fan nozzle can unchoke, changing $\Phi_c(M_{19})$ even before any change in the component pressure ratios is considered.

The low-pressure turbine pressure ratio obeys

$$\pi_{tL} = \pi_{tL,R} \sqrt{\frac{\tau_{tL}}{\tau_{tL,R}} \frac{\Phi_h(M_{9R})}{\Phi_h(M_9)}},$$

with the isentropic-efficiency relation

$$\tau_{tL} = 1 - \eta_{tL} \left(1 - \pi_{tL}^{(\gamma_h-1)/\gamma_h} \right).$$

If the core exit remains choked in both states, its mass-flow-function ratio is one. This removes one source of variation but does not make every engine variable constant. Unchoking restores direct feedback from exhaust conditions into the turbine relation.

The retained normalized equations neglect changes in some fuel-mass factors used in deriving reference flow and power ratios. Fuel is recovered later for thrust and consumption. Thus fuel mass appearing in final exhaust momentum does not make the preceding matching equations exact fuel-inclusive component matching.

15.5 Shaft relations and the unknowns

The high-pressure spool relation is

$$\tau_{cH} = 1 + H \frac{\tau_{fR}}{\tau_f} (\tau_{cH,R} - 1), \quad \pi_{cH} = [1 + \eta_{cH}(\tau_{cH} - 1)]^{\gamma_c/(\gamma_c - 1)}.$$

The factor τ_{fR}/τ_f accounts for the high-pressure compressor's changing inlet temperature. Omitting it incorrectly treats that compressor as though it still ingested the inlet state ahead of the fan.

The low-pressure spool gives

$$\tau_f = 1 + \frac{1 - \tau_{tL}}{1 - \tau_{tL,R}} H \frac{1 + \alpha_R}{1 + \alpha} (\tau_{fR} - 1),$$

$$\pi_f = [1 + \eta_f(\tau_f - 1)]^{\gamma_c/(\gamma_c - 1)}.$$

An increasing bypass ratio spreads available low-spool work over more air, reducing the rise available per kilogram. These relations use component isentropic efficiencies, unlike the polytropic inputs of several design-point modes.

Nine quantities are mutually dependent: the fan, high-pressure compressor and low-pressure turbine pressure and temperature ratios, bypass ratio, and the two exit Mach numbers. The program starts from reference fan temperature ratio and low-pressure turbine pressure ratio, evaluates the dependent state, and repeatedly updates the fan and low-turbine relations.

Its convergence measure is the largest successive change in those nine quantities, each divided by the larger of the new magnitude and one. A value below 10^{-9} ends iteration, with a limit of 1000 updates. This is a fixed-point change criterion. It is not a dimensional shaft-power residual, nor proof of uniqueness or experimental agreement. Invalid intermediate states can terminate the calculation before the iteration limit.

15.6 A transparent reference-closure example

A useful original check of the equations is a deliberately consistent algebraic reference. Set both gas gammas to 1.4, all component efficiencies to one, $\tau_{fR} = 1.2$, $\tau_{cH,R} = 2$, $\tau_{tL,R} = 5/6$, and $\alpha_R = 3$. Define pressure ratios from these temperatures: $\pi_{fR} = 1.892929$, $\pi_{cH,R} = 11.313708$, and $\pi_{tL,R} = 0.528282$. Choose fixed high-turbine pressure ratio $0.8^{3.5} = 0.457947$, no pressure losses, static flight and both reference exit Mach numbers equal to one.

At the unchanged flight and turbine-temperature state, $H = 1$. The high-compressor equation gives

$$\tau_{cH} = 1 + (1.2/1.2)(2 - 1) = 2.$$

The bypass mass-flow-function ratio is one and the pressure-ratio factor is one, so $\alpha = 3$. The low-spool equation then gives

$$\tau_f = 1 + \frac{1 - 5/6}{1 - 5/6} \frac{1 + 3}{1 + 3} (1.2 - 1) = 1.2.$$

Both nozzle branches remain sonic: the bypass is exactly at the critical pressure ratio, while the product of core pressure ratios exceeds it. The low-turbine update also returns its reference value. Every normalized update therefore reproduces the declared point.

This is a constructed demonstration of reference consistency, not a measured engine or a numerical fixture generated by the App. It isolates an important user responsibility: reference pressure ratios, temperature ratios and efficiencies must describe the same state. Changing only π_{fR} while leaving its temperature ratio and efficiency inconsistent removes this fixed point. Passing the input range checks alone does not guarantee coherent reference data.

15.7 Recovering dimensional performance

After matching, the fuel ratio follows the main-burner balance:

$$f = \frac{\tau_\lambda - \tau_r \tau_f \tau_{cH}}{\eta_b Q / (c_{pc} T_0) - \tau_\lambda}.$$

A positive denominator and positive heating are required. Total air flow is recovered as

$$\frac{\dot{m}_a}{\dot{m}_{aR}} = \frac{1 + \alpha}{1 + \alpha_R} \frac{p_0 \pi_r \pi_d \pi_f \pi_{cH}}{p_{0R} \pi_{rR} \pi_{dR} \pi_{fR} \pi_{cH,R}} \sqrt{\frac{T_{t4,R}}{T_{t4}}}.$$

Current inlet recovery includes the specified Mach-dependent empirical factor. The stored reference recovery is used directly in the denominator, so it must represent the intended complete reference recovery. A raw maximum inlet recovery is not interchangeable with a recovered reference value at supersonic flight.

The core exit temperature is obtained from $T_{t4} \tau_{tH} \tau_{tL}$ and its convergent-nozzle pressure ratio. The bypass total temperature is $T_0 \tau_r \tau_f$. Apply the nozzle equations to obtain both static temperatures and velocities, then the separate-stream momentum balance of Chapter 13, including each pressure term.

Specific thrust is based on total incoming air; fuel ratio is based on core air. Consequently,

$$F = \dot{m}_a F_s, \quad \text{TSFC} = \frac{f}{(1 + \alpha) F_s}.$$

The speed outputs are estimated mechanical ratios from temperature-scaled compressor work. For example,

$$\frac{N_f}{N_{fR}} = \sqrt{\frac{T_0 \tau_r}{T_{0R} \tau_{rR}} \frac{\pi_{f^c}^e - 1}{\pi_{fR}^e - 1}}.$$

The high-spool expression additionally includes τ_f / τ_{fR} and uses high-compressor pressure ratios. These are not corrected map speeds, and they do not establish blade stress or a permissible operating-speed limit.

15.8 Refusal and interpretation

A nozzle total pressure at or below ambient admits no positive forward-flow solution under this closure. A low-turbine pressure ratio outside zero to one contradicts expansion; a fan temperature ratio at or below one contradicts the required compressing branch. Nonpositive fuel, insufficient heating value, nonpositive net thrust and nonconverged iteration also cause refusal.

The reference bypass ratio must be positive and reference exit Mach numbers lie above zero and at most one. This off-design mode does not offer a zero-bypass turbojet limit through a zero reference bypass input. That boundary differs from some design-point modes and is intentional in interpreting the accepted input domain.

A successful result establishes a fixed point within the specified approximation. It does not supply component surge prediction, a mixed-flow operating line, cooling and bleed schedules, shaft transients or installed inlet interaction. The App omits kinetic-decomposition thermal and propulsive efficiencies when either active nozzle has pressure thrust; overall efficiency still follows thrust power divided by fuel power.

Use this model to understand how a declared reference engine redistributes flow and work as flight or thermal level changes. Use the independent component mode to inspect a particular transformation. Neither should be described as a general replacement for a complete engine performance deck.

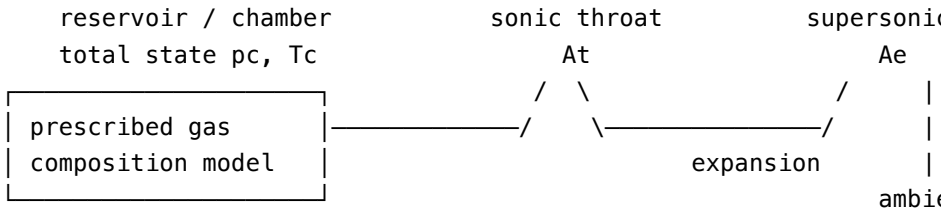
16. Rocket chamber and nozzle performance

A rocket carries the material that becomes its exhaust. There is no captured-air momentum subtraction in the thrust-chamber calculation. The chamber supplies a high-pressure, high-enthalpy reservoir; the nozzle converts part of that enthalpy into directed kinetic energy. Ambient pressure changes the external pressure force and can change the physical flow regime, even though the simplest attached-flow calculation retains the same internal isentropic solution.

This chapter develops the explicit constant-property model shared by ThrustOne's thrust-chamber performance and stand-alone nozzle performance modes. The gas temperature, molecular weight and ratio of specific heats are supplied inputs. They are not inferred from a propellant name or mixture ratio. Equilibrium and frozen variable-property chemistry are separate calculations developed in Chapter 17.

16.1 Reservoir state, throat and exit

The chamber pressure p_c and temperature T_c in this model act as nozzle-inlet total conditions. A sufficiently large, slowly moving chamber makes static and total conditions nearly equal, but the distinction is physical rather than merely a naming convention. A finite-velocity chamber plane requires the separate conversion of Chapter 18.



$$\varepsilon = A_e / A_t$$

$$F = \dot{m} V_e + (p_e - p_a) A_e$$

The assumed flow is steady, one-dimensional, adiabatic and isentropic, with constant $\gamma > 1$ and gas constant R . In the explicit-property model, frozen means the effective gas properties remain fixed; it does not mean a species-resolved NASA calculation has been performed. There is no wall-friction, boundary-layer displacement, shock or separated-flow reconstruction in this equation chain.

Molecular weight $\overline{W}$ is entered in kg/kmol. The App uses

$$R = \frac{R_u}{\overline{W}}, \quad R_u = 8314.462618 \text{ J/(kmol K)}, \quad c_p = \frac{\gamma R}{\gamma - 1}.$$

Using a mol-based gas constant with a kmol-based molecular weight introduces a factor of one thousand in R . All subsequent temperature-to-velocity and choked-flow results would then be wrong.

16.2 Deriving the critical mass flow

Continuity gives $\dot{m} = \rho V A$. Substitute $\rho = p/(RT)$, $V = M\sqrt{\gamma RT}$, and the static-to-total relations from Chapter 2:

$$\frac{\dot{m}}{A} = \frac{p_c}{\sqrt{RT_c}} \sqrt{\gamma} M \left(1 + \frac{\gamma - 1}{2} M^2 \right)^{-\frac{\gamma+1}{2(\gamma-1)}}.$$

The mass flux has its maximum at $M = 1$. At the choked throat,

$$\dot{m}_i = \frac{p_c A_t}{\sqrt{RT_c}} \Gamma(\gamma), \quad \Gamma(\gamma) = \sqrt{\gamma} \left(\frac{2}{\gamma + 1} \right)^{\frac{\gamma+1}{2(\gamma-1)}}.$$

The subscript i marks the ideal reference. Pressure times area has units of force; division by a velocity has units of mass flow. This dimensional check also explains the usefulness of characteristic velocity:

$$c_i^* = \frac{p_c A_t}{\dot{m}_i} = \frac{\sqrt{RT_c}}{\Gamma(\gamma)}.$$

Characteristic velocity has units of m/s but is not the gas velocity at the throat or at the exit. It expresses the relation between chamber pressure, throat area and mass flow. Its star belongs to the conventional name; it should not be read as a claim that it equals sonic speed.

At fixed R, T_c, γ , ideal mass flow is proportional to $p_c A_t$. Temperature raises the specific energy available for expansion but reduces density and choked mass flux at fixed chamber pressure. These two effects must be distinguished when interpreting a hotter chamber.

16.3 Geometry selects the expansion branch

Define expansion ratio $\epsilon = A_e/A_t$. Equating the exit and sonic mass-flow expressions produces

$$\epsilon = \frac{1}{M_e} \left[\frac{2}{\gamma + 1} \left(1 + \frac{\gamma - 1}{2} M_e^2 \right) \right]^{\frac{\gamma+1}{2(\gamma-1)}}.$$

For $\epsilon > 1$, this relation has subsonic and supersonic roots. The rocket performance modes select the supersonic root. An area ratio of one gives the sonic state. A subsonic root is mathematically meaningful elsewhere, such as a chamber contraction calculation, but it is not an alternative exhaust branch silently selected by these modes.

Once M_e is known,

$$T_e = \frac{T_c}{1 + (\gamma - 1)M_e^2/2}, \quad \frac{p_e}{p_c} = \left(1 + \frac{\gamma - 1}{2}M_e^2\right)^{-\gamma/(\gamma-1)}.$$

The velocity can be reconstructed either from Mach or from energy:

$$V_{e,i} = M_e \sqrt{\gamma R T_e} = \sqrt{\frac{2\gamma}{\gamma - 1} R T_c \left[1 - \left(\frac{p_e}{p_c}\right)^{(\gamma-1)/\gamma}\right]}.$$

The second form follows from $V_e^2/2 = c_p(T_c - T_e)$. It shows that an arbitrarily large expansion ratio cannot produce an arbitrarily large velocity. Within the constant-property model, velocity approaches $\sqrt{2c_p T_c}$ as exit pressure approaches zero, while the required area diverges. The zero-temperature mathematical limit is not an attainable material or chemistry model.

The App inverts area ratio by a bracketed search restricted to the requested branch. Near $\epsilon = 1$, the two roots merge and area changes only weakly with Mach, as discussed in Chapter 2. It treats an area ratio within 10^{-12} above one as sonic. Iteration failure is reported rather than replacing a failed root by an arbitrary large Mach.

16.4 Equivalent nozzle input choices

Area ratio is not the only way to specify the same ideal expansion. ThrusOne also accepts exit Mach, exit static pressure, chamber-to-exit pressure ratio, or an altitude at which the nozzle is ideally expanded. Each is converted into the supersonic area ratio before performance is calculated.

A specified exit pressure gives

$$M_e^2 = \frac{2}{\gamma - 1} \left[\left(\frac{p_c}{p_e}\right)^{(\gamma-1)/\gamma} - 1 \right].$$

A real positive Mach is not sufficient: this expansion-nozzle interface requires $M_e \geq 1$. Thus an exit pressure below chamber pressure can still be rejected if the implied Mach is subsonic. Equivalently, the accepted chamber-to-exit ratio must reach the sonic critical ratio.

The optimum-altitude input sets exit pressure equal to the standard-atmosphere pressure at that design altitude. The separate environment input sets the ambient pressure used for the current thrust calculation. Designing for one altitude and evaluating at another is therefore a meaningful and supported distinction. Choosing an optimum altitude does not continuously resize the nozzle whenever the environment changes.

Vacuum is an environment choice, $p_a = 0$. It is not a finite-area nozzle choice with $p_e = 0$. A finite expansion ratio leaves positive exit pressure and consequently retains a positive vacuum pressure-thrust term.

16.5 Thrust coefficient separates geometry from scale

The ideal thrust balance is

$$F_i = \dot{m}_i V_{e,i} + (p_e - p_a) A_e.$$

Divide by $p_c A_t$, substitute $\dot{m}_i = p_c A_t / c_i^*$, and define

$$C_{F,i} = C_{F,m} + C_{F,p}, \quad C_{F,m} = \frac{V_{e,i}}{c_i^*}, \quad C_{F,p} = \left(\frac{p_e}{p_c} - \frac{p_a}{p_c} \right) \epsilon.$$

Both contributions are dimensionless. The momentum coefficient depends on the gas exponent and expansion. The pressure coefficient also depends on the external pressure relative to chamber pressure. No captured-air term belongs in this chamber model; whole-vehicle drag remains a separate problem.

If $p_e > p_a$, the nozzle is underexpanded and pressure thrust is positive. If $p_e = p_a$, pressure thrust vanishes, but momentum thrust remains. If

$p_e < p_a$, the negative pressure term reduces total thrust. A negative pressure contribution does not by itself make total thrust negative.

For a fixed attached nozzle state,

$$C_{F,vac} = C_{F,m} + \frac{p_e}{p_c} \epsilon, \quad C_{F,vac} - C_{F,i} = \frac{p_a}{p_c} \epsilon.$$

This is a comparison of the same geometry in different environments. It is not a comparison between two nozzles separately optimized for each environment.

16.6 Why ideal expansion is optimal

Consider varying exit area along a family of ideal, attached expansions while holding reservoir state and choked throat flow fixed. Differentiate the thrust expression:

$$dF_i = \dot{m}_i dV_e + A_e dp_e + (p_e - p_a) dA_e.$$

The inviscid momentum relation along the expansion is $dp = -\rho V dV$. Continuity gives $\dot{m}_i = \rho V A$, so the first two terms cancel:

$$dF_i = (p_e - p_a) dA_e.$$

Adding exit area improves ideal thrust while the exit pressure exceeds ambient. Once it falls below ambient, further area reduces thrust in this family. The stationary point occurs at $p_e = p_a$.

This result does not include nozzle mass, length, cooling, flight trajectory or separation. It is an ideal local thrust optimum under specified constraints. In vacuum the derivative remains positive for finite positive exit pressure, so there is no finite-area optimum from this criterion alone. A practical vacuum nozzle needs additional constraints that this performance model does not optimize.

16.7 Effective exhaust velocity and specific impulse

Define effective exhaust velocity $c_{eff} = F/\dot{m}$. It includes pressure thrust and therefore generally differs from local exit velocity. Specific impulse is

$$I_{sp} = \frac{F}{\dot{m}g_0} = \frac{c_{eff}}{g_0}, \quad c_{eff} = c^*C_F,$$

where $g_0 = 9.80665 \text{ m/s}^2$ is the fixed standard-gravity conversion, not the local gravitational acceleration at the operating altitude. Specific impulse is reported in seconds. A change in local gravity does not change the same engine's reported I_{sp} under this definition.

These relations divide performance into two useful questions: how much mass crosses a given pressure–throat scale, and how effectively that scale produces thrust. They are definitions and bookkeeping relations, not proof that combustion and nozzle losses are physically independent.

16.8 The App's two performance factors

The explicit-property solver accepts a characteristic-velocity factor η_{c^*} and a thrust-coefficient factor η_{C_F} , each in $(0, 1]$. It applies them as

$$c^* = \eta_{c^*}c_i^*, \quad C_F = \eta_{C_F}(C_{F,m} + C_{F,p}),$$

$$\dot{m} = \frac{p_c A_t}{\eta_{c^*}c_i^*}, \quad F = \eta_{C_F}C_{F,i}p_c A_t, \quad I_{sp} = \eta_{c^*}\eta_{C_F} \frac{c_i^*C_{F,i}}{g_0}.$$

At fixed chamber pressure and throat area, reducing η_{c^*} increases the inferred mass flow. This follows from the definition; it should not be described as an increase in combustion quality. At fixed input mass flow, the same change instead reduces the resolved throat area. Always state which scale is held fixed before discussing an efficiency trend.

The thrust-coefficient factor multiplies the sum of momentum and pressure terms, including the vacuum result. The separately reported $cf_momentum$ and $cf_pressure$ remain the ideal reference components. They must be multiplied by the factor before their sum is compared to the reported final cf .

Crucially, these empirical performance factors do not reconstruct a new exit static state. The reported exit Mach, temperature, pressure and velocity remain the ideal thermodynamic reference, and $exit_velocity$ equals $exit_velocity_ideal$. Combining that reference velocity with the corrected mass flow in the raw momentum equation will generally not recover the corrected thrust. Use $F = C_F p_c A_t$ and $F/\dot{m} = c^* C_F$ for the corrected performance. A physically resolved lossy nozzle would need additional state equations beyond these two factors.

16.9 Original numerical example

Choose an illustrative gas with $\gamma = 1.25$, $R = 400 \text{ J/(kg K)}$, and $T_c = 3200 \text{ K}$. Its corresponding molecular weight is $20.786156545 \text{ kg/kmol}$. Set $p_c = 5 \text{ MPa}$, $A_t = 0.002 \text{ m}^2$, ambient pressure 100 kPa and exit Mach 3. These are teaching inputs, not a named propellant or measured motor.

First, the static-to-total temperature factor is $1 + 0.25(3^2)/2 = 2.125$. It gives $T_e = 1505.882353 \text{ K}$ and $p_e = 115391.902 \text{ Pa}$. The area–Mach relation gives $\epsilon = 5.831866839$, so $A_e = 0.011663734 \text{ m}^2$. The ideal exit velocity is 2603.165494 m/s .

Next, $\Gamma = 0.658064792$, and $c_i^* = 1719.239297 \text{ m/s}$. The pressure–throat scale is $p_c A_t = 10000 \text{ N}$. Therefore ideal mass flow is 5.816525959 kg/s . The momentum coefficient is 1.514137967 , and the positive pressure coefficient is 0.017952705 . Their sum is 1.532090672 , giving thrust 15320.906717 N and specific impulse 268.596360 s .

The exit pressure slightly exceeds ambient; the pressure term adds approximately 179.53 N to the momentum thrust. In vacuum the coefficient becomes 1.648728009 . The change in force is $p_a A_e$, approximately 1166.37 N , with the same ideal internal nozzle state.

Now retain the same pressure, temperature and geometry, but set $\eta_{c^*} = 0.95$ and $\eta_{C_F} = 0.97$. The reported mass flow becomes 6.122658904 kg/s, thrust 14861.279516 N, and specific impulse 247.511545 s. The exit reference velocity remains 2603.165494 m/s. These numbers illustrate the exact performance-factor convention; they are direct analytical arithmetic, not App-generated evidence of experimental agreement.

16.10 Scale choices and the two mode names

The same dimensionless nozzle can be assigned a throat area or throat diameter. For a circular throat, $A_t = \pi d_t^2/4$. Alternatively, the user can specify mass flow, sea-level thrust or vacuum thrust. The first requires $A_t = \dot{m}c^*/p_c$. A thrust target requires $A_t = F_{target}/(C_F p_c)$, with the coefficient evaluated in the target environment.

The sea-level-thrust scale uses 101325 Pa for sizing even when the separately selected current environment differs. Vacuum-thrust scale uses zero ambient pressure. After resolving area, the calculation evaluates the current environment. Therefore a nozzle sized to produce a chosen sea-level thrust need not display that same force when the current environment is vacuum.

Scale-free mode reports specific performance without absolute thrust, mass flow or areas. Its internal unit-area evaluation is a normalization device, not a declaration that the user has designed a one-square-metre throat.

The current thrust-chamber and stand-alone nozzle mode names share this explicit-property performance chain. The nozzle selection does not infer a chamber's chemical state, and the thrust-chamber selection does not add an injector, feed system or cooling calculation. Scale choices are resolved before either result is returned.

16.11 Back pressure and the boundary of the result

The model warns when exit pressure falls below ambient. It also adds a possible-separation warning when $p_e < p_a/2.5$. This is a fixed screening

threshold, not a geometry-resolved separation calculation or a universal operating boundary. No separation location, side load or corrected post-separation thrust is returned.

The solver continues the attached supersonic calculation while its coefficient stays positive. If back pressure makes that coefficient nonpositive, it rejects the state. A positive coefficient alone does not establish that the nozzle can physically sustain the assumed supersonic attached branch. In particular, the explicit model does not solve startup, an internal normal shock or an unchoked alternative when back pressure is high.

Report overexpanded results as conditional ideal predictions and retain their warnings. Do not use the presence of a numeric result to erase the flow-regime assumptions. Chapters 17–20 add thermochemistry, chamber geometry, contours and parameter studies while preserving this distinction between a computed reference and a complete physical engine.

17. Equilibrium and frozen rocket expansion

A hot chamber mixture can contain dissociated species as well as stable combustion products. As pressure and temperature change through a nozzle, the preferred chemical composition changes. Whether the gas can approach that composition depends on reaction times compared with the available flow time. An equilibrium calculation assumes continuous chemical adjustment; a frozen calculation suppresses adjustment. Neither assumption supplies a finite-rate chemical history.

ThrustOne provides three thermochemical rocket paths: composition frozen from the chamber, equilibrium maintained through the nozzle, and equilibrium followed by freezing at a user-specified area. They use formation-inclusive NASA9 gas properties and the bounded gas-phase equilibrium model introduced in Chapters 5–6. This chapter follows their energy, entropy and mass-flow closures and explains why their throat definitions must differ.

17.1 Two independent chamber decisions

First decide how chamber temperature is obtained. In a specified-temperature calculation, chamber pressure and temperature are inputs. The calculation determines compatible properties or equilibrium composition at that state; it does not prove that a supplied feed can produce that temperature.

Alternatively, the chamber temperature can be obtained from a constant-pressure enthalpy balance. With negligible chamber velocity,

$$h_c = h_{in} + q,$$

where h_{in} is the gaseous inlet mixture's specific enthalpy and q is heat added per unit inlet mass. The solver searches the supplied temperature bracket for equilibrium products with the target enthalpy. Setting $q = 0$ means adiabatic conversion of the prescribed gaseous feed. Chemical energy is already included in species formation enthalpies; adding a separate heating value would count it twice.

The second decision concerns chamber composition. The frozen rocket mode permits either equilibrium at the chamber temperature and pressure or a directly specified frozen mixture. Solving chamber temperature from the equilibrium enthalpy balance requires chamber equilibrium to be enabled. The equilibrium and specified-freeze-area nozzle modes also require an equilibrium chamber.

These decisions should not be collapsed into the phrase combustion calculation. Specifying a hot mixture, equilibrating a specified hot mixture, and solving an adiabatic reaction temperature are different problems. The current gaseous enthalpy closure excludes liquid vaporization, injector pressure losses, two-phase feed properties and condensed combustion products.

17.2 What is preserved along each path

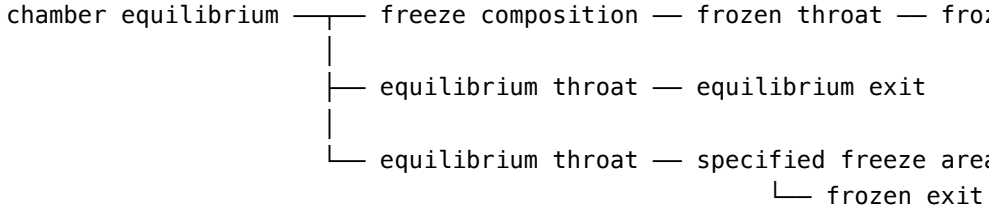
For steady adiabatic flow with no shaft work,

$$h + \frac{V^2}{2} = h_c.$$

For the ideal reversible expansion,

$$s = s_c.$$

These are specific enthalpy in J/kg and specific entropy in J/(kg K), referenced consistently to the chamber state. All paths preserve elemental inventory. A frozen path additionally preserves every species amount relative to the total composition. An equilibrium path preserves elements while allowing the species distribution and total mole count to change.



The frozen and equilibrium paths can have different throat mass fluxes even for the same chamber pressure, temperature, elemental inventory and physical throat area. Thus comparing exit velocity alone does not compare thrust at a fixed throat scale. Both mass flow and pressure thrust must be retained.

17.3 Frozen throat from energy and sound speed

For fixed composition $\mathbf{x}_c$, define frozen sound speed

$$a_f^2(T) = \gamma_f(T) R_c T, \quad \gamma_f(T) = \frac{c_p(T)}{c_p(T) - R_c}.$$

Although species fractions and R_c remain fixed, heat capacity and gamma can vary with temperature. At the frozen sonic throat, $V_* = a_f(T_*)$. Substitution into energy gives the scalar equation

$$h(T_*, \mathbf{x}_c) + \frac{a_f^2(T_*, \mathbf{x}_c)}{2} = h_c.$$

The App brackets this temperature between the common lower data limit and chamber temperature. It then finds the throat pressure from entropy conservation. With a reference pressure p_r ,

$$s(T, p, \mathbf{x}_c) = s(T, p_r, \mathbf{x}_c) - R_c \ln(p/p_r),$$

so

$$p_* = p_r \exp \left[\frac{s(T_*, p_r, \mathbf{x}_c) - s_c}{R_c} \right].$$

This expression includes the mixture's entropy convention consistently. It does not replace the variable-property expansion by a power law using a locally evaluated gamma. Mass flow follows from $\dot{m} = \rho_* a_f A_t$. The frozen chamber-to-nozzle mode requires ambient pressure below this critical static pressure before proceeding with its choked expansion assumption.

17.4 Frozen expansion from an already moving inlet

The shared frozen-nozzle solver also has a stand-alone scramjet-family entry. Its temperature and pressure are static inlet quantities, and its inlet Mach is at least one. Consequently its conserved total enthalpy is

$$h_t = h_i + \frac{M_i^2 a_{f,i}^2}{2},$$

not simply h_i . Its mass flow is $\rho_i M_i a_{f,i} A_i$. A rocket calls this same solver using the sonic throat just calculated; a stand-alone nozzle can start farther downstream at a supersonic state.

At any trial temperature, entropy determines pressure and energy determines velocity:

$$p(T) = p_r \exp[(s(T, p_r) - s_i)/R], \quad V(T) = \sqrt{2[h_t - h(T)]}.$$

Continuity then supplies the implied area,

$$A(T) = \frac{\dot{m}}{\rho(T)V(T)}.$$

For each requested downstream area, the solver inverts this relation on the supersonic branch, decreasing temperature as area increases. It rejects a state that leaves this branch or requires a temperature below the property-data range. No extrapolation is used to manufacture a large-area exit.

The displayed profile uses linearly increasing area along the specified length. Length assigns an axial coordinate; it does not add friction, reaction residence time or a wall-pressure solution. Two lengths with the same inlet and exit areas have the same ideal endpoint thermodynamics in this model.

17.5 Constructing an equilibrium isentrope

For equilibrium expansion, composition is a function of both temperature and pressure, subject to the fixed elemental inventory. At each trial temperature T , the App solves

$$\mathbf{x} = \mathbf{x}_{eq}(T, p; \mathbf{b}), \quad s(T, p, \mathbf{x}) = s_c.$$

The pressure root cannot be found by treating composition as constant in an entropy pressure correction. Each pressure trial requires its own equilibrium composition and thermodynamic properties. The implementation brackets pressure progressively below chamber pressure and searches in logarithmic pressure, preserving positive pressure.

Once the entropy root is found,

$$V(T) = \sqrt{2[h_c - h(T, p(T), \mathbf{x}(T))]}, \quad j(T) = \rho(T)V(T).$$

Here j is mass flux in $\text{kg}/(\text{m}^2 \text{ s})$. The chamber endpoint has zero velocity and zero flux. Moving down the isentrope initially increases velocity

fast enough to increase flux; farther into expansion, decreasing density eventually dominates. The interior maximum determines the critical flux.

The chemistry calculation remains the included ideal-gas species model. No equilibrium with unlisted ions, liquids or solids is implied. An equilibrium residual belongs to that declared system; it does not establish that omitted species are physically negligible under every possible input.

17.6 Why frozen Mach is not the equilibrium choking test

For a reversible equilibrium path with fixed elemental composition, the thermodynamic relation is $dh = dp/\rho$. Differentiating energy gives $V dV = -dp/\rho$. Introduce the equilibrium sound speed through

$$a_{eq}^2 = \left(\frac{\partial p}{\partial \rho} \right)_{s,b,eq}.$$

Then the derivative of mass flux along that path is

$$\frac{dj}{d\rho} = V + \rho \frac{dV}{d\rho} = V - \frac{a_{eq}^2}{V}.$$

At an interior stationary flux, $V = a_{eq}$. This is the chemical-equilibrium choking condition. By contrast, $\sqrt{\gamma_f RT}$ holds local species fixed during the perturbation. It generally represents a different derivative because chemical rearrangement contributes to the equilibrium response.

The App therefore maximizes ρV on its computed equilibrium isentrope instead of imposing $V/a_f = 1$. It reports `throat_frozen_mach` and `exit_frozen_mach` using local frozen sound speed. A throat value different from one is not by itself a failed choking calculation. It must not be relabeled equilibrium Mach.

The search first samples the allowed temperature interval, requires a single sampled interior peak, and refines its bracket. An endpoint maximum

or multiple sampled critical candidates is rejected. This guards the intended branch but does not prove that an arbitrarily complicated isentrope is globally unimodal. The current solver is explicitly restricted to this single-maximum regime.

17.7 Area selects the downstream equilibrium state

Once critical flux j_* is known, $\dot{m} = j_* A_t$. For exit expansion ratio ϵ , the target is

$$j_e = \frac{j_*}{\epsilon}.$$

The same flux below its maximum can occur on two sides of the throat. The nozzle selects the lower-temperature downstream branch, searching between its minimum allowed temperature and the throat. At $\epsilon = 1$, it returns the throat directly. It checks the area/mass-flow residual after inversion.

The user-specified minimum nozzle temperature bounds the equilibrium path search. If the critical peak is outside the interval, or the desired exit flux requires a still colder state, the result is refused. Increasing area without extending physically supported thermodynamic data cannot be resolved by changing an iteration count.

Ambient pressure must lie below critical static pressure, but this condition alone is not a complete test for attached supersonic flow throughout a long nozzle. The thermochemical modes do not reconstruct shocks or post-separation states. Their final thrust expression remains conditional on the ideal expansion path.

17.8 A specified freezing area

In the partially frozen model, equilibrium is followed to $\epsilon_f = A_f/A_t$, where $1 < \epsilon_f \leq \epsilon_e$. The transition transfers the equilibrium temperature, pressure, velocity, mass flow and full composition to the frozen solver. No heat or impulse is introduced at that interface. Total enthalpy and entropy

remain continuous, and the downstream species fractions are held at their transition values.

The frozen continuation requires $V_f/a_{f,f} > 1$. Being downstream of the equilibrium critical flux does not automatically satisfy this separate frozen-branch condition. A selected point too close to the equilibrium throat may therefore be refused, with a request for a farther downstream freeze area.

When freeze area equals exit area, there is no downstream frozen segment. The endpoint is the equilibrium exit, provided the frozen-Mach eligibility condition passes. Freezing exactly at the throat is not accepted by this interface; the input must be strictly greater than one. The chamber-frozen model is a separate path with its own throat and mass flow, not a shortcut obtained by assigning zero downstream equilibrium length.

The freeze location is prescribed. No reaction mechanism, Damköhler-number criterion, residence-time estimate or chemical relaxation equation predicts it here. Varying it is a sensitivity study of an assumption, not a simulation of where a real gas ceases to equilibrate.

17.9 Original inert-limit example

An inert monatomic ideal gas makes the relation between these paths transparent. Take $\gamma = 5/3$, $R = 8314.462618/39.948 = 208.132137$ J/(kg K), $c_p = 2.5R$, chamber temperature 2400 K, chamber pressure 1 MPa and throat area 0.001 m². These define an analytical constant-property teaching limit; the numbers are not represented as a run of the full NASA9 equilibrium solver.

Because no species conversion is available in this constructed gas, equilibrium and frozen thermodynamic derivatives coincide. The throat temperature is 1800 K, throat pressure 487139.290 Pa and critical mass flux 1027.476053 kg/(m² s). Thus mass flow is 1.027476053 kg/s and characteristic velocity 973.258692 m/s.

Choose exit Mach 2. The area–Mach relation gives $\epsilon = 1.53125$; the exit temperature is 1028.571429 K, pressure 120242.511 Pa and veloc-

ity 1194.651329 m/s. The equilibrium flux-maximum construction and frozen sonic construction reduce to the same result because both use the same constant-composition isentrope. An arbitrary freeze point between throat and exit cannot change this gas's composition or endpoint state.

This example explains what should disappear in the inert limit: the distinction between frozen and equilibrium sound speeds, the chemical shift in critical flux, and sensitivity to the chosen freeze location. It does not demonstrate how much recombination occurs in a reacting hydrogen-oxygen mixture. That requires an actual equilibrium calculation with its declared species and temperature bounds.

17.10 Interpreting reactive comparisons

Dissociation stores part of the mixture energy in chemical form. As equilibrium shifts toward more strongly bound species, energy may become available for thermal and kinetic changes. Frozen expansion cannot redistribute that stored energy through changing composition. This explains why equilibrium and frozen performance can differ without an external heat source.

Do not turn that explanation into a universal ranking of every reported quantity. Exit temperature, pressure, mass flow, velocity and thrust are coupled. Comparisons must retain the same chamber closure, elemental inventory, throat area, exit area and environment. In particular, comparing a specified-temperature chamber in one path with an adiabatic enthalpy-closed chamber in another changes two assumptions simultaneously.

Nor do equilibrium and chamber-frozen results constitute rigorous upper and lower bounds on every real engine. Finite-rate chemistry, viscous losses, heat transfer, two-phase effects and separated flow add physical mechanisms absent from both ideal calculations. The partially frozen path is an intermediate chemical assumption, not a correction for those other losses.

17.11 Reading the outputs

For all thermochemical rocket paths, thrust is evaluated from

$$\begin{aligned} F &= \dot{m}V_e + (p_e - p_a)A_e, \\ c^* &= \frac{p_c A_t}{\dot{m}}, \\ C_F &= \frac{F}{p_c A_t}, \\ I_{sp} &= \frac{F}{\dot{m}g_0}. \end{aligned}$$

The explicit-property empirical factors of Chapter 16 are not automatically applied to these paths. Their results follow the computed gas states and ideal momentum balance. Unlike the explicit-property solver's positive-coefficient check, these routines check final finiteness without imposing a general positive-thrust acceptance rule. A finite output is therefore not an assertion of useful propulsion at that back pressure.

The stand-alone frozen nozzle reports `gross_thrust`. It lacks a complete upstream engine control volume and cannot subtract captured-air momentum on its own. The rocket wrapper can identify this same exhaust force as chamber thrust, while a full airbreathing engine must add the appropriate inlet accounting.

Energy residuals measure consistency of total enthalpy; entropy residuals measure the isentropic closure; area residuals measure flow compatibility. They answer different questions. A displayed zero equilibrium residual when chamber equilibrium is disabled indicates that no equilibrium solve was performed, not that the prescribed composition was proved to be an equilibrium mixture.

Profiles show composition changes on the equilibrium segment and fixed fractions after freezing. Their axial coordinate is generated from the prescribed linear area variation, not from a solved nozzle wall contour. A completed value confirms that the chosen mathematical path reached its requested endpoint. It does not establish chemical-rate accuracy, attached

flow or measured hardware performance.

18. Chamber and convergent geometry

A chamber-volume calculation and a chamber-flow calculation answer different questions. The first assigns space upstream of the throat. The second relates a moving gas at a finite-area plane to a total state and a choked throat. Neither, by itself, predicts how an injector vaporizes propellant or how completely combustion occurs.

ThrustOne separates these tasks into three modes: equivalent chamber sizing, cylindrical chamber plus convergent-section sizing, and finite contraction from an explicitly static chamber plane. This chapter develops their equations and the boundaries between them. The separation prevents a geometrical length from being mistaken for a combustion-performance prediction, or a static pressure from being silently reused as a total pressure.

18.1 Characteristic length is a volume measure

Let A_t be throat area and V_c the chamber volume measured from the injector face to the throat plane. Characteristic length is

$$L^* = \frac{V_c}{A_t}.$$

It has units of length, but it is not generally the physical injector-to-throat

distance. Different chamber radii and convergent shapes can have the same volume and hence the same L^* . The input is a prescribed volume criterion; the App does not derive its required value from propellant chemistry.

Define contraction ratio

$$\chi = \frac{A_c}{A_t} > 1,$$

where A_c is the cylindrical chamber cross-section. For circular sections,

$$r_t = \sqrt{A_t/\pi}, \quad r_c = r_t\sqrt{\chi}.$$

An equivalent cylinder occupying the entire prescribed volume would have length

$$L_{eq} = \frac{V_c}{A_c} = \frac{L^*}{\chi}.$$

This is precisely the equivalent-sizing interpretation. It does not reserve part of the volume for a convergent section. Adding a convergent volume to that equivalent cylinder without shortening the cylinder would exceed the original L^* requirement.

18.2 Why volume is not residence time

As a rough background estimate, a uniform gas occupying volume V_c would have an inventory $\rho_c V_c$ and inventory-to-flow time $t_{inv} = \rho_c V_c / \dot{m}$. Substituting $V_c = L^* A_t$, $\rho_c = p_c / (RT_c)$ and $\dot{m} = p_c A_t / c^*$ gives

$$t_{inv} \approx \frac{L^* c^*}{RT_c}.$$

The units are seconds. This derivation explains why a volume normalized by throat area can be useful when comparing similar chambers. It also exposes the assumptions: approximately uniform density, a compatible pressure convention and a steady mass flow.

Actual spray penetration, vaporization, mixing, recirculation and reaction times need not equal this inventory time. Spatially varying density and a finite convergent volume further weaken the uniform approximation. The current sizing modes do not calculate these processes or certify a minimum combustion residence time. A supplied L^* must therefore carry its own engineering rationale outside the calculation.

18.3 Allocating a conical convergent section

The explicit geometry mode uses a circular cylinder followed by a straight conical frustum terminating at the throat. Let θ_c be the convergent half-angle measured from the axis. The required convergent length is

$$L_{conv} = \frac{r_c - r_t}{\tan \theta_c}, \quad 0 < \theta_c < \frac{\pi}{2}.$$

The full included angle is $2\theta_c$. Confusing the included angle with the half-angle changes both length and allocated volume.

To derive the volume rather than memorize a frustum formula, measure x from the start of convergence and write

$$r(x) = r_c - \frac{r_c - r_t}{L_{conv}}x.$$

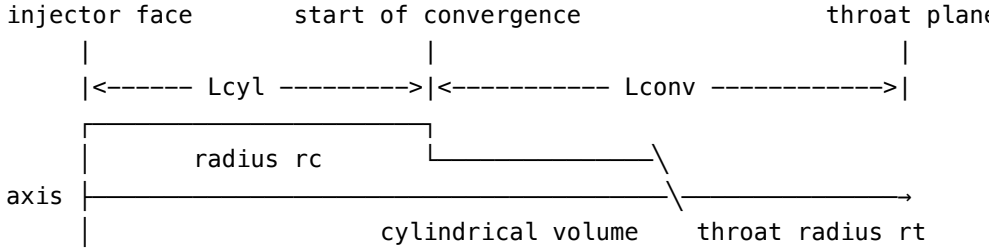
Integrating the circular cross-section gives

$$V_{conv} = \int_0^{L_{conv}} \pi r(x)^2 dx = \frac{\pi L_{conv}}{3} (r_c^2 + r_c r_t + r_t^2).$$

The cylinder receives the remaining volume:

$$V_{cyl} = L^* A_t - V_{conv}, \quad L_{cyl} = \frac{V_{cyl}}{A_c}, \quad L_{total} = L_{cyl} + L_{conv}.$$

The cylindrical volume fraction is $V_{cyl}/(L^* A_t)$. It is a geometric fraction, not the fraction of heat release occurring in the cylinder.



$$V_{cyl} + V_{conv} = L^* A_t$$

The sketch identifies volume boundaries, not a manufactured wall contour. The current profile joins cylinder and frustum directly without a fillet or throat rounding. Those omissions matter for mechanical and thermal design but do not change the declared elementary volume allocation.

18.4 The minimum admissible characteristic length

The convergent section alone must fit inside the prescribed volume. Therefore

$$L^* \geq \frac{V_{conv}}{A_t}.$$

Substitute $r_c = r_t \sqrt{\chi}$ and $A_t = \pi r_t^2$:

$$L_{min}^* = \frac{r_t(\sqrt{\chi} - 1)(\chi + \sqrt{\chi} + 1)}{3 \tan \theta_c} = \frac{r_t(\chi^{3/2} - 1)}{3 \tan \theta_c}.$$

This compact relation shows three effects. A larger throat at the same contraction and angle requires a larger absolute L^* just to contain the convergent volume. Increasing contraction ratio raises that minimum. A shallower half-angle also raises it because the convergent section becomes longer.

At equality, the cylinder has zero length and the whole volume is the frustum. The implementation accepts that geometric limit. Below it, a negative cylinder length would be required, so the input is rejected. It does not clip the length to zero and silently change the requested volume.

This also explains why scaling a chamber is not always simple. If every linear dimension is multiplied by k , volume scales as k^3 , throat area as k^2 , and L^* as k . Holding L^* fixed while increasing throat diameter does not preserve geometric similarity.

18.5 Original volume-allocation example

Choose throat radius 25 mm, contraction ratio 4, characteristic length 0.6 m and convergent half-angle 45° . These are illustrative geometric inputs, not a recommended chamber design. The throat area is 0.0019634954 m^2 ; chamber radius is 50 mm and chamber area 0.0078539816 m^2 .

The convergent length is 25 mm. Its volume is 0.0001145372 m^3 , or 0.114537 L . The total required volume is 0.0011780972 m^3 , or 1.178097 L . Thus the cylinder receives 1.063560 L and has length 0.1354167 m . Injector-to-throat length is 0.1604167 m .

The equivalent-cylinder result for the same volume would instead be $L_{eq} = 0.6/4 = 0.15 \text{ m}$. It is neither the actual cylindrical segment length nor the complete cylinder-plus-convergent length. The tapered section contains less volume per unit axial length than the full chamber cylinder, which is why allocating it increases total physical length beyond the equivalent value.

For this throat and angle, $L_{min}^* = 0.0583333 \text{ m}$. Requesting $L^* = 0.05 \text{ m}$ would provide only 0.098175 L , less than the convergent section's 0.114537 L . Refusal is the correct geometric outcome. Reducing

a numerical tolerance would not solve that incompatibility; the requested geometry or volume must change.

18.6 A finite-area chamber plane has velocity

Now consider the separate flow problem. Inputs are throat area, contraction ratio, constant γ , specific gas constant R , and chamber-plane static pressure p_c and temperature T_c . Assume an isentropic subsonic contraction to a choked throat. The chamber Mach is the subsonic solution of

$$\chi = \frac{1}{M_c} \left[\frac{2}{\gamma + 1} \left(1 + \frac{\gamma - 1}{2} M_c^2 \right) \right]^{\frac{\gamma + 1}{2(\gamma - 1)}}.$$

The selected root lies strictly between zero and one. A supersonic root would describe a different branch and is not an alternative chamber solution in this mode. The implementation solves the logarithm of the area relation, which avoids directly raising large factors during the search. Contraction ratio must be greater than one; the exactly sonic chamber plane is outside this interface.

Once M_c is obtained,

$$V_c = M_c \sqrt{\gamma R T_c}, \quad T_{tc} = T_c \left(1 + \frac{\gamma - 1}{2} M_c^2 \right),$$

$$p_{tc} = p_c \left(1 + \frac{\gamma - 1}{2} M_c^2 \right)^{\gamma/(\gamma - 1)}.$$

The total pressure is higher than the supplied static pressure because the static plane contains kinetic energy. This is an isentropic state conversion. It must not be confused with a combustion-induced total-pressure loss or with a correction between two different heat-addition chamber models.

At large contraction ratio, M_c approaches zero and static and total states approach each other. At a contraction ratio close to one, kinetic energy

is substantial and treating static pressure as reservoir pressure produces a meaningful flow error.

18.7 Mass continuity links the two planes

The chamber-plane flow is

$$\dot{m}_c = \frac{p_c}{RT_c} V_c A_c.$$

Using the recovered total state, the choked-throat expression is

$$\dot{m}_* = A_t p_{tc} \sqrt{\frac{\gamma}{RT_{tc}}} \left(\frac{2}{\gamma + 1} \right)^{\frac{\gamma+1}{2(\gamma-1)}}.$$

These must agree because the area–Mach equation was derived from that same isentropic continuity relation. The reported mass residual is $\dot{m}_c - \dot{m}_*$, in kg/s; its internal acceptance uses the residual relative to throat flow. The separate area residual is dimensionless. Neither residual measures combustion completeness or real nozzle discharge behavior.

The current mode does not insert its total state into other rocket calculations automatically. Its inputs belong to an independent static-plane problem. Transferring its results to a performance calculation requires deliberately using the recovered total pressure and temperature, with compatible gas properties and scale. Reusing its original static inputs as total conditions would erase the very distinction the mode computes.

18.8 Original static-to-total example

Use the same throat radius 25 mm for scale, but specify a separate flow case with $\gamma = 1.4$, $R = 300 \text{ J/(kg K)}$, static temperature 2000 K and static pressure 3 MPa. Choose a contraction ratio 2.96352; this was constructed from $M_c = 0.2$ using the area relation, so the subsonic inverse is transparent.

The temperature factor is 1.008, giving total temperature 2016 K. The total pressure is 3.084843363 MPa and the chamber velocity 183.303028 m/s. Continuity gives mass flow 5.333071369 kg/s. The throat calculation using the recovered total state gives the same analytical flow.

If static pressure and temperature were mistakenly inserted directly as the throat reservoir state, the inferred mass flow would be smaller by the factor

$$\frac{\dot{m}_*}{\dot{m}_{wrong}} = \frac{p_{tc}}{p_c} \sqrt{\frac{T_c}{T_{tc}}} = 1.008^{\gamma/(\gamma-1)-1/2} = 1.008^3.$$

The ratio is about 1.02419. Even this modest chamber Mach therefore produces a roughly 2.4% change relative to that mistaken flow estimate. This original example tests the meaning of the input planes; it is not a measured finite-contraction combustion correction.

18.9 What the App combines, and what it keeps separate

Equivalent sizing and cylinder-plus-convergent sizing obtain their absolute throat scale from the resolved explicit-property rocket definition. Scale-free input is rejected for these dimensions. Because the current routing first resolves the base performance point, an invalid base rocket input can prevent geometry output even when the elementary geometry formulas themselves would be meaningful. The geometric mode does not repair that upstream input.

Equivalent sizing retains the base performance results and adds equivalent dimensions. The explicit convergent allocation returns its volume and length metrics with a simple radius profile. Neither applies a finite-contraction total-pressure correction to the base thrust calculation. The static-plane finite-contraction mode uses its own independent inputs and does not require that base performance route.

For the explicit geometry, radians enter the mathematical kernel; the UI

half-angle in degrees is converted before evaluation. Volumes are m^3 and lengths m. Reporting a value in litres or millimetres requires the corresponding unit conversion, not a new geometric model.

The finite-contraction mode warns that finite-velocity heat-addition losses can matter at small contraction ratios, but does not calculate them. The volume modes exclude evaporation, combustion completion and injector design. Wall thickness, cooling passages, throat erosion, manufacturing fillets and structural life are also outside the result. The correct deliverable is a clearly defined ideal state or volume allocation, with enough information to use it consistently in the next calculation.

19. Nozzle reference contours

An expansion ratio fixes the inlet-to-exit area relationship of an ideal nozzle. It does not fix the wall shape between those planes. A long cone and a shorter bell can share the same throat and exit areas while having different wall angles, lengths and real flow losses. The one-dimensional performance calculation cannot distinguish those effects unless an additional model supplies them.

ThrustOne's contour mode constructs reference geometry: a straight cone or a circular throat turn followed by a quadratic parabolic arc. The bell's endpoint angles are explicit inputs. No method-of-characteristics solution, empirical angle chart or thrust optimization selects them. This chapter develops the construction so that the resulting curve can be interpreted as geometry rather than as an implied aerodynamic optimum.

19.1 Coordinates and scale

Use axial coordinate x , radius r , and the throat plane as $x = 0$. For circular throat area A_t and expansion ratio ϵ ,

$$r_t = \sqrt{A_t/\pi}, \quad r_e = r_t\sqrt{\epsilon}.$$

A radius profile determines area through $A(x) = \pi r(x)^2$. The plotted curve is the gas-side reference radius, not wall thickness or an outer structural envelope. The model includes no cooling-channel allowance or manufacturing tolerance.

Absolute scale is required for the App's exported contour coordinates. The scale-free performance option therefore cannot supply this profile. Scaling every radius and axial coordinate by a common factor preserves shape and expansion ratio, while all areas scale by the square of that factor.

19.2 The straight conical section

For a cone with half-angle θ measured from the axis,

$$L_{cone} = \frac{r_e - r_t}{\tan \theta}, \quad r(x) = r_t + x \tan \theta, \quad 0 \leq x \leq L_{cone}.$$

This is the divergent section only. It begins at the throat radius but does not include a smooth throat arc or the upstream convergent chamber. In the App, points are spaced uniformly in axial coordinate. The resulting area is quadratic in x , even though radius is linear.

At fixed end radii, a smaller angle increases length. A very large angle shortens the cone but does not establish that a physically attached, efficient exhaust follows it. The numerical domain is an angle strictly between zero and 90° , with finite positive throat area. It is a geometric domain rather than a recommended design-angle range.

The conical implementation permits expansion ratio one through the underlying performance definition. In that limiting case its length is zero and its points coincide at the throat radius. This is a degenerate profile, not a finite divergent nozzle. The bell interface instead requires expansion ratio strictly greater than one.

19.3 Meaning of the conical divergence factor

The contour output includes

$$\lambda_{cone} = \frac{1 + \cos \theta}{2}.$$

A simple background interpretation comes from averaging axial projection over an idealized spherical outflow cap. If momentum per solid angle and speed are uniform, the mean axial projection is

$$\frac{\int_0^\theta \cos \phi \sin \phi d\phi}{\int_0^\theta \sin \phi d\phi} = \frac{\sin^2 \theta}{2(1 - \cos \theta)} = \frac{1 + \cos \theta}{2}.$$

This derivation explains the geometric projection factor and its assumptions. It does not establish a resolved velocity distribution on a real conical nozzle's exit plane. At 15° the factor is 0.9829629; it tends to one as angle tends to zero, while cone length diverges for nonzero radius change.

The App reports this factor separately and does not multiply it into the already prescribed thrust-coefficient efficiency. Applying it again without knowing what that supplied efficiency represents could double-count a loss. A reference contour calculation is not an instruction to replace every nozzle loss by this single projection factor.

19.4 Bell length and the throat-turn circle

The bell uses a length fraction $f_L > 0$ relative to a 15° reference cone:

$$L = f_L \frac{r_e - r_t}{\tan 15^\circ}.$$

The input is a multiplier: 0.8 means 80% of that reference length. The implementation does not impose an upper bound of one. A value greater than one is geometrically possible if the remaining constraints permit it; the word fraction does not itself force a shorter contour.

Let the throat-turn radius be $r_a = k_a r_t$, with positive supplied ratio k_a . Starting at $(0, r_t)$, parameterize the circular arc by its tangent angle ϕ :

$$x(\phi) = r_a \sin \phi, \quad r(\phi) = r_t + r_a(1 - \cos \phi), \quad 0 \leq \phi \leq \theta_n.$$

Differentiating gives $dr/dx = \tan \phi$. The throat tangent is axial, and the join point N has prescribed tangent angle θ_n :

$$x_n = r_a \sin \theta_n, \quad r_n = r_t + r_a(1 - \cos \theta_n).$$

The circular segment is downstream of the throat. It does not construct an upstream rounding or join to the convergent geometry of Chapter 18.

19.5 The tangent intersection fixes the parabola

Let the exit be $E = (L, r_e)$, with tangent slope $m_e = \tan \theta_e$. The join tangent has slope $m_n = \tan \theta_n$. Their intersection $Q = (x_q, r_q)$ satisfies

$$r_q = r_n + m_n(x_q - x_n) = r_e + m_e(x_q - L).$$

Solving these two line equations gives

$$x_q = \frac{r_e - r_n + m_n x_n - m_e L}{m_n - m_e}, \quad r_q = r_n + m_n(x_q - x_n).$$

With N, Q, E as control points, the quadratic curve is

$$\mathbf{B}(t) = (1 - t)^2 \mathbf{N} + 2(1 - t)t \mathbf{Q} + t^2 \mathbf{E}, \quad 0 \leq t \leq 1.$$

Its derivative is $2[(1 - t)(\mathbf{Q} - \mathbf{N}) + t(\mathbf{E} - \mathbf{Q})]$. Therefore its initial tangent lies along NQ , and its exit tangent along QE . These tangents have exactly the two prescribed slopes. The curve generally does not pass through Q ; Q controls direction and shape.

A nondegenerate quadratic curve of this form is a parabolic arc in its own orientation. It need not be representable as a simple quadratic radius function $r = ax^2 + bx + c$ in the nozzle's chosen axes. The parametric representation avoids imposing that extra restriction.

The arc and parabola share position and tangent at N . They are not required to share curvature. A tangent-continuous join is not automatically a curvature-continuous or aerodynamically optimized join.

19.6 Monotonicity and admissible endpoint angles

The model requires

$$0 \leq \theta_e < \theta_n < 90^\circ, \quad x_n < x_q < L, \quad r_n < r_q \leq r_e.$$

The ordered control-point coordinates keep the quadratic segment moving downstream and outward. Its derivatives are convex combinations of positive axial increments and nonnegative radial increments, so it cannot double back between sampled points under these constraints. At zero exit angle, $r_q = r_e$, permitting a horizontal exit tangent.

An equivalent way to understand the length restriction is to compare the endpoint secant slope

$$s = \frac{r_e - r_n}{L - x_n}$$

with the endpoint tangent slopes. A useful monotonic construction requires $m_e < s < m_n$. For a positive exit angle this becomes

$$x_n + \frac{r_e - r_n}{m_n} < L < x_n + \frac{r_e - r_n}{m_e}.$$

Too short a bell asks for a mean outward slope larger than its initial tangent. Too long a bell asks for a mean slope smaller than its exit tangent. Either makes the chosen tangent intersection incompatible with the required control-point order. When $m_e = 0$, there is no finite upper bound from this secant test, although the other positivity constraints still apply.

Changing the throat-turn radius changes N , and therefore changes these length bounds. It cannot be varied independently while assuming that the previously acceptable parabola remains valid.

19.7 Original bell construction

Choose throat radius 20 mm, expansion ratio 9, throat-turn ratio 0.5, initial angle 35° , exit angle 5° and length multiplier 0.8. The exit radius is 60 mm. The 15° reference cone is 149.282032 mm long; the selected bell length is 119.425626 mm.

The circular arc ends at

$$N = (5.735764, 21.808480) \text{ mm.}$$

The tangent intersection is

$$Q = (51.833490, 54.086454) \text{ mm,}$$

and the exit is $E = (119.425626, 60)$ mm. These satisfy both coordinate-order conditions. At $t = 0.5$, the parabolic segment passes through $(57.207092, 47.495347)$ mm. This midpoint is neither the tangent intersection nor necessarily the point halfway along physical arc length.

The secant-derived length bounds are approximately 60.278908 and 442.266841 mm. Selecting 40 mm while retaining the same throat arc, end radius and angles violates the lower bound. The correct response is to reject that shape, not to draw an inward-turning or backward-running curve and label it a bell.

These coordinates were obtained directly from the displayed construction for an original example. They describe a reference curve and do not carry a measured thrust efficiency or wall-pressure prediction.

19.8 Sampling and profile interpretation

The bell profile allocates roughly one fifth of its requested points to the circular arc, with at least two arc points. It uses the remaining count for the quadratic segment and avoids duplicating the shared join. Arc points are uniform in angle; quadratic points are uniform in parameter t . Neither spacing is generally uniform in x or physical arc length.

Increasing point count makes a plotted polyline smoother and changes the discretization of exported geometry. It does not improve the underlying aerodynamic physics, because no aerodynamic field is solved. Point-count limits are 5–1000 for the bell and 2–1000 for the conical profile. Each row reports axial coordinate, radius and local area ratio $(r/r_t)^2$.

The end area of the constructed bell is the requested expansion ratio, and its end slopes are fixed analytically. A smooth-looking polyline alone cannot establish curvature quality near the join, manufacturability, absence of internal shocks or acceptable heat flux. These are separate physical or geometric questions beyond the supplied construction.

19.9 Geometry versus optimized nozzle design

A thrust-optimized contour requires an objective, constraints and a flow model connecting wall shape to performance. Depending on the design problem, those constraints might include length, exit area, chamber conditions, pressure distribution or integrated wall force. The present mode does not solve that optimization. Its explicit length and angle inputs already choose the geometry before the points are generated.

The familiar use of a parabola as an approximation to a designed bell does not imply that every parabola with plausible angles is optimal. Likewise, an 80% reference length is a geometric convention, not an 80% efficiency. The underlying performance remains the one-dimensional reference calculated from the same throat, expansion ratio, gas properties and supplied efficiency factors.

Changing the contour angles while keeping those performance inputs fixed therefore does not automatically change the App's predicted thrust. That behavior follows the current separation of geometry and performance; it should not be read as evidence that real nozzle thrust is independent of wall shape.

Use the profile to inspect endpoint geometry, length allocation and admissibility. Preserve the declared boundary when handing the curve onward: it excludes wall pressure, boundary layers, separation position, side loads, cooling, thickness and a method-of-characteristics flowfield. A reference

contour can support later engineering work without pretending that those additional analyses have already been performed.

20. Rocket performance studies

A parameter study is defined as much by what stays fixed as by what changes. Raising chamber pressure at fixed throat area is different from resizing a throat to preserve thrust. Changing mixture ratio at fixed chamber temperature is different from predicting a new adiabatic temperature. Curves without these constraints are easy to misinterpret even when every individual point is calculated correctly.

ThrustOne provides altitude, area-ratio and chamber-pressure studies using the explicit-property rocket model, plus a gaseous mixture-ratio study using a selected thermochemical expansion path. Each is a sequence of bounded point calculations. The program does not turn these scans into a trajectory, feed-system model or unrestricted optimization.

20.1 The baseline is resolved once

For the three explicit-property studies, the initial rocket definition first resolves the environment, nozzle input and absolute scale as described in Chapter16. The resulting throat area and effective gas properties become the baseline. Each scan row copies that resolved input and changes one quantity.

Study	Quantity changed per row	Important quantities retained
Altitude	Ambient pressure from standard atmosphere	Chamber state, gas properties, throat and exit geometry
Area ratio	Exit-to-throat area ratio	Chamber state, throat area, gas properties, ambient pressure
Chamber pressure	Chamber total pressure	Chamber temperature, gas properties, throat and exit geometry, ambient pressure

A baseline sized for a specified thrust is therefore not resized at every scan point. Similarly, a baseline nozzle defined by an optimum altitude is converted once into area ratio. The altitude study subsequently evaluates that fixed nozzle at changing back pressure.

The top-level metrics retain the resolved baseline performance and add scan counts. The row profiles contain the changing study results. Reading a baseline metric as the last scan point would mix two meanings. Scale-free mode suppresses absolute thrust, mass flow and area fields in the rows, preserving only the specific and dimensionless interpretation.

20.2 Altitude changes external pressure force

Under the fixed attached-flow assumptions of Chapter 16, altitude changes ambient pressure p_a , while exit pressure, velocity and choked mass flow stay fixed. With the supplied thrust-coefficient factor η_{C_F} ,

$$F(p_a) = \eta_{C_F} [p_c A_t C_{F,vac,i} - p_a A_e].$$

Hence

$$\frac{dF}{dp_a} = -\eta_{C_F} A_e.$$

The force response is linear in ambient pressure, but not generally linear in altitude because atmosphere pressure is not linear in height. The calculation uses the bounded standard-atmosphere relation; it does not use weather, vehicle speed or aerodynamic drag to modify chamber thrust.

This study therefore isolates back-pressure sensitivity. It does not predict vehicle acceleration, required thrust, propellant depletion or altitude reached. Those require vehicle mass, trajectory and operating schedules absent from this model.

20.3 Area ratio changes the internal expansion

Changing $\epsilon = A_e/A_t$ at fixed throat area changes both exit area and the supersonic area–Mach solution. Exit temperature, pressure and velocity must be recomputed. Mass flow remains fixed in this explicit-property study because chamber state, throat area and characteristic-velocity factor remain fixed.

For ideal attached expansions, Chapter 16 derived

$$dF_i = (p_e - p_a) dA_e.$$

A positive exit-pressure excess rewards more area; a negative excess penalizes it. At nonzero ambient pressure, the ideal stationary point occurs at matched exit pressure. In vacuum, no finite-area maximum follows from this relation alone. A discrete scan can still show diminishing gains without establishing a practical optimum.

The area-ratio study does not generate a new wall contour at every point, estimate nozzle weight or apply a separation correction. The larger exit might imply a longer or differently shaped nozzle in hardware, but that geometry is not inferred by this curve.

20.4 Chamber-pressure scaling and the meaning of throttle

Let $k_e = p_e/p_c$ and hold the explicit gas state and expansion ratio fixed. Both k_e and the ideal momentum coefficient remain constant. Then

$$F = \eta_{C_F} A_t [p_c (C_{F,m} + k_e \epsilon) - p_a \epsilon], \quad \dot{m} = \frac{p_c A_t}{\eta_{c^*} c_i^*}.$$

Mass flow is proportional to chamber pressure. Thrust has a linear pressure dependence with a fixed negative ambient-force intercept. Specific impulse follows

$$I_{sp} = \frac{\eta_{c^*} \eta_{C_F} c_i^*}{g_0} \left[C_{F,m} + k_e \epsilon - \frac{p_a \epsilon}{p_c} \right].$$

At vacuum back pressure it is constant in this explicit-property scan. At positive ambient pressure, increasing chamber pressure reduces the relative pressure penalty. This is an algebraic pressure sweep, not a complete throttling model: chamber temperature, combustion efficiency, mixture ratio, injector pressure drop and pump behavior do not automatically change with pressure.

The finite coefficient becomes nonpositive below a model-specific pressure threshold. A failed row there should not be interpreted as a solved low-throttle operating state. Even above that threshold, a physical nozzle may not sustain the assumed attached supersonic branch.

20.5 Original pressure-study example

Reuse the original gas and geometry of Chapter16: $\gamma = 1.25$, $R = 400$ J/(kg K), chamber temperature 3200 K, throat area 0.002 m², expansion ratio 5.831866839 and unity performance factors. At chamber pressure 5 MPa, the vacuum coefficient is 1.648728009, characteristic velocity 1719.239297 m/s and exit pressure 115.391902 kPa.

At ambient pressure 100 kPa, thrust is 15320.906717 N. Reducing back pressure to 50 kPa raises thrust by $50000A_e = 583.186684$ N, giving 15904.093401 N. Mass flow stays 5.816525959 kg/s. These are pressure levels for a transparent arithmetic example; they are not assigned invented standard-atmosphere altitudes.

Now instead retain ambient pressure 100 kPa and raise chamber pressure to 10 MPa. Mass flow doubles to 11.633051918 kg/s and exit pressure doubles to 230.783804 kPa. Thrust becomes

$$F = 20000(1.648728009) - 1166.373368 \approx 31808.186802 \text{ N}.$$

This exceeds twice the original thrust by 1166.373368 N because the ambient-force penalty is subtracted once, not twice. Calling the result simply twice the thrust would overlook that fixed force term.

The example is direct arithmetic from the earlier analytical state, not a numerical dataset generated by the App. It demonstrates which outputs are proportional and which retain an ambient offset.

20.6 Mixture ratio is a mass ratio

The thermochemical study defines

$$MR = \frac{m_{oxidizer}}{m_{fuel}}.$$

The supported scan selects H₂, CH₄ or CO fuel with O₂ oxidizer. On a basis of one kilogram of fuel, the feed amounts are

$$n_f = \frac{1}{\overline{W}_f}, \quad n_o = \frac{MR}{\overline{W}_o},$$

in kmol when molecular weights are kg/kmol. The total feed mass is $1 + MR$ kg on this bookkeeping basis. These amounts set the elemental inventory; they are not a mass-flow schedule for the sized nozzle.

For an illustrative molecular-weight approximation of 2 kg/kmol for H₂ and 32 kg/kmol for O₂, a mass ratio 8 gives 0.5 kmol fuel and 0.25 kmol oxidizer: a 2:1 molecular ratio. Entering 8 as a mole ratio would describe an entirely different mixture. The App uses its property database molecular weights rather than these deliberately rounded teaching values.

Stoichiometry identifies elemental proportions for a chosen complete reaction. It does not alone identify the maximum specific impulse. Temperature, molecular weight, dissociation, recombination and nozzle expansion all contribute to that objective.

20.7 What changes in each chemical sample

Every mixture sample uses chamber equilibrium and the same selected nozzle model: frozen, equilibrium or specified-area freezing. Geometry and back pressure stay fixed. The selected freeze area, when applicable, remains a prescribed assumption at every sample; a sample can fail if that location does not admit the required supersonic frozen continuation.

With specified chamber temperature, that temperature is held constant while feed composition and equilibrium products change. This examines composition sensitivity at an imposed thermal state. It is not an adiabatic combustion-temperature prediction.

With enthalpy closure, the inlet temperature and added heat per unit mixture mass are retained. Each mixture has a new inlet specific enthalpy and elemental inventory, so the chamber temperature is solved afresh. Holding heat per unit mixture mass fixed does not mean a fixed total heater power: the sized nozzle's mass flow can change.

Formation-inclusive properties carry chemical energy throughout this calculation. No separate fuel heating value should be added on top of the equilibrium enthalpy balance. The scan remains gaseous and excludes liquid vaporization, condensed phases and finite-rate chemistry. It is not interpolation between tabulated propellant performance rows.

20.8 Sampling, failed rows and maxima

Explicit-property scans use equally spaced coordinates between finite endpoints, with 2–1000 samples. Their implementation permits decreasing or identical endpoints; those choices give reverse ordering or repeated points rather than a new physical branch. The mixture scan specifically requires a positive increasing mass-ratio interval and 2–101 points.

Each attempted point retains its coordinate and a status. Successful rows contain numerical metrics; failed rows do not acquire fabricated zero thrust or interpolated properties. Failure may indicate invalid input, an unsupported flow branch, insufficient thermodynamic range or a solver limitation. It is not automatically evidence of a physical discontinuity.

The mixture scan reports the largest specific impulse among its successful samples. If the largest value occurs at an endpoint, the selected interval does not establish a surrounding maximum. If neighboring samples fail, the missing region remains unknown. Even an interior sampled maximum is not proof of a continuous or global optimum.

If all mixture points fail, scan counts remain available but no best-sample pair is reported. The three explicit-property studies do not themselves return an optimized design. Their role is to expose trends under the declared fixed conditions.

20.9 Warning propagation and result limits

The current explicit-property scan records thrown point failures, but it does not append each successful point's individual warning list to the aggregate result. Its baseline warnings therefore must not be treated as a complete per-row overexpansion or separation assessment. A sweep can move into lower exit-to-ambient pressure ratios after a benign baseline. Inspect a selected point through the corresponding single-point calculation before interpreting it as an attached-flow prediction.

Likewise, successful mixture rows carry metrics rather than a full collection of each point's warnings. General scan warnings describe the common chemistry and scope. Neither a status value of one nor a smooth

curve establishes freedom from nozzle separation, acceptable materials, combustion stability or practical feed-system operation.

These limits do not prevent useful comparison. They determine what the comparison supports: an ideal pressure trend, an explicit expansion trade-off, or a bounded chemical sensitivity under fixed geometry. Retain the input state, scan interval, failed rows and chosen objective whenever presenting a conclusion. Chapter34 develops the general interpretation of discrete studies across all engine families.

21. Solid-propellant equilibrium pressure

A burning solid surface supplies gas while the nozzle removes it. In a steady lumped balance, chamber pressure adjusts until these two mass rates agree. Pressure also changes the prescribed surface regression rate, creating feedback between gas generation and discharge. This chapter derives the equilibrium of that simplified model and explains what its sensitivity can and cannot establish.

The App's equilibrium pressure mode takes an explicit burning area, a reference burn rate and pressure exponent, solid density, characteristic velocity and throat area. It does not identify a propellant formulation or infer a burn law from chemistry. Its output is conditional on the supplied law and the assumption of choked discharge.

21.1 Surface regression and generated mass

Burn rate r_b is the speed at which the burning surface recedes normal to itself, measured in m/s. It is not the gas velocity in the port. For burning area A_b and solid density ρ_p , a small regression distance dw consumes volume $A_b dw$. With $dw/dt = r_b$, gas generation is

$$\dot{m}_g = \rho_p A_b r_b.$$

This relation assumes that the consumed solid mass enters the represented

gas inventory. It does not resolve condensed particles, incomplete conversion or separate solid and gas velocities. The burning area is the active surface area, not necessarily the frontal area, port area or exterior grain area.

For the equilibrium mode, A_b is supplied directly and remains fixed during the calculation. Grain dimensions do not determine it. The transient mode instead derives active area from its selected geometry, as developed in Chapter 22. Passing the same input structure to both modes does not make their area definitions interchangeable.

21.2 A reference-pressure burn law

The implemented law is

$$r_b(p) = r_{ref} \left(\frac{p}{p_{ref}} \right)^n, \quad 0 < n < 1.$$

The ratio inside the power is dimensionless, r_{ref} has units of m/s, and p_{ref} is the pressure at which that rate is specified. This representation avoids burying pressure units inside a fitted coefficient. The App evaluates it using the logarithm of the pressure ratio.

An equivalent form is $r_b = ap^n$, with $a = r_{ref}/p_{ref}^n$. The numerical value and units of a depend on the pressure unit used. A coefficient fitted using MPa cannot be inserted into a Pa-based expression without conversion. For example, converting pressure from MPa to Pa multiplies the corresponding coefficient by 10^{-6n} , before any additional conversion of regression-rate units.

Reference pressure is a representation choice if the reference rate is transformed consistently. Replacing p_{ref} by kp_{ref} while representing the same physical curve requires replacing r_{ref} by $k^n r_{ref}$. Changing reference pressure alone changes the law rather than merely relabeling it.

A power law is normally an approximation over a declared pressure and material-temperature range. The current input has no fitted validity interval or grain-temperature correction. The program therefore cannot certify

that a mathematically accepted pressure lies within the range of the supplied empirical data.

21.3 Choked discharge and the pressure balance

With supplied characteristic velocity $c^* > 0$, the nozzle mass-flow model is

$$\dot{m}_n = \frac{pA_t}{c^*}.$$

This uses chamber pressure in the reservoir sense adopted by the lumped model. No finite chamber-velocity correction is inferred. The supplied c^* is not recomputed from the input gas temperature and molecular weight in this equilibrium mode.

At equilibrium, gas storage is stationary:

$$\rho_p A_b r_{ref} \left(\frac{p_{eq}}{p_{ref}} \right)^n = \frac{p_{eq} A_t}{c^*}.$$

Collecting the pressure terms gives

$$p_{eq}^{1-n} = \frac{\rho_p A_b r_{ref} c^*}{A_t p_{ref}^n}.$$

A dimensionless version makes the scaling especially clear. Define

$$S = \frac{\rho_p A_b r_{ref} c^*}{A_t p_{ref}}, \quad \frac{p_{eq}}{p_{ref}} = S^{1/(1-n)}.$$

The numerator of S has units of force and so does its denominator. The equilibrium is positive and unique for positive parameters with $0 < n < 1$. The implementation evaluates the logarithm of the dimensional expression before exponentiation to reduce avoidable overflow in intermediate

products. An extreme result can still exceed finite numerical range and is not accepted.

21.4 Choking is an additional requirement

An algebraic intersection is not enough. The discharge law assumes a choked throat. For nonzero ambient pressure, the App requires

$$\frac{p_{eq}}{p_a} \geq \left(\frac{\gamma + 1}{2} \right)^{\gamma/(\gamma-1)}.$$

At vacuum ambient pressure, this pressure-ratio check is automatically satisfied. At finite back pressure, an equilibrium below the critical ratio is rejected; the program does not replace the nozzle relation with an unchoked discharge equation and find a different equilibrium.

Gamma affects this admissibility test. In the current equilibrium formula it does not separately alter the mass flow after c^* has been supplied. Treating gamma and c^* as independently calibrated inputs is therefore part of the model. Giving them mutually inconsistent physical interpretations can produce a numerical result without a coherent gas model.

Choking does not establish the absence of shocks or separation downstream, nor does it qualify a complete nozzle. This mode has no exit area or wall contour from which to calculate those effects.

21.5 Sensitivity to area and burn-law inputs

For fixed exponent and reference pressure, logarithmic differentiation gives

$$d \ln p_{eq} = \frac{1}{1-n} (d \ln \rho_p + d \ln A_b + d \ln r_{ref} + d \ln c^* - d \ln A_t).$$

The pressure elasticity with respect to burning area is $1/(1-n)$, while the elasticity with respect to throat area is its negative. As n approaches

one, a small fractional input change causes a much larger fractional pressure change. This is sensitivity of the supplied algebraic model, not a numerical defect that should be hidden by rounding.

For a finite change in burning area by a factor k ,

$$\frac{p_{eq,new}}{p_{eq,old}} = k^{1/(1-n)}.$$

Using the differential approximation for a large change can be inaccurate. For example, at $n = 0.5$, a 10% increase in area gives a 21% pressure increase, because $1.1^2 = 1.21$. The first-order estimate would give 20%.

Exponent sensitivity depends on the position relative to the reference state:

$$\frac{\partial \ln(p_{eq}/p_{ref})}{\partial n} = \frac{\ln S}{(1-n)^2},$$

when S is held fixed. Raising the exponent increases equilibrium pressure if $S > 1$, decreases it if $S < 1$, and leaves the reference intersection unchanged if $S = 1$. A universal statement that larger exponent always means larger equilibrium pressure would therefore be incorrect.

21.6 Original normalized example

Choose a purely illustrative aggregate supply parameter $S = 1.5$ and exponent $n = 0.5$. No real propellant coefficients are assigned. The equilibrium ratio is

$$P = \frac{p_{eq}}{p_{ref}} = 1.5^2 = 2.25.$$

Normalize mass flow by $\dot{m}_{ref} = A_t p_{ref}/c^*$. The discharge is $\dot{m}_n/\dot{m}_{ref} = P = 2.25$. Generation is $\dot{m}_g/\dot{m}_{ref} = SP^n =$

$1.5\sqrt{2.25} = 2.25$. The two rates agree, and the burn rate is 1.5 times its reference value.

Increasing burning area by 10% changes S to 1.65 and gives $P = 2.7225$, a 21% increase. Increasing throat area by 10% instead gives $S = 1.5/1.1$ and $P = 2.25/1.21 \approx 1.85950$. The same percentage change in different areas has opposite effects because one supplies gas and the other removes it.

Now let ambient pressure equal reference pressure and choose $\gamma = 1.4$. The critical ratio is approximately 1.89293. The initial 2.25 pressure ratio passes the model's choking test, but the enlarged-throat result 1.85950 does not. The latter algebraic number must therefore be rejected under the choked-only model. It is not a valid prediction of an unchoked motor state.

21.7 Limited pressure-feedback interpretation

For a fixed gas volume V , fixed gas temperature and constant R , a background lumped pressure equation is

$$\frac{dp}{dt} = \frac{RT}{V}(\dot{m}_g - \dot{m}_n).$$

At the equilibrium point, differentiating the two mass rates gives

$$\left. \frac{d(\dot{m}_g - \dot{m}_n)}{dp} \right|_{eq} = (n - 1) \frac{A_t}{c^*}.$$

For $0 < n < 1$, a small positive pressure perturbation increases discharge faster than generation, producing negative feedback under these assumptions. A linearized perturbation obeys

$$\frac{d\delta p}{dt} = -\frac{RT(1 - n)A_t}{Vc^*}\delta p.$$

This calculation explains a local relaxation tendency in a fixed-volume, instantaneous burn-law model. It is not a combustion-stability assessment. Acoustic modes, delayed surface response, erosive burning, temperature feedback, structural motion and changing burning area can alter real behavior. The App does not compute those mechanisms or report a qualified stability margin.

The transient model also permits chamber free volume and burning area to change. Its pressure equation contains a volume-growth contribution, so this fixed-volume linearization must not be used as a complete explanation of its time history.

21.8 Thrust and supplied coefficients

The equilibrium mode reports

$$F = C_F p_{eq} A_t,$$

using the supplied positive thrust coefficient. It does not recompute C_F from ambient pressure, expansion ratio or nozzle geometry. Ambient pressure enters the choking test; it does not independently subtract a pressure force from this supplied-coefficient thrust expression.

A meaningful use of C_F therefore requires knowing the conditions it represents. If a coefficient already includes a particular pressure-thrust contribution, adding another one externally can double-count it. Conversely, using one constant coefficient across different back pressures does not constitute an altitude-performance model.

Gas temperature and molecular weight are used in the transient gas-storage relation, but not in this equilibrium algebra beyond the separately supplied c^* . Initial pressure, free volume and time-step settings likewise do not alter the equilibrium root. The shared input validation nevertheless requires their values to remain in the common input domain. A field's presence is not evidence that it contributes to every mode's equations.

21.9 From an equilibrium point to a history

The result describes one frozen geometry and one supplied burn law. It contains no elapsed burn time or propellant depletion. As a grain regresses, its active surface can change; applying the equilibrium relation repeatedly then gives a sequence of quasi-steady reference pressures. A real lumped transient additionally stores gas and may lag that sequence.

Chapter 22 develops the implemented established-burning history with geometric regression and discharged mass. Keep the roles separate: the equilibrium mode locates the steady mass-balance intersection; the transient mode follows a finite gas inventory under changing geometry. Neither supplies ignition, tail-off, a qualified material dataset or a complete motor design.

22. Established-burning transients

A steady pressure balance suppresses gas storage. A transient chamber can accumulate or lose gas while the burning surface changes, so its pressure need not equal the instantaneous equilibrium value. ThrustOne's transient solid-rocket mode follows this lumped storage problem from an already pressurized, burning state. It does not calculate ignition or the discharge after propellant burnout.

The model retains the reference burn law and supplied nozzle coefficients of Chapter 21. It adds two idealized regression geometries, a prescribed constant gas temperature and a finite initial free volume. The state variables are regression distance and cumulative discharged mass; chamber gas mass is reconstructed from the exact volume swept by regression.

22.1 Initial inventory and state variables

Let w be regression distance in metres, with $w = 0$ at the initial surface, and D cumulative discharged mass in kilograms. With initial chamber pressure p_i , free volume V_i , gas temperature T and specific gas constant R , initial gas mass is

$$m_{g,i} = \frac{p_i V_i}{RT}.$$

The App obtains $R = 8314.462618/\overline{W}$, with molecular weight in kg/kmol. Temperature and molecular weight remain fixed throughout this history. There is no gas energy equation predicting cooling or a changing combustion temperature.

If $V_b(w)$ is the consumed solid volume, gas mass and free volume become

$$m_g(w, D) = m_{g,i} + \rho_p V_b(w) - D, \quad V_g(w) = V_i + V_b(w).$$

Pressure is reconstructed algebraically:

$$p(w, D) = \frac{[m_{g,i} + \rho_p V_b(w) - D]RT}{V_i + V_b(w)}.$$

The initial free volume is an independent input. It must represent the intended gas space consistently with the chosen grain geometry; the solver does not derive an initial casing volume and subtract the grain to obtain it.

22.2 End-burning geometry

For an end burner of outer radius r_o , optional inner radius r_i , and axial grain length L_g , the active face area is

$$A_b = \pi(r_o^2 - r_i^2).$$

The model exposes one annular or circular end face, with the other surfaces effectively inhibited. Regression proceeds axially:

$$V_b(w) = \pi(r_o^2 - r_i^2)w, \quad 0 \leq w \leq L_g.$$

Thus $dV_b/dw = A_b$, and the area remains constant before burnout. A nonzero inner radius does not activate inner-bore burning in this mode. It simply removes material from the end face and initial propellant volume.

Constant burning area does not force constant chamber pressure. Initial storage, free-volume growth and the pressure-dependent regression law can still produce a transient. Calling the geometry neutral describes its area history, not a guarantee of a perfectly flat thrust curve.

22.3 Internal-bore geometry

For internal-bore burning, only the cylindrical inner surface regresses; end faces and outer surface do not burn. The instantaneous port radius is $r_i + w$, and

$$A_b(w) = 2\pi(r_i + w)L_g,$$

$$V_b(w) = \pi[(r_i + w)^2 - r_i^2]L_g = \pi(2r_iw + w^2)L_g.$$

Burnout occurs at $w = r_o - r_i$. A strictly positive initial bore radius is required for this mode. Its area increases with regression, creating a progressive gas-generation tendency. Free volume increases simultaneously, so the pressure response cannot be inferred from area alone.

Both geometries begin with propellant volume $\pi(r_o^2 - r_i^2)L_g$. Remaining propellant mass is that initial volume minus $V_b(w)$, multiplied by density. These are ideal geometric inventories; no sliver, cracking, deformation or erosion correction is inferred.

22.4 Evolution equations and pressure storage

The integrated equations are

$$\frac{dw}{dt} = r_{ref}(p/p_{ref})^n, \quad \frac{dD}{dt} = \frac{pA_t}{c^*}.$$

They are coupled through the reconstructed pressure. To understand their physical content, differentiate $pV_g = m_gRT$ at constant R, T . Since $\dot{V}_g = A_b r_b$ and $\dot{m}_g = \rho_p A_b r_b - \dot{m}_n$,

$$\frac{dp}{dt} = \frac{RT}{V_g}(\rho_p A_b r_b - \dot{m}_n) - \frac{p}{V_g} A_b r_b.$$

The final term is the pressure effect of newly available volume. It is absent from a fixed-volume chamber equation. At an instant when generated and discharged mass rates are equal, pressure still decreases if the gas volume is growing. An instantaneous mass-flow equilibrium is therefore not exactly a constant-pressure condition in this model.

The equations assume immediate conversion of the consumed solid into the represented gas inventory and instantaneous response to the pressure-dependent burn law. They contain no ignition delay, surface thermal memory or acoustic pressure variation.

22.5 Why the implementation integrates discharged mass

Integrating regression determines consumed volume from the exact geometry. Integrating discharged mass then gives the gas inventory by subtraction. This retains the bookkeeping identity

$$m_{g,i} + m_{burned} - D - m_g = 0$$

at the reconstructed state, up to arithmetic error. It avoids separately integrating gas generation and remaining solid mass with inconsistent geometric approximations.

This identity is useful but limited. A small mass residual does not prove that a large time step has accurately followed the pressure history. The integration can preserve its inventory equation while approximating the timing of regression poorly. Conservation and temporal accuracy answer different questions.

The App uses classical fourth-order Runge–Kutta updates for w , D . Intermediate states must have nonnegative regression and discharge, positive

gas mass and finite positive pressure. An invalid tentative update is retried with a smaller step, for a bounded number of halvings. This is a validity recovery mechanism, not a local truncation-error estimator or an automatic accuracy guarantee.

22.6 Original inventory example

Consider a dimensionless teaching snapshot. Let initial gas mass be one mass unit, initial free volume one volume unit, and choose units so $RT = 1$. Suppose regression has consumed 0.2 volume units of solid, whose density is 10 mass units per volume unit, while cumulative discharge is 1.5 mass units.

Burned mass is 2 units. Gas mass is $1 + 2 - 1.5 = 1.5$, free volume is 1.2, and pressure is $1.5/1.2 = 1.25$ pressure units. The inventory residual is exactly zero in this algebraic example.

If the added free volume were omitted, the same gas inventory would incorrectly give pressure 1.5. If discharge were incorrectly set to 3.2, the reconstructed gas mass would be negative; the state must be rejected rather than assigning a small positive pressure. These numbers illustrate book-keeping only and are not propellant properties or an App-generated time history.

For an internal bore, a regression of $w = 0.1r_i$ increases active area by 10%, but the consumed cross-sectional area is $\pi r_i^2(1.1^2 - 1) = 0.21\pi r_i^2$. Replacing that finite swept area by the initial circumference times regression would give $0.20\pi r_i^2$, undercounting the consumed volume. The exact quadratic expression used by the model avoids that discrepancy.

22.7 Burnout and validity stops

The simulation starts only if initial pressure meets the choked-nozzle criterion. It advances until the requested duration, burnout, loss of the accepted choked regime, or its step limit. These stopping conditions have different meanings.

When an update crosses the geometric web limit, the App bisects the time increment to locate burnout and sets regression to the limiting value. The final burnout row has zero remaining propellant, burn rate and active burning area. Chamber pressure and nozzle discharge can still be positive at that instant. The calculation then stops; it does not integrate the subsequent gas blowdown.

If a candidate endpoint no longer meets the choking criterion, that candidate is not appended. The preceding accepted history is retained. This stop is not a precisely interpolated unchoking event. The reported final time is the last accepted point, not necessarily the instant at which the critical pressure ratio was crossed.

A limit of 10000 accepted steps also bounds work. A requested nominal duration/time-step ratio above that limit is rejected initially; step reductions can nevertheless reach the accepted-step limit before the requested duration. Such an interrupted history is retained with an explanatory reason. It must not be presented as a complete burn.

22.8 Impulse and sampled peaks

With constant supplied C_F and c^* ,

$$F = C_F p A_t = C_F c^* \dot{m}_n.$$

Integrating gives

$$I = \int F dt = C_F c^* D.$$

The App uses cumulative discharged mass to obtain total impulse, rather than independently integrating a displayed thrust polyline. The result includes only the simulated interval. At burnout, unexpelled gas can remain, so it excludes any later tail-off impulse.

Peak pressure and peak thrust are maxima among stored history points. They are not continuously optimized extrema between samples. The re-

ported burn time is the simulated elapsed time; if the requested duration is reached with remaining propellant, it is not the complete material consumption time.

The completed indicator is true when the requested duration is reached or burnout is reached. The separate burnout_reached field distinguishes these cases. Always read them with remaining propellant mass and the stop reason. A completed requested interval can legitimately end before burnout.

22.9 Relating histories to equilibrium predictions

One can use the instantaneous burning area in Chapter 21's equilibrium relation as a reference curve. A history close to that curve suggests rapid storage adjustment relative to geometric change under the model assumptions. A difference can arise from initial gas inventory, growing free volume or finite response; it is not automatically an integration failure.

The equilibrium mode's explicit burnArea does not control the transient surface. The selected geometry does. Conversely, changing grain geometry while leaving equilibrium burning area unchanged does not change that separate equilibrium calculation. This input distinction is essential when comparing the two modes.

The prescribed gas temperature, characteristic velocity and thrust coefficient remain independent model inputs. No energy balance enforces mutual thermochemical consistency, and ambient pressure affects regime validity rather than recalculating the supplied thrust coefficient. Neither a smooth pressure history nor a small mass residual resolves that modeling limitation.

The useful result is a bounded established-burning history with explicit inventory, regression and stopping semantics. It excludes ignition, tail-off, erosive burning, star grains, structural response, two-phase flow and motor qualification. Those exclusions should remain attached to any use of the plotted history.

23. Shock waves

A shock compresses a supersonic gas through an irreversible transition. Static pressure and temperature rise while total pressure falls. These statements are compatible: static pressure measures the local mechanical state, whereas total pressure measures the pressure recoverable by an isentropic stagnation from that state. A shock destroys some of that recoverability without removing total enthalpy from an adiabatic, stationary flow.

ThrustOne provides independent normal- and oblique-shock calculations for a calorically perfect gas. Their results are ratios and downstream Mach numbers. They do not locate a shock inside a complete inlet, determine a shock train or predict a detached bow-shock shape.

23.1 Conservation across a normal shock

Place a thin stationary control volume across a plane shock normal to the incoming velocity. Let states 1 and 2 denote upstream and downstream. The area is unchanged and there is no heat or shaft work crossing the control volume. Conservation gives

$$\rho_1 V_1 = \rho_2 V_2 = j,$$

$$p_1 + \rho_1 V_1^2 = p_2 + \rho_2 V_2^2,$$

$$c_p T_1 + V_1^2/2 = c_p T_2 + V_2^2/2.$$

The pressure term in momentum cannot be omitted. Equating momentum flux alone would incorrectly force equal velocities when mass flux is conserved. The energy equation states that total temperature is unchanged; static temperature is free to rise as velocity falls.

Introduce the density ratio $d = \rho_2/\rho_1$. Continuity gives $V_2 = V_1/d$. With $V_1^2 = \gamma R T_1 M_1^2$, momentum becomes

$$\frac{p_2}{p_1} = 1 + \gamma M_1^2 (1 - 1/d).$$

The ideal-gas relation gives $T_2/T_1 = (p_2/p_1)/d$. Substituting these expressions into energy gives the unchanged-state root $d = 1$ and the compressive shock root

$$d = \frac{(\gamma + 1)M_1^2}{2 + (\gamma - 1)M_1^2}.$$

The unchanged-state solution is not a shock. Selecting the compressive branch for $M_1 > 1$ gives

$$\frac{p_2}{p_1} = 1 + \frac{2\gamma}{\gamma + 1}(M_1^2 - 1), \quad \frac{T_2}{T_1} = \frac{p_2/p_1}{d}.$$

Finally, use the new velocity and sound speed to obtain

$$M_2^2 = \frac{1 + (\gamma - 1)M_1^2/2}{\gamma M_1^2 - (\gamma - 1)/2}.$$

The downstream normal-shock state is subsonic. The App requires upstream Mach strictly greater than one and gamma greater than one; the exactly sonic unchanged-state limit is not an accepted normal-shock input.

23.2 Entropy and total-pressure loss

For constant heat capacities,

$$\frac{s_2 - s_1}{R} = \frac{\gamma}{\gamma - 1} \ln(T_2/T_1) - \ln(p_2/p_1).$$

Because total temperature is conserved, the same entropy difference can be written using the two stagnation states:

$$s_2 - s_1 = -R \ln(p_{t2}/p_{t1}).$$

Thus positive entropy production implies total-pressure recovery below one. Combining the static jump relations gives

$$\frac{p_{t2}}{p_{t1}} = \left[\frac{(\gamma + 1)M_1^2}{2 + (\gamma - 1)M_1^2} \right]^{\gamma/(\gamma-1)} \left[\frac{\gamma + 1}{2\gamma M_1^2 - (\gamma - 1)} \right]^{1/(\gamma-1)}.$$

Total-pressure loss is not a heat-loss fraction. Total enthalpy remains constant across this ideal adiabatic shock. A downstream compressor must nevertheless perform more work to recover a given delivery pressure because the gas has lost stagnation-pressure capability.

In the limit $M_1 \rightarrow 1^+$, pressure, density and temperature ratios approach one and entropy generation approaches zero. For very large upstream Mach at fixed gamma, density ratio approaches $(\gamma + 1)/(\gamma - 1)$, while pressure and temperature can continue increasing. Real high-temperature gases may then require variable properties and chemistry outside this component model.

23.3 Original normal-shock example

Choose $M_1 = 2.5$ and $\gamma = 1.4$. The density ratio is 10/3, pressure ratio 7.125 and temperature ratio 2.1375. Downstream Mach is 0.512989176.

Total-pressure recovery is 0.499014812, corresponding to $(s_2 - s_1)/R = 0.695119500$.

For illustrative upstream static values of 40 kPa and 240 K, the downstream pressure is 285 kPa and temperature 513 K. Both stagnation temperatures are 540 K, since $240[1 + 0.2(2.5)^2] = 540$. Static heating therefore does not contradict conservation of total temperature.

The velocity ratio is 0.3, fixed by continuity. Total pressure is approximately halved despite the static pressure rising by more than a factor of seven. These values are direct analytical arithmetic for an original teaching case, not a measured inlet or an App-generated verification dataset.

23.4 Foundation for an oblique shock

An oblique shock retains the same normal compression physics but also has a tangential velocity component. Let β be the shock angle measured from the upstream flow and θ the flow deflection. The upstream normal Mach is $M_{n1} = M_1 \sin \beta$. In inviscid flow the tangential velocity is unchanged across the shock; the normal component follows the normal-shock jump relations.

The downstream normal Mach is obtained using M_{n1} , and the full downstream Mach follows from

$$M_2 = \frac{M_{n2}}{\sin(\beta - \theta)}.$$

Pressure, density, temperature and total-pressure recovery are the normal-shock ratios evaluated at M_{n1} . They must not be evaluated at the full upstream Mach unless the shock is normal. The angle relation below closes the problem and exposes the two attached branches.

23.5 Deriving the deflection relation

Resolve velocity perpendicular and parallel to the shock. Tangential continuity gives $V_{t2} = V_{t1}$; normal mass continuity gives $V_{n2} = V_{n1}/d$,

where d is the density jump evaluated at $M_1 \sin \beta$. Therefore

$$\tan(\beta - \theta) = \frac{V_{n2}}{V_{t2}} = \frac{\tan \beta}{d}.$$

Substitute the normal density ratio and use the tangent difference identity. After collecting terms,

$$\tan \theta = 2 \cot \beta \frac{M_1^2 \sin^2 \beta - 1}{M_1^2 (\gamma + \cos 2\beta) + 2}.$$

The admissible compressive shock angle lies between the Mach angle $\mu = \sin^{-1}(1/M_1)$ and 90° . A smaller angle would give a subsonic normal upstream component and cannot support this compressive shock. The deflection is measured between upstream and downstream flow directions, not between the shock and the wall normal.

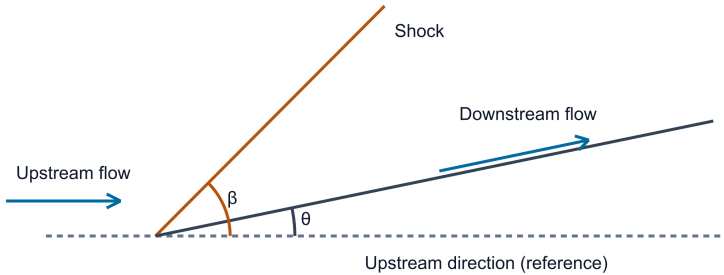


Figure 6: Original shock and flow-angle schematic

23.6 Two roots and the attached limit

For fixed upstream Mach and gamma, deflection is zero at both the Mach angle and 90° . Between them it reaches a maximum. For a positive deflection below that maximum, the smaller shock angle is the weak branch and the larger angle is the strong branch. They are distinct states with different

compression and entropy production, not two numerical approximations to one state.

A weak oblique shock often leaves supersonic downstream flow, but weakness does not guarantee downstream Mach above one near the attachment limit. Selecting weak solely because a subsequent component requires supersonic flow is insufficient; inspect the actual downstream result. The surrounding flow and downstream conditions determine which physical configuration can occur. This isolated component does not solve that global selection problem.

The App maximizes the angle relation over the physical interval, then brackets the requested branch on the corresponding side of the maximum. It uses bounded golden-section search for the peak and bisection for the root. A requested deflection above the maximum is refused; a detached-shock substitute is not returned. A small angular tolerance merges the roots at the computed peak.

At exactly zero deflection, weak returns the unchanged Mach-wave limit with all property ratios one. Strong returns the normal-shock state at 90° . This explicit distinction matters even though the specified turning angle is identical. Angles in the mathematical routines are radians; the App's displayed deflection and shock angles are degrees.

23.7 Original weak–strong comparison

Choose upstream Mach 2 and gamma 1.4. Select a 45° shock angle to construct a convenient weak-branch teaching case. Substitution into the angle relation gives deflection 14.743562836° . Its normal upstream Mach is $\sqrt{2}$, so the pressure ratio is 13/6, density ratio 12/7 and temperature ratio 91/72. Downstream Mach is 1.456323838 and total-pressure recovery 0.954577001.

Solving the same angle relation on its larger-angle branch gives shock angle 80.055319827° . The strong result has downstream Mach 0.641200943, pressure ratio 4.360819743, density ratio 2.621888917, temperature ratio 1.663235888 and total-pressure recovery 0.734902406.

Both states turn the flow through the same angle and satisfy the same conservation laws. The strong shock produces more static compression and more stagnation-pressure loss. Using the full upstream Mach 2 in a normal-shock formula for the weak case would wrongly predict a pressure ratio 4.5 instead of 2.166666667: the normal component, not the full velocity, drives the compression.

These are direct arithmetic and a scalar angle-root calculation for an original example. They are not generated by calling the App or presented as a measured inlet dataset.

23.8 Interpreting component outputs

The reported pressure, temperature and density ratios are downstream divided by upstream. To recover dimensional states, multiply them by compatible upstream static values. Total-pressure recovery multiplies upstream total pressure. Static and total ratios must not be interchanged.

The normal and oblique modes do not determine capture area, inlet drag, wall heat transfer or downstream combustor compatibility. Several isolated shock calculations can explain a proposed compression sequence, but they do not automatically establish a self-consistent installed inlet. Reflections, boundary-layer interactions and detached structures require additional closure.

Finally, constant gamma is an explicit assumption. A high predicted temperature is a reason to examine its applicability, not evidence that equilibrium chemistry has already been included. The result is an ideal jump across the selected compressive branch, with the irreversible loss that follows from those assumptions.

24. Friction, heating and quasi-one-dimensional flow

Friction and heat addition can both drive compressible flow toward a sonic limit, but they do not conserve the same quantities. Fanno flow preserves total enthalpy while wall drag removes axial momentum. Rayleigh flow preserves momentum flux while external heating changes total enthalpy. The common appearance of choking does not make these models interchangeable.

ThrustOne exposes each idealized component and a more general quasi-one-dimensional segment calculation. The special cases provide useful physical reference paths; the segment calculation also admits area variation and mass injection. Its additional terms must be interpreted with their own units and control-volume conventions.

24.1 Fanno assumptions and wall force

Consider steady flow in a constant-area duct with no heat addition, shaft work or mass injection. For fixed gas composition and constant heat capacities,

$$\dot{m} = \rho V A = \text{constant}, \quad c_p T + V^2/2 = c_p T_t = \text{constant}.$$

The differential momentum balance is

$$dp + \rho V dV = -\frac{f_D}{D} \frac{\rho V^2}{2} dx.$$

Here f_D is the Darcy friction factor and D the duct diameter, or the compatible hydraulic diameter for the intended model. The Darcy convention satisfies $f_D = 4f_F$, where f_F is the Fanning factor. Entering a Fanning value into a Darcy field underestimates the specified friction parameter by a factor of four.

The App's isolated Fanno input is the product $f_D L/D$. It does not infer friction factor from Reynolds number or roughness. A fitted or estimated factor must therefore be supplied consistently with its convention and intended regime.

24.2 Deriving the remaining choking length

Continuity gives $d\rho/\rho = -dV/V$, and the ideal-gas law gives $dp/p = dT/T - dV/V$. Constant total enthalpy supplies $c_p dT = -V dV$. Substitution into momentum yields

$$(1 - M^2) \frac{dV}{V} = \frac{\gamma M^2}{2} \frac{f_D}{D} dx.$$

Thus wall friction accelerates subsonic flow and decelerates supersonic flow under these assumptions. Both move toward Mach one. Friction does not necessarily reduce local velocity; its momentum loss combines with changes in pressure and density.

Since $M = V/\sqrt{\gamma RT}$, the energy relation gives

$$\frac{dM}{M} = \left[1 + \frac{\gamma - 1}{2} M^2 \right] \frac{dV}{V}.$$

Eliminating velocity and integrating to the sonic state produces

$$\mathcal{F}(M) = \frac{1 - M^2}{\gamma M^2} + \frac{\gamma + 1}{2\gamma} \ln \left[\frac{(\gamma + 1)M^2}{2 + (\gamma - 1)M^2} \right].$$

This is the remaining dimensionless friction length to choking. For a segment,

$$\frac{f_D L}{D} = \mathcal{F}(M_1) - \mathcal{F}(M_2).$$

The App finds M_2 on the same side of Mach one as the inlet. A prescribed friction length larger than the available value is rejected. It does not cross the sonic point by switching branches. At the choking limit it returns Mach one within its numerical tolerance; zero friction leaves the state unchanged.

24.3 Fanno static and total ratios

A star denotes the sonic state on the same Fanno line. Constant total temperature and continuity give

$$\frac{T}{T^*} = \frac{\gamma + 1}{2 + (\gamma - 1)M^2}, \quad \frac{p}{p^*} = \frac{1}{M} \sqrt{\frac{\gamma + 1}{2 + (\gamma - 1)M^2}},$$

$$\frac{\rho}{\rho^*} = \frac{1}{M} \sqrt{\frac{2 + (\gamma - 1)M^2}{\gamma + 1}}.$$

Isentropically stagnating each local state yields

$$\frac{p_t}{p_t^*} = \frac{1}{M} \left[\frac{2 + (\gamma - 1)M^2}{\gamma + 1} \right]^{\frac{\gamma+1}{2(\gamma-1)}}.$$

The ratio between states 2 and 1 is obtained by dividing their respective starred expressions. The starred quantities are normalization references, not an assertion that the requested segment actually reaches choking. Total pressure decreases downstream with positive friction while total temperature stays constant.

24.4 Original friction-length example

For $\gamma = 1.4$, choose a supersonic segment from Mach 2 to Mach 1.5. Direct evaluation gives $\mathcal{F}(2) = 0.304996503$ and $\mathcal{F}(1.5) = 0.136050217$. The required Darcy friction length is 0.168946285.

For an illustrative supplied $f_D = 0.02$, this corresponds to $L/D = 8.44731426$. The total available distance to choking from the same inlet is $L^*/D = 15.2498251$. A request for $f_D L/D = 0.4$ exceeds that capacity and must be refused under the fixed inlet conditions.

The Mach numbers and friction factor here are teaching values. The arithmetic demonstrates the component closure; it does not establish an experimentally appropriate friction factor or a real duct's flow capacity.

24.5 Rayleigh mass and momentum constraints

Rayleigh flow instead has constant area, no wall friction and no mass injection, but permits heat transfer. Mass and momentum conservation imply

$$\rho V = j = \text{constant}, \quad p + \rho V^2 = p(1 + \gamma M^2) = \text{constant}.$$

Energy changes according to $q = c_p(T_{t2} - T_{t1})$, with q in J/kg. The isolated App component specifies the ratio of downstream to upstream total temperature. It does not obtain this heat from a reacting fuel mixture.

Using the sonic state on the same Rayleigh line as reference gives

$$\frac{p}{p^*} = \frac{1 + \gamma}{1 + \gamma M^2}, \quad \frac{T}{T^*} = \frac{(1 + \gamma)^2 M^2}{(1 + \gamma M^2)^2}.$$

Combining static and total temperature relations gives

$$\mathcal{R}(M) = \frac{T_t}{T_t^*} = \frac{(1 + \gamma)M^2[2 + (\gamma - 1)M^2]}{(1 + \gamma M^2)^2}.$$

If the specified total-temperature ratio is τ_q , the downstream target is $\mathcal{R}(M_2) = \tau_q \mathcal{R}(M_1)$. Its maximum is one at the sonic state. Heating raises the target toward that maximum; the solver rejects a target above it rather than continuing through thermal choking.

24.6 Heating does not imply monotonic static temperature

Positive heat raises total temperature, but static temperature also reflects changing kinetic energy. Differentiating the starred static-temperature expression shows that its maximum occurs at $M = 1/\sqrt{\gamma}$, rather than at Mach one. A subsonic stream approaching choking can therefore have increasing total temperature while static temperature falls over the final part of the path.

The App selects the downstream Rayleigh root on the branch implied by upstream Mach. An exactly sonic inlet is assigned to the subsonic search branch for subsequent cooling. This is a numerical branch convention, not a general physical selection rule at a sonic junction.

On the supersonic branch, $\mathcal{R}(M)$ approaches $1 - 1/\gamma^2$ as Mach tends to infinity. Sufficiently strong cooling can request a target below that limiting value, so no finite supersonic state exists on the selected line. The current bounded search then refuses to establish a root; positivity of the requested temperature ratio alone does not ensure a solution.

24.7 Original thermal-choking example

For $\gamma = 1.4$ and upstream Mach 2, $\mathcal{R}(M_1) = 0.793388430$. A downstream Mach 1.5 corresponds to 0.909275657, requiring total-temperature ratio 1.146066192. If the inlet total temperature were 1000 K, the target would be 1146.066192 K, with heat per mass $146.066192c_p$ J/kg when c_p is entered in J/(kg K).

The largest supported heating ratio on this line is $1/\mathcal{R}(2) = 1.260416667$. A ratio 1.3 therefore crosses the thermal limit. That input must not be interpreted as a completed segment with a chosen downstream subsonic branch. These values are direct analytical arithmetic, not App-generated validation data.

24.8 General segment inputs and units

The quasi-one-dimensional solver advances static pressure p , temperature T , axial velocity V , mass flow $\dot{m}$ and area A along x . The inlet must satisfy $\dot{m} = \rho V A$; these five supplied quantities are not independent. A helper can synchronize mass flow from the other state inputs, but the solver itself rejects an inconsistent inlet rather than silently replacing it.

Each segment specifies positive length and exit area. Area varies linearly from the area at that segment's entrance, so $A' = (A_{out} - A_{in})/L$. The following distributed sources are constant inputs within the segment:

Quantity	Meaning	Units
Q'	External heat supplied per unit axial length	W/m
q_m	Injected mass flow per unit axial length	kg/(s m)
h_{tj}	Total specific enthalpy of injected material	J/kg
V_j	Axial injection velocity	m/s
f_D	Darcy wall-friction factor	Dimensionless

Heat per length is a power density along the duct, not heat per unit gas mass. Its integral over length is watts; dividing by an appropriate mass flow produces a specific-energy change. When mass is injected, that mass flow changes along the segment and cannot be treated as a single fixed denominator without accounting for energy carried by the injection.

The solver uses the circular equivalent diameter $D = 2\sqrt{A/\pi}$ for wall drag. It does not accept an independent wetted perimeter or arbitrary-section hydraulic diameter. A rectangular or annular duct interpreted through this area-only diameter is therefore an additional approximation.

24.9 Mass, momentum and energy with injection

Mass conservation is

$$\dot{m}' = q_m.$$

For momentum, include injected axial momentum, pressure force on a changing-area wall, and wall drag. The flux form is

$$(pA + \dot{m}V)' = pA' + q_m V_j - F'_w, \quad F'_w = \frac{f_D \rho V^2 A}{2D}.$$

Expanding the derivative and cancelling pA' yields

$$Ap' + \dot{m}V' = q_m(V_j - V) - F'_w \equiv F'.$$

The $-q_m V$ term is essential: added mass must acquire the stream velocity. A zero axial injection velocity does not mean zero momentum effect. The quantity F' is an effective force per length, in N/m, rather than the total force on the engine.

The total-enthalpy flux balance is

$$(\dot{m}h_t)' = Q' + q_m h_{tj}, \quad h_t = h + V^2/2.$$

Using $\dot{m}' = q_m$ gives

$$\dot{m}h'_t = Q' + q_m(h_{tj} - h_t) \equiv Q'_{eff}.$$

Injection can increase total energy flux while reducing the stream's specific total enthalpy if the added material has lower h_{tj} . This is dilution, not energy destruction. For constant composition, $dh = c_p(T)dT$, so

$$c_p T' + VV' = Q'_{eff}/\dot{m}.$$

No chemical source is introduced by calling the injection fuel. A different species inventory requires a composition-aware mixing or reacting model, developed in Chapter 29.

24.10 The velocity differential equation

Differentiate continuity in logarithmic form with fixed gas constant:

$$\frac{q_m}{\dot{m}} = \frac{p'}{p} - \frac{T'}{T} + \frac{V'}{V} + \frac{A'}{A}.$$

Substitute $p' = (F' - \dot{m}V')/A$ and $T' = (Q'_{eff}/\dot{m} - VV')/c_p$. Collecting velocity terms gives

$$V' = \frac{V}{1 - M^2} \left[\frac{q_m}{\dot{m}} - \frac{A'}{A} - \frac{F'}{Ap} + \frac{Q'_{eff}}{\dot{m}c_p T} \right].$$

Every bracketed term has units of inverse length. The factor $1 - M^2$ reveals the sonic singularity of this formulation. A numerator that happens to approach zero does not authorize the current solver to construct a transonic continuation; that would require a separate critical-point compatibility treatment.

With no sources, the relation reduces to $(1 - M^2)dV/V = -dA/A$. A convergent passage accelerates subsonic flow; a divergent passage accelerates supersonic flow. With constant area, no injection and no heat, the wall-force term recovers Fanno flow. With constant area, no injection and no drag, the heat term recovers Rayleigh flow. These reductions identify meaningful comparisons without pretending that the three numerical interfaces have identical stopping behavior.

24.11 Original injection bookkeeping example

Take an illustrative inlet mass flow 2 kg/s and total specific enthalpy 400 kJ/kg. A one-metre segment supplies 10 kW/m external heat and 0.2 kg/(s

m) injection with total enthalpy 300 kJ/kg. The exit mass flow must be 2.2 kg/s. The inlet energy flux is 800 kW; external heat adds 10 kW and injection adds 60 kW. Therefore any completed adiabatic-wall calculation with those declared sources must have

$$h_{t,out} = \frac{800 + 10 + 60}{2.2} = 395.454545 \text{ kJ/kg.}$$

Total energy flux increases while specific total enthalpy decreases. Adding 10 kJ/kg directly to the inlet enthalpy would be wrong: the supplied heat has different units, and the added mass carries its own energy.

This is a necessary flux balance, not a complete outlet-state solution. Pressure, temperature and velocity still depend on area, momentum injection and the selected flow branch. In particular, changing V_j at fixed injection total enthalpy changes injected static enthalpy and the momentum balance. The same total-energy target can therefore accompany different admissible outlet states.

24.12 Property and composition restrictions

The constant-property variant uses supplied gamma and gas constant, with $h = c_p T$. Its injection total enthalpy must be nonnegative. The frozen NASA9 variant instead evaluates $c_p(T)$, formation-inclusive enthalpy and frozen sound speed from one fixed mixture; its gas constant remains fixed because composition does not evolve.

Active injection with a separately specified injection composition is rejected in this solver. The frozen calculation cannot silently combine different compositions while keeping one mixture property law. For same-composition frozen injection, the model checks that $h_{tj} - V_j^2/2$ can be inverted to a supported static temperature. Negative formation-inclusive total enthalpy is not automatically invalid in that convention.

External heat can be negative to represent cooling, and injection velocity is a signed axial quantity. Injected mass rate and friction factor must be

nonnegative. These input domains do not make the model a resolved injector calculation: no penetration, mixing length or transverse momentum field is predicted.

24.13 Integration and sonic stopping

The App applies a fixed-step fourth-order Runge–Kutta method within each segment, with at most 10000 total requested steps. It preserves the subsonic or supersonic side selected by the initial Mach. Intermediate and candidate states must remain finite and positive, and $|1 - M^2|$ must exceed 10^{-4} .

If a step approaches that sonic exclusion region, crosses branches, produces an invalid state or exceeds its conservation-residual criteria, the solver returns the previous accepted points with `completed=false` and a stop reason. It does not append a synthetic Mach-one endpoint or adaptively locate the exact choking coordinate. The last plotted position is therefore a retained valid station, not necessarily the true critical station.

Inlet continuity is checked to a tighter relative tolerance than accepted downstream steps. Frozen-property steps also check integrated energy flux against accumulated heat and injection enthalpy. These checks expose gross inconsistency; they do not replace an accuracy estimate for the spatial discretization. A smaller step can resolve a numerical difficulty but cannot create a physically missing branch beyond choking.

The isolated Fanno component can return its analytic sonic endpoint, whereas this general differential solver stops short of its singularity. The Rayleigh component may return a numerical state close to the thermal limit. Those endpoint differences must be respected when comparing results. Agreement should be assessed at compatible pre-sonic states with the same gas, area and source conventions.

A completed segment supplies a one-dimensional state history under these assumptions. It does not provide engine thrust by itself, infer a shock train, or guarantee that downstream back pressure can support the computed branch. The subsequent full-engine chapters add their own inlet, combustor and nozzle closures around these local balances.

25. Ramjet cycle and complete subsonic flowpath

A ramjet uses incoming flight motion to provide compression before heat addition. It has no shaft-driven compressor or turbine in the ideal cycle considered here. The inlet slows the captured stream, the combustor raises its total enthalpy, and the nozzle accelerates the exhaust. Net thrust is the difference between exhaust force and captured-air momentum, with any nozzle pressure term retained.

ThrustOne offers a design-point cycle and a spatial subsonic-combustion flowpath. The first specifies a combustor outlet total temperature and pressure-loss factors. The second integrates prescribed area, heat, drag and injection sources before attaching a pressure-specified nozzle. They answer different questions and should not be described as interchangeable ways of solving the same fixed hardware.

25.1 Inlet total conditions

For ambient static temperature T_0 , pressure p_0 , flight Mach M_0 and cold-gas gamma γ_c ,

$$\tau_r = 1 + \frac{\gamma_c - 1}{2} M_0^2, \quad T_{t2} = T_0 \tau_r, \quad p_{t2} = p_0 \tau_r^{\gamma_c/(\gamma_c - 1)} \pi_d.$$

The total-pressure recovery π_d represents inlet loss. In the nonideal

design-point mode it combines the supplied recovery with the empirical Mach-dependent factor discussed in Chapter 4. The ideal selection uses unity recovery. Neither option determines a shock position, capture geometry or inlet starting margin.

At zero flight Mach, an ideal inlet provides no ram pressure rise. A conventional ramjet therefore cannot obtain its normal propulsion cycle merely by adding heat to static ambient air in this ideal pressure chain. The current cycle also requires a positive nozzle pressure ratio above one and positive net specific thrust.

25.2 Combustor fuel balance

Let T_{t4} be the prescribed combustor exit total temperature. On a unit captured-air mass basis, with effective cold and hot heat capacities, the nonideal energy balance is

$$c_{pc}T_{t2} + \eta_b f Q = (1 + f)c_{ph}T_{t4}.$$

Solving for fuel-to-air mass ratio gives

$$f = \frac{c_{ph}T_{t4} - c_{pc}T_{t2}}{\eta_b Q - c_{ph}T_{t4}}.$$

The denominator must remain positive, and the numerator cannot require negative fuel. This is a prescribed effective sensible-enthalpy and heating-value model. It does not infer equilibrium species or gaseous feed enthalpies from chemical data.

The ideal selection uses one cold-gas property set throughout, unity efficiencies and pressure recoveries, and neglects fuel mass in exhaust momentum. Its fuel balance becomes

$$f_i = \frac{c_p(T_{t4} - T_{t2})}{Q}.$$

Fuel energy is retained while its added exhaust mass is neglected. This is a deliberate cycle approximation, not an exact finite-fuel mass balance. The ideal mode also requires complete expansion to ambient pressure.

25.3 Specified-pressure nozzle and thrust

Let $r_e = p_0/p_9$. With combustor and nozzle pressure ratios π_b and π_n ,

$$\frac{p_{t9}}{p_9} = r_e \tau_r^{\gamma_c/(\gamma_c-1)} \pi_d \pi_b \pi_n.$$

The nozzle uses the hot-gas gamma and gas constant $R_h = c_{ph}(\gamma_h - 1)/\gamma_h$ to recover static temperature and velocity:

$$T_9 = T_{t4}(p_9/p_{t9})^{(\gamma_h-1)/\gamma_h}, \quad V_9 = \sqrt{2c_{ph}(T_{t4} - T_9)}.$$

For the nonideal cycle, net thrust per unit captured-air flow is

$$F_s = (1 + f)V_9 - V_0 + \frac{(1 + f)R_h T_9}{V_9}(1 - r_e).$$

The last term follows by eliminating exit area through continuity. Positive excess exit pressure adds thrust; overexpansion subtracts it. The nozzle geometry is not supplied or solved as a fixed-area back-pressure problem. A prescribed exit pressure can imply a supersonic expansion without proving that an installed nozzle sustains that branch.

Specific fuel consumption is f/F_s . An optional air mass flow converts specific thrust to net force. Overall efficiency is thrust power divided by fuel chemical power when fuel is positive; kinetic-decomposition thermal and propulsive efficiencies are reported only for full expansion.

25.4 Ideal limiting relation

With one gas, unity recoveries and full expansion, the inlet and nozzle pressure ratios cancel so that $M_9 = M_0$. Nevertheless exhaust velocity

exceeds flight velocity if heat has raised the nozzle temperature. Specifically,

$$T_9 = T_{t4}/\tau_r, \quad \frac{V_9}{V_0} = \sqrt{\frac{T_{t4}}{T_0\tau_r}},$$

and, neglecting fuel mass,

$$F_{s,i} = V_0 \left[\sqrt{\frac{T_{t4}}{T_0\tau_r}} - 1 \right].$$

Equal inlet and exit Mach numbers do not imply equal velocities because their sound speeds differ. Positive heating requires $T_{t4} > T_0\tau_r$. Increasing flight Mach at fixed maximum combustor temperature eventually reduces the available temperature rise; a larger ram pressure rise is not an unlimited performance benefit.

25.5 Original ideal-cycle example

Choose an illustrative constant gas with $\gamma = 1.4$, $c_p = 1000$ J/(kg K), ambient temperature 250 K and flight Mach 2. Set combustor total temperature 1800 K and heating value 40 MJ/kg. These inputs are an ideal teaching case, not a fuel-property recommendation.

The inlet temperature factor is 1.8 and its total temperature 450 K. The gas constant is 285.714286 J/(kg K), giving ambient sound speed 316.227766 m/s and flight velocity 632.455532 m/s. Full ideal expansion gives exit temperature 1000 K and exit Mach 2, so exhaust speed 1264.911064 m/s.

The ideal specific thrust is 632.455532 N per kg/s of captured air. The approximate fuel ratio is $1000(1800 - 450)/(40 \times 10^6) = 0.03375$. These values follow directly from the limiting equations. Restoring fuel mass, different hot-gas properties or pressure losses changes the model and therefore the result; it should not be treated as a rounding correction to this ideal calculation.

25.6 A captured stream and a subsonic combustor inlet

The spatial full-flowpath mode requires a supersonic freestream and a positive capture area. It first computes captured air flow,

$$\dot{m}_a = \frac{p_0}{RT_0} V_0 A_{capture}, \quad V_0 = M_0 \sqrt{\gamma RT_0}.$$

Its inlet total temperature follows ram heating, and total pressure is multiplied by the directly supplied recovery. Unlike the nonideal design-point cycle, this inlet does not add the empirical Mach recovery factor. Identical numerical recovery inputs therefore need not describe identical recovered total pressures in the two modes.

The selected inlet closure specifies exit Mach, static temperature, static pressure, or an area constraint with recovery. A temperature input gives $M_i^2 = 2(T_t/T_i - 1)/(\gamma - 1)$; a pressure input gives the corresponding isentropic Mach relation. Given Mach,

$$T_i = \frac{T_t}{1 + (\gamma - 1)M_i^2/2}, \quad p_i = p_t (T_i/T_t)^{\gamma/(\gamma-1)}, \quad A_i = \frac{\dot{m}_a RT_i}{p_i V_i}.$$

The inlet must decelerate the freestream into a positive subsonic Mach and produce static pressure no lower than ambient. No shock sequence is reconstructed to prove how that specified recovery and state are achieved.

For contraction/recovery closures, the area is capture area divided by contraction ratio. The solver checks its sonic mass-flow capacity, then selects the subsonic root compatible with captured flow and recovered total conditions. In the current implementation, both named closure routes use the same supplied contraction and recovery relation; recovery is not inferred from a separate shock solution.

25.7 Distributed heat and injected mass

The combustor is one or more constant-property segments governed by Chapter 24's equations. They permit area change, Darcy drag, external heat per length and injected mass carrying specified momentum and total enthalpy. The calculation follows the subsonic branch and refuses to integrate through its sonic singularity.

This is not a resolved combustion-chemistry calculation. A heat input in W/m is an imposed energy source; a mass source in kg/(s m) is an imposed material addition. Neither alone supplies a fuel heating value or establishes combustion efficiency. If the user represents fuel through both mass injection and heat, the chosen conventions must avoid counting its chemical energy twice.

In particular, the design-point fuel ratio from Section 25.2 is not automatically transferred to this spatial model. The full-flowpath output has captured and exit mass flows, but does not report a chemically inferred fuel-to-air ratio or TSFC. An energy residual checks the supplied flux balance, not combustion completeness.

If the combustor stops early, the returned exit fields describe its last accepted point. The downstream nozzle is not calculated and no completed full-engine thrust is supplied. More heat cannot be accommodated merely by allowing a plotted line to jump from a subsonic to a supersonic combustor state.

25.8 A pressure-specified nozzle after the combustor

At a completed combustor exit, the wrapper reconstructs total conditions from its static state:

$$T_{tn} = T_b + V_b^2/(2c_p), \quad p_{tn} = p_b[1 + (\gamma - 1)M_b^2/2]^{\gamma/(\gamma-1)}.$$

The nozzle recovery reduces total pressure while total temperature is retained. The specified ambient-to-exit pressure ratio determines p_e . The

shared nozzle component then obtains exit temperature and velocity from the available enthalpy drop. Finally,

$$A_e = \frac{\dot{m}_e R T_e}{p_e V_e}.$$

Exit area is a result. The full ramjet route ignores separately supplied nozzle segments and uses this pressure closure instead. A nozzle length input positions the final station but does not determine the area solution or internal flow distribution.

Only inlet and exit nozzle states are reported in a separate station profile. Connecting them visually does not turn them into a solved continuous nozzle. A supersonic exit would require a suitable throat and expansion in hardware, but the current wrapper does not construct that throat or solve its back-pressure operating regimes.

25.9 Original nozzle and momentum example

Consider a constructed nozzle interface with total temperature 1000 K, total pressure 200 kPa before a recovery of 0.9, mass flow 2 kg/s, $\gamma = 1.4$, $R = 300$ J/(kg K), and exit pressure 100 kPa equal to ambient. The available nozzle total pressure is 180 kPa and $c_p = 1050$ J/(kg K).

The isentropic endpoint is $T_e = 845.405363$ K, $V_e = 569.779552$ m/s and $M_e = 0.956202148$. Continuity yields $A_e = 0.008902447$ m². This example deliberately gives a subsonic nozzle exit: a ramjet nozzle result is not forced to be supersonic when the prescribed pressure ratio is below the critical ratio.

Gross thrust is 1139.559104 N. If the upstream bookkeeping gives captured air flow 1.8 kg/s at 500 m/s, captured momentum is 900 N and net thrust is 239.559104 N. The 0.2 kg/s difference between exit and captured flow would need to be supplied consistently by the upstream mass sources. These selected interface values illustrate endpoint and force accounting; they do not claim a fully integrated combustor history.

The nozzle area cannot be prescribed independently while retaining all these endpoint quantities. Choosing a different area would require changing a pressure, velocity, mass-flow or upstream state constraint. That is why this model is described as pressure-specified rather than fixed-geometry ambient matching.

25.10 Full-flowpath force and diagnostics

The completed engine uses

$$F = \dot{m}_e V_e - \dot{m}_a V_0 + (p_e - p_0)A_e, \quad F_s = F/\dot{m}_a.$$

Captured air flow remains the denominator even when injection increases exit mass. Gross thrust omits the captured-momentum subtraction. The current full-flowpath routine checks finiteness but does not require positive net thrust, unlike the design-point cycle. A finite negative result represents drag under the declared model, not useful propulsion.

The retained flow profile ends at the combustor. Its mass-flow field and final static state remain useful for interpreting the nozzle inlet. The final overall exit fields are then replaced by the nozzle result. The energy residual retained from the upstream calculation belongs to the integrated combustor source balance, not to a separately integrated nozzle flowfield.

Neither capture area nor a specified recovery predicts spillage, installation drag, inlet unstart or shock–boundary-layer interaction. Nozzle pressure specification likewise does not prove that the combustor and an installed nozzle can share a stable operating point. The calculation connects a bounded subsonic source-term flowpath to a thermodynamically consistent endpoint expansion; that is the scope of the completed result.

26. Frozen variable-property flow and state reconstruction

A flow state can be specified through temperature and pressure, density and velocity, or conserved flux quantities. These forms describe the same physical variables only when they share a composition, an enthalpy reference and a branch. ThrustOne's frozen variable-property flow mode reconstructs an inlet from seven input forms, then advances that fixed composition through the source-term equations of Chapter 24.

Frozen composition does not require constant heat capacity. NASA9 enthalpy, heat capacity and sound speed are evaluated at the local temperature. The distinction is essential for the reconstruction from momentum and total enthalpy, where an incorrect reference energy can select an entirely different state or leave no admissible root.

26.1 Seven forms, one physical inlet

The supported forms are temperature–pressure–Mach, temperature–pressure–velocity, temperature–density–Mach, temperature–density–velocity, pressure–density–Mach, pressure–density–velocity, and momentum/energy flux reconstruction. Every form also requires a positive area and a declared mixture.

For the first six, the ideal-gas relation closes the missing thermodynamic variable:

$$p = \rho RT, \quad T = p/(\rho R).$$

A supplied velocity is used directly. A supplied Mach number instead gives $V = Ma_f(T)$, with a_f obtained from the frozen mixture properties. Finally,

$$\dot{m} = \frac{pVA}{RT}.$$

Mass flow is an output of these forms. It is not another independent condition that can be chosen arbitrarily alongside pressure, temperature, velocity and area. The properties and reconstructed state must stay finite, positive and within their supported temperature range.

These are seven inlet representations, not seven chemical models or seven flow-regime predictions. Composition remains fixed after reconstruction unless a different, explicitly mixing or reacting mode is selected.

26.2 What F and G mean

The final input form supplies

$$F = pA + \dot{m}V, \quad G = h + V^2/2,$$

along with mass flow, area and an explicit subsonic or supersonic branch. F has units of newtons and is the pressure-plus-momentum flux at the selected plane. It is not net engine thrust. G has units of J/kg and is total specific enthalpy, not energy flow in watts.

The associated energy flux is $\dot{m}G$. Subtracting incoming momentum and including external pressure forces is still necessary before a full control volume yields thrust. A positive F therefore says nothing by itself about whether a complete engine produces positive propulsion.

NASA enthalpy includes formation reference energies. Consequently G can legitimately be negative. The velocity contribution is nonnegative,

but it need not exceed a negative mixture enthalpy. Rejecting every negative total enthalpy would remove physically representable states from this convention.

26.3 Reduce momentum to a velocity–temperature curve

Let $q_F = F/\dot{m}$, a quantity with units of m/s. Continuity gives $pA/\dot{m} = RT/V$. Substituting into the momentum definition yields

$$q_F = V + RT/V, \quad T(V) = \frac{V(q_F - V)}{R}.$$

Positive temperature requires $0 < V < q_F$. The temperature curve reaches its maximum $q_F^2/(4R)$ at $V = q_F/2$. If even that maximum is below the mixture's minimum supported temperature, there is no admissible reconstruction.

For any admissible temperature, the two momentum-compatible velocities are

$$V_{\pm} = \frac{q_F \pm \sqrt{q_F^2 - 4RT}}{2}.$$

The low-speed expression is evaluated in the cancellation-resistant form

$$V_- = \frac{2RT/q_F}{1 + \sqrt{1 - 4RT/q_F^2}}, \quad V_+ = q_F - V_-.$$

These two momentum roots do not alone identify the requested flow state. They construct the allowable velocity intervals associated with the property's temperature limits. Energy then selects roots within those intervals.

26.4 Energy closure and the sonic partition

Substitute the momentum temperature into total enthalpy:

$$E(V) = h[T(V)] + V^2/2 - G = 0.$$

The App evaluates mixture properties at each trial temperature, brackets roots and partitions the search around a frozen sonic point where needed. To see why that point is significant, differentiate at fixed composition:

$$E'(V) = c_p \frac{q_F - 2V}{R} + V.$$

On the momentum curve, $q_F = V + RT/V$, so

$$E'(V) = \frac{c_p T}{V} - \frac{c_v V}{R}.$$

The stationary condition is $V^2 = (c_p/c_v)RT = a_f^2$. The energy curve therefore turns at the frozen sonic state, separating the subsonic and supersonic branches. This gives a physical basis for the branch partition rather than treating two roots as an arbitrary numerical nuisance.

The current implementation accepts exactly one nonsonic root on the requested branch within the valid temperature domain. It excludes roots within its small sonic-Mach tolerance and rejects no-root or nonunique outcomes. It does not silently choose whichever root happens to be discovered first.

Once velocity is selected, pressure follows from

$$p = \frac{\dot{m}RT}{AV}.$$

Area controls dimensional pressure for this fixed $F/\dot{m}$, G reconstruction. It does not change the velocity-temperature root when the composition and other flux inputs stay fixed.

26.5 Original two-root illustration

Use an analytical constant-property gas to make the roots transparent: $R = 250 \text{ J/(kg K)}$, $c_p = 1000 \text{ J/(kg K)}$, hence $\gamma = 4/3$, with the deliberate reference $h = c_p T$. Set $q_F = 1000 \text{ m/s}$ and $G = 900000 \text{ J/kg}$. This is a teaching limit, not an assertion that these energies use the NASA9 formation reference.

Substitution gives

$$G = 4V(1000 - V) + V^2/2 = 4000V - 3.5V^2.$$

There are two roots. The low-speed state has velocity 308.013016 m/s, temperature 852.563991 K and Mach 0.577784920. The high-speed state has velocity 834.844127 m/s, temperature 551.517642 K and Mach 1.947091314.

For mass flow 2 kg/s and area 0.01 m², their pressures are 138397.396899 Pa and 33031.174530 Pa respectively. Both have $F = 2000 \text{ N}$ and $G = 900000 \text{ J/kg}$. Flux values alone do not say which branch the user intends.

The two states can also be viewed as compatible constant-area adiabatic jump endpoints. Entropy direction would matter if interpreting them as a shock, but this reconstruction does not impose an upstream-to-downstream shock event. Its branch input selects a state, not an irreversible process connecting the roots.

These numbers were calculated directly from the quadratic. A NASA9 calculation must use its actual $h(T)$, including the reference term. Adding a constant enthalpy offset to both the property law and supplied G preserves the state; changing only one of them does not.

26.6 Advancing the reconstructed state

The variable-property flow mode passes the inlet and fixed mixture to the same quasi-one-dimensional solver developed in Chapter 24. Area, drag, external heat and same-composition mass injection can change the local

state. The gas constant remains fixed; local heat capacity, enthalpy and sound speed vary with temperature.

A separately specified injection composition is not accepted with active injection in this frozen flow. Different-composition mixing belongs to the appropriate composition-aware mode. Similarly, an explicit heat source is not a finite-rate chemical mechanism. Renaming a segment combustor does not create species evolution.

At each accepted point, the profile includes static pressure, temperature, velocity, area, mass flow, Mach, heat capacity, gamma and specific enthalpy. The enthalpy field is static h ; add $V^2/2$ to obtain G . Comparing that static field directly with an inlet total-enthalpy input would create a false apparent energy discrepancy.

26.7 Partial histories and energy diagnostics

The general solver retains the initial subsonic or supersonic side and stops before its sonic singularity or an invalid state. The frozen-flow wrapper reports the final accepted point even when completed is false. The exit fields then describe a partial segment, not the requested endpoint.

Its energy diagnostic is the dimensional flux difference

$$\mathcal{E} = \dot{m}_e(h_e + V_e^2/2) - \dot{m}_i(h_i + V_i^2/2) - \int_0^{x_e} (Q' + q_m h_{tj}) dx.$$

It has units of watts. For a partial history, the source integral includes only the length actually reached, including a partially completed segment. It must not include all the heat specified beyond the retained endpoint.

The wrapper exposes this value as `flow_energy_residual`; it is not the same normalized quantity as the general solver's internal relative checks. A small dimensional residual depends on the flow scale and does not certify chemistry, temporal response or an installed flowpath.

26.8 Choosing the appropriate frozen calculation

Use the standalone frozen nozzle of Chapter 17 when the problem is an adiabatic, isentropic, sonic-or-supersonic expansion through nondecreasing area. That solver reconstructs states from conserved entropy and total enthalpy. Use the present source-term segment when heating, friction, general area change or permitted mass injection must be represented.

The nozzle can begin at a sonic throat. The general differential flow solver excludes its sonic singularity. A comparison that starts both calculations at exactly Mach one is therefore not a comparison of compatible numerical domains. Compare common nonsonic states and consistent source assumptions instead.

Neither calculation predicts a shock train, composition relaxation, cooling system or automatic transition between branches. Their common strength is explicit thermodynamic and flow accounting within a fixed-composition model, with the reference state and root choice made visible.

27. Calibrated variable-property inlets

An inlet model can conserve energy while requiring an empirical description of irreversibility. ThrustOne's calibrated inlet uses one explicit reference state to set the relationship between entropy production and static-pressure rise. It then combines that relationship with frozen dry-air properties to reconstruct other compression states.

The word calibrated does not mean the App contains measured data for an unspecified inlet. The reference altitude, flight Mach, exit Mach and contraction ratio are user inputs. Their consistency and relevance determine what the resulting model represents. The calculation does not find shock locations, boundary layers, capture spillage or inlet starting limits.

27.1 The property and freestream model

The mixture is fixed at 0.79 N₂ and 0.21 O₂ by mole, using NASA9 properties. It does not react or dissociate as it heats. Its gas constant is constant, but heat capacity, enthalpy and frozen sound speed depend on temperature.

For an atmosphere state T_0, p_0 ,

$$V_0 = M_0 a_f(T_0), \quad \rho_0 = p_0 / (R T_0), \quad q_0 = \rho_0 V_0^2 / 2,$$

$$h_t = h(T_0) + V_0^2/2, \quad s_0 = s(T_0, p_0).$$

The supported geometric altitude range is 0–70 km. The calculation uses its bundled atmosphere and property data, not measured weather. The actual flight state must be supersonic. Property-temperature limits can constrain high-energy cases even when altitude is within the declared range.

Three freestream input forms are available. Altitude and Mach directly determine the state. At specified altitude and dynamic pressure, Mach follows from $M_0 = \sqrt{q_0/q_{M=1}}$, because the local thermodynamic state is fixed. At specified Mach and dynamic pressure, altitude is bracketed within the allowed atmosphere range and solved numerically. An unbracketed pressure request is rejected rather than assigned an extrapolated altitude.

27.2 Recovering the reference exit

Apply the same freestream construction to the reference inputs, denoted by subscript R. Given reference exit Mach M_{eR} , find its static temperature from

$$h(T_{eR}) + \frac{M_{eR}^2 a_f(T_{eR})^2}{2} = h_{tR}.$$

The reference exit Mach must be positive and lower than the reference flight Mach. Temperature is bracketed between freestream temperature and the stagnation temperature obtained by enthalpy inversion. This avoids treating a locally evaluated gamma as a global calorically perfect relation.

Define contraction ratio $C = A_{capture}/A_e$. Continuity implies

$$\rho_e V_e = C \rho_0 V_0.$$

Thus reference pressure is recovered from

$$p_{eR} = C_R \rho_{0R} V_{0R} \frac{RT_{eR}}{V_{eR}}.$$

Reference contraction must exceed one and this reconstructed pressure must exceed reference ambient pressure. The entropy change is then evaluated from the two actual property states.

27.3 The calibrated entropy law

The implemented relationship is

$$\Delta s = K \ln(p_e/p_0), \quad K = \frac{s_{eR} - s_{0R}}{\ln(p_{eR}/p_{0R})}.$$

The coefficient K has units of J/(kg K), not a dimensionless efficiency. The model requires a finite positive coefficient, so a reversible reference with zero entropy change is outside this lossy calibration interface. A negative-entropy reference is rejected.

This law is a supplied reference approximation. It is not derived from a universal shock train or wall-friction correlation. One reference fixes one parameter; it does not demonstrate accuracy over all Mach numbers, altitudes or contractions. Changing the reference changes the model even if the current flight inputs stay identical.

The calibration and the current-flow closure are distinct steps. The App reconstructs the coefficient from the saved reference each time, then applies it to the selected current state. It does not overwrite the reference with each newly computed point.

27.4 Deriving the current compression path

At a trial exit temperature, write the ideal-gas entropy pressure correction for the same frozen mixture:

$$s(T, p_e) = s(T, p_0) - R \ln(p_e/p_0).$$

Set this equal to $s_0 + K \ln(p_e/p_0)$, obtaining

$$\ln(p_e/p_0) = \frac{s(T, p_0) - s_0}{R + K}.$$

Energy and continuity then give

$$V_e = \sqrt{2[h_t - h(T)]}, \quad M_e = V_e/a_f(T), \quad C = \frac{\rho_e V_e}{\rho_0 V_0}.$$

The accepted state must have temperature above freestream temperature, positive remaining kinetic energy and pressure above freestream pressure. Those conditions describe adiabatic deceleration and compression on the modeled path. They do not determine the physical devices or shocks producing it.

Absolute capture area is absent from this independent calculation: continuity provides an area ratio through mass fluxes. A dimensional mass flow requires an additional capture-area scale outside this mode.

27.5 Total-pressure recovery without a local-gamma power law

Because total enthalpy and composition are unchanged, isentropically stagnating the upstream and downstream states gives the same total temperature. Their entropy difference therefore satisfies

$$\Delta s = -R \ln(p_{te}/p_{t0}), \quad \pi_d = \exp(-\Delta s/R).$$

Together with the calibrated law,

$$\pi_d = (p_e/p_0)^{-K/R}.$$

This power describes the fitted entropy relation, not a calorically perfect static-to-total formula. Static pressure can increase while total-pressure recovery falls, consistent with irreversible compression.

As an original dimensionless illustration, suppose $K/R = 0.1$ and the chosen path state has static-pressure ratio 10. Then $\Delta s/R = 0.1 \ln 10$ and recovery is $10^{-0.1} \approx 0.794328$. Conversely, a recovery of 0.8 requires pressure ratio $0.8^{-10} \approx 9.313226$ under this particular calibration. These relationships alone do not establish that the corresponding temperature and kinetic energy are admissible for a given flight state; those closures must still be solved.

27.6 Five exit constraints

Exit Mach uses the energy inversion of Section 27.2, followed by the calibrated pressure relation. Exit temperature directly selects a point on that path. Specified exit pressure sets its logarithmic pressure rise and obtains temperature by entropy inversion:

$$s(T, p_0) = s_0 + (R + K) \ln(p_e/p_0).$$

Specified total-pressure recovery first gives

$$\ln(p_e/p_0) = -\frac{R}{K} \ln \pi_d,$$

then uses the same temperature inversion. This branch requires recovery strictly between zero and one. The lossless endpoint is not an accepted calibration limit merely because other inlet models permit unity recovery.

The fifth constraint is contraction ratio. It requires inversion of the computed mass-flux ratio $C(M_e)$. Multiple exits can satisfy the same contraction because velocity and density change in competing directions along the compression path. The root choice becomes part of the input, not an optional plotting preference.

27.7 Higher- and lower-Mach roots

For contraction inversion, the App samples candidate exit Mach values from 0.05 to 0.999 times flight Mach, detects sign-change brackets and

refines the selected bracket. The higher-Mach choice takes the highest detected bracket; the lower-Mach choice takes the lowest. It reports the number of detected brackets.

Higher and lower are relative labels. They do not necessarily mean supersonic and subsonic. The returned Mach itself establishes that distinction. For directly specified exit temperature, pressure, Mach or recovery, the high/low selection is not used and `root_count` is set to one; it is not a survey of every possible inverse solution.

A root near a tangency can fail to produce a sign change in the sampled grid. Likewise, a root outside the scanned Mach interval is not discovered. The implementation rejects a contraction without distinguishable brackets rather than inventing a limiting solution. A reported root count therefore describes the search result, not a mathematical proof that no other roots exist in every possible domain.

27.8 Interpreting a result and a refusal

The energy residual describes $h_e + V_e^2/2 - h_t$, in J/kg. Entropy change and recovered total-pressure ratio describe the calibrated irreversible path. Contraction is a mass-flux ratio. These quantities should not be relabeled thrust, captured mass flow or measured inlet efficiency.

An impossible reference can fail before the current-state calculation begins. A current pressure or recovery can require a temperature outside available property data or leave no positive kinetic energy. A contraction can fail the bracketing condition. Each refusal concerns a specific combination of assumptions and inputs; it does not prove that every physical inlet at that flight condition is impossible.

The fixed dry-air chemistry remains a major boundary at high temperatures. The mode does not automatically substitute equilibrium air, predict dissociation losses or match a reacting combustor. A full-engine calculation using this inlet adds further interface constraints, described in Chapter 29. The independent result remains a reference-calibrated compression state with explicit thermodynamic and inverse-root assumptions.

28. Constant-property scramjet flowpaths

A scramjet retains supersonic flow through its modeled heat-addition region. This avoids requiring the entire captured stream to be decelerated to subsonic conditions, but it does not remove the coupling between heat, area, momentum and choking. Adding sufficient heat at fixed area can still drive the flow toward a sonic limit.

ThrustOne's constant-property scramjet modes assemble the source-term equations of Chapter24 into a combustor, a nozzle, or a complete inlet-combustor-nozzle path. Their gas constant and gamma remain fixed. They do not solve chemical rates, molecular mixing or an automatic change to subsonic combustion.

28.1 Three boundaries for three questions

The combustor-only mode begins from an explicitly reconstructed static flow state and advances the combustor segments. The nozzle-only mode begins from its own reconstructed state and advances the nozzle segments. Neither has a freestream capture calculation from which to determine full-engine net thrust.

The full-engine mode first constructs a compressed supersonic inlet state from flight conditions and a specified total-pressure recovery. It then advances combustor segments followed by nozzle segments as one con-

nected flowpath. The last accepted state of one segment becomes the initial state of the next.

Full engine: freestream → prescribed inlet → source-term combustor -

Combustor: supplied inlet → combustor exit

Nozzle: supplied inlet

At least one segment is required in the selected combined list. The code does not infer a missing physical component merely because the mode is named full engine. The actual supplied segment list determines what is integrated.

28.2 Supersonic inlet closure

For positive capture area and ambient state, captured air flow is $\dot{m}_a = \rho_0 V_0 A_{capture}$. Total temperature is obtained by ram heating; total pressure is multiplied by the directly supplied recovery. The inlet is adiabatic, so it retains total enthalpy while losing total pressure.

Exit Mach, temperature or pressure can specify a state as in Chapter25, but here the selected branch must remain supersonic and slower than the freestream. The static exit pressure must not fall below ambient. The implementation does not reconstruct oblique shocks to derive the specified loss.

For an area-contraction constraint, recovered total state and area determine the mass-flow curve. The sonic capacity must accommodate captured flow, and a decelerating supersonic root must exist between Mach one and flight Mach. Otherwise the input is refused. A physically possible subsonic root does not rescue this supersonic-flow mode.

The recovery-named and contraction-named closures share the same current area/recovery equation. This conventional inlet is separate from the NASA9 reference-calibrated model of Chapter27. Their property and loss laws must not be mixed without explicitly changing the model.

28.3 Heat addition and area compensation

In a constant-area segment with no injection or drag, Chapter24 gives

$$V' = \frac{V}{1 - M^2} \frac{Q'}{\dot{m} c_p T}.$$

For supersonic flow, positive heat decreases velocity while increasing total enthalpy. Density, pressure and sound speed also change, driving Mach toward one. The added energy is not simply converted into immediate axial acceleration inside the combustor.

Area divergence contributes the term $-A'/A$ inside the same bracket. On the supersonic branch, divergence tends to accelerate the flow and can counter part of the heat-addition tendency. This is why area schedule and heat schedule must be considered together. It does not mean that arbitrary expansion can guarantee an acceptable combustor state: pressure, temperature, finite length and downstream conditions still constrain the result.

Darcy drag and mass injection add the forces and energy-dilution terms already derived in Chapter24. Injection with low axial velocity can reduce stream momentum even while its mass and total enthalpy increase the downstream fluxes. The fixed-property model does not infer those source values from a detailed injector or combustion mechanism.

28.4 What the nozzle accepts

Nozzle segments in the full-engine and nozzle-only modes must have zero heat source and zero mass injection. Their area variation and nonnegative Darcy friction remain available. Thus a nozzle can be adiabatic yet lose total pressure through drag. It should only be called isentropic when the relevant dissipative sources are absent.

The model does not impose monotonically increasing area on every constant-property nozzle segment. A decreasing-area supersonic segment decelerates the stream and may approach choking. The branch

guard prevents continuation through the sonic region; it does not redesign the area schedule.

Unlike the full ramjet wrapper of Chapter 25, this full scramjet integrates supplied nozzle segments. Exit pressure is an output of that integration. Ambient pressure is used in the final force balance but does not adjust the internal solution to match an installed nozzle back pressure. A mismatch is therefore not solved by relocating a shock or changing the inlet state.

28.5 Original energy and force bookkeeping

Take an illustrative completed path with captured air flow 1 kg/s and no mass injection. If inlet total temperature is 600 K, constant $c_p = 1000$ J/(kg K), and the combustor receives 200 kW total heat, outlet total enthalpy is 800 kJ/kg. An adiabatic nozzle retains this total enthalpy. The actual exit temperature and velocity must still satisfy its momentum and geometry solution.

For a separate illustrative endpoint force calculation, let exit mass flow 1.1 kg/s, velocity 1200 m/s, exit pressure 30 kPa, ambient pressure 20 kPa and exit area 0.02 m². Gross thrust is

$$F_g = 1.1(1200) + (30000 - 20000)(0.02) = 1520 \text{ N}.$$

With captured air flow 1 kg/s at 900 m/s, net thrust is 620 N. The additional 0.1 kg/s must be represented by the upstream source bookkeeping; it is not automatically counted as a fuel species or assigned a heating value by this calculation.

These are transparent flux and force examples, not a claimed spatial solver run. They separate the energy constraint from the endpoint momentum accounting. A proposed endpoint must still be attainable through a completed flowpath before it can be interpreted as its prediction.

28.6 Incomplete histories are not completed engines

The general solver uses finite-step integration and retains the selected Mach branch. When a step enters its sonic exclusion region, becomes invalid or exceeds its conservation checks, it returns the previous valid history. The last state remains available for diagnosis, and completed is false.

The scramjet wrapper then omits gross and full-engine thrust. A pressure and velocity shown in the exit fields of such a result describe the retained stopping point, not the intended physical exit. Combining those fields manually with the originally requested exit area would produce an inconsistent force estimate.

The stop does not automatically mean the real engine changes to ram-jet mode. A shock train, isolator response, combustor unstart or mode transition would require a different coupled flow solution. Increasing the integration resolution can address a discretization problem; it cannot authorize crossing a physically singular branch in this model.

28.7 Completed outputs and their conventions

On a completed path,

$$F_g = \dot{m}_e V_e + (p_e - p_a)A_e, \quad F = F_g - \dot{m}_a V_0.$$

The full-engine specific thrust uses captured air flow, even if injected mass increases exit flow. Component-only modes report gross exhaust force without a captured-air subtraction. This quantity is not the net axial force on the nozzle wall or a complete component control-volume force balance; its inlet momentum and pressure terms would also be needed for that interpretation.

The output energy residual compares inlet and exit total-enthalpy flux with integrated heat and injected total enthalpy. Its denominator scales with the magnitudes of those terms. A small residual demonstrates con-

sistency with the specified sources, not chemical accuracy or thrust agreement with a real engine.

The wrapper does not derive fuel consumption, combustion efficiency or specific impulse from unnamed mass and heat sources. It also does not require positive net thrust in the completed calculation. A negative value is possible under unfavorable supplied conditions and should not be silently described as successful propulsion.

28.8 Choosing a more detailed path

Use the fixed-property modes when the intended question is how prescribed area, heat, drag and mass sources influence an assumed constant-gas flowpath. Use frozen variable-property flow when temperature-dependent properties matter but composition is held fixed. Use the composition-aware and finite-rate paths of Chapter29 when injected species or reaction evolution are part of the actual question.

Changing the model is more than replacing γ with a local value. It changes the enthalpy reference, sound speed, species accounting and sometimes the inlet closure. Comparisons must therefore retain common boundary conditions and clearly state which physical assumptions were changed.

The constant-property scramjet remains a bounded one-dimensional model. Capture area does not predict spillage or installed drag, source terms do not predict mixing, and a completed supersonic integration does not prove that downstream back pressure permits stable hardware operation. These distinctions are the conditions under which its plots and force results remain useful.

29. Mixing and finite-rate scramjet flowpaths

A reacting flowpath must carry chemical composition as part of its state. Temperature alone cannot identify its stored energy: two mixtures at the same temperature can have different formation enthalpies, gas constants and heat capacities. Injection introduces a second complication. Even with every chemical rate set to zero, adding a different gas changes the mixture and its thermodynamic properties. This chapter develops a conservation formulation that handles both effects without adding an artificial combustion heat source.

The three relevant models are the standalone finite-rate quasi-one-dimensional flowpath, the calibrated-inlet nonreacting mixed-flow engine, and the calibrated-inlet finite-rate reacting engine. They share the spatial flow solver. The full-engine modes additionally establish captured-air conditions and calculate exhaust force after the complete supplied geometry has been traversed. Their common mathematical structure does not make them a multidimensional mixing or combustion simulation.

29.1 Control boundary and state variables

Consider a stationary duct with axial coordinate x , cross-sectional area $A(x)$, and a positive axial bulk velocity V . Each cross section has one pressure p , one temperature T and one composition. Side injection contributes

29.2 Species transport and injection

Let ω_i be the net chemical production rate in $\text{kmol}/(\text{m}^3 \text{ s})$. A slice of length dx has volume $A dx$, so its chemical contribution to species flow is $A\omega_i dx$. Let $q_m = d\dot{m}_{inj}/dx$ denote injected mass per unit length, in $\text{kg}/(\text{s m})$, with injected mole fractions $X_{i,inj}$ and mean molecular weight $\bar{W}_{inj}$. The balance is

$$\frac{dN_i}{dx} = A\omega_i + q_m \frac{X_{i,inj}}{\bar{W}_{inj}}.$$

Both terms have units $\text{kmol}/(\text{s m})$. Using a mass fraction in place of the mole fraction in this expression would be incorrect. The equivalent mass-flow source is $q_m Y_{i,inj}$, where $Y_{i,inj} = X_{i,inj} W_i / \bar{W}_{inj}$.

Concentrations required by the kinetic laws follow from the transported molar flows:

$$C_i = \frac{N_i}{AV}.$$

This relation is also a useful conceptual link between residence time and reaction. A slower flow contains more material in a fixed duct volume at the same molar throughput. Reaction rates depend on concentrations and temperature; they do not depend on axial distance alone. Chapter 7 explains how elementary, reverse and pressure-dependent rates contribute to ω_i .

For a chemically balanced mechanism,

$$\sum_i W_i \omega_i = 0, \quad \frac{d\dot{m}}{dx} = q_m.$$

Total molar flow need not be conserved. Dissociation can increase the number of moles without creating mass. Consequently, a change in $\sum_i N_i$ is not itself evidence of a conservation failure.

The App accepts a composition for each injecting segment. If that composition is omitted, it uses the initial flow composition, not the local composition reached after earlier reactions or injections. This default matters in a multisegment calculation. A later segment with unspecified injection composition does not automatically sample the gas currently passing through it.

29.3 Momentum and energy balances

The pressure force exerted by a gradually varying wall produces a $p dA/dx$ contribution to the derivative of momentum flux. With Darcy friction factor f_D and the circular-equivalent diameter $D = 2\sqrt{A/\pi}$, the implemented balance is

$$\frac{d\mathcal{F}}{dx} = p \frac{dA}{dx} - \frac{f_D \rho V^2 A}{2D} + q_m V_{inj}.$$

The injection velocity here is its prescribed axial component. The equation accounts for its axial momentum contribution; it does not solve the transverse jet, penetration distance or shear layer. The friction convention is the Darcy convention discussed in Chapter 24. Substituting a Fanning coefficient without conversion changes the intended drag.

Write Q' for external heat supplied per unit length, in W/m, and $h_{t,inj}$ for injected total specific enthalpy. Then

$$\frac{d\mathcal{E}}{dx} = Q' + q_m h_{t,inj}.$$

No extra lower-heating-value term belongs in this equation when formation enthalpy is already included. In an adiabatic reacting segment without injection, $\mathcal{E}$ remains constant. Chemical conversion changes the relation between h , T and composition, and the reconstruction finds the resulting thermal and kinetic state. Adding a fuel heating value again would count chemical energy twice.

Injected total enthalpy includes kinetic energy:

$$h_{t,inj} = h_{inj} + \frac{V_{inj}^2}{2}.$$

The App checks that the corresponding static enthalpy can be inverted within the selected property model when injection is positive. Supplying an arbitrary positive number is not enough to establish consistency: enthalpy values depend on the database reference convention. A negative formation-inclusive enthalpy is not automatically unphysical.

Finally,

$$\frac{d\tau}{dx} = \frac{1}{V}.$$

The integrated residence time is therefore $\int dx/V(x)$. Dividing total length by inlet velocity is only an approximation. Heating, injection, area variation and reactions can all change velocity along the path.

29.4 Closing the local state

After an integration step, the species flows define composition and mass flow. At the local area, the solver recovers a state satisfying

$$pA + \dot{m}V = \mathcal{F}, \quad h(T, \mathbf{X}) + \frac{V^2}{2} = \frac{\mathcal{E}}{\dot{m}}, \quad \dot{m} = \rho AV.$$

The ideal-gas mixture relation provides density. Property evaluation supplies h , heat capacity and the frozen sound speed. These equations close the thermodynamic state without an independent temperature evolution equation.

The initial state selects a subsonic or supersonic branch using the frozen Mach number. Subsequent reconstruction retains that branch. The model rejects an initial state too close to Mach one and stops if it cannot continue an admissible state. It does not pass through a sonic singularity by switching branches, insert a shock, or solve an isolator adjustment. In this

context a reported Mach number is based on the local frozen sound speed; it is not an equilibrium acoustic derivative for an infinitely fast reacting mixture.

Within each segment, area varies linearly between its entrance and specified exit area. The source coefficients supplied for that segment prescribe the distributed forcing. Making many short segments allows a piecewise description, but it does not add radial resolution. Every cross section still has one uniform mixed state.

29.5 Nonreacting mixing is a variable-composition problem

The calibrated nonreacting engine sets chemical rates to zero while retaining species transport and property reconstruction. If the injected gas differs from the entering air, its mole fractions change downstream. Thus nonreacting does not mean constant composition, constant heat capacity or constant gas constant.

As an original bookkeeping example, suppose 1 kg/s of an inert carrier with molecular weight 28 kg/kmol receives 0.10 kg/s of a second inert species with molecular weight 32 kg/kmol. The species molar flows after injection are

$$\begin{aligned}N_1 &= 1/28 = 0.0357143 \text{ kmol/s,} \\N_2 &= 0.10/32 = 0.003125 \text{ kmol/s.}\end{aligned}$$

The second species has mass fraction 0.090909 but mole fraction approximately 0.080460. The mixture molecular weight is approximately 28.3218 kg/kmol. Treating 0.090909 as its mole fraction would change the gas constant and therefore the recovered state.

For a separate energy illustration, take an entering energy flux of 1 MW, injected total enthalpy of 0.5 MJ/kg and no wall heat. The added energy flux is 50 kW. The downstream total specific enthalpy is

$$h_{t,out} = \frac{1.05 \text{ MW}}{1.10 \text{ kg/s}} = 0.954545 \text{ MJ/kg}.$$

Total energy throughput increased, but energy per unit mass decreased. This is not an energy loss. If the injection has axial velocity 100 m/s, its direct axial momentum contribution is 10 N. These balances alone do not determine the outlet temperature: the area, entering momentum flux, property model and selected branch are still required. The numbers illustrate independent conservation arithmetic, not a computed complete engine history.

29.6 Finite-rate evolution and conservation diagnostics

With reactions enabled, the chemical source is recomputed from the reconstructed temperature and concentrations. A faster intrinsic rate does not guarantee complete conversion. The gas may leave the duct before it approaches equilibrium, cool during expansion, or enter conditions where the reverse reaction matters. Equilibrium and finite-rate calculations therefore answer different questions.

For each element k , let a_{ki} be the number of its atoms in species i . The elemental molar flow is

$$B_k = \sum_i a_{ki} N_i.$$

A balanced chemical mechanism satisfies $\sum_i a_{ki} \omega_i = 0$. Changes in B_k must then equal the integrated injection of that element. ThrustOne reports elemental balance diagnostics for the supported element set together with mass, momentum and energy residuals. These checks detect failures of conservation accounting. Small residuals do not establish that supplied rate coefficients describe a real fuel or that a uniform-flow approximation captures physical mixing.

The spatial integrator uses an Euler predictor and a Heun corrector. Their difference estimates local error. Rejected steps are shortened; accepted steps may grow, subject to the segment's scheduled maximum step. This method differs from the fixed-step Runge–Kutta treatment used by some constant-property flow modes. Its tolerance controls a scaled numerical error estimate, not a guaranteed relative error in every trace-species concentration.

Chemical timescales can become much shorter than the convective timescale. An explicit method may then require very small steps. Hitting a step or attempt limit is a numerical termination condition, not proof that the physical flow cannot exist. Conversely, repeated reconstruction failure near a sonic boundary may reflect the limits of the selected steady branch. Increasing an iteration allowance cannot supply missing shock physics.

29.7 From calibrated inlet to full engine

The full-engine modes use the calibrated dry-air inlet of Chapter 27. Flight density and velocity determine captured flow:

$$\dot{m}_a = \rho_0 V_0 A_{capture}.$$

The inlet-exit area is capture area divided by contraction ratio. The wrapper checks that the reconstructed inlet-exit mass flow agrees with captured air. Its inlet mixture is 0.79 nitrogen and 0.21 oxygen by mole. The selected inlet-exit root must be supersonic; a low-Mach selection is not silently exchanged for a different root.

These full-engine modes require NASA9 properties through inlet and downstream flow. The standalone reacting-flow mode can use its supported NASA7 option, but the full-engine wrapper rejects that substitution because the inlet and downstream enthalpy references must remain consistent.

The supplied combustor and nozzle segment lists are concatenated into one flowpath. Reactions can continue through the nozzle list. Unlike the

constant-property scramjet nozzle wrapper, this implementation does not impose zero heat or zero injection merely because a segment appears in the nozzle list. Its specified sources remain active. The list name describes the intended region; it does not override its equations.

On a completed path, the exit mass flow must equal captured air plus all integrated injection. With ambient pressure p_a ,

$$F_g = \dot{m}_e V_e + (p_e - p_a) A_e, \quad F_n = F_g - \dot{m}_a V_0.$$

Specific thrust uses captured-air flow. The reported injected-mass specific impulse uses the sum of every injected stream:

$$I_{sp, inj} = \frac{F_n}{g_0 \dot{m}_{inj}}.$$

If injection includes oxidizer, diluent or air, none of those contributions is reclassified as fuel. For illustration, a net force of 600 N with total injection of 0.20 kg/s gives about 305.915 s. Using only a 0.05 kg/s fuel portion would give about 1,223.659 s, a different normalization rather than a better result. The App reports the former definition. Without injection it does not construct an injected-mass specific impulse.

29.8 Reading a completed or interrupted result

A completed calculation means the integration traversed the supplied segment lengths and passed its conservation checks. It does not mean the exit pressure matches ambient or that thrust is positive. Fixed downstream geometry is not adjusted for back pressure. External drag, spillage, shock trains and automatic operating-mode transition remain outside this calculation.

An interrupted calculation retains its valid prefix. The final retained row is the last admissible computed state, not necessarily the precisely located physical termination point. The full-engine wrapper withholds engine

thrust for an incomplete path. A prefix ending at high temperature or pressure must not be presented as the intended nozzle exit.

Profiles may be reduced to roughly five hundred retained samples for output. This is a presentation reduction of the accepted history; it is not the internal step count. A smooth exported curve cannot establish that an unresolved steep chemical transition was accurately integrated.

The most informative comparison is therefore specific: hold inlet, geometry, injection and enthalpy conventions fixed; compare zero-rate mixing with the chosen finite-rate mechanism; then explain changes through species, residence time and energy redistribution. Such a comparison isolates the consequences of the supplied chemistry within this model. It does not establish a validated combustion mechanism or resolve transport processes that the equations do not contain.

30. TBCC with an explicit air split

A turbine-based combined-cycle concept uses different propulsion processes over a broad flight envelope. A turbine branch can supply compression at flight conditions where a ramjet lacks sufficient ram compression. At other conditions a ramjet branch can avoid passing all captured flow through rotating machinery. This architectural motivation does not, by itself, determine how much air enters each branch or whether both branches can operate together.

The ThrustOne explicit-split model answers a deliberately narrower question: given a common freestream, a total air flow and a prescribed fraction sent to each of two independently calculated cycles, what are their combined thrust and fuel flow? It is a separate-exhaust accounting model. It does not solve a common inlet, a mixer, a shared nozzle or a transition controller.

30.1 The two control volumes

Define total captured-air flow as $\dot{m}_a$ and the ramjet fraction as λ . The branch air flows are

$$\dot{m}_{a,r} = \lambda \dot{m}_a, \quad \dot{m}_{a,t} = (1 - \lambda) \dot{m}_a, \quad 0 \leq \lambda \leq 1.$$

The subscripts r and t identify ramjet and turbojet. Each branch has its

own fuel stream and exhaust:

```

                                +--> turbojet + fuel t -> exhaust t
common freestream ----|
total captured air      +--> ramjet   + fuel r -> exhaust r
                                prescribed air allocation
```

There is no downstream recombination plane in this diagram. Adding thrusts is not the same operation as mixing exhaust states. A physical mixer would require static-pressure compatibility, area, momentum and energy closure as in Chapter 14; none of those quantities can be inferred from the sum of two thrust numbers.

The air-flow split is an input, not an unknown obtained from pressure matching. Once it is specified, each branch can be solved using its own design-point cycle equations. Chapter 9 gives the turbojet station and shaft balances, and Chapter 25 gives the ramjet design-point calculation. This model calls those cycle calculations; it does not substitute the spatial ramjet flowpath or finite-rate scramjet model.

30.2 A common flight condition

For both branches to belong to one operating point, they must receive the same prescribed freestream Mach number, static temperature and static pressure. ThrustOne takes those three inputs from the turbojet portion of the form. It replaces the corresponding ramjet flight inputs with them before solving the active ramjet branch. The shared-flight fields remain relevant even at the all-ramjet endpoint.

A common flight condition is necessary but not sufficient for a physically matched inlet. It says where the vehicle is operating; it does not establish that the two branch entrances can simultaneously accept their assigned flows. The explicit-split formulation contains no equation equating their inlet-exit static pressures or enforcing a common capture-area geometry.

Care is also needed with user-supplied gas properties. Identical Mach number and static temperature imply identical dimensional speed only when the freestream sound-speed convention agrees. For a calorically

perfect gas,

$$V_0 = M_0 \sqrt{\gamma R T_0}, \quad R = c_p \frac{\gamma - 1}{\gamma}.$$

The branch property assumptions should therefore represent the same entering air if the result is interpreted as one physical flight state. Sharing the three flight inputs does not automatically reconcile independently chosen caloric approximations.

30.3 Recovering dimensional thrust and fuel flow

Let s_t and s_r be branch net specific thrusts, each normalized by its own entering air flow. Their units are N/(kg/s), equivalently m/s. Let f_t and f_r be the corresponding fuel-to-air mass ratios. Then

$$F_t = s_t \dot{m}_{a,t}, \quad F_r = s_r \dot{m}_{a,r},$$

$$\dot{m}_{f,t} = f_t \dot{m}_{a,t}, \quad \dot{m}_{f,r} = f_r \dot{m}_{a,r}.$$

The combined outputs are

$$F = F_t + F_r, \quad \dot{m}_f = \dot{m}_{f,t} + \dot{m}_{f,r}.$$

Because each branch result is already a net thrust, its entering-air momentum debit is already present. With compatible freestream conventions, those debits sum to $\dot{m}_a V_0$. Subtracting the total captured-air momentum a second time would understate the result. Likewise, pressure-thrust contributions already included by a branch cycle must not be independently added again.

Dividing by total air flow gives

$$s = (1 - \lambda)s_t + \lambda s_r,$$

$$f_{overall} = (1 - \lambda)f_t + \lambda f_r.$$

The second expression is a useful derived bookkeeping quantity; the App reports dimensional combined fuel flow rather than requiring this additional metric. Both relations are air-flow-weighted sums. They should not be confused with an unweighted average of the branch values.

30.4 Why TSFC is not averaged by air fraction

Combined thrust-specific fuel consumption is

$$\text{TSFC} = \frac{\dot{m}_f}{F} = \frac{(1 - \lambda)f_t + \lambda f_r}{(1 - \lambda)s_t + \lambda s_r}.$$

Its SI units are kg/(N s). Multiplication by 3,600 converts it to kg/(N h). Air-flow weighting the two branch TSFC values generally gives the wrong answer because their thrust per unit air differs.

When both branch thrusts are positive, an equivalent expression is

$$\text{TSFC} = \frac{F_t}{F} \text{TSFC}_t + \frac{F_r}{F} \text{TSFC}_r.$$

It is a thrust-weighted average. Under those conditions the combined value lies between the branch values. This bound is a statement about a positive weighted mixture, not a claim that the combined hardware has no installation penalties.

If the two branches use different fuel heating values, minimum TSFC need not mean minimum chemical-energy input per unit thrust. A separate energy accounting would use

$$\dot{Q}_{fuel} = \dot{m}_{f,t} H_t + \dot{m}_{f,r} H_r.$$

The explicit-split result does not calculate a combined thermal or overall efficiency. One should not invent such an efficiency by averaging branch efficiencies or by assigning one heating value to unlike fuels.

30.5 Original worked accounting example

Consider a hypothetical pair of already solved branch cycles at one compatible freestream condition:

Quantity	Turbojet	Ramjet
Net specific thrust	500 N/(kg/s)	800 N/(kg/s)
Fuel-to-air ratio	0.020	0.040

These deliberately simple values illustrate the combining equations; they are not a reference engine or an independently confirmed pair of Thrust-
One cycle solutions.

Set total air flow to 80 kg/s and ramjet fraction to 0.25. The turbojet receives 60 kg/s and the ramjet receives 20 kg/s. Therefore

$$F_t = 500(60) = 30,000 \text{ N}, \quad F_r = 800(20) = 16,000 \text{ N}.$$

The fuel flows are 1.20 and 0.80 kg/s. Combined thrust is 46,000 N, fuel flow is 2.00 kg/s, and total-air specific thrust is 575 N/(kg/s). The TSFC is

$$\frac{2}{46,000} = 4.347826 \times 10^{-5} \text{ kg}/(\text{N s}) = 0.156522 \text{ kg}/(\text{N h}).$$

The branch TSFC values are 0.000040 and 0.000050 kg/(N s). Their air-fraction average would be 0.0000425, which differs from the correct result. The ramjet supplies about 34.78% of thrust although it receives only 25% of air, explaining why thrust weighting is different.

For a contrast, keep both branch states fixed and change the ramjet fraction to 0.75. Branch air flows become 20 and 60 kg/s. Combined thrust becomes 58,000 N and fuel flow becomes 2.80 kg/s. Specific thrust rises to 725 N/(kg/s), while TSFC rises to approximately 0.0000482759 kg/(N s). This example improves thrust per unit captured air but worsens fuel mass per unit thrust. Calling one split “better” requires an objective.

Doubling total air flow with the same branch assumptions and fraction doubles thrust and fuel flow. It leaves specific thrust and TSFC unchanged. This proportionality reflects the design-point accounting model. It does not establish that a fixed physical inlet, compressor or nozzle can pass twice as much air.

30.6 What a split sweep can and cannot optimize

If branch specific thrusts and fuel ratios remain fixed while only the allocation changes,

$$\frac{ds}{d\lambda} = s_r - s_t.$$

Specific thrust is linear in the split. Its maximum is at an endpoint unless the branch values are equal. For positive combined thrust, differentiating the TSFC ratio gives

$$\frac{d \text{TSFC}}{d\lambda} = \frac{f_r s_t - f_t s_r}{[s_t + \lambda(s_r - s_t)]^2}.$$

The numerator is independent of the split. TSFC is therefore monotonic, or constant, under these fixed-branch assumptions. An interior optimum cannot arise from simple allocation alone. It would require another dependence or constraint: changing branch efficiencies, admissible flow limits, installation penalties, temperature restrictions, or a prescribed thrust requirement, for example.

This result prevents a common interpretation error. A smooth curve generated by moving a split slider is not a computed transition schedule. The

model has not solved how geometry, inlet recovery or operating limits evolve during a transition. It has repeated the explicit accounting at different allocations.

30.7 Endpoints, failures and reporting

At $\lambda = 0$, only the turbojet branch is solved. At $\lambda = 1$, only the ramjet branch is solved. ThrustOne deliberately skips a zero-flow branch so that an irrelevant cycle failure does not invalidate the selected endpoint. Metrics for an inactive branch need not exist; absence does not mean a hidden calculation returned zero.

For any positive assigned branch flow, that branch must solve successfully. An invalid active cycle is not replaced by the other branch, and its mass allocation is not automatically changed. Even a very small positive allocation invokes its cycle. Endpoint behavior therefore distinguishes a truly inactive branch from an active branch operating outside its supported conditions.

The wrapper requires finite positive total air flow, a finite split in $[0,1]$, finite positive resultant thrust, and finite nonnegative combined fuel flow. Passing these checks establishes an admissible result for this synthesis. It does not certify inlet matching, stable simultaneous operation or hardware feasibility.

The reported warnings identify the independence of the cycles and the absence of common-inlet/back-pressure matching, mixing, variable geometry, channel gating and automatic mode switching. The wrapper does not aggregate all child-cycle warnings. Consequently, the absence of a particular branch warning from the combined warning list should not be read as an additional acceptance test. The branch assumptions and any known branch limitations remain applicable.

30.8 Relation to a physical combined-cycle engine

A transition analysis would need to connect the assigned flow split to realizable flow areas and compatible pressures. It would also need to estab-

lish acceptable rotating-machine conditions while the ramjet path changes its demand, account for inactive-path drag and thermal exposure, and determine a stable evolution through the changeover. Those are coupled physical questions, not missing arithmetic in the weighted sum.

The current mode is useful before those questions are addressed: it makes normalization explicit, separates branch contributions, and shows the trade between thrust and fuel consumption under declared cycle assumptions. Its proper conclusion is a conditional one: these independently calculated branch performances produce this combined accounting result at this prescribed split. It does not establish that an integrated engine will realize that operating point.

31. The ideal ejector–ramjet

An ejector transfers momentum and energy from a primary stream to an entrained secondary stream. The primary material is carried on the vehicle; the secondary material is drawn from the surrounding air. Unlike the explicit allocation of Chapter 30, the entrainment ratio in this model is an output of a coupled mixing calculation. No compressor shaft is needed, but the absence of rotating machinery does not make the transfer reversible.

ThrustOne implements one specific branch: a supersonic primary stream and a subsonic secondary stream enter a constant-area mixing passage at equal static pressure, and the fully mixed passage exit is sonic. An idealized downstream heater and fully expanded nozzle complete the performance calculation. This is a local ejector–ramjet submodel, not a complete rocket-based combined-cycle engine.

31.1 Stations, assumptions and closure

Use p and s for the primary and secondary streams, I for their common-pressure entrance plane, E for the sonic mixed exit, and N for the final nozzle exit. A subscript t denotes a total state. The flow sequence is

```
primary total state -> supersonic primary at I --+
                                     +-> sonic mixed E ->
ambient secondary -> subsonic secondary at I ----+
                               equal static pressure      constant-area mix
```

Both streams have the same constant gas properties, including γ

and gas constant R . The mixer is steady, adiabatic and free of wall friction and shock waves. There is no mixing before the specified entrance plane. Downstream heat addition occurs in an ideal zero-Mach limit without an imposed total-pressure loss, and its fuel mass is neglected. The final nozzle expands isentropically to ambient pressure.

The inputs specify the two total states, primary critical area A_{p*} , total mixer area A , common γ , ambient conditions, secondary freestream Mach number and final total temperature. The unknown common inlet pressure p_I determines both inlet Mach numbers, their area allocation and the entrainment ratio. Conservation then determines the sonic exit state. One remaining momentum equation selects p_I .

This ordering is essential. The inlet pressure cannot be specified independently along with an arbitrary entrainment ratio and still be expected to satisfy the same sonic-exit problem. A hand calculation may try a pressure, but a trial with a nonzero momentum residual is not a converged operating point.

31.2 The permitted pressure interval

For a calorically perfect gas, define

$$\Theta(M) = 1 + \frac{\gamma - 1}{2} M^2, \quad \frac{p_t}{p} = \Theta(M)^{\gamma/(\gamma-1)}.$$

At a trial p_I ,

$$M_{pI} = \sqrt{\frac{2}{\gamma - 1} \left[\left(\frac{p_{tp}}{p_I} \right)^{(\gamma-1)/\gamma} - 1 \right]},$$

with the analogous expression using p_{ts} for M_{sI} . Define the sonic static-to-total pressure ratio

$$r_* = \left(\frac{2}{\gamma + 1} \right)^{\gamma/(\gamma-1)}.$$

The selected branches require

$$p_{ts}r_* < p_I < \min(p_{ts}, p_{tp}r_*).$$

The lower limit excludes sonic or supersonic secondary entry. The upper limits exclude zero secondary velocity and subsonic primary entry. These are strict inequalities. An empty interval means the specified total pressures cannot support the requested pair of branches.

A pressure inside the interval is only a candidate. It must also yield a positive remaining area for the secondary stream and satisfy the downstream requirements. Pressure admissibility alone does not establish an ejector solution.

31.3 Area allocation and entrainment

The primary critical area establishes its mass-flow scale. Its inlet area follows from the supersonic area relation:

$$\frac{A_{pI}}{A_p^*} = \frac{1}{M_{pI}} \left[\frac{2}{\gamma + 1} \Theta(M_{pI}) \right]^n, \quad n = \frac{\gamma + 1}{2(\gamma - 1)}.$$

Define area fractions $\phi_p = A_{pI}/A$ and $\phi_s = 1 - \phi_p$. Both must be positive. If the primary stream alone occupies the whole passage, a positive secondary mass flow cannot be accommodated on this branch.

The ideal mass-flow expression at a station is

$$\dot{m} = A \frac{p_t}{\sqrt{T_t}} \sqrt{\frac{\gamma}{R}} M \Theta(M)^{-n}.$$

Taking the secondary-to-primary ratio cancels the common gas constant and gives the entrainment ratio

$$\beta = \frac{\dot{m}_s}{\dot{m}_p} = \frac{p_{ts}}{p_{tp}} \frac{\phi_s}{\phi_p} \frac{M_{sI}}{M_{pI}} \sqrt{\frac{T_{tp}}{T_{ts}}} \left[\frac{\Theta(M_{pI})}{\Theta(M_{sI})} \right]^n.$$

This is a mass-flow ratio. It is not an area ratio, a fuel-to-air ratio or a freely assigned bypass ratio. In particular, a large secondary area does not necessarily imply equally large entrainment: its total pressure, temperature and Mach number also determine the mass flux.

31.4 Energy and mass at the sonic exit

Adiabatic mixing with the same constant heat capacity gives

$$\dot{m}_p c_p T_{tp} + \dot{m}_s c_p T_{ts} = (\dot{m}_p + \dot{m}_s) c_p T_{tE}.$$

Therefore

$$T_{tE} = \frac{T_{tp} + \beta T_{ts}}{1 + \beta}, \quad T_E = \frac{2T_{tE}}{\gamma + 1}.$$

The second relation uses the imposed sonic exit. The mixed total temperature lies between the two entering total temperatures; the static exit temperature is lower because some energy remains kinetic.

Primary choked flow is

$$\dot{m}_p = A_p^* p_{tp} \sqrt{\frac{\gamma}{RT_{tp}}} \left(\frac{2}{\gamma + 1} \right)^n.$$

At the sonic mixed exit, $\dot{m}_E = A p_E \sqrt{\gamma/(RT_E)}$. Equating this with $(1 + \beta)\dot{m}_p$ gives

$$p_E = (1 + \beta) \frac{A_p^*}{A} p_{tp} \sqrt{\frac{T_E}{T_{tp}}} \left(\frac{2}{\gamma + 1} \right)^n.$$

No dimensional R is required to calculate this pressure or the entrainment ratio because it cancels. This is why the submodel can report useful nondimensional performance without requiring a separately supplied gas constant or a dimensional primary mass flow.

31.5 Momentum selects the inlet pressure

For an ideal gas, $\rho V^2 = \gamma p M^2$, so pressure plus axial momentum flux through area A_i is $p_i A_i (1 + \gamma M_i^2)$. The constant-area mixer momentum equation is

$$p_I A [\phi_p (1 + \gamma M_{pI}^2) + \phi_s (1 + \gamma M_{sI}^2)] = p_E A (1 + \gamma).$$

The area cancels. ThrustOne reports the dimensionless residual

$$r_F = \frac{p_E (1 + \gamma)}{p_I [\phi_p (1 + \gamma M_{pI}^2) + \phi_s (1 + \gamma M_{sI}^2)]} - 1.$$

A zero residual closes the selected problem. It does not prove reversibility. In particular, an adiabatic mixer can redistribute momentum and generate entropy while conserving total enthalpy. “No prescribed total-pressure-loss factor” should not be interpreted as preservation of either entering total pressure.

The sonic exit total pressure is recovered from its local state:

$$p_{tE} = p_E \left(\frac{\gamma + 1}{2} \right)^{\gamma/(\gamma-1)}.$$

Its value is an output, not an average of the two entering total pressures.

31.6 Ideal heating and thrust augmentation

The final total temperature T_{tN} is prescribed. The implementation requires it to be at least T_{tE} , so this stage represents heating or zero additional heating. It does not solve fuel chemistry or calculate the fuel mass

needed to reach that temperature. The ideal heater retains p_{tE} . The final nozzle Mach number follows from p_{tE}/p_a and must be supersonic on the supported branch.

For comparison, imagine the primary stream alone expanding to ambient pressure. Denote its resulting speed by V_{p0} . Let V_0 be the secondary freestream speed and V_N the mixed final exhaust speed. With no final pressure thrust and negligible heater fuel mass,

$$F = (\dot{m}_p + \dot{m}_s)V_N - \dot{m}_s V_0.$$

There is no captured-air momentum debit for the onboard primary stream. Dividing by the primary-alone thrust $\dot{m}_p V_{p0}$ gives

$$\mathcal{A}_F = (1 + \beta) \frac{V_N}{V_{p0}} - \beta \frac{V_0}{V_{p0}}.$$

Velocity ratios follow from $V = M\sqrt{\gamma RT_t/\Theta(M)}$, so R again cancels. An augmentation above one means increased force relative to this particular primary-only reference. It is not a fuel efficiency or specific impulse. Additional downstream heating can contribute energy without a corresponding fuel-mass penalty in this ideal model.

31.7 Original accounting example and a failed trial

To isolate the balances, consider an illustrative candidate with γ 1.4, entrainment ratio 2, primary total temperature 2,400 K and secondary total temperature 300 K. These numbers are not asserted to form a solved geometry. The mixed total temperature is

$$T_{tE} = \frac{2400 + 2(300)}{3} = 1000 \text{ K}, \quad T_E = 1000/1.2 = 833.333 \text{ K}.$$

Per unit primary mass flow, the incoming total-enthalpy factor is $c_p[2400 + 2(300)] = 3000c_p$. The outgoing factor is

$3c_p(1000) = 3000c_p$, demonstrating energy closure. Prescribing a final total temperature of 900 K would violate the heating-only condition; 1,200 K would require an external specific heat input of $c_p(1200 - 1000)$ per unit mixed mass, with fuel mass still unmodeled.

For a separate momentum trial, take $p_I=100$ kPa, $\gamma=1.4$, primary Mach 2, secondary Mach 0.5 and area fractions 0.2 and 0.8. The entering pressure-plus-momentum flux per unit total area is

$$100[0.2(1 + 1.4 \times 4) + 0.8(1 + 1.4 \times 0.25)] = 240 \text{ kPa}.$$

A sonic exit would need $p_E=100$ kPa for momentum closure. If the independently recovered mass/energy exit pressure were 110 kPa, the residual would be 0.10. The positive temperatures and admissible Mach branches would not rescue that trial. It would still have a ten-percent momentum imbalance and require a different trial pressure.

Finally, illustrative speed ratios $V_N/V_{p0}=0.6$ and $V_0/V_{p0}=0.1$ with $\beta=2$ give augmentation 1.6. The exhaust speed is lower than the primary-alone speed, but three times the primary mass leaves the nozzle. Increasing entrainment cannot be assessed while holding these speed ratios arbitrarily fixed: the coupled mixing solution normally changes them.

31.8 Numerical and physical boundaries

The solver samples the admissible inlet-pressure interval to seek a residual sign change, then bisection a detected bracket. Invalid candidates break the continuity of the scan. Multiple detected sign-change brackets are rejected rather than silently selecting one; a sampled near-zero residual can be accepted directly. This finite search is not a mathematical proof that every possible root has been enumerated. Failure to detect a bracket means the implemented search did not find a solution on its permitted states.

The secondary freestream Mach input is distinct from the secondary Mach at the mixer entrance. Its total temperature and pressure must agree with

the ambient static state through isentropic relations; the narrow input allowance accommodates rounded values, not an independently pressurized secondary supply. Final static pressure must equal ambient. Both the primary-alone reference expansion and final mixed expansion must be supersonic.

The model excludes a choked secondary inlet, an unchoked mixed exit, shocks, wall friction, finite-rate mixing and combustion, and additional fuel mass. It does not predict a mixing length, injector geometry, rocket feed system or mode transition. Its internal trial evaluator can report a nonzero residual for a hand-iteration worksheet; that output must be distinguished from the automatic solver's converged result.

The App mode `rbcc.ideal_ejector` is consequently best read as an ideal entrainment and augmentation calculation under explicit branch assumptions. The broader combined-cycle label supplies architectural context. It does not expand the equations into the other operating modes of a vehicle engine.

32. The ideal air-turboramjet

An air-turboramjet uses a supplied hot primary stream to drive a turbine, whose shaft powers a fan in the captured-air stream. The streams then mix and may receive additional heat before expansion. Its defining transfer is mechanical: energy removed from the primary stream raises the total pressure and temperature of the air. This differs from the direct momentum exchange in the ejector of Chapter 31.

The ThrustOne model begins with the primary total pressure and temperature already specified. It does not derive that state from a gas generator, propellant combination, pump or cooling circuit. Its output describes the ideal downstream use of that supplied stream.

32.1 Flow sequence and the zero-Mach mixing limit

Let a denote captured air, p the primary stream, f the fan exit, q the primary turbine exit, m the mixed state and N the final exhaust. The topology is

captured air $\rightarrow$ ideal fan -----+
 ↑ shaft power + $\rightarrow$ slow mixing $\rightarrow$ heating $\rightarrow$ no
supplied primary $\rightarrow$ ideal turbine ---+

Both streams use the same calorically perfect gas properties. Fan and turbine processes are isentropic, shaft transfer is lossless, and the incoming air total state is the isentropic total state of the prescribed freestream. Mixing and downstream heating are idealized as occurring at negligible Mach number. At the mixing plane, static and total pressures therefore coincide

in the approximation.

This is not the finite-velocity, momentum-conserving mixer of Chapter 14. It has no solved area, entrance velocity or mixing-length requirement. The zero-Mach assumption removes kinetic mixing and Rayleigh total-pressure penalties. The primary turbine is assigned the fan-exit total pressure at its outlet so the streams meet at a common pressure.

32.2 Fan work and available turbine expansion

Define the fan pressure ratio as $\pi_f = p_{tf}/p_{ta} > 1$, and let $k = (\gamma - 1)/\gamma$. Isentropic compression gives

$$T_{tf} = T_{ta} \pi_f^k, \quad w_f = c_p T_{ta} (\pi_f^k - 1).$$

The primary turbine expands from p_{tp} to $p_{tq} = \pi_f p_{ta}$, giving

$$T_{tq} = T_{tp} \left(\frac{\pi_f p_{ta}}{p_{tp}} \right)^k,$$

$$w_t = c_p T_{tp} \left[1 - \left(\frac{\pi_f p_{ta}}{p_{tp}} \right)^k \right].$$

The fan-exit pressure must be lower than the supplied primary total pressure. Otherwise the assumed turbine expansion cannot supply positive shaft work. Both pressure availability and a positive fan temperature rise are needed; a nominally hot primary stream alone does not guarantee a solution.

Let $\beta = \dot{m}_a/\dot{m}_p$. Equating turbine power to fan power,

$$\dot{m}_p c_p (T_{tp} - T_{tq}) = \dot{m}_a c_p (T_{tf} - T_{ta}),$$

so

$$\beta = \frac{T_{tp}}{T_{ta}} \frac{1 - (\pi_f p_{ta}/p_{tp})^k}{\pi_f^k - 1}.$$

The bypass ratio is therefore an output of shaft balance, not a selected air split. Gas constant and common heat capacity cancel from this expression. Their cancellation relies on the shared-property assumptions; it does not authorize substituting two unrelated gases while keeping this equation unchanged.

32.3 Mixing and conservation of the original energy

The mixed total temperature is

$$T_{tm} = \frac{\beta T_{tf} + T_{tq}}{1 + \beta}.$$

Using the work balance to eliminate the temperatures changed by the machinery gives a particularly useful equivalent expression:

$$\beta T_{tf} + T_{tq} = \beta T_{ta} + T_{tp}, \quad T_{tm} = \frac{\beta T_{ta} + T_{tp}}{1 + \beta}.$$

The fan and turbine redistribute energy between the streams; their internal shaft transfer does not create energy in the combined control volume. The machinery changes pressure and establishes the air throughput even though the combined incoming and mixed total enthalpies agree.

ThrustOne reports two residuals in temperature units:

$$r_W = T_{tq} - T_{tp} + \beta(T_{tf} - T_{ta}),$$

$$r_M = \beta T_{tf} + T_{tq} - (1 + \beta)T_{tm}.$$

Multiplying r_W by $\dot{m}_p c_p$ converts its scale to a power imbalance. A small residual confirms these algebraic balances, not fan efficiency, turbine cooling adequacy or a verified primary-stream source.

32.4 Downstream heating and the reported fuel ratio

The target exhaust total temperature must not be below the mixed total temperature. Equality describes no additional heating. Define the supplied dimensionless fuel-energy parameter

$$\mathcal{H} = \frac{\eta_b H}{c_p T_0},$$

where H is fuel heating value and η_b is the assumed heat-release efficiency. The implemented simplified relation is

$$f = \frac{T_{tN} - T_{tm}}{T_0 \mathcal{H}}.$$

The App uses this f as added fuel per captured-air mass in exhaust-mass accounting. It is a reduced heat-addition convention, not the exact mass-inclusive energy balance for the whole primary-plus-air mixture.

To see the distinction, neglect fuel sensible enthalpy and retain all mass in a constant- c_p balance. With $f = \dot{m}_f / \dot{m}_a$, the exact algebra under those additional assumptions would give

$$f \eta_b H = c_p [(1 + 1/\beta + f)T_{tN} - (1 + 1/\beta)T_{tm}],$$

or

$$f = \frac{(1 + 1/\beta)(T_{tN} - T_{tm})}{T_0 \mathcal{H} - T_{tN}}.$$

This last equation is background for interpreting the approximation; it is not an alternative option implemented by the current mode. The App's

simpler expression neglects the primary-to-air correction and the extra fuel's sensible-energy demand in this heating estimate. Those approximations become more consequential when beta is small or final temperature is not small relative to $T_0\mathcal{H}$. The reported fuel ratio should not be promoted to a complete gas-generator or propellant-system requirement.

32.5 Expansion and two force normalizations

The nozzle total pressure is $p_{tN} = \pi_f p_{ta}$, retained through ideal slow mixing and heating. With final static pressure equal to ambient,

$$M_N = \sqrt{\frac{2}{\gamma - 1} \left[\left(\frac{\pi_f p_{ta}}{p_0} \right)^k - 1 \right]}.$$

The current mode admits subsonic and supersonic exhaust solutions. Unlike the ejector model, it does not require a supersonic nozzle exit. The branch is determined by the pressure ratio rather than independently chosen. The exact sonic boundary is classified with the non-subsonic branch in the implementation; it represents Mach one, not a finite supersonic margin.

Neglecting final pressure thrust, the exhaust mass per unit primary mass is $1 + \beta(1 + f)$. If V_{p0} denotes the speed obtained by expanding the supplied primary stream alone to ambient pressure, augmentation is

$$\mathcal{A}_F = [1 + \beta(1 + f)] \frac{V_N}{V_{p0}} - \beta \frac{V_0}{V_{p0}}.$$

Only captured air incurs the incoming momentum debit. The primary and added fuel are onboard streams. Specific thrust per captured-air flow is

$$s_a = (1 + f + 1/\beta)V_N - V_0.$$

The two outputs use different denominators and cannot be numerically compared as though both were specific thrust. Velocity ratios follow the

common-gas total-to-static relations. The dimensional scale is supplied by the ambient sound speed and freestream Mach through $V_0 = a_0 M_0$.

32.6 Original shaft and mixing example

Choose illustrative temperatures $T_{ta}=300$ K, $T_{tp}=1,800$ K and fan temperature ratio 1.2. Suppose the turbine temperature ratio is 0.8. The fan exit is 360 K and the turbine exit is 1,440 K. The shaft balance gives

$$\beta = \frac{1800 - 1440}{360 - 300} = 6.$$

For gamma 1.4, consistent pressure ratios can be constructed as $\pi_f = 1.2^{3.5}$ and $p_{tq}/p_{tp} = 0.8^{3.5}$; these ensure the temperatures describe isentropic machines. The example is an original subsystem balance, not a complete freestream/nozzle solution.

The mixed total temperature is

$$T_{tm} = \frac{6(360) + 1440}{7} = 514.286 \text{ K.}$$

Using the original incoming temperatures gives $[6(300) + 1800]/7$, exactly the same value. The residuals vanish because the turbine temperature drop of 360 K per primary mass equals the fan temperature rise of 60 K multiplied by six times as much air.

For a separate illustration of the heating convention, let $T_0 \mathcal{H} = 120,000$ K and prescribe $T_{tN}=1,200$ K. The implemented relation gives $f=0.00571429$. The mass-inclusive background expression above would give approximately 0.00673401. The difference arises from different energy bookkeeping assumptions, not numerical precision. Neither number supplies the propellant consumption needed to create the primary stream.

32.7 Limiting behavior and admissibility

As fan-exit pressure approaches primary total pressure from below, available turbine work vanishes and β tends toward zero for a fixed fan ratio above one. Specific thrust per captured-air mass then becomes a poorly conditioned normalization because its denominator shrinks. The App requires a finite positive β and does not continue to a primary-only endpoint as a separate mode.

As fan pressure ratio approaches one from above while positive turbine expansion remains available, required work per unit air approaches zero and the ideal formula can imply very large β . This is a limitation of the ideal unbounded-throughput model. No fan area, speed or flow map constrains the inferred air mass flow. The exact ratio one is excluded rather than evaluated by division by zero.

The freestream Mach number must be positive. Although the architecture is motivated partly by low-speed operation, a static-takeoff result at exactly zero Mach is not supported by this implementation's velocity-ratio normalization. The air total state must be consistent with its prescribed ambient static state and Mach, allowing only a narrow rounding tolerance. Final pressure must equal ambient, and returned net specific thrust must be finite and positive.

The model assumes no primary fuel remains to react during mixing, no gas-generator chemistry, no pumps, no cooling, no component maps and no automatic transition. It cannot calculate the cost of maintaining the supplied primary pressure and temperature. Its proper role is to expose how that given primary stream can ideally power compression, combine with air and contribute to a fully expanded exhaust under the stated approximations.

33. Numerical solutions as engineering results

A numerical answer is more than a value that stopped changing. It represents an equation, a physical branch, a set of input assumptions and a termination condition. Two calculations can print the same temperature while answering different questions; two calculations can print slightly different temperatures while representing the same solution within meaningful precision. This chapter explains how to read the numerical evidence accompanying the models developed earlier.

ThrustOne uses several kinds of numerical calculation: scalar root searches, coupled state reconstruction, interpolation and spatial or temporal integration. Their convergence indicators are not interchangeable. A small root residual is different from a small integration error estimate, and neither is a statement that the underlying physical approximation is valid for a particular engine.

33.1 Three different questions

A useful interpretation separates three questions. First, do the inputs define a problem admitted by the model? Second, did the numerical method solve that problem to its stated termination criterion? Third, does the model represent the intended physical situation closely enough to support the conclusion being drawn?

For example, a fully expanded ideal nozzle can have a tightly converged

exit Mach number. If the actual task concerns a fixed nozzle under varying back pressure, convergence does not supply the missing shock or separation response. Conversely, failure to finish an explicit reacting-flow integration may reflect chemical stiffness rather than the nonexistence of a physical flow.

The dependencies can be pictured as

```

specified inputs -> admissible model/branch -> numerical solution ->
                    |                               |
                    rejected definition         residual or stop reason
  
```

A rejection before solving has no numerical answer to interpret. An interrupted integration may have valid intermediate states but no completed exit. A completed result still carries the physical limits of its model.

33.2 Residuals and units

Suppose a scalar unknown x is defined by $R(x)=0$. The residual R measures failure to satisfy the equation; its units follow the equation. A residual of 0.01 K in an enthalpy inversion rewritten in temperature units is not comparable with 0.01 N in a momentum balance or 0.01 in a normalized pressure ratio.

Scaling makes residuals easier to interpret:

$$r = \frac{R}{S},$$

where S is a stated reference scale. However, a reference that approaches zero can make a relative residual misleading or undefined. A useful combined acceptance form is

$$|R| \leq \epsilon_{abs} + \epsilon_{rel} S.$$

The absolute allowance protects quantities whose correct value vanishes. The relative allowance expresses proportional accuracy away from zero.

These are explanatory forms, not a claim that every ThrustOne solver uses one shared formula or tolerance.

Consider an energy balance with incoming and outgoing powers near 2 MW and a difference of 0.1 W. Relative to transported energy, the discrepancy is five parts in one hundred million. If the intended output is a net power of only 0.2 W obtained by subtracting those large flows, the same discrepancy is half the net result. The acceptable accuracy depends on what the user is trying to infer, not just the size of the operands.

The reacting-flow diagnostics in Chapter 29 distinguish mass-relative residual, elemental-relative residual and dimensional energy and momentum residuals. The turbine–fan residuals in Chapter 32 use temperature units. Their labels are part of the meaning of the result.

33.3 A small residual need not mean a small state error

Near a simple root x^* , a first-order expansion gives

$$R(x) \approx R'(x^*)(x - x^*), \quad x - x^* \approx \frac{R(x)}{R'(x^*)}.$$

If the derivative is small, a small residual can coexist with a appreciable state error. This is conditioning: it describes sensitivity inherent in the equation, not merely the quality of the algorithm.

The area–Mach relation provides a physical example. At the sonic minimum, the first derivative of area ratio with respect to Mach vanishes. Its local expansion for a calorically perfect gas is

$$\frac{A}{A^*} \approx 1 + \frac{2}{\gamma + 1}(M - 1)^2.$$

Thus a small positive area excess delta corresponds approximately to

$$M - 1 \approx \pm \sqrt{\frac{\gamma + 1}{2}} \delta.$$

For $\gamma=1.4$ and $\delta=0.0001$, this gives M approximately 0.98905 or 1.01095. An area ratio only one ten-thousandth above its minimum produces two distinct states roughly one percent from sonic speed. Specifying more digits in the area ratio does not remove the need to choose a branch.

At $\delta=0$ the roots merge. Below zero there is no real area–Mach state. A solver must distinguish these cases rather than interpreting every failure near the minimum as a request for more iterations.

33.4 Brackets and physical branches

For a continuous residual, opposite signs at two admissible endpoints guarantee at least one zero between them. Bisection repeatedly halves the interval while preserving that sign change. After n halvings, an initial width W becomes $W/2^n$. This is a transparent numerical guarantee about a continuous bracket; it does not prove the root is physically desired or unique.

As an original arithmetic example, use $R(x)=x^2-2$ on $[1,2]$. The first midpoint 1.5 has positive residual 0.25, so the root remains in $[1,1.5]$. The next midpoint 1.25 has negative residual -0.4375 , leaving $[1.25,1.5]$. After twenty halvings the bracket width is below one millionth. The argument works because the residual is continuous and each retained interval contains a sign change.

A pole can imitate a sign change without containing a zero: $R(x)=1/(x-1)$ is negative below one and positive above one, but has no root there. An implementation must not bridge an invalid state and call the resulting interval a physical bracket. The ejector search described in Chapter 31 breaks its scan across inadmissible candidates for this reason.

A finite scan can also miss a root that touches zero without changing sign, or two nearby crossings within a single sampled interval. Detected bracket

counts describe what the search found. They should not be reported as a universal proof of the number of mathematical solutions.

For oblique shocks, branch selection has a physical meaning: weak and strong attached roots describe different compression states. For frozen flux reconstruction, the retained subsonic or supersonic state is part of the supplied closure. Choosing whichever root converges first would change the problem.

33.5 Coupled equations and conservation checks

Several models calculate one quantity from a balance and then use it in another. Each stage may be locally converged while the assembled result is inconsistent if conventions differ. Typical causes include total versus static temperature, fuel-to-core-air versus fuel-to-total-air normalization, and mixing enthalpies with incompatible reference states.

A conservation check is valuable because it recombines the final quantities into the original balance. For a steady duct with injection,

$$\dot{m}_{out} - \dot{m}_{in} - \int q_m dx = 0,$$

$$\mathcal{E}_{out} - \mathcal{E}_{in} - \int (Q' + q_m h_{t,inj}) dx = 0.$$

These checks ask whether the computed solution obeys the imposed book-keeping. They do not show that an injection profile, heat-transfer coefficient or kinetic mechanism was correctly chosen for an experiment. Conservation is necessary for the model; it is not a replacement for the model's assumptions.

Likewise, the zero turbine–fan work residual in Chapter 32 follows the equations used to calculate bypass ratio. It is useful for detecting implementation or arithmetic inconsistency, but it is not independent evidence for the realism of ideal machinery.

33.6 Interpolation and inverse maps

A component map defines supported data coordinates. Interpolation estimates between those coordinates according to a chosen shape rule. It does not establish performance outside the measured or supplied domain. Extrapolation introduces an additional model even when software can produce a smooth line.

The map operations discussed in Chapter 11 preserve their declared domain and branch restrictions. An inverse query can have more than one answer when the mapped quantity turns around. A target value alone may therefore be insufficient; a branch or interval is needed to close the inverse problem.

A visually smooth interpolant is not evidence that every derivative represents measured physics. Derivative-sensitive conclusions, such as a predicted optimum close to a sparse endpoint, require particular care. A maximum among supplied samples, a maximum of the interpolant and the actual maximum of a component are three different quantities.

33.7 Integration accuracy and stiffness

A trajectory solves differential equations over time or distance. A local error estimate controls the discrepancy of one step, not the uncertainty of the complete physical prediction. Errors can accumulate or be amplified by sensitive dynamics. Conversely, some stable balances constrain the trajectory even when individual local changes are large.

The reacting-flow method of Chapter 29 compares an Euler predictor with a Heun corrector. The constant-property duct and established-burning transient use other stepping arrangements. Their step counts and tolerances should not be compared as though they were one universal accuracy scale.

Stiffness arises when the equations contain widely separated timescales. For illustration, the decay equation $y' = -ky$ integrated by explicit Euler gives

$$y_{n+1} = (1 - k\Delta t)y_n.$$

A positive monotonically decaying solution requires a sufficiently small step; if $k\Delta t$ exceeds two, even linear stability is lost. This analytical example explains why a rapidly relaxing chemical component can force short explicit steps while the bulk flow changes slowly. It is not the exact stability bound for the App's multicomponent Heun system.

Reaching an accepted-step limit therefore has a different meaning from reaching an imposed sonic boundary. The first is a computational stop. The second concerns the supported branch of the steady equations. Both can leave a valid prefix, but neither produces a completed downstream solution.

33.8 Events, prefixes and displayed samples

An event is a condition such as burnout, unchoking or approach to Mach one. Event handling can retain the last admissible state, interpolate to a boundary, or solve for a precise event location. These approaches yield different meanings for the final row.

The established-burning model of Chapter 22 interpolates its burnout boundary but retains a previous valid state for certain other stops. The duct and reacting-flow chapters state their own prefix behavior. A reader should not assume that the last printed coordinate is an exactly located physical endpoint simply because its label says “final.”

A further distinction is output sampling. A solver may accept many internal steps but retain fewer rows for export. Connecting those rows produces a readable chart; it does not disclose every internal step or prove the resolution of a steep transition. Conversely, more plotted points can be interpolation rather than additional solved information.

A useful report preserves the original input, model and branch, completion flag, stop explanation, meaningful residuals and units. It presents interrupted results as partial histories and reserves final-engine performance for a completed path where the selected model requires one.

33.9 Precision and defensible conclusions

Digits should reflect the information available. A calculator may determine a root to many digits while using rounded gas properties, ideal efficiencies or a prescribed empirical coefficient with much larger uncertainty. Printing all numerical digits does not improve those inputs.

The appropriate conclusion connects the result to its question: a root on the specified branch, a conservation-consistent state, a completed history under the supplied sources, or a best sampled point under fixed assumptions. When a result fails, the useful next action is to identify whether the input, physical branch, numerical search or unsupported model assumption caused the failure. Repeatedly tightening tolerances cannot repair a problem whose required physics is absent.

34. Parameter studies and defensible comparisons

A parameter study is a sequence of individually defined problems. Its value comes from the relationship between them: what changes, what remains fixed, what is recalculated, and which results satisfy the chosen constraints. A dense plot without those definitions is difficult to interpret even when every point has been calculated correctly.

The engine-specific chapters supply the equations and limitations. This chapter explains how to use their outputs to compare alternatives without turning a sampled trend into an unsupported design claim.

34.1 Define the question before the axis

Suppose the question is how rocket thrust changes with chamber pressure. At least three studies are possible: hold throat area fixed; resize throat area to preserve mass flow; or resize it to preserve target thrust. All vary chamber pressure, but they answer different questions.

For fixed gas properties and choked flow,

$$\dot{m} = \frac{p_c A_t}{c^*}.$$

Holding A_t and c^* fixed makes mass flow proportional to pressure. Holding mass flow fixed instead requires A_t to vary inversely with pressure.

The resulting exhaust size, pressure thrust and interpretation are different. A curve labeled only “thrust versus chamber pressure” conceals this distinction.

A useful study definition states the model, varied inputs, held inputs, re-computed states, constraints and objective. For a turbine cycle, pressure ratio may be a design input or an off-design output. The same axis cannot be treated as independent in both problems without changing the closure.

34.2 Fixed inputs are not fixed outputs

Changing an input can alter several dependent quantities. In a chemistry calculation at fixed chamber temperature, changing mixture ratio changes composition and properties. In an enthalpy-closed chamber, changing the mixture may also change chamber temperature. Those two curves cannot be interpreted as though they differ only by numerical method.

Similarly, increasing the target turbine-inlet temperature can change fuel ratio, turbine work balance and nozzle state. Holding the displayed efficiency inputs constant does not hold the cycle outputs constant. A sensitivity statement should identify the dependent path rather than describing every quantity as independently adjustable.

The rocket scans in Chapter 20 preserve their declared baseline scale. An original target-thrust sizing operation is not repeated at every scan point merely because it was used to create the baseline. Once throat area has been resolved and held fixed, the scan is a fixed-scale study. Its changing thrust is part of the answer.

34.3 Design families versus one engine off design

A design-point sweep usually compares a family of cycles that each satisfy the same type of station equations with different prescribed design inputs. It does not automatically represent one piece of hardware moving through an operating envelope.

An off-design calculation retains reference constraints, such as normal-

ized flow or area relationships, and solves dependent quantities. The legacy turbojet sweep consequently rejects compressor pressure ratio as an independent off-design axis: that quantity is determined by the reference model. It also rejects unsupported combinations rather than silently routing every mode through the same constant-property calculation.

A fair comparison makes this distinction visible:

```
reference inputs + chosen study variable
      |
    selected model closure
      |
recomputed dependent states and outputs
      |
admissibility, constraints, objective
```

Changing the reference point between rows creates another study. It may be useful, but it is no longer the original fixed-reference off-design question.

34.4 Sampling and axis construction

For n uniformly spaced samples between $x_{\min}$ and $x_{\max}$,

$$x_i = x_{\min} + \frac{i}{n-1}(x_{\max} - x_{\min}), \quad i = 0, \dots, n-1.$$

The endpoints are included. A two-axis study forms pairs of samples, so its cost grows as the product of the axis counts. The legacy turbojet sweep accepts two to one thousand points per axis but limits the combined number to one thousand. Those limits govern that executor, not every study mode in the App.

Uniform sampling in x is not uniform sampling in $\log x$. If a parameter spans several orders of magnitude, those distributions emphasize different regions. A plot must identify the actual sample locations; a logarithmic display axis does not retroactively convert a linearly sampled calculation into a logarithmic sample set.

Some existing scan implementations allow descending or repeated end-point values; others impose increasing bounds. The relevant mode's input rules control admissibility. Reversing an axis changes row ordering, not the governing physics. Repeated values do not add independent information.

34.5 Successful, failed and unattempted rows

A failed row is information about a specified problem, not a zero performance value. Substituting zero thrust for a failure can create false minima, slopes and averages. Connecting a curve across an invalid interval can imply continuous operability where none has been established.

Cancellation creates a different category: planned rows that were never attempted. A study with one hundred planned points, thirty successful points, five failures and sixty-five unattempted points has not examined its full requested range. A maximum among the thirty successes is only a maximum within that evaluated subset.

The legacy turbojet executor retains each attempted row with either a result or an error, and separately records planned count and cancellation. Other mode-specific study outputs have their own semantics. Chapter 20 records an additional caveat: successful per-point warnings are not propagated by the rocket scan loops. The top-level baseline warning list therefore cannot certify every row against a condition such as overexpansion.

The appropriate interpretation keeps four questions separate: did the row calculate, did it satisfy the model's admissibility conditions, did it satisfy the user's additional constraints, and is it suitable for the intended comparison? Numerical success answers only part of this sequence.

34.6 A sampled maximum is not a continuous optimum

Consider the original illustrative objective

$$J(x) = 100 - (x - 2.3)^2, \quad 0 \leq x \leq 4.$$

At samples $x=0,1,2,3,4$, the values are 94.71, 98.31, 99.91, 99.51 and 97.11. The best sampled point is $x=2$. The actual stationary maximum is $x=2.3$, where $J=100$. Calling $x=2$ the optimum would confuse the grid with the underlying function.

Now impose a constraint $x \leq 1.8$. The continuous constrained optimum moves to $x=1.8$, with $J=99.75$. Among the original admissible samples, the best is $x=1$, with $J=98.31$. Refining the unconstrained peak would not answer this constrained problem. The first useful refinement is near the admissible boundary.

For a smooth function with an interior maximum, Taylor expansion gives an estimate of grid loss. If the nearest sample lies a distance d from the true maximum,

$$J(x^*) - J(x^* + d) \approx -\frac{1}{2}J''(x^*)d^2.$$

This explains why finer sampling can help near a well-behaved peak. It is not an error bound when curvature is unknown, a branch changes, or failed rows interrupt the neighborhood.

A sampled rocket mixture-ratio maximum should therefore be reported with its sample spacing, evaluated range and success count. If every sample fails, there is no best result. The absence of a best-value field is preferable to a fabricated optimum.

34.7 Changing the objective changes the answer

Maximum thrust, maximum specific impulse and minimum fuel consumption are different objectives. A change that raises thrust can increase fuel use faster than it raises force. Chapter 30 gives an explicit example in which a larger ramjet allocation improves specific thrust but worsens TSFC.

Consider two hypothetical candidates: A produces 10 kN while consuming 0.30 kg/s of fuel; B produces 12 kN while consuming 0.40 kg/s. B has greater thrust, while A has lower TSFC: 0.000030 versus approximately 0.00003333 kg/(N s). If a mission requires at least 11 kN, A is infeasible despite its better consumption ratio. If the thrust requirement is 9 kN, selecting B solely because it is larger does not establish a benefit.

Ratios also conceal scale. A small engine and a large engine can have identical specific thrust while delivering very different total forces. Comparisons must preserve the denominator: core air, total captured air, primary stream, fuel alone or all injected mass. Chapters 3, 13, 29, 31 and 32 explain the mode-specific conventions.

A set of nondominated alternatives can be more informative than one scalar winner. Such a comparison is an interpretation of results, not an additional optimizer implemented by every App study mode. The constraints and priority among objectives remain the user's engineering decision.

34.8 Sensitivity and meaningful changes

For an output y depending on input x , the derivative dy/dx describes a local change. A dimensionless sensitivity is

$$S_x^y = \frac{x}{y} \frac{dy}{dx},$$

when x and y are nonzero. It approximates the percentage output change caused by a one-percent input change near that state. It becomes unsuitable when y is close to zero, where an absolute sensitivity is clearer.

A central difference,

$$\frac{dy}{dx} \approx \frac{y(x+h) - y(x-h)}{2h},$$

requires both neighboring problems to lie on the same admissible branch. Shrinking h indefinitely is not always helpful: subtraction can amplify

numerical noise, and the printed precision may hide the change. Crossing a sonic boundary or map limit does not produce an ordinary smooth derivative.

Sensitivity is also conditional on the fixed quantities. For an ideal fixed-throat rocket at fixed exhaust state, changing ambient pressure gives $dF/dp_a = -A_e$. If exit area is resized at each point, that derivative no longer describes the study. The short phrase “all else equal” should be replaced by the actual held variables whenever ambiguity matters.

34.9 Preserve the comparison, not only the plot

A reusable study record contains the original model and inputs, reference state, units, axes and sample values, completion information, failed-row explanations and selected objective. The result table should remain associated with that definition. A figure alone rarely contains enough information to reconstruct the calculation.

The legacy turbojet executor copies the baseline input before applying each point’s axes and retains the request with its rows. This prevents an earlier sample from becoming the next sample’s unintended baseline. It is an important distinction from a manual workflow that repeatedly edits whatever state the previous solve left on screen.

Interpretation should retain the same discipline. Report “highest specific impulse among the successful samples at fixed throat area and this chamber closure,” rather than “best engine.” Report a partial range when execution was cancelled. Report an excluded region when the model rejected it, without claiming that all physical engines are impossible there.

The strongest comparison is the one whose conclusion remains true after its assumptions are stated. Additional rows are useful when they resolve a peak, boundary or ambiguity that matters to the intended decision. They add little when they merely reproduce a trend already implied by the equations. A purposeful study connects every calculated point to a question that the current model can answer.

Appendix A. Symbols, units and reference conventions

Symbols are defined again where an equation first uses them. This index collects recurring meanings; it does not override a chapter's local definition. In particular, distinguish a mass-specific quantity from a molar quantity before combining thermochemical expressions.

A.1 Flow, state and geometry

Symbol	Meaning	SI unit or convention
A	Flow cross-sectional area	m ²
A _t	Geometric throat area	m ²
A*	Critical reference area for the stated flow	m ² ; not automatically the physical throat in every branch
A _b	Burning surface area	m ²
D	Diameter or stated hydraulic/equivalent diameter	m
L	Length	m
L*	Chamber characteristic length V _c /A _t	m
V _c	Chamber volume	m ³

Symbol	Meaning	SI unit or convention
V, u	Axial flow speed when used in flow equations	m/s; u can instead mean specific internal energy in thermochemistry
x	Axial distance; also a locally defined generic variable	m for distance
t, τ	Time or integrated residence time	s
p	Static absolute pressure	Pa
p_t	Total pressure at a specified composition convention	Pa
T	Static absolute temperature	K
T_t	Total temperature under the stated thermodynamic closure	K
ρ	Density	kg/m ³
M	Mach number V/a	Dimensionless
a	Sound speed	m/s; frozen or equilibrium convention must be stated
γ	c_p/c_v	Dimensionless; may vary with state
ϵ	Nozzle exit-to-throat area ratio when so defined	Dimensionless

A.2 Energy and thermochemistry

Symbol	Meaning	SI unit or convention
h	Mass-specific static enthalpy	J/kg
h_t	Total specific enthalpy $h+V^2/2$	J/kg

Symbol	Meaning	SI unit or convention
u	Specific internal energy in thermodynamic equations	J/kg
s	Specific entropy in thermodynamic equations	J/(kg K)
c _p , c _v	Mass-specific heat capacities	J/(kg K)
R	Mass-specific gas constant	J/(kg K)
R _u	Universal gas constant in a molar convention	J/(kmol K) when amounts are kmol
W _i	Species molecular weight	kg/kmol
X _i	Species mole fraction	Dimensionless; sums to one
Y _i	Species mass fraction	Dimensionless; sums to one
N _i	Species molar flow in spatial reacting flow	kmol/s
C _i	Species molar concentration	kmol/m ³
omega _i	Net species molar production per volume	kmol/(m ³ s)
H, h _{PR}	Fuel heating value in simplified cycle balances	J/kg of fuel
Q'	Distributed external heat input per axial length	W/m
q _m	Distributed injected mass flow per axial length	kg/(s m)
E, script E	Total energy flux in flux-state reconstruction	W when defined as mass flow times total enthalpy
G	Total specific enthalpy in an F/G state formulation	J/kg; distinguish from Gibbs functions elsewhere

Molar heat capacities and enthalpies carry units per kmol rather than per kilogram. Multiplication or division by W_i performs the conversion. A numerical universal gas constant expressed per mol must not be combined unchanged with kmol-based concentrations and enthalpies.

Formation-inclusive enthalpy carries a chemical reference. It is not generally interchangeable with c_{pT} measured from an arbitrary zero. Adding a heating value on top of formation-inclusive reaction energy can count the same chemical contribution twice. Chapters 5–8 and 29 explain the relevant conventions.

A.3 Force, power and normalization

Symbol	Meaning	SI unit or convention
$\dot{m}$	Mass flow	kg/s
$\dot{m}_a$	Captured-air flow, or the locally defined air stream	kg/s
$\dot{m}_f$	Fuel flow	kg/s
F_g	Gross exhaust force under the stated pressure convention	N
F_n	Net thrust after the stated incoming momentum debit	N
$F, \text{script } F$	Thrust, or $pA + \dot{m}V$ when explicitly a momentum function	N; equal units do not imply equal meanings
$s, F/\dot{m}_{\text{ref}}$	Specific thrust	N/(kg/s), equivalent to m/s
I_{sp}	Specific impulse with a stated mass denominator	s
g_0	Standard gravity used for impulse normalization	9.80665 m/s ²
TSFC	Fuel flow divided by net thrust	kg/(N s) before display conversion

Symbol	Meaning	SI unit or convention
c^*	Characteristic exhaust velocity $p_{cA_t}/\dot{m}$	m/s
C_F	Thrust coefficient $F/(p_{cA_t})$	Dimensionless
f	Fuel-to-air mass ratio	Dimensionless; air basis must be stated
beta, alpha	Locally defined bypass or entrainment mass ratio	Dimensionless; can instead denote angles in shock/geometry chapters
lambda	Prescribed ramjet air fraction in explicit TBCC allocation	Dimensionless, between zero and one
pi	A stated total-pressure ratio	Dimensionless; numerator and denominator are local
eta	A stated efficiency or correction factor	Dimensionless; factors are not interchangeable
f_D	Darcy friction factor	Dimensionless; not a Fanning factor

A.4 Subscripts and symbols that need particular care

The subscript 0 identifies a reference freestream in most cycle equations. The subscript e or N commonly identifies an exit, while t identifies a total state. In A_t , however, t denotes a throat. Read the complete symbol and its context rather than assigning one universal meaning to a letter.

A superscript star often identifies a sonic reference in a calorically perfect flow. For equilibrium expansion, the critical mass-flux condition must be defined along the specified thermochemical path; a reported frozen Mach number need not equal one there. Chapter 17 explains that distinction.

The symbols beta and theta are angles in oblique-shock geometry, usually expressed in degrees at an interface but used consistently in mathematical functions. A beta used as an entrainment ratio is dimensionless without an angular interpretation. Likewise, a temperature ratio tau and a residence

time tau are distinguished by definition and units.

Pressure ratios differ among controls. An ambient-to-exit static-pressure ratio is the inverse of an exit-to-ambient ratio. An inlet total-pressure recovery is downstream total pressure divided by its upstream reference. A fan or compressor pressure ratio is greater than one for compression; a turbine exit-to-inlet pressure ratio is less than one for expansion. The word “ratio” alone does not specify the direction.

A.5 Useful unit identities

One newton is one kg m/s^2 . One watt is one joule per second. Thus mass flow times specific enthalpy is power, and mass flow times velocity is force. Pressure times area is also force, which allows pressure and momentum terms to be added in a control-volume balance.

Specific thrust and speed have the same dimensions, but specific thrust is not necessarily exhaust velocity. Pressure thrust, inlet momentum and different mass-flow denominators can all contribute. Similarly, an impulse measured in seconds is a force divided by a weight-flow rate, not a physical burn duration.

To convert TSFC from $\text{kg}/(\text{N s})$ to $\text{kg}/(\text{N h})$, multiply by 3,600. To express the latter per kilonewton-hour, multiply by a further 1,000. Keep the full unit visible whenever comparing values from different tables.

Temperature differences in kelvin and degrees Celsius have equal numerical size, but absolute temperatures do not. Pressure-ratio and gas-state equations require absolute pressure and absolute temperature. Unit conversion belongs before applying those equations, not after an invalid ratio has been formed.

Appendix B. Performance normalization

A performance ratio is defined by its numerator, denominator and control boundary. This appendix collects the conventions developed in the chapters so that outputs from different models can be compared without silently changing their basis.

B.1 Recover dimensional quantities first

For a reported specific thrust s referred to mass flow $\dot{m}_{dot_ref}$,

$$F = s\dot{m}_{ref}.$$

To change the reference flow from one stream to another, recover F and divide by the new denominator. Do not rescale a fuel ratio in the same way unless its own denominator has been identified.

For example, a turbofan with bypass ratio $\beta=4$ has total air flow five times its core air flow. A force of 25 kN with core flow 10 kg/s corresponds to 2,500 N/(kg/s) per core flow but 500 N/(kg/s) per total air flow. These are descriptions of one force. If fuel flow is 0.5 kg/s, fuel-to-core-air ratio is 0.05, while fuel-to-total-air ratio is 0.01. TSFC is $0.5/25,000=0.000020$ kg/(N s), regardless of which air basis is used consistently in the intermediate calculation.

B.2 Reading common outputs

Output or family	Numerator	Denominator or reference	Interpretation
Turbojet net specific thrust	Branch net force	Entering air flow	Includes the branch's incoming momentum debit
Turbofan overall specific thrust	Combined net force	Total core plus bypass air	Do not confuse with separately reported core-based quantities
Turbofan thrust per core flow	Combined net force	Core air flow	Equals total-air specific thrust multiplied by $1+\beta$
Explicit-split TBCC specific thrust	Sum of independent branch net forces	Total assigned air flow	Branch net forces already include their own inlet debits
Full flowpath captured-air specific thrust	Completed-engine net force	Captured-air flow	Partial paths do not establish the intended exit performance
Rocket specific impulse	Rocket force	Propellant mass flow times g_0	Mass denominator includes the propellant flow represented by the mode
Calibrated-engine injected-mass impulse	Net force	All injected mass flow times g_0	Air or oxidizer injection is not relabeled as fuel
Ejector augmentation	Combined net force	Primary-alone ambient-expanded force	Dimensionless; not a fuel efficiency

Output or family	Numerator	Denominator or reference	Interpretation
ATR augmentation	Combined net force	Primary-alone ambient-expanded force	Includes primary, air and simplified additional-fuel accounting
ATR air-based specific thrust	Combined net force	Captured secondary air flow	Different from its primary-based augmentation
TSFC	Fuel mass flow	Net force	The fuel definition follows the selected cycle

The table is a reading aid, not a statement that every listed quantity exists in every result. A component, thermodynamic property calculation, shock or geometry construction may have no meaningful engine thrust at all.

B.3 Gross, net and pressure force

At an exhaust plane, the common gross expression is

$$F_g = \dot{m}_e V_e + (p_e - p_a) A_e.$$

A full airbreathing control volume subtracts the momentum of its captured air. A rocket does not acquire the same ambient-air debit merely because it operates in the atmosphere. A standalone duct momentum function $pA + \dot{m}V$ uses absolute pressure and describes an internal flux. It is not automatically either gross or net engine thrust.

Pressure thrust must be treated once. When a branch solver has already included it, a combined calculation should not add it again. When the nozzle is explicitly fully expanded, its exit pressure term vanishes under that model. This does not establish that a fixed physical nozzle would remain fully expanded at another ambient condition.

B.4 Fuel and injected mass

Fuel-to-air ratios conventionally divide fuel by a specified air flow. A mixed-flow afterburner may introduce a second fuel ratio with a different intermediate basis. The correct total fuel flow is obtained by following the mass bookkeeping in its chapter, not by adding differently normalized ratios.

In finite-rate mixed-flow engines, total injected mass can include several species and streams. The metric `specific_impulse_injected` uses all of it. To construct a fuel-only metric externally, one would need a defensible identification of fuel mass; its denominator is not supplied merely by seeing that injection exists.

The ideal ATR heating convention is reduced, as explained in Chapter32. Its additional fuel ratio does not account for the propellant expenditure needed to create the supplied primary stream. Comparing that ratio with a complete propulsion system's onboard propellant consumption would mix boundaries.

B.5 Useful identities and their conditions

If thrust and mass flow use one consistent propellant basis,

$$I_{sp} = \frac{F}{\dot{m}_p g_0}, \quad c_{eff} = \frac{F}{\dot{m}_p} = g_0 I_{sp}.$$

For a rocket model defining $c^* = p_c A_t / \dot{m}_p$ and $C_F = F / (p_c A_t)$,

$$I_{sp} = \frac{C_F c^*}{g_0}.$$

Correction factors can change the definitions of the effective mass flow or force used in these identities. Chapter16 explains the current performance-factor convention. One should not reconstruct corrected force from an ideal exit velocity and corrected mass without checking how those factors were applied.

TSFC and fuel-only impulse are reciprocally related only if they refer to the same positive force and fuel flow. A rocket propellant impulse, a fuel-only airbreathing impulse and an all-injected-mass impulse do not become interchangeable because each is measured in seconds.

B.6 Comparing two reports

Before comparing magnitudes, recover the dimensional force and all relevant mass flows. Identify whether ambient pressure force and incoming momentum are already included. Then choose a common denominator and retain the model's physical constraints.

A higher ratio can arise from a smaller denominator rather than a larger useful output. In the ATR limit of vanishing air flow, air-based specific thrust can become large even while the usefulness of that normalization deteriorates. A large primary-normalized augmentation may include extra downstream heating. The denominator is part of the engineering conclusion.

Appendix C. Domain and closure guide

This appendix summarizes the boundaries explained in the main text. It distinguishes an admissible mathematical state, a completed numerical calculation and a physically appropriate application. It is not an additional certification or a universal validity range for real engines.

C.1 State and property choices

Model choice	What is held or solved	Boundary that matters	Chapters
Constant-property gas	Fixed heat capacity and gamma	Thermal range and chemistry are approximations	2,9,16,24
Frozen mixture	Composition fixed while properties vary with temperature	“Frozen” does not require constant gamma	5,17,26
Equilibrium at T,p	Composition satisfying chemical equilibrium at prescribed temperature and pressure	No chemical residence time predicted	6

Model choice	What is held or solved	Boundary that matters	Chapters
Equilibrium with enthalpy closure	Temperature and composition adjusted to meet energy at pressure	Formation enthalpy references must agree	6,17
Specified-area freezing	Equilibrium path changes to frozen composition at the supplied area	Freeze location is prescribed, not kinetically predicted	17
Finite-rate reactor	Species evolve under imposed reactor constraints	The included rigid reactor is externally isothermal	7–8
Finite-rate spatial flow	Species and fluxes evolve along a supplied geometry	Uniform cross-sectional state; no diffusion or shock train	29

A property polynomial is valid only over its supported temperature interval. A mixture inversion must remain within the common usable range of the required species. Failure outside that range is not an invitation to extrapolate without another property model.

C.2 Cycle and geometry closures

Calculation	Closure	What must not be inferred	Chapters
Turbojet/turbofan design point	Specified cycle inputs and component balances	Fixed hardware operability over an unrestricted envelope	9–15

Calculation	Closure	What must not be inferred	Chapters
Reference-based off design	Selected reference, area/flow or normalized constraints	Complete map matching for all components	12,15
Rocket explicit-gas performance	Supplied gas properties, pressure and scale	Feed-system or chamber chemistry prediction	16
Chemical rocket expansion	Declared chamber and expansion chemistry	Liquid vaporization, condensed phases or injector dynamics	17
Chamber and convergent sizing	Supplied characteristic length and geometric assumptions	Cooling, structure or combustion stability	18
Conical/bell contour	Reference geometry with declared endpoints and slopes	Optimized characteristics, wall pressure or separation loads	19
Solid equilibrium pressure	Supplied burn law plus choked discharge	Qualified propellant behavior or ignition	21
Solid transient	Established burning and supported grain regression	Complete ignition and tail-off histories	22

A geometry result can be dimensionally consistent and still be unsuitable for a physical design. The current contour and chamber modules intentionally supply reference shapes and volumes, not a coupled thermal, structural or reacting-flow design.

C.3 Flow branches and termination

Normal shocks require an upstream supersonic state. Oblique shocks require an attached solution on the chosen weak or strong branch. A weak branch is not a guarantee that the downstream flow remains supersonic. Beyond the attached turning limit, the current attached-shock equations do not become a detached-shock solver.

Fanno and Rayleigh relations approach sonic limits under their respective assumptions. A general source-term duct combines area, friction, heat and injection but still retains a chosen branch. An integration that stops near its sonic boundary does not calculate a transition through it. The final row may be a last valid state rather than a precisely located event.

Frozen flux reconstruction can admit two thermodynamic states for the same supplied flux quantities. The branch selection is an input to the problem. Calibrated inlets have their own high/low root conventions; a high root does not universally mean supersonic. A full calibrated scramjet additionally requires a supersonic inlet exit and rejects an incompatible selection.

Nozzle closure differs by mode. A specified exit pressure determines an expansion state and can imply an exit area. A supplied geometry determines the area and generally leaves an exit pressure that differs from ambient. The full scramjet geometry is not automatically adjusted for ambient back pressure. These modes should not be compared as though they solve the same external condition.

C.4 Combined-cycle boundaries

The explicit TBCC mode adds two independent design-point branch results at a prescribed allocation. A zero-flow branch is skipped. Active branches must solve; the allocation is not adjusted to repair one of them. Common freestream inputs do not establish pressure matching in a shared inlet.

The ideal ejector requires supersonic primary entry, subsonic secondary entry, equal inlet static pressure and a sonic mixed exit. Its final expansion

is fully expanded and supersonic. Alternative secondary-choked or exit-unchoked operation is outside that branch.

The ideal ATR starts from a supplied primary total state. It requires positive fan work and sufficient primary expansion to supply it. Its nozzle can be subsonic or supersonic, but exactly zero freestream Mach is excluded by the current formulation. Its heating estimate is explicitly simplified. None of these combined-cycle modes derives a gas generator, pump system, cooling network, installation drag or automatic transition control.

C.5 Interpreting warning and completion states

A warning can identify an approximation or a risky interpretation without invalidating the algebra. A rejection means the current solve did not return an admissible result. A partial history contains accepted states before termination; it is not a completed engine. A completed result satisfies the selected solver's completion criteria but retains all physical assumptions.

Study outputs require additional care. Failed samples are not zeros, cancelled samples are not evaluated, and successful samples may still require the user's constraints. Some rocket scan outputs do not propagate each successful row's warnings. A baseline warning list therefore does not describe every sampled state.

Residual magnitudes must be read with units and scales. A small conservation residual establishes consistency with the imposed balance, not the accuracy of empirical inputs. More iterations cannot supply missing physics. The useful response to a failure is to identify which assumption, branch or numerical condition failed before changing the study.

C.6 Energy and geometry interpretation checks

Chapters 8, 10 and 12 develop three useful interpretation checks: use internal energy for the rigid isothermal reactor heat load, use the augmented total temperature throughout an afterburning nozzle, and retain the pressure-ratio orientation in fixed-throat exit-area reconstruction. The current App

follows these closures. Numerical convergence alone does not establish that a selected physical constraint describes a real engine.

This reference guide should be used together with the relevant chapter. It is deliberately explicit about where the current implementation ends, so that a broad family title does not imply a broader solved problem.

Appendix D. Directory of the 49 calculation modes

This directory connects each current mode to its primary theoretical treatment. Chapters 1–4 establish shared conventions; Chapters 33–34 explain numerical results and comparisons. A listed mode can contain explicit options or branches: the directory counts modes, not every selectable sub-case. The directory describes theoretical coverage; it does not certify the software or the physical suitability of a model.

Mode identifier	Calculation	Main chapter(s)
turbojet.variable_cp_design	Turbojet · Variable-specific-heat gaseous-fuel design point	10
turbojet.design_point	Turbojet · Design point	9
turbojet.design_point_ab	Turbojet · Afterburning design point	10
turbojet.component	Turbojet · Component calculation	11
turbojet.off_design	Turbojet · Off-design point	12
turbojet.dual_spool_afterburning_off_design	Turbojet · Dual-spool afterburning off-design point	12

Mode identifier	Calculation	Main chapter(s)
turbofan.variable_cp_design	Turbofan · Variable-specific-heat single-spool design point (separate-flow or optional mixed-flow)	13–14
turbofan.component	Turbofan · Turbofan component calculation	15
turbofan.separate_design	Turbofan · Single-spool separate-flow design point	13
turbofan.separate_off_design	Turbofan · Dual-spool separate-flow off-design point	15
turbofan.mixed_constant_cp	Turbofan · Constant-specific-heat dual-spool mixed-flow	14
liquid_rocket.equilibrium_thrust_chamber	Liquid Rocket · Thermochemical chamber and equilibrium nozzle	17
liquid_rocket.partial_frozen_thrust_chamber	Liquid Rocket · Thermochemical nozzle · Specified freeze area	17
liquid_rocket.thermochemical_frozen_nozzle	Liquid Rocket · Thermochemical chamber and frozen nozzle	17
liquid_rocket.finite_contraction	Liquid Rocket · Finite contraction · Static chamber plane	18
liquid_rocket.convergent_chamber_geometry	Liquid Rocket · Cylindrical chamber and convergent-section sizing	18
liquid_rocket.thrust_chamber	Liquid Rocket · Thrust-chamber performance	16

Mode identifier	Calculation	Main chapter(s)
liquid_rocket.nozzle_performance	Liquid Rocket · Standalone nozzle performance	16
liquid_rocket.chamber_sizing	Liquid Rocket · Equivalent chamber sizing	18
liquid_rocket.nozzle_contour_design	Liquid Rocket · Nozzle contour · Conical or bell	19
liquid_rocket.altitude_characteristic	Liquid Rocket · Altitude performance	20
liquid_rocket.area_ratio_sweep	Liquid Rocket · Area-ratio sweep	20
liquid_rocket.throttle_characteristic	Liquid Rocket · Chamber-pressure sweep	20
liquid_rocket.mixture_ratio_sweep	Liquid Rocket · Gaseous-propellant mixture-ratio sweep	20
ramjet.full_engine	Ramjet · Full flowpath · Subsonic combustion	25
ramjet.design_point	Ramjet · Full-engine design point	25
ramjet.normal_shock	Ramjet · Normal shock	23
ramjet.rayleigh	Ramjet · Rayleigh flow · Constant-area heating	24
ramjet.oblique_shock	Ramjet · Oblique shock · Explicit weak/strong branch	23
ramjet.fanno	Ramjet · Fanno flow · Constant-area friction	24
ramjet.one_d_flow	Ramjet · Quasi-one-dimensional flow segment	24
solid_rocket.equilibrium	Solid Rocket · Equilibrium Chamber Pressure	21
solid_rocket.transient	Solid Rocket · Grain Regression Transient	22

Mode identifier	Calculation	Main chapter(s)
tbcc.explicit_split	TBCC · Explicit Air Split	30
rbcc.ideal_ejector	RBCC · Ideal Ejector-Ramjet Submodel	31
atr.ideal_air_turboramjet	ATR · Ideal Air-Turboramjet Submodel	32
scramjet.calibrated_mixing_engine	Scramjet · Calibrated-Inlet Full Engine · Nonreacting Mixed Flow	29
scramjet.calibrated_reacting_engine	Scramjet · Calibrated-Inlet Full Engine · Finite-Rate Reacting Flow	29
scramjet.finite_rate_flow	Scramjet · Finite-Rate · Quasi-One-Dimensional Reacting Flow	29
scramjet.variable_cp_flow	Scramjet · Seven-State Variable-Specific-Heat · Frozen Flow Segment	26
scramjet.calibrated_variable_inlet	Scramjet · Reference Calibration · Variable-Specific-Heat Inlet	27
scramjet.finite_rate_reactor	Scramjet · Finite-Rate · Constant-Temperature, Constant-Volume Reactor	7–8
scramjet.gas_equilibrium_tp	Scramjet · Specified Temperature and Pressure · Gas-Phase Equilibrium	6
scramjet.gas_equilibrium_hp	Scramjet · Constant-Pressure Energy Closure · Gas-Phase Equilibrium	6

Mode identifier	Calculation	Main chapter(s)
scramjet.variable_cp_frozen_nozzle	Scramjet · Variable-Specific-Heat · Frozen Nozzle	5, 17, 26
scramjet.frozen_gas_properties	Scramjet · Frozen Composition · Variable-Specific-Heat Properties	5
scramjet.full_engine	Scramjet · Full Engine · Constant-Specific-Heat Flowpath	28
scramjet.combustor_only	Scramjet · Combustor · Specified Heat Source	28
scramjet.nozzle_only	Scramjet · Nozzle · Frozen Constant-Specific-Heat Flow	28

Property and chemistry modes support several engine families even when grouped under one family in the interface. Likewise, a component or flow-segment calculation is not a complete engine. Read the chapter's control boundary and closure before transferring its result into another mode.

For a combined task, follow dependencies: properties before variable-property reconstruction, reconstruction before spatial flow, and completed flow before full-engine force. The separate user guide describes interface navigation; this directory identifies the explanatory content.